

# LIFEBOAT TRAINING STANDARDS



# About this handbook

The purpose of the Lifeboat Training Standards is to be the definitive reference document for the Operational Competence Framework (OCF). It specifies the structure and components of the OCF as well as the individual training plans for all roles associated with the crewing, operation and support of lifeboats. It also details the knowledge or skills required to conduct each task.

Often, the training standards refer to Standard Operating Procedures (SOPs), Emergency Operating Procedures (EOPs), RNLI policies or other guidance: it is important to check these references to ensure that the correct standards are being followed. When training for a given role, it is also important to refer to the relevant pass out that a trainee is working towards.

These training standards will be regularly reviewed as part of our continuous improvement. Please refer to Horizon for the latest version.

Please give any feedback or comments to your area lifesaving manager (ALM) or assessor trainer (AT).

## November 2023

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Version: 1.0  
Date produced: November 2023  
RNLI Classification: Protected

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**PLEASE REFER TO HORIZON FOR THE  
LATEST VERSION OF THIS HANDBOOK**

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## Foreword

By Andrew Reed

This edition of the Lifeboat Training Standards is an evolution of a series of standards documents which, over many years, have helped to make the RNLI not only one of the world's most effective search and rescue (SAR) organisations, but one of the safest too. While the standards will be familiar to many of you, the new OCF in which they sit is wholly new. You are encouraged to become familiar with these standards and other supporting documentation to help ensure that the RNLI remains the outstanding service that it is.

**Andrew Reed - OBE FCM**  
Head of Learning and  
Organisational Development



Photo: RNLI



Photo: RNLI / Nathan Williams



## Foreword

By Alex Evans

The fundamentals of seamanship and SAR have not changed, and we will still operate lifeboats and plant and machinery in much the same way as before. Yet, whilst those fundamentals remain the same, the review of training standards has brought about some major changes. Under the new OCF roles have changed, as have training plans. We are introducing a new system to maintain currency and have revised the process for pass outs and revalidations for crew, specialist roles and command roles.

If you are conducting an activity and, especially, if you are training someone else in an activity, it is essential that you review the training standards in advance to ensure you are applying best practice and training to the most up-to-date standards.

Under the new Lifeboat Training Standards, the lifeboat trainer assessor (LTA) role remains central and is evolving. In addition to an increased emphasis on training, LTAs will ensure that crew are safe to go afloat, to go on a service, are ready for their next pass out, and will also have the option to conduct tier 1 crew pass outs. But that does not mean the training needs at station sit solely with an LTA - it is important to recognise that everyone can be a trainer to some degree, especially helms, coxswains, commanders, mechanics, head launchers and those with greater experience.

The system of training and for maintaining currency has been simplified. It is based on sound training principles and has been developed with, and tested by, experienced operators on the coast. If you have any questions, please speak to your ALM and your AT.

These training standards are central to our Safe System of Training (SST) and will be regularly reviewed as part of our continuous improvement. Any major changes will be communicated via Horizon notices but smaller detail changes may not be. Please refer to Horizon for the latest version of these standards.

There are many handbooks which have been in use for a number of years and which make reference to the old CoBT system. This is not a problem. As I mentioned, the fundamentals of seamanship and SAR, as well as aspects of operating and maintaining lifeboats, plant and machinery remain, and you



should continue to use these handbooks until they are replaced.

Alongside this document, we have issued supporting and introductory documents. These include:

- *An Introduction to the Operational Competence Framework (OCF)*
- *Lifeboat Pass Out and Revalidation Handbook*
- *Lifeboat Currency Activity and Training Handbook*

Together they are intended to keep you safe, keep you effective and ensure we can continue to save lives at sea.

**Alex Evans**

Lifesaving Training Manager

## The RNLI Safe System of Training (SST)

The key components of our SST are below:

### We must:

Train as we mean to operate.  
 Reduce risks to as low as reasonably practicable.  
 Balance training risks against the risks of not training.  
 Communicate with colleagues, subject matter experts and other interested parties.  
 Ensure boats, buildings, equipment and plant are maintained and safe.  
 Ensure that trainees are motivated, committed and have everything they need to learn.

### Our trainers, assessors, those in command roles and with specialist responsibilities must:

Be suitably qualified and experienced.  
 Strive for continual professional development.  
 Be aware of, identify and address 'normalisation of deviation'.  
 Have the courage to call out unsafe practice.

### In designing, developing and delivering training we must:

Conduct thorough learning needs analyses.  
 Measure performance.  
 Be consistent.  
 Risk assess all activities.  
 Properly brief, evaluate and debrief activities.  
 Ensure that all activities are properly recorded.  
 Comply with the RNLI's assurance systems.

### The outcome of the SST is a system which must:

Be demonstrably safe.  
 Be achievable in a reasonable time for a new joiner from a non-maritime background.  
 Take account of an individual's existing knowledge and skills.  
 Require no more than an acceptable level of commitment from volunteers.  
 Be simple to record and administer.

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## Section 1 - Introduction to the OCF

# Section 1 - Introduction to the OCF

## Governance

The Operational Competence Framework (OCF) is owned and maintained by Learning and Organisational Development (L&OD) and was developed in cooperation with Lifesaving Operations (LSO), Engineering and Supply (E&S) and other departments.

## Purpose

The purpose of the OCF is to give structure to training, exercising and progression through operational roles. It is important to follow the guidance in this document so that the RNLI can meet its training policy commitments and national and international regulatory requirements. It is also an essential element of the verification and assurance processes in place.

## Review Process

The OCF is a living document, subject to continuous review and improvement in response to:

- RNLI policy changes
- New equipment
- Incidents, accidents and hazard observations
- Revised Standard Operating Procedures (SOP), Emergency Operating Procedures (EOP) and other procedures
- Changes to legislation

This process will be led and coordinated by L&OD. The online version of the OCF is the authoritative version; printed copies may not be up to date.



Photo: RNLI / Nigel Millard



### Recording Competence

The RNLI will record all activity contributing to competence. This includes attendances at training events and courses. The recording system will be the basis of competence reporting and will be subject to periodic audit and verification. Information will be retained in accordance with general data protection regulation (GDPR) and other relevant regulations.

### Accreditation and Accredited Prior Learning

The OCF specifies the RNLI's competence standards for all operational roles and activities. Where an individual can evidence existing competence in a given area, it may not be necessary to train them in that skill set. For example, a qualified and current Yachtmaster® will likely understand IRPCS, buoyage, tides and tidal streams. However, where specific RNLI SOPs exist or equipment or procedures are unique to the RNLI, then everyone must be appropriately trained. The standard to be achieved is that laid down in the appropriate pass out.









## Section 2 - Structure of the OCF

## Section 2 - Structure of the OCF

### Training Plans

A training plan comprises all the activities, knowledge and skills for a specific role, in which a trainee must be competent. Training plans are structured such that trainees have a clearly laid out training pathway so that knowledge, skills and experience are gained in a logical order. The stages of a training pathway, for afloat crew, are as follows (note there are variations between SAR unit types and for shore roles):

**Stage 1:** Safety and Induction

**Stage 2:** Safe to go Afloat

**Stage 3:** Safe to go on a Service

**Stage 4:** Tier 1 Crew

**Stage 5:** Tier 2 Crew

### Currency Programmes

Crew will be considered fully trained on completion of their Tier 2 Pass Out. However, from the point they complete their Tier 1 Pass Out, crew will begin maintaining currency via a Currency Programme.

A Currency Programme is a structured set of activities designed to ensure that individuals regularly practise the tasks they are required to perform on service and that they do so in a team environment.

Each station will plan their training and exercising to meet their needs and to provide sufficient opportunities for trainees and fully trained crew to build and maintain their knowledge and skills. Each crew member will be responsible for ensuring that they complete at least the minimum number of currency activities required to maintain competence in their fully trained role(s).

### Safety, Health and Environment (SHE) Tasks

Almost all practical skills can be embedded in exercises. However, there are some tasks which cannot be routinely practised as part of a practical exercise. These tasks are generally the knowledge based SHE competences and will require periodic requalification. Learning Zone will notify individuals when they are required to refresh these tasks. They may be undertaken online individually or, in some cases, can be conducted collectively at the station.

The process and methodology of training plans and currency programmes are described in Section 3. The content and stages of progression of training plans for each role are illustrated in Section 4.

### Structure of the OCF

The competences in the OCF are organised into:

- Activities
- Task Groups
- Tasks

Activities, task groups and tasks have a three-figure reference system. The first number refers to the activity, the second to the task group and the third to the task. At the top level of the OCF are activities, which are broad areas of related competences. Examples include:

|            |                            |
|------------|----------------------------|
| Activity 3 | Launch and Recovery        |
| Activity 4 | Seamanship                 |
| Activity 5 | Search and Rescue          |
| Activity 6 | Operational Communications |



## Tasks and Task Groups

Each activity comprises several task groups. A task group is a discrete set of knowledge and skill within an activity. Task groups are further broken down into individual tasks. When referring to a particular competence, it is usually the task level which is quoted. Examples of tasks include:

| Activity 4 - Seamanship |               |       |                  |
|-------------------------|---------------|-------|------------------|
| Task Group              |               | Task  |                  |
| 4.1                     | Watchkeeping  | 4.1.1 | Lookout          |
|                         |               | 4.1.2 | Watchkeeping     |
|                         |               | 4.1.3 | IRPCS (Command)  |
| 4.2                     | Rope Handling | 4.2.1 | Rope Handling    |
|                         |               | 4.2.2 | Capstans         |
| 4.3                     | Anchoring     | 4.3.1 | Anchoring        |
|                         |               | 4.3.2 | Manage Anchoring |

Note that there is no requalification interval for most tasks. Exceptions include SHE tasks, emergency and survival procedures and clinical.

## Format and Contents of Activities

Each task is described in Section 6 and includes the standards to be met.

The Lifeboat Training Standards do not, in most cases, specify how an individual must be trained, nor do they contain the specific knowledge required. Hence it is essential for trainees, trainers and assessors to refer to the relevant policies, operational procedures, manuals, handbooks, course material or other reference material. These can all be accessed on either Learning Zone or Horizon. In addition, many of the activities in the *Currency Activity and Training Handbook* contain references to supporting material.

Training criteria require trainees to do one or more of the following with respect to the task under consideration:

- **Identify:** Recognise, show or otherwise indicate something and its location.
- **State:** List objects, criteria or factors.
- **Describe:** Give a simple overview of the object, process or task.
- **Explain:** Give a clear account of a task or process, including causes, context and related factors.
- **Demonstrate:** Practically complete the specified task to the required standard, following the correct procedures and under the specified conditions.

A given task may require one or more of the above. For example, a trainee may be required to both describe and demonstrate or to identify and explain. In a small number of cases, it may be a requirement to complete a particular course. Where this is the case, it is specified in the text.







## Section 3 - Training, Assessment and Verification



## Section 3 - Training, Assessment and Verification

### Introduction

This section provides guidance for trainees, trainers and assessors on the principles of RNLI training and assessment as well as information on the roles, responsibilities and procedures in place.

The purpose of the OCF is to:

- Enable new joiners to the RNLI to achieve competence in their role as efficiently and effectively as possible.
- To allow fully trained crew to maintain their skills.
- To allow crew to broaden their skills and, where appropriate, progress into additional roles with greater responsibility.

Role specific training plans have been designed to guide individuals through their training in a logical order, building up a solid foundation of knowledge and skills before progressing to more challenging tasks. They are also designed to ensure that individuals are safe ashore and afloat, and that crew do not take on operational activity until they are ready.

### Training Plans

#### Safety and Induction (Stage 1)

The first stage in all training plans is a set of SHE tasks, common to all roles. These should be completed first by any new-joiner to ensure that they are aware of the risks and hazards in and around the station and are able to take the appropriate actions. Completion of these tasks should be a priority. Most are self-taught via eLearning and may be completed in the station or by individuals in their own time. Those which are solely briefings, presentations or films, and do not require individual completion of questions, can be delivered collectively to all station personnel. This may be followed by group discussions on the material. On completion of these SHE tasks individuals are deemed to have completed their safety induction and are safe in and around the lifeboat station.

#### Safe to go Afloat (Stage 2)

Those in afloat roles must complete PPE and SAR unit Layout tasks and be assessed by a SAR unit commander or lifeboat trainer assessor (LTA) before conducting any other training afloat. They must also have an in-date health assessment. Individuals may go afloat for acquaint purposes (not in Republic of Ireland), prior to completion of these tasks, in

accordance with current policy. Trainees at this stage should not be involved in operational tasking.

#### Safe to go on a Service (Stage 3)

Once safe to go afloat for training, the next set of tasks is intended to ensure trainees meet the minimum competence requirement to undertake operational activity. These tasks ensure that trainees are safe on board, competent to deal with emergencies and their own survival, can participate in launch and recovery and do not require close supervision by other crew members. Trainees who have completed this stage can be considered for operational tasking if the launch authority and SAR unit commander think the circumstances are within their capability. Individuals at this stage should not be considered for challenging services or those likely to require a greater level of competence than they possess.

#### Diversion from Training to Service

If a SAR unit conducting training with trainees who have not reached this stage is diverted to an operational task, it is the decision of the SAR unit commander whether to continue with the trainees on board, return for a crew change or advise that another SAR unit responds. This decision will depend on the nature of the tasking, the conditions, the availability of other SAR units and the competence of the crew, both trainees and fully trained crew.

#### Tier 1 and Tier 2 Crew (Stages 4 and 5)

The purpose of introducing a two-tier crew structure is to provide a clear sequence to training and development and to break down training into more manageable stages. Tier 1 includes the basic skills and knowledge that all crew require on a frequent basis, and which can be achieved relatively quickly, irrespective of previous maritime experience. Completion of tier 1 is achieved by completion of a tier 1 crew pass out.

Tier 2 is slightly more advanced and includes helming the SAR unit and an introduction to navigation. Tier 2 finalises crew training and is also achieved by completion of a tier 2 crew pass out.



### Training Methods and Media

Training may be delivered by various media. These include:

- **Self-directed learning.** This includes working through your training plan on Learning Zone, reading handbooks, manuals, EOPs, SOPs and LOPs and other RNLI material.
- **Online training courses.** These may be completed individually or, in some cases, as part of a planned group activity at the station. Group activity will still need to be recorded in Learning Zone.
- **One-to-one instruction at station.** Delivered by a visiting member of the training team, the local LTA, your SAR unit commander or by suitably experienced crew members.
- **Group instruction at station or a regional hub location.** Delivered by a visiting member of the training team, the local LTA, your SAR unit commander or by suitably experienced crew members.
- **Attendance at residential courses.** Courses include, amongst others, seamanship, boat handling and command. All afloat crew must attend the Crew Emergency Procedures (CEP) course.
- **Shadowing and observation.** Trainees may be paired-up with experienced crew members to understand the practical application of skills, procedures and safe ways of working.

### What does “Trained in” mean?

When a station records a trainee as having been “Trained in” a given task it means that the station is satisfied that the individual has the necessary knowledge or skills to conduct that task to the specified RNLI standard. In addition, the station is confident that, if asked to conduct that task during a pass out, or during an exercise or operation, the individual is capable of doing so.

### Pass Outs

A pass out is an assessment comprising all the operational skills and knowledge a trainee has learned to date conducted by an authorised assessor. The purpose of a pass out is to demonstrate that the individual can put together separate elements of skill and knowledge and apply them in an operational scenario. An individual is ready to conduct a crew pass out once they have completed all the tasks at the relevant tier and the station believes they are ready in all respects. If in doubt, the station may

consult with the AT and / or ALM who may be able to advise on an individual's preparedness and likelihood of passing.

Guidance and the specific content of all pass outs is published separately in the *Lifeboat Pass Out and Revalidation Handbook*, and is available to stations and trainees to aid their preparation. This guidance also specifies revalidation criteria for those roles where periodic revalidation is required.

### Timescales

Timescales in which trainees should complete the various stages of the OCF have not been set. However, LOMs and ALMs must monitor trainees and ensure that they are attending, are committed and are making reasonable progress with their training plan. Crew should complete their overall training plan within two years in accordance with policy.

### Fitness Standards

Most physical tasks in the OCF must be actively demonstrated. The guiding principle is that:

**An individual is fit enough for their role if they are fit enough to correctly carry out all the tasks in their role without undue difficulty.**

It is the responsibility of individuals to inform their LOM if they become unwell or unfit such that they cannot carry out their role. It is also the responsibility of fellow crew members to raise concerns they may have over the ability of their colleagues to carry out their role. This should be done diplomatically but also with complete honesty, as it concerns the safety of all on board. All SAR unit crew are required to have an in-date health assessment.

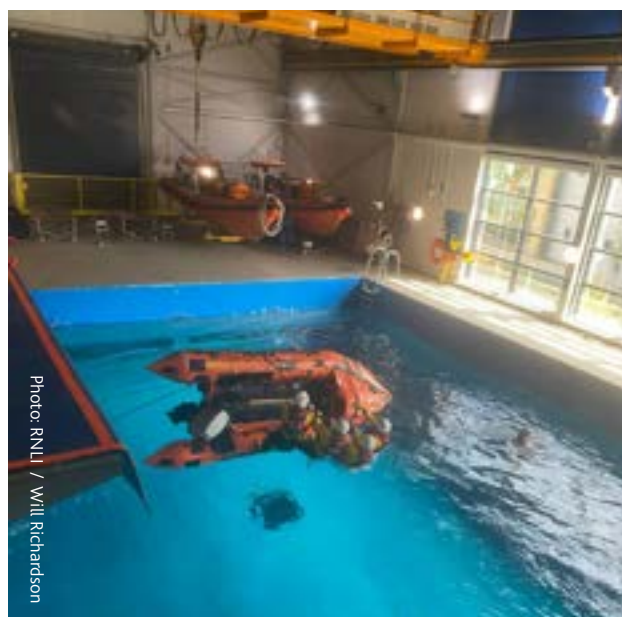


Photo: RNLI / Will Richardson

## Section 3 - Training, Assessment and Verification

The most physically demanding task for lifeboat crew is likely to be the CEP course. Completion of the CEP is mandatory before trainees can be selected for a shout and must be completed at intervals of no more than five years for all crew. The CEP requires a basic level of fitness, as well as a degree of water confidence to complete the capsize and sea survival drills.

### Currency Programmes

#### Purpose

In contrast to previous competence-based systems in the RNLI, under the OCF most individual tasks (formerly units and sub-units) do not have expiry dates. Instead, competence is maintained and assured by ensuring crew regularly carry out all the operational tasks appropriate to their role. This is referred to as maintaining competence by doing.

Once an individual has passed their tier 1 pass out they will begin a programme of activities to maintain currency in all the key skills.

| Currency Activities (Afloat) |                                       |                                       |
|------------------------------|---------------------------------------|---------------------------------------|
| 1                            | MOB, Person Recovery and Pyrotechnics |                                       |
| 2                            | Towing                                |                                       |
| 3                            | Mooring and Berthing                  |                                       |
| 4                            | Helming and SAR Unit Handling         |                                       |
| 5a                           | Anchoring and Veering (ILB)           |                                       |
| 5b                           | Anchoring (ALB and IRH)               |                                       |
| 6                            | Emergency Procedures                  |                                       |
| 7                            | Search and Rescue                     |                                       |
| 8                            | Navigation                            |                                       |
| 9                            | Daughter Boat and Breeches Buoy (ALB) |                                       |
| 10                           | Helicopter Operations (Optional)      |                                       |
| 11                           | ALB Machinery Breakdown Drills (MBD)  | Electrical and Electronic Systems     |
| 12                           |                                       | Hydraulic Systems                     |
| 13                           |                                       | Main Machinery and Propulsion Systems |
| 14                           |                                       | Main Engine Failure                   |
| 15                           |                                       | Control Failure                       |

#### Currency Activities (Shore)

|   |                                 |
|---|---------------------------------|
| 1 | PPE, Equipment and Pyrotechnics |
| 2 | Launch SAR Unit - High Water    |
| 3 | Launch SAR Unit – Low Water     |
| 4 | Recover SAR Unit – High Water   |
| 5 | Recover SAR Unit – Low Water    |
| 6 | Emergency Procedures            |

Detailed instructions and guidance are published separately in the *Lifeboat Currency Activity and Training Handbook*. The activities outline what crew should achieve, however, stations are strongly encouraged to customise and build on the activities by adding scenarios and content appropriate to their location and circumstances.

Given their unique capabilities and operational parameters, this is particularly the case with IRH.



So, although much of the wording in the currency activities refers to ILB and ALB, IRH stations should interpret the activities appropriately to ensure their crew routinely practise the full range of applicable activities.

For crew who are still developing, for example working towards tier 2 or to become a helm, navigator, coxswain or IRH commander, the currency programme will run concurrently with the training plan for their next role.

### Recording of Competence

It is essential, for assurance purposes, that the RNLI is able to track and report on levels of competence at the individual, station and regional levels. Completion of tasks in a training plan and completion of exercises must be recorded.

In all cases, launches for exercises or services are also to be recorded on the appropriate recording system (currently LSAR).

### Casualty Care and First Aid

Practising first aid and casualty care regularly will help to ensure that skills are maintained, and that crew are able to deliver appropriate care when the need arises. Therefore, it is essential that crew periodically practise clinical skills, using the scenario cards provided. An exercise could be conducted as a separate event or could be incorporated into a scenario alongside other lifeboat activities. Scenarios could last for 10 minutes, focussing on basic assessment and equipment familiarisation, or could last for several hours covering the full range of procedures. The important thing is to ensure that crew not only refresh their knowledge and skills, but that they retain the self-confidence to act quickly and correctly when required.



Photo: RNLI / Nigel Millard





Photo: Hamish Miller



## Section 4 - Role Specific Training Plans



## Section 4 - Role Specific Training Plans

| Inshore Lifeboat (ILB)          |
|---------------------------------|
| D Class                         |
| B Class                         |
| Inshore Rescue Hovercraft (IRH) |
| All-weather Lifeboat (ALB)      |
| Thames                          |
| D Class                         |
| B Class                         |
| E Class                         |
| Mechanic                        |
| Launch and Recovery             |
| Launch Authority                |
| Shore Crew                      |
| Launch Vehicle Driver           |
| Plant Operator                  |
| Head launcher                   |
| Boarding Boat Helm              |
| Other Roles                     |
| Lifeboat Operations Manager     |
| Lifeboat Trainer Assessor       |
| Lifeboat RWC Operator           |
| First Aider                     |
| Casualty Carer                  |
| Press Officer                   |





# Section 4 - Role Specific Training Plans

## D Class

| Safety and Induction          |                                     |
|-------------------------------|-------------------------------------|
| Stage 1                       | 1.2.1 Hazard Awareness              |
|                               | 1.2.2 Manual Handling               |
|                               | 1.2.3 Hazardous Substances          |
|                               | 1.2.4 Noise and Vibration           |
|                               | 1.2.5 Fire Safety                   |
|                               | 1.2.6 Personal Safety and Wellbeing |
|                               | 1.2.7 Safe to the Shout             |
|                               | 1.10.1 Working at Height            |
|                               | 8.1.1 First Aid 1                   |
| Safety and Induction Complete |                                     |

| D Class Navigator  |                                              |
|--------------------|----------------------------------------------|
| Navigator          | 1.4.2 Roles and Responsibilities (Navigator) |
|                    | 4.1.3 IRPCS (Command)                        |
|                    | 5.1.2 Locate and Assist 2                    |
|                    | 5.2.1 Plan SAR                               |
|                    | 7.1.2 Navigation                             |
|                    | 7.1.3 Navigation Advanced                    |
|                    | 7.2.1 E Navigation                           |
|                    | 7.2.2 E Navigation Advanced                  |
| Navigator Pass Out |                                              |

| D Class Crew            |                                         |
|-------------------------|-----------------------------------------|
| Stage 2                 | 1.1.1 PPE                               |
|                         | 1.3.1 SAR Unit Layout                   |
| Safe to go Afloat       |                                         |
| Stage 3                 | 1.5.1 SAR Unit Firefighting             |
|                         | 1.6.1 Emergency and Survival Procedures |
|                         | 1.7.1 Pyrotechnics                      |
|                         | 3.1.1 Launch Procedures                 |
|                         | 3.1.2 Recovery Procedures               |
|                         | 4.1.1 Lookout                           |
|                         | 4.2.1 Rope Handling                     |
| Safe to go on a Service |                                         |
| Stage 4                 | 1.4.1 Roles and Responsibilities 1      |
|                         | 4.3.1 Anchoring                         |
|                         | 4.4.1 Veering Down                      |
|                         | 4.5.1 Towing                            |
|                         | 4.6.1 Stability Introduction            |
|                         | 4.7.3 Mooring and Berthing              |
|                         | 5.1.1 Locate and Assist 1               |
|                         | 6.1.1 Operate VHF                       |
|                         | 7.4.2 Local Knowledge (Afloat)          |
| Tier 1 Crew Pass Out    |                                         |
| Stage 5                 | 1.8.1 Risk Assessment                   |
|                         | 4.1.2 Watchkeeping                      |
|                         | 4.6.2 Hydrology and Hull Types          |
|                         | 4.7.1 Helming 1                         |
|                         | 4.10.1 Helicopter Operations            |
| Tier 2 Crew Pass Out    |                                         |

| D Class Helm     |                                            |
|------------------|--------------------------------------------|
| Command          | 1.4.3 Roles and Responsibilities (Command) |
|                  | 1.5.2 Manage SAR Unit Firefighting         |
|                  | 1.6.2 Manage Emergency Procedures          |
|                  | 3.1.3 Manage Launch and Recovery           |
|                  | 4.3.2 Manage Anchoring                     |
|                  | 4.4.2 Manage Veering Down                  |
|                  | 4.5.2 Manage Towing                        |
|                  | 4.7.2 Helming 2                            |
|                  | 4.10.2 Manage Helicopter Operations        |
|                  | 5.3.1 Manage SAR                           |
|                  | 6.1.2 SAR Communication                    |
|                  | 7.4.3 Local Knowledge (Command)            |
|                  | 9.1.2 Operate ILB Main Machinery           |
| Command Pass Out |                                            |

## Section 4 - Role Specific Training Plans

### B Class

| Safety and Induction          |                                     |
|-------------------------------|-------------------------------------|
| Stage 1                       | 1.2.1 Hazard Awareness              |
|                               | 1.2.2 Manual Handling               |
|                               | 1.2.3 Hazardous Substances          |
|                               | 1.2.4 Noise and Vibration           |
|                               | 1.2.5 Fire Safety                   |
|                               | 1.2.6 Personal Safety and Wellbeing |
|                               | 1.2.7 Safe to the Shout             |
|                               | 1.10.1 Working at Height            |
|                               | 8.1.1 First Aid 1                   |
| Safety and Induction Complete |                                     |

| B Class Navigator  |                                              |
|--------------------|----------------------------------------------|
| Navigator          | 1.4.2 Roles and Responsibilities (Navigator) |
|                    | 4.1.3 IRPCS (Command)                        |
|                    | 5.1.2 Locate and Assist 2                    |
|                    | 5.2.1 Plan SAR                               |
|                    | 7.1.2 Navigation                             |
|                    | 7.1.3 Navigation Advanced                    |
|                    | 7.2.1 E Navigation                           |
|                    | 7.2.2 E Nav Advanced                         |
|                    | 7.3.1 Radar                                  |
| Navigator Pass Out |                                              |

| B Class Crew            |                                         |
|-------------------------|-----------------------------------------|
| Stage 2                 | 1.1.1 PPE                               |
|                         | 1.3.1 SAR Unit Layout                   |
| Safe to go Afloat       |                                         |
| Stage 3                 | 1.5.1 SAR Unit Firefighting             |
|                         | 1.6.1 Emergency and Survival Procedures |
|                         | 1.7.1 Pyrotechnics                      |
|                         | 3.1.1 Launch Procedures                 |
|                         | 3.1.2 Recovery Procedures               |
|                         | 4.1.1 Lookout                           |
|                         | 4.2.1 Rope Handling                     |
| Safe to go on a Service |                                         |
| Stage 4                 | 1.4.1 Roles and Responsibilities 1      |
|                         | 4.3.1 Anchoring                         |
|                         | 4.4.1 Veering Down                      |
|                         | 4.5.1 Towing                            |
|                         | 4.6.1 Stability Introduction            |
|                         | 4.7.3 Mooring and Berthing              |
|                         | 5.1.1 Locate and Assist 1               |
|                         | 6.1.1 Operate VHF                       |
|                         | 7.4.2 Local Knowledge (Afloat)          |
| Tier 1 Crew Pass Out    |                                         |
| Stage 5                 | 1.8.1 Risk Assessment                   |
|                         | 4.1.2 Watchkeeping                      |
|                         | 4.6.2 Hydrology and Hull Types          |
|                         | 4.7.1 Helming 1                         |
|                         | 4.10.1 Helicopter Operations            |
|                         | 7.1.1 Nav Introduction                  |
| Tier 2 Crew Pass Out    |                                         |

| B Class Helm     |                                            |
|------------------|--------------------------------------------|
| Command          | 1.4.3 Roles and Responsibilities (Command) |
|                  | 1.5.2 Manage SAR Unit Firefighting         |
|                  | 1.6.2 Manage Emergency Procedures          |
|                  | 3.1.3 Manage Launch and Recovery           |
|                  | 4.3.2 Manage Anchoring                     |
|                  | 4.4.2 Manage Veering Down                  |
|                  | 4.5.2 Manage Towing                        |
|                  | 4.7.2 Helming 2                            |
|                  | 4.10.2 Manage Helicopter Operations        |
|                  | 5.3.1 Manage SAR                           |
|                  | 6.1.2 SAR Communication                    |
|                  | 7.4.3 Local Knowledge (Command)            |
|                  | 9.1.2 Operate ILB Main Machinery           |
| Command Pass Out |                                            |

# Section 4 - Role Specific Training Plans

## Inshore Rescue Hovercraft (IRH)

| Stage 1                       | Safety and Induction |                               |
|-------------------------------|----------------------|-------------------------------|
|                               | 1.2.1                | Hazard Awareness              |
|                               | 1.2.2                | Manual Handling               |
|                               | 1.2.3                | Hazardous Substances          |
|                               | 1.2.4                | Noise and Vibration           |
|                               | 1.2.5                | Fire Safety                   |
|                               | 1.2.6                | Personal Safety and Wellbeing |
|                               | 1.2.7                | Safe to the Shout             |
|                               | 1.10.1               | Working at Height             |
|                               | 8.1.1                | First Aid 1                   |
| Safety and Induction Complete |                      |                               |

| Stage 2              | IRH Crew                |                                   |
|----------------------|-------------------------|-----------------------------------|
|                      | 1.1.1                   | PPE                               |
|                      | 1.3.1                   | SAR Unit Layout                   |
| Stage 3              | Safe to go Afloat       |                                   |
|                      | 1.5.1                   | SAR Unit Firefighting             |
|                      | 1.6.1                   | Emergency and Survival Procedures |
|                      | 1.7.1                   | Pyrotechnics                      |
|                      | 3.1.1                   | Launch Procedures                 |
|                      | 3.1.2                   | Recovery Procedures               |
|                      | 4.1.1                   | Lookout                           |
|                      | 4.2.1                   | Rope Handling                     |
|                      | Safe to go on a Service |                                   |
| Stage 4              | 1.4.1                   | Roles and Responsibilities 1      |
|                      | 4.3.1                   | Anchoring                         |
|                      | 4.6.1                   | Stability Introduction            |
|                      | 4.7.3                   | Mooring and Berthing              |
|                      | 5.1.1                   | Locate and Assist 1               |
|                      | 5.1.3                   | Mud Rescue                        |
|                      | 6.1.1                   | Operate VHF                       |
|                      | 7.4.2                   | Local Knowledge (Afloat)          |
| Stage 5              | Tier 1 Crew Pass Out    |                                   |
|                      | 1.8.1                   | Risk Assessment                   |
|                      | 4.1.2                   | Watchkeeping                      |
|                      | 4.10.1                  | Helicopter Operations             |
|                      | 7.1.1                   | Navigation Introduction           |
| Tier 2 Crew Pass Out |                         |                                   |

| Pilot          | IRH Pilot |                            |
|----------------|-----------|----------------------------|
|                | 4.7.4     | Hovercraft Handling        |
|                | 7.4.3     | Local Knowledge (Command)  |
|                | 9.1.3     | Operate IRH Main Machinery |
| Pilot Pass Out |           |                            |

| Command          | IRH Commander |                                        |
|------------------|---------------|----------------------------------------|
|                  | 1.4.2         | Roles and Responsibilities (Navigator) |
|                  | 1.4.3         | Roles and Responsibilities (Command)   |
|                  | 1.5.2         | Manage SAR Unit Firefighting           |
|                  | 1.6.2         | Manage Emergency Procedures            |
|                  | 3.1.3         | Manage Launch and Recovery             |
|                  | 4.1.3         | IRPCS (Command)                        |
|                  | 4.3.2         | Manage Anchoring                       |
|                  | 4.10.2        | Manage Helicopter Operations           |
|                  | 5.1.2         | Locate and Assist 2                    |
|                  | 5.2.1         | Plan SAR                               |
|                  | 5.3.1         | Manage SAR                             |
|                  | 5.3.2         | Manage Mud Rescue                      |
|                  | 6.1.2         | SAR Communication                      |
|                  | 7.1.2         | Navigation                             |
|                  | 7.1.3         | Navigation Advanced                    |
|                  | 7.2.1         | E Navigation                           |
|                  | 7.2.2         | E Navigation Advanced                  |
| Command Pass Out |               |                                        |



## Section 4 - Role Specific Training Plans

### All-weather Lifeboat (ALB)

| Stage 1                       | Safety and Induction |                               |
|-------------------------------|----------------------|-------------------------------|
|                               | 1.2.1                | Hazard Awareness              |
|                               | 1.2.2                | Manual Handling               |
|                               | 1.2.3                | Hazardous Substances          |
|                               | 1.2.4                | Noise and Vibration           |
|                               | 1.2.5                | Fire Safety                   |
|                               | 1.2.6                | Personal Safety and Wellbeing |
|                               | 1.2.7                | Safe to the Shout             |
|                               | 1.10.1               | Working at Height             |
|                               | 8.1.1                | First Aid 1                   |
| Safety and Induction Complete |                      |                               |

| Stage 2 | ALB Crew                |                                   |
|---------|-------------------------|-----------------------------------|
|         | 1.1.1                   | PPE                               |
|         | 1.3.1                   | SAR Unit Layout                   |
| Stage 3 | Safe to go Afloat       |                                   |
|         | 1.5.1                   | SAR Unit Firefighting             |
|         | 1.6.1                   | Emergency and Survival Procedures |
|         | 1.7.1                   | Pyrotechnics                      |
|         | 3.1.1                   | Launch Procedures                 |
|         | 3.1.2                   | Recovery Procedures               |
|         | 4.1.1                   | Lookout                           |
|         | 4.2.1                   | Rope Handling                     |
|         | Safe to go on a Service |                                   |

| Stage 4              | 1.3.2 | ALB Systems and Watertight Integrity |
|----------------------|-------|--------------------------------------|
|                      | 1.4.1 | Roles and Responsibilities 1         |
|                      | 4.2.2 | Capstans                             |
|                      | 4.3.1 | Anchoring                            |
|                      | 4.5.1 | Towing                               |
|                      | 4.6.1 | Stability Introduction               |
|                      | 4.7.3 | Mooring and Berthing                 |
|                      | 5.1.1 | Locate and Assist 1                  |
|                      | 6.1.1 | Operate VHF                          |
|                      | 7.4.2 | Local Knowledge (Afloat)             |
| Tier 1 Crew Pass Out |       |                                      |

| Stage 5              | 1.8.1  | Risk Assessment          |
|----------------------|--------|--------------------------|
|                      | 4.1.2  | Watchkeeping             |
|                      | 4.6.2  | Hydrology and Hull Types |
|                      | 4.7.1  | Helming 1                |
|                      | 4.8.1  | Breeches Buoy            |
|                      | 4.9.1  | Daughter Craft           |
|                      | 4.10.1 | Helicopter Operations    |
|                      | 7.1.1  | Navigation Introduction  |
| Tier 2 Crew Pass Out |        |                          |

| Mechanic          | ALB Afloat Mechanic |                                |
|-------------------|---------------------|--------------------------------|
|                   | 1.5.2               | Manage SAR Unit Firefighting   |
|                   | 1.9.1               | Restricted and Confined Spaces |
|                   | 6.1.2               | SAR Communication              |
|                   | 6.2.1               | Operate MF Radio               |
|                   | 6.2.2               | LRC Qualification              |
|                   | 6.2.3               | LRC Satellite Module           |
|                   | 9.1.1               | Operate ALB Main Machinery     |
|                   | 9.2.1               | Operate Auxiliary Systems      |
|                   | 9.3.1               | Diagnose Faults                |
|                   | 9.3.2               | Rectify Faults                 |
|                   | 9.3.3               | Mechanical Emergencies         |
| Mechanic Pass Out |                     |                                |

| Navigator          | ALB Navigator |                                        |
|--------------------|---------------|----------------------------------------|
|                    | 1.4.2         | Roles and Responsibilities (Navigator) |
|                    | 4.1.3         | IRPCS (Command)                        |
|                    | 5.1.2         | Locate and Assist 2                    |
|                    | 5.2.1         | Plan SAR                               |
|                    | 7.1.2         | Navigation                             |
|                    | 7.1.3         | Navigation Advanced                    |
|                    | 7.2.1         | E Navigation                           |
|                    | 7.2.2         | E Navigation Advanced                  |
|                    | 7.3.1         | Radar                                  |
| Navigator Pass Out |               |                                        |

| Command          | ALB Coxswain |                                      |
|------------------|--------------|--------------------------------------|
|                  | 1.4.3        | Roles and Responsibilities (Command) |
|                  | 1.5.2        | Manage SAR Unit Firefighting         |
|                  | 1.6.2        | Manage Emergency Procedures          |
|                  | 3.1.3        | Manage Launch and Recovery           |
|                  | 4.3.2        | Manage Anchoring                     |
|                  | 4.5.2        | Manage Towing                        |
|                  | 4.7.2        | Helming 2                            |
|                  | 4.10.2       | Manage Helicopter Operations         |
|                  | 5.3.1        | Manage SAR                           |
|                  | 6.1.2        | SAR Communication                    |
|                  | 6.2.1        | Operate MF Radio                     |
|                  | 6.2.2        | LRC Qualification                    |
|                  | 6.2.3        | LRC Satellite Module                 |
|                  | 7.4.3        | Local Knowledge (Command)            |
|                  | 9.1.1        | Operate ALB Main Machinery           |
| Command Pass Out |              |                                      |

## Section 4 - Role Specific Training Plans

### Thames D Class

| Safety and Induction          |                                     |
|-------------------------------|-------------------------------------|
| Stage 1                       | 1.2.1 Hazard Awareness              |
|                               | 1.2.2 Manual Handling               |
|                               | 1.2.3 Hazardous Substances          |
|                               | 1.2.4 Noise and Vibration           |
|                               | 1.2.5 Fire Safety                   |
|                               | 1.2.6 Personal Safety and Wellbeing |
|                               | 1.2.7 Safe to the Shout             |
|                               | 1.10.1 Working at Height            |
|                               | 8.1.1 First Aid 1                   |
| Safety and Induction Complete |                                     |

| D Class Crew |                       |
|--------------|-----------------------|
| Stage 2      | 1.1.1 PPE             |
|              | 1.3.1 SAR Unit Layout |

| Safe to go Afloat       |                                         |
|-------------------------|-----------------------------------------|
| Stage 3                 | 1.5.1 SAR Unit Firefighting             |
|                         | 1.6.1 Emergency and Survival Procedures |
|                         | 1.6.3 Dynamic Water Survival            |
|                         | 1.7.1 Pyrotechnics                      |
|                         | 3.1.1 Launch Procedures                 |
|                         | 3.1.2 Recovery Procedures               |
|                         | 4.1.1 Lookout                           |
|                         | 4.2.1 Rope Handling                     |
| Safe to go on a Service |                                         |

| Stage 4              | 1.4.1 Roles and Responsibilities 1 |
|----------------------|------------------------------------|
|                      | 4.3.1 Anchoring                    |
|                      | 4.4.1 Veering Down                 |
|                      | 4.5.1 Towing                       |
|                      | 4.6.1 Stability Introduction       |
|                      | 4.7.3 Mooring and Berthing         |
|                      | 5.1.1 Locate and Assist 1          |
|                      | 6.1.1 Operate VHF                  |
|                      | 7.4.2 Local Knowledge (Afloat)     |
| Tier 1 Crew Pass Out |                                    |

| Stage 5              | 1.8.1 Risk Assessment          |
|----------------------|--------------------------------|
|                      | 4.1.2 Watchkeeping             |
|                      | 4.6.2 Hydrology and Hull Types |
|                      | 4.7.1 Helming 1                |
|                      | 4.10.1 Helicopter Operations   |
|                      | 6.3.1 Tetra Communications     |
|                      | 7.1.1 Navigation Introduction  |
| Tier 2 Crew Pass Out |                                |

| D Class Navigator  |                                              |
|--------------------|----------------------------------------------|
| Navigator          | 1.4.2 Roles and Responsibilities (Navigator) |
|                    | 4.1.3 IRPCS (Command)                        |
|                    | 5.1.2 Locate and Assist 2                    |
|                    | 5.2.1 Plan SAR                               |
|                    | 7.1.2 Navigation                             |
|                    | 7.1.3 Navigation Advanced                    |
| Navigator Pass Out |                                              |

| D Class Helm     |                                              |
|------------------|----------------------------------------------|
| Command          | 1.4.2 Roles and Responsibilities (Navigator) |
|                  | 1.5.2 Manage SAR Unit Firefighting           |
|                  | 1.6.2 Manage Emergency Procedures            |
|                  | 3.1.3 Manage Launch and Recovery             |
|                  | 4.3.2 Manage Anchoring                       |
|                  | 4.4.2 Manage Veering Down                    |
|                  | 4.5.2 Manage Towing                          |
|                  | 4.7.2 Helming 2                              |
|                  | 4.10.2 Manage Helicopter Operations          |
|                  | 5.3.1 Manage SAR                             |
|                  | 6.1.2 SAR Communication                      |
|                  | 7.4.3 Local Knowledge (Command)              |
|                  | 9.1.2 Operate ILB Main Machinery             |
| Command Pass Out |                                              |

# Section 4 - Role Specific Training Plans

## Thames B Class

| Safety and Induction          |                                     |
|-------------------------------|-------------------------------------|
| Stage 1                       | 1.2.1 Hazard Awareness              |
|                               | 1.2.2 Manual Handling               |
|                               | 1.2.3 Hazardous Substances          |
|                               | 1.2.4 Noise and Vibration           |
|                               | 1.2.5 Fire Safety                   |
|                               | 1.2.6 Personal Safety and Wellbeing |
|                               | 1.2.7 Safe to the Shout             |
|                               | 1.10.1 Working at Height            |
|                               | 8.1.1 First Aid 1                   |
| Safety and Induction Complete |                                     |

| B Class Navigator  |                                              |
|--------------------|----------------------------------------------|
| Navigator          | 1.4.2 Roles and Responsibilities (Navigator) |
|                    | 4.1.3 IRPCS (Command)                        |
|                    | 5.1.2 Locate and Assist 2                    |
|                    | 5.2.1 Plan SAR                               |
|                    | 7.1.2 Navigation                             |
|                    | 7.1.3 Navigation Advanced                    |
|                    | 7.2.1 E Navigation                           |
|                    | 7.2.2 E Navigation Advanced                  |
|                    | 7.3.1 Radar                                  |
| Navigator Pass Out |                                              |

| B Class Crew            |                                         |
|-------------------------|-----------------------------------------|
| Stage 2                 | 1.1.1 PPE                               |
|                         | 1.3.1 SAR Unit Layout                   |
| Safe to go Afloat       |                                         |
| Stage 3                 | 1.5.1 SAR Unit Firefighting             |
|                         | 1.6.1 Emergency and Survival Procedures |
|                         | 1.6.3 Dynamic Water Survival            |
|                         | 1.7.1 Pyrotechnics                      |
|                         | 3.1.1 Launch Procedures                 |
|                         | 3.1.2 Recovery Procedures               |
|                         | 4.1.1 Lookout                           |
|                         | 4.2.1 Rope Handling                     |
| Safe to go on a Service |                                         |
| Stage 4                 | 1.4.1 Roles and Responsibilities 1      |
|                         | 4.3.1 Anchoring                         |
|                         | 4.4.1 Veering Down                      |
|                         | 4.5.1 Towing                            |
|                         | 4.6.1 Stability Introduction            |
|                         | 4.7.3 Mooring and Berthing              |
|                         | 5.1.1 Locate and Assist 1               |
|                         | 6.1.1 Operate VHF                       |
|                         | 7.4.2 Local Knowledge (Afloat)          |
| Tier 1 Crew Pass Out    |                                         |
| Stage 5                 | 1.8.1 Risk Assessment                   |
|                         | 4.1.2 Watchkeeping                      |
|                         | 4.6.2 Hydrology and Hull Types          |
|                         | 4.7.1 Helming 1                         |
|                         | 4.10.1 Helicopter Operations            |
|                         | 6.3.1 Tetra Communications              |
|                         | 7.1.1 Navigation Introduction           |
| Tier 2 Crew Pass Out    |                                         |

| B Class Helm  |                                              |
|---------------|----------------------------------------------|
| Helm          | 1.4.2 Roles and Responsibilities (Navigator) |
|               | 1.5.2 Manage SAR Unit Firefighting           |
|               | 1.6.2 Manage Emergency Procedures            |
|               | 3.1.3 Manage Launch and Recovery             |
|               | 4.3.2 Manage Anchoring                       |
|               | 4.4.2 Manage Veering Down                    |
|               | 4.5.2 Manage Towing                          |
|               | 4.7.2 Helming 2                              |
|               | 4.10.2 Manage Helicopter Operations          |
|               | 5.3.1 Manage SAR                             |
|               | 6.1.2 SAR Communication                      |
|               | 7.4.3 Local Knowledge (Command)              |
|               | 9.1.2 Operate ILB Main Machinery             |
| Helm Pass Out |                                              |

| B Class Commander |                                       |
|-------------------|---------------------------------------|
| Command           | 3.4.1 Authorising Launches            |
|                   | 5.4.1 Manage Operational Services     |
|                   | 9.3.1 Diagnose Faults                 |
|                   | 9.3.2 Rectify Faults                  |
|                   | 10.1.1 Maintain and Service Equipment |
| Command Pass Out  |                                       |



## Section 4 - Role Specific Training Plans

### Thames E Class

| Stage 1                       | Safety and Induction |                               |
|-------------------------------|----------------------|-------------------------------|
|                               | 1.2.1                | Hazard Awareness              |
|                               | 1.2.2                | Manual Handling               |
|                               | 1.2.3                | Hazardous Substances          |
|                               | 1.2.4                | Noise and Vibration           |
|                               | 1.2.5                | Fire Safety                   |
|                               | 1.2.6                | Personal Safety and Wellbeing |
|                               | 1.2.7                | Safe to the Shout             |
|                               | 1.10.1               | Working at Height             |
|                               | 8.1.1                | First Aid 1                   |
| Safety and Induction Complete |                      |                               |

| Navigator          | E Class Navigator |                                        |
|--------------------|-------------------|----------------------------------------|
|                    | 1.4.2             | Roles and Responsibilities (Navigator) |
|                    | 4.1.3             | IRPCS (Command)                        |
|                    | 5.1.2             | Locate and Assist 2                    |
|                    | 5.2.1             | Plan SAR                               |
|                    | 7.1.2             | Navigation                             |
|                    | 7.1.3             | Navigation Advanced                    |
|                    | 7.2.1             | E Navigation                           |
|                    | 7.2.2             | E Navigation Advanced                  |
|                    | 7.3.1             | Radar                                  |
| Navigator Pass Out |                   |                                        |

| Stage 2 | E Class Crew            |                                      |
|---------|-------------------------|--------------------------------------|
|         | 1.1.1                   | PPE                                  |
|         | 1.3.1                   | SAR Unit Layout                      |
| Stage 3 | Safe to go Afloat       |                                      |
|         | 1.5.1                   | SAR Unit Firefighting                |
|         | 1.6.1                   | Emergency and Survival Procedures    |
|         | 1.6.3                   | Dynamic Water Survival               |
|         | 1.7.1                   | Pyrotechnics                         |
|         | 3.1.1                   | Launch Procedures                    |
|         | 3.1.2                   | Recovery Procedures                  |
|         | 4.1.1                   | Lookout                              |
|         | 4.2.1                   | Rope Handling                        |
|         | Safe to go on a Service |                                      |
| Stage 4 | 1.3.2                   | ALB Systems and Watertight Integrity |
|         | 1.4.1                   | Roles and Responsibilities 1         |
|         | 4.3.1                   | Anchoring                            |
|         | 4.4.1                   | Veering Down                         |
|         | 4.5.1                   | Towing                               |
|         | 4.6.1                   | Stability Introduction               |
|         | 4.7.3                   | Mooring and Berthing                 |
|         | 5.1.1                   | Locate and Assist 1                  |
|         | 6.1.1                   | Operate VHF                          |
|         | 7.4.2                   | Local Knowledge (Afloat)             |
|         | Tier 1 Crew Pass Out    |                                      |
| Stage 5 | 1.8.1                   | Risk Assessment                      |
|         | 4.1.2                   | Watchkeeping                         |
|         | 4.6.2                   | Hydrology and Hull Types             |
|         | 4.7.1                   | Helming 1                            |
|         | 4.10.1                  | Helicopter Operations                |
|         | 6.3.1                   | Tetra Communications                 |
|         | 7.1.1                   | Navigation Introduction              |
|         | Tier 2 Crew Pass Out    |                                      |

| Helm          | E Class Helm |                                      |
|---------------|--------------|--------------------------------------|
|               | 1.4.3        | Roles and Responsibilities (Command) |
|               | 1.5.2        | Manage SAR Unit Firefighting         |
|               | 1.6.2        | Manage Emergency Procedures          |
|               | 3.1.3        | Manage Launch and Recovery           |
|               | 4.3.2        | Manage Anchoring                     |
|               | 4.4.2        | Manage Veering Down                  |
|               | 4.5.2        | Manage Towing                        |
|               | 4.7.2        | Helming 2                            |
|               | 4.10.2       | Manage Helicopter Operations         |
|               | 5.3.1        | Manage SAR                           |
|               | 6.1.2        | SAR Communication                    |
|               | 7.4.3        | Local Knowledge (Command)            |
|               | 9.1.2        | Operate ILB Main Machinery           |
| Helm Pass Out |              |                                      |

| Command          | E Class Commander |                                |
|------------------|-------------------|--------------------------------|
|                  | 1.9.1             | Restricted and Confined Spaces |
|                  | 3.4.1             | Authorising Launches           |
|                  | 5.4.1             | Manage Operational Services    |
|                  | 9.3.1             | Diagnose Faults                |
|                  | 9.3.2             | Rectify Faults                 |
|                  | 9.3.3             | Mechanical Emergencies         |
|                  | 10.1.1            | Maintain and Service Equipment |
| Command Pass Out |                   |                                |

## Section 4 - Role Specific Training Plans

### Mechanic

| Safety and Induction          |                                     |
|-------------------------------|-------------------------------------|
| Stage 1                       | 1.2.1 Hazard Awareness              |
|                               | 1.2.2 Manual Handling               |
|                               | 1.2.3 Hazardous Substances          |
|                               | 1.2.4 Noise and Vibration           |
|                               | 1.2.5 Fire Safety                   |
|                               | 1.2.6 Personal Safety and Wellbeing |
|                               | 1.2.7 Safe to the Shout             |
|                               | 1.10.1 Working at Height            |
|                               | 8.1 First Aid 1                     |
| Safety and Induction Complete |                                     |

| ILB Mechanic      |                                       |
|-------------------|---------------------------------------|
| Stage 2           | 1.1.1 PPE                             |
|                   | 1.4.1 Roles and Responsibilities 1    |
|                   | 1.5.1 SAR Unit Firefighting           |
|                   | 1.8.1 Risk Assessment                 |
|                   | 1.9.1 Restricted and Confined Spaces  |
|                   | 9.1.2 Operate ILB Main Machinery      |
|                   | 9.3.1 Diagnose Faults                 |
|                   | 9.3.2 Rectify Faults                  |
|                   | 10.1.1 Maintain and Service Equipment |
| Mechanic Pass Out |                                       |

| ALB Station Mechanic |                                       |
|----------------------|---------------------------------------|
| Stage 2              | 1.1.1 PPE                             |
|                      | 1.3.1 SAR Unit Layout                 |
|                      | 1.5.1 SAR Unit Firefighting           |
|                      | 1.8.1 Risk Assessment                 |
|                      | 1.9.1 Restricted and Confined Spaces  |
|                      | 9.1.1 Operate ALB Main Machinery      |
|                      | 9.2.1 Operate Auxiliary Systems       |
|                      | 9.3.1 Diagnose Faults                 |
|                      | 9.3.2 Rectify Faults                  |
|                      | 10.1.1 Maintain and Service Equipment |
| Mechanic Pass Out    |                                       |

| IRH Mechanic      |                                       |
|-------------------|---------------------------------------|
| Stage 2           | 1.1.1 PPE                             |
|                   | 1.4.1 Roles and Responsibilities 1    |
|                   | 1.5.1 SAR Unit Firefighting           |
|                   | 1.8.1 Risk Assessment                 |
|                   | 1.9.1 Restricted and Confined Spaces  |
|                   | 9.1.3 Operate IRH Main Machinery      |
|                   | 9.3.1 Diagnose Faults                 |
|                   | 9.3.2 Rectify Faults                  |
|                   | 10.1.1 Maintain and Service Equipment |
| Mechanic Pass Out |                                       |

| Launch and Recovery Mechanic |                                       |
|------------------------------|---------------------------------------|
| Stage 2                      | 1.1.1 PPE                             |
|                              | 1.4.1 Roles and Responsibilities 1    |
|                              | 1.5.1 SAR Unit Firefighting           |
|                              | 1.8.1 Risk Assessment                 |
|                              | 1.9.1 Restricted and Confined Spaces  |
|                              | 9.3.1 Diagnose Faults                 |
|                              | 9.3.2 Rectify Faults                  |
|                              | 10.1.1 Maintain and Service Equipment |
|                              | Mechanic Pass Out                     |

# Section 4 - Role Specific Training Plans

## Launch and Recovery

| Stage 1                       | Safety and Induction |                               |
|-------------------------------|----------------------|-------------------------------|
|                               | 1.2.1                | Hazard Awareness              |
|                               | 1.2.2                | Manual Handling               |
|                               | 1.2.3                | Hazardous Substances          |
|                               | 1.2.4                | Noise and Vibration           |
|                               | 1.2.5                | Fire Safety                   |
|                               | 1.2.6                | Personal Safety and Wellbeing |
|                               | 1.2.7                | Safe to the Shout             |
|                               | 1.10.1               | Working at Height             |
|                               | 8.1.1                | First Aid 1                   |
| Safety and Induction Complete |                      |                               |

| Stage 2                   | Launch Authority |                                    |
|---------------------------|------------------|------------------------------------|
|                           | 1.1.1            | PPE                                |
|                           | 1.4.1            | Roles and Responsibilities 1       |
|                           | 1.4.3            | Roles and Responsibilities Command |
|                           | 1.8.1            | Risk Assessment                    |
|                           | 3.4.1            | Authorising Launches               |
|                           | 5.4.1            | Manage Operational Services        |
|                           | 6.1.1            | Operate VHF                        |
|                           | 7.1.1            | Navigation Introduction            |
|                           | 7.4.1            | Local Knowledge (Ashore)           |
|                           | 7.4.3            | Local Knowledge (Command)          |
| Launch Authority Pass Out |                  |                                    |

| Stage 2             | Shore Crew (SC) |                              |
|---------------------|-----------------|------------------------------|
|                     | 1.1.1           | PPE                          |
|                     | 1.4.1           | Roles and Responsibilities 1 |
|                     | 1.7.1           | Pyrotechnics                 |
|                     | 3.2.1           | Launch Procedures            |
|                     | 3.2.2           | Recovery Procedures          |
|                     | 4.2.1           | Rope Handling                |
|                     | 6.1.1           | Operate VHF                  |
|                     | 7.1.1           | Navigation Introduction      |
|                     | 7.4.1           | Local Knowledge (Ashore)     |
| Shore Crew Pass Out |                 |                              |

| Stage 3                        | Launch Vehicle Driver (LVD) |                                |
|--------------------------------|-----------------------------|--------------------------------|
|                                | 1.5.1                       | SAR Unit Firefighting          |
|                                | 1.8.1                       | Risk Assessment                |
|                                | 1.9.1                       | Restricted and Confined Spaces |
|                                | 3.3.1                       | Plant Operator                 |
|                                | 7.2.1                       | Electronic Navigation          |
| Launch Vehicle Driver Pass Out |                             |                                |

| Stage 3                 | Plant Operator |                                |
|-------------------------|----------------|--------------------------------|
|                         | 1.9.1          | Restricted and Confined Spaces |
|                         | 3.3.1          | Plant Operator                 |
| Plant Operator Pass Out |                |                                |

| Stage 3                | Head Launcher |                                      |
|------------------------|---------------|--------------------------------------|
|                        | 1.4.3         | Roles and Responsibilities (Command) |
|                        | 1.5.1         | SAR Unit Firefighting                |
|                        | 1.8.1         | Risk Assessment                      |
|                        | 3.2.3         | Manage Launch and Recovery           |
|                        | 3.3.1         | Plant Operator                       |
|                        | 7.4.3         | Local Knowledge (Command)            |
| Head Launcher Pass Out |               |                                      |

| Stage 3       | Boarding Boat Helm |                            |
|---------------|--------------------|----------------------------|
|               | 1.1.1              | PPE                        |
|               | 1.3.1              | SAR Unit Layout            |
|               | 1.7.1              | Pyrotechnics               |
|               | 1.8.1              | Risk Assessment            |
|               | 3.2.1              | Launch Procedures          |
|               | 3.2.2              | Recovery Procedures        |
|               | 4.2.1              | Rope Handling              |
|               | 4.7.1              | Helming 1                  |
|               | 4.7.2              | Helming 2                  |
|               | 4.7.3              | Mooring and Berthing       |
|               | 6.1.1              | Operate VHF                |
|               | 7.1.1              | Navigation Introduction    |
|               | 7.4.2              | Local Knowledge (Afloat)   |
|               | 9.1.2              | Operate ILB Main Machinery |
| Helm Pass Out |                    |                            |

| Stage 4           | Helm SAR Pass Out |                                   |
|-------------------|-------------------|-----------------------------------|
|                   | 1.6.1             | Emergency and Survival Procedures |
|                   | 4.1.1             | Lookout                           |
|                   | 4.1.2             | Watchkeeping                      |
|                   | 4.3.1             | Anchoring                         |
|                   | 5.1.1             | Locate and Assist 1               |
|                   | 5.2.1             | Plan SAR                          |
|                   | 6.1.2             | SAR Communications                |
| Helm SAR Pass Out |                   |                                   |

Note: Use of boarding boats as SAR units is strictly limited to authorised locations.



## Section 4 - Role Specific Training Plans

### Other Roles

| Stage 1                       | Safety and Induction |                               |
|-------------------------------|----------------------|-------------------------------|
|                               | 1.2.1                | Hazard Awareness              |
|                               | 1.2.2                | Manual Handling               |
|                               | 1.2.3                | Hazardous Substances          |
|                               | 1.2.4                | Noise and Vibration           |
|                               | 1.2.5                | Fire Safety                   |
|                               | 1.2.6                | Personal Safety and Wellbeing |
|                               | 1.2.7                | Safe to the Shout             |
|                               | 1.10.1               | Working at Height             |
|                               | 8.1.1                | First Aid 1                   |
| Safety and Induction Complete |                      |                               |

| Manager          | Lifeboat Operations Manager |                                    |
|------------------|-----------------------------|------------------------------------|
|                  | 1.1.1                       | PPE                                |
|                  | 1.4.1                       | Roles and Responsibilities 1       |
|                  | 1.4.3                       | Roles and Responsibilities Command |
|                  | 1.8.1                       | Risk Assessment                    |
|                  | 3.4.1                       | Authorising Launches               |
|                  | 5.4.1                       | Manage Operational Services        |
|                  | 6.1.1                       | Operate VHF                        |
|                  | 7.1.1                       | Navigation Introduction            |
|                  | 7.4.1                       | Local Knowledge (Ashore)           |
|                  | 7.4.3                       | Local Knowledge (Command)          |
| Manager Pass Out |                             |                                    |

| Assessor     | Lifeboat Trainer Assessor (LTA) |                         |
|--------------|---------------------------------|-------------------------|
|              | 30.1.1                          | RNLI Trainer Theory     |
|              | 30.1.2                          | RNLI Trainer Practical  |
|              | 30.2.1                          | RNLI Assessor Theory    |
|              | 30.2.2                          | RNLI Assessor Practical |
| LTA Pass Out |                                 |                         |

| Operator          | Lifeboat RWC Operator |                                   |
|-------------------|-----------------------|-----------------------------------|
|                   | 1.1.1                 | PPE                               |
|                   | 1.6.1                 | Emergency and Survival Procedures |
|                   | 1.7.1                 | Pyrotechnics                      |
|                   | 6.1.1                 | Operate VHF                       |
|                   | 7.1.1                 | Navigation Introduction           |
|                   | 7.4.1                 | Local Knowledge (Ashore)          |
|                   | *                     | RWC Operator                      |
| Operator Pass Out |                       |                                   |

| Tier 1 Crew / SC + First Aider |             |
|--------------------------------|-------------|
| 8.2.1                          | First Aid 2 |

| .1 | Tier 1 Crew / SC + Casualty Carer |                                         |
|----|-----------------------------------|-----------------------------------------|
|    | 8.3.1                             | Casualty Care                           |
|    | 8.4.1                             | Automated External Defibrillation (AED) |

| Press Officer | Press Officer |                             |
|---------------|---------------|-----------------------------|
|               | 15.1.1        | Media Awareness             |
|               | 15.2.1        | Camera Operations           |
|               | 15.3.1        | Understanding the Media     |
|               | 15.4.1        | Key Messages                |
|               | 15.5.1        | Press Releases              |
|               | 15.6.1        | Incident Handling           |
|               | 15.7.1        | Media Interview Logistics   |
|               | 15.8.1        | Basic Photography and Video |
|               | 15.9.1        | Interview Skills            |
|               | 15.10.1       | Video Production            |
|               | 15.11.1       | Social Media                |

\*Note: Please refer to the *Lifeguard Operational Competence Framework* for RWC training and assessment standards and to Local Operating Procedures (LOPs) for any additional requirements or operational limitations..

### Passage Coxswain, Relief Coxswain and Relief Mechanic

#### Passage Coxswain

To be eligible to act as a passage coxswain, an individual must be a passed out coxswain in the class of ALB under consideration and should have a minimum of 20 hours in command in that class of ALB. They must also hold the relevant maritime qualifications specified in GU1054 (*Requirements for External Maritime Qualifications*).

#### Relief Roles

##### Coxswain

To be eligible to act as a relief coxswain, an individual must be a passed out coxswain in the class of ALB under consideration and should have a minimum of 20 hours in command in that class of ALB.

##### Carriage launch

In addition to the above, to be eligible to act as a relief coxswain at a carriage-launch station a coxswain must either have been originally passed out on a carriage-launched ALB in their primary role or must have completed Task 3.1.3 (Manage Launch and Recovery) at their destination station.

##### Slipway launch

In addition to the above, to be eligible to act as a relief coxswain at a slipway-launch station a coxswain must either have been originally passed out on a slipway-launched ALB in their primary role or must have completed Task 3.1.3 (Manage Launch and Recovery) at their destination station.

##### Afloat mechanic

To be eligible to act as a relief ALB afloat mechanic, an individual must be a passed out afloat mechanic in the class of ALB under consideration and should have a minimum of 20 hours experience afloat in the afloat mechanic in that class of ALB.

##### Station mechanic

To be eligible to act as a relief station mechanic, an individual must be a passed out station mechanic in the class of ALB under consideration and should have a minimum of 20 hours experience afloat in the afloat mechanic role in that class of ALB.







Photo: RNLI Galway





## Section 5 - Currency Activities



## Section 5 - Currency Activities

### Purpose

The purpose of currency activities is to maintain the operational competence of lifeboat crew and shore crew by ensuring they undertake a broad range of different activities. They are not tests or assessments but are intended to refresh, consolidate and embed existing skills and knowledge. They are also intended to help build crew cohesion and collective competence.

### Responsibilities

#### LOM

It is the responsibility of the LOM, supported by the Lifeboat Training Coordinator (LTC), to ensure there is a viable exercise programme, with sufficient opportunities for all crew members to maintain competence in their roles.

#### Individual Crew

It is the responsibility of individual crew to:

- ensure they understand the attendance and activity requirements for their role(s)
- manage their own attendance
- monitor progress through the currency programme for each role they hold.

#### ALM

It is the responsibility of the ALM to ensure that stations are monitoring currency effectively and taking appropriate management action when necessary.

### Maintaining Currency

If an individual believes they are falling behind and will not be able to complete enough activity to maintain currency, they should notify their LTC and / or LOM who will advise how to get their attendance back on track. If this is not feasible then consideration should be given to reducing the number of roles held.

In addition to individuals monitoring their own progress, the LTC should periodically monitor overall station progress and identify anyone who appears to be behind. If there is reasonable mitigation for lack of progress, and it is likely that they can catch up, then the LTC should continue to monitor progress. Guidance on action to be taken for individuals who fall behind is contained in the *Lifeboat Currency Activity and Training Handbook*.



Photo: Stephen Duncombe

### Conduct of Exercises

The SAR unit commander will, at all times, be in command of an exercise. For exercises overseen by an LTA or visiting member of the training team, and for machinery breakdown drills organised by an ALB mechanic, the SAR unit commander will agree on the conduct of the activity to ensure maximum benefit is gained.

As currency activities are intended to refresh existing skills, it is not necessary for every individual to carry out every element of every task. This would not be achievable in most cases.

The templates in the *Lifeboat Currency Activity and Training Handbook* are a guide and an aide memoire for the individual running the exercise. They provide an outline of what is to be achieved, however, stations are strongly encouraged to tailor the exercises to their own needs by adding realistic scenarios representative of local conditions and circumstances.

In all cases, the authoritative references for the conduct of operational activity are the relevant policies, SOPs and EOPs. The individual in charge of the exercise must refamiliarise themselves with these prior to the start of the exercise.

As stated, currency activities are not tests or assessments. However, if significant shortcomings or areas for improvement are observed, either by individuals or the crew overall, they should be addressed as follows:

- Minor mistakes by individuals may be addressed by re-briefing on the spot or soon afterwards.
- Repeated errors by multiple crew members should be addressed by additional training provided to the wider station crew.
- Major mistakes by an individual should be discussed with the AT or LTA who may recommend appropriate additional training.
- Mistakes which fall far short of expected standards or are unequivocally dangerous should be addressed by the station in consultation with the AT and ALM. Action may include being stood down from their role or reverted to an earlier stage in their training plan.

### Maintaining Currency through Services

Where the activities performed during a service call broadly cover the same content as a particular currency activity or activities, and crew carry out their roles correctly, they are judged to have refreshed their knowledge in those areas. The SAR unit commander may then (at their own discretion) mark those present as having completed the activity or activities during the service call. This also applies for assurance or AT visits.

### How Many Exercises?

A key principle of the Operational Competence Framework is that an individual must remain current in all roles held. Hence, the more roles held, the more frequently an individual will need to exercise. The *Lifeboat Currency Activity and Training Handbook* specifies which activities individuals must complete for a given role. Each activity is to be completed at intervals no greater than 18 months. Completion may include service calls, assurance and AT visits.

The date of completion of a currency activity by each crew member should be recorded in LSAR. Instructions on recording exercise activity is published separately.

Although aimed principally at fully trained crew, trainee crew will also benefit from taking part. Doing so will allow them to observe more experienced crew and help build their experience and confidence. Trainees must not be given tasks beyond their competence, but could be buddied up with a fully trained crew member to guide them through tasks. Such peer-to-peer learning benefits both the trainee and the experienced crew and is a good way to build trust and to develop strong bonds within the crew.

### Crew Falling Behind with Currency Programmes

Crew who are not making adequate progress through their currency programme(s) should consult with station management to address matters as early as possible. Early action should help to ensure that no one goes overdue for currency.



Photo: RNLI Beth Brooks

## Section 5 - Currency Activities

### Assurance

ALMs, as part of their assurance, should monitor the overall progress of crew through their currency programme. Where it is clear that an individual is falling behind and activities are likely to become overdue, the ALM should determine what remedial action has been taken and what additional measures might be introduced.

### Exercise Deficits

If an individual exceeds the 18 month period for one or more activities they will appear as “overdue” on the station crew dashboard. At this point the station can either:

- Grant an extension of three months in which the individual should complete any outstanding activities. If the required activities are not undertaken within this period, the station should take remedial action as described below.

or

- Take immediate remedial action as described below.

The decision whether to grant an extension period is for the station to make, in consultation with the ALM. The decision should be based on the reasons behind the shortfall and the likelihood that the individual will make up the deficit in the time available. Any individual placed in an extension period will remain annotated as “overdue”.

### Remedial Action

Stations, in discussion with their ALM, will decide appropriate remedial action given the individual circumstances of the person in question. Remedial action may include (but not limited to):

- Relinquishing one or more roles.
- Reverting to an earlier stage in one or more roles (e.g. from helm to navigator or crew, or from head launcher to shore crew).

When deciding what action to take, LOMs and ALMs should take into account any extenuating circumstances such as maternity, working offshore or time away at university, and aim to be as fair and flexible as possible. It is in the interests of both the individual and the RNLI to keep people motivated, engaged and operationally competent by whatever means are available.

If either of the above steps are taken, individuals may recover their previous status after a period of consolidation and successful completion of the appropriate pass out. However, LOMs must be satisfied that the reasons for previous shortfalls have been addressed and are unlikely to reoccur.

If the above steps are not feasible, or if the individual remains unable to meet the necessary attendance and commitment requirements, then the LOM may, in consultation with the ALM, ask the individual to stand down from their operational role. They may continue to support the RNLI in non-operational roles.







Photo: RNLI / Mike Lang









## Section 6 - Operational Competence Standards



## Section 6 - Operational Competence Standards

Photo: RNLI / Nigel Millard



## Section 6 - Operational Competence Standards

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## Activity 1 - Safety and Induction

Photo: Stephen Duncombe





## Tasks

### 1.1 Personal Protective Equipment (PPE)

1.1.1 Personal Protective Equipment (PPE)

### 1.2 Working Safely

1.2.1 Hazard Awareness

1.2.2 Manual Handling

1.2.3 Hazardous Substances

1.2.4 Noise and Vibration

1.2.5 Fire Safety

1.2.6 Personal Safety and Wellbeing

1.2.7 Safe to the Shout

### 1.3 SAR Unit Layout and Equipment

1.3.1 SAR Unit Layout

1.3.2 ALB Systems and Watertight Integrity

### 1.4 Roles and Responsibilities

1.4.1 Roles and Responsibilities 1

1.4.2 Roles and Responsibilities (Navigator)

1.4.3 Roles and Responsibilities (Command)

### 1.5 SAR Unit Firefighting

1.5.1 SAR Unit Firefighting

1.5.2 Manage SAR Unit Firefighting

### 1.6 Emergency and Survival Procedures

1.6.1 Emergency and Survival Procedures

1.6.2 Manage Emergency Procedures

1.6.3 Dynamic Water Survival

### 1.7 Pyrotechnics

1.7.1 Pyrotechnics

### 1.8 Risk Assessment

1.8.1 Risk Assessment

### 1.9 Restricted and Confined Spaces

1.9.1 Restricted and Confined Spaces

### 1.10 Working at Height

1.10.1 Working at Height

# Activity 1 - Safety and Induction

## Task 1.1.1

## Personal Protective Equipment (PPE)

### a Personal Protective Equipment

#### Training guidelines

#### PPE

##### • Procedures.

Training criteria. Explain the risks present and why PPE is needed:

- Protection from the environment e.g. thermal protection.
- Protection from the risks and adverse effects associated with tasks.

Training criteria. Explain the operation, performance features and limitations of equipment.

Training criteria. Explain the selection, use and storage of PPE.

Training criteria. Identify written operating procedures, including permits to work (if appropriate):

- In accordance with the current SOP or guidelines.
- To include all functions and features of PPE equipment.
- To include when and where certain PPE is required.
- To include hazards of wearing PPE e.g. Gloves acting as snag hazards during duties and bulky equipment in confined spaces.
- To include all PPE required by the role.

Training criteria. Explain factors which can affect the protection provided by the PPE such as other protective equipment, personal factors, working conditions, inadequate fitting, and defects, damage and wear:

- In appropriate donning leading to leaks, exposure, PPE damage, insufficient protection.
- Increased personal risk to health.
- Reduced capacity or working time in environment available.

Training criteria. Explain recognising defects in PPE and arrangements for reporting loss or defects:

- As per location procedures.

##### • PPE usage.

Training criteria. Demonstrate donning, wearing and removing the equipment.

Training criteria. Demonstrate inspection of and / or testing of the PPE before use (if appropriate):

- In accordance with the current SOP or guidelines.
- To include all functions and features of PPE equipment required for the role.

##### • PPE Maintenance.

Training criteria. Explain how to maintain PPE such as cleaning and the replacement of any components.

Training criteria. Demonstrate safe storage of equipment:

- In accordance with the current SOP or guidelines.
- As per location procedures.

## Task 1.2.1

## Hazard Awareness

### a Hazard awareness

#### Training guidelines

- **Identification of hazards.**

Training criteria. State the main hazards you may be exposed to in your role:

- Slips and trips.
- Hazardous substances.
- Driving.
- Noise and vibration.
- Excessive heat or cold.
- Biohazards.
- Moving machinery and equipment.
- Pyrotechnics.
- Lines under load.
- Radars and radio antennae.
- Aggressive individuals and members of the public.

Training criteria. Explain the importance of weighing the risk versus the benefit of conducting any hazardous activity:

- It is important to determine whether the hazards and potential risks to you and to others justify undertaking that activity.
- Have all reasonable measures to negate or minimise hazards been taken?

task continues



# Activity 1 - Safety and Induction

## Task 1.2.1

## Hazard Awareness

### b Incident reporting

#### Training guidelines

##### • Types of incident.

Training criteria. State the types of incident which should be reported:

- Hazard observation.
- Near miss.
- Dangerous occurrence.
- Injury or illness.
- Equipment or property damage.
- Road traffic incident.
- Environmental.

##### • Reporting incidents

Training criteria. Explain why it is important that incidents must be reported:

- There may have been similar incidents elsewhere and hence a pattern of incidents.
- Action may be required to prevent similar incidents.

Training criteria. Explain why it is equally important to report hazard observations:

- Even if nothing happened, a hazard observation may trigger action to prevent an incident in the future.
- Do not walk by or turn a blind eye to potential hazards.

Training criteria. Explain how incidents are to be reported and to whom:

- Serious incidents are to be reported immediately to the Central Operations and Information Room (COIR).
- All other incidents are to be reported via the link on Compass / RNL12.

## Task 1.2.2

## Manual Handling

### a Risk awareness

#### Training guidelines

#### Identification of risks

- **Activities with the potential to cause manual handling injuries.**

Training criteria. Describe the activities with the potential to cause manual handling injuries:

- Lifting and moving heavy objects.
- Repetitive movements.
- Receiving deliveries.
- Equipment changes.
- Casualty handling.
- Moving awkwardly shaped objects.

- **Potential injuries caused by improper manual handling techniques.**

Training criteria. Describe the injuries caused by poor manual handling techniques:

- Injury to back, neck, limbs, body, head.
- Cuts and bruises.
- Strains.
- Breaks.
- Slips, trips and falls.

- **Reducing the risks.**

Training criteria. Explain methods of reducing or removing manual handling risks:

- Assess task and load using 'LITE' methodology (Load, Individual capacity, Task, Environment).
- Use appropriate lifting equipment / mechanical aids.
- Avoid the need to move object or undertake task in the first instance.
- Follow operating procedures if present which will have considered and reduced risks.
- Frequency and height of lift.
- Awkward postures.
- Size, shape and weight of load.
- Working environment: floor surface, obstacles, confined space.
- Individual capability: more than one person to lift.

task continues

# Activity 1 - Safety and Induction

## Task 1.2.2

## Manual Handling

### b Procedure awareness

#### Training guidelines

#### Manual handling guidance

- **Locating information on good manual handling practices.**

Training criteria. State where to find the information on good manual handling practices:

- Online SHE training and information.
- Handbook and manual resources.
- Embedded as part of RNLI procedures.

- **Good practice.**

Training criteria. Explain common and safe manual handling techniques:

- Mechanical means where possible.
- Testing of weight before moving.
- Planning the route and method of putting the object down.
- Single person lift / move techniques.
- Multiple person lift / move techniques.



## Task 1.2.3

## Hazardous Substances

### a Hazardous substances

#### Training guidelines

#### Control of substances hazardous to health and environment

##### • Responsibilities.

Training criteria. State personal responsibilities and the RNLI's responsibilities regarding hazardous substances:

- Personal responsibilities include: follow procedures, remove risks where possible, report incidents, use approved substances only.
- RNLI responsibilities include: reduce risk to people and environment where possible, provide training and information on hazardous substances that may be encountered, be legally compliant to legislation and report on incidents where required.

##### • Encountering hazardous substances.

Training criteria. Identify hazardous substances you may encounter relative to role and location:

- Those registered and found in environment or encountered by the specific role - gas, liquid, solid, other.

Training criteria. State the risks associated with hazardous substances relative to your role and location:

- As currently evidenced and recorded for the mediums that apply.
- Personal risks - illness, injury, length and type of exposure.
- Environmental - Contamination, pollution, workplace safety.

Training criteria. State where you can find more information regarding hazardous substances you may encounter:

- Control of substances hazardous to health (COSHH) Assessment / summary sheet.
- Safety data sheet or equivalent.
- Packaging information and warnings.

Training criteria. Describe hazardous substance storage and signage requirements relevant to role and location:

- Signage in line with current regulations e.g. warning signs, PPE requirements, glow in the dark signage, vehicle signage.
- Storage in line with current regulations e.g. fuel storage, chemical cupboards, storage of incompatible chemicals, containers and bunding.

##### • Procedures.

Training criteria. Describe the control measures and actions to take when encountering hazardous substances to ensure personal and environmental safety:

- In accordance with the current operating procedures and risk assessments.
- Be aware of hazards of mixing chemicals.
- COSHH assessment.
- Safety data sheet / summary sheet or equivalent procedures.
- Controlling the risks and access.
- Disposal of and environmental protection including spill procedures.

Training criteria. Describe the procedures in the event of emergencies or contamination / exposure to hazardous substances:

- In accordance with the current SOP, EOP.
- Reportable Incidents to be entered into the database and escalated if severe.

# Activity 1 - Safety and Induction

## Task 1.2.4

## Noise and Vibration

### a Noise hazard awareness

#### Training guidelines

#### Noise

- **Awareness.**

Training criteria. Describe the sources of noise hazards:

- Environmental - environments with high noise levels such as engine rooms or wind.
- Equipment - equipment use generating noise risks.

Training criteria. Describe the ways of minimising or removing noise risks:

- Recognising possible sources and avoiding.
- Following RNLI operating procedures with risk mitigations embedded in them.
- Not undertaking unauthorised or non-risk assessed activities.
- Informing the RNLI of any pre-existing conditions that may be affected.

Training criteria. State the possible effects of exposure and how to report it:

- Illness or injury.
- Report using Incident reporting system.

task continues

## Task 1.2.4

## Noise and Vibration

### b Vibration hazard awareness

#### Training guidelines

#### Vibration

- **Awareness.**

Training criteria. Describe the sources of vibration risks:

- Environmental - workplace environment or afloat on vessels to whole body.
- Equipment - equipment and power tool use to hands and arms.

Training criteria. Describe the ways of minimising or removing vibration risks:

- Recognising possible sources and avoiding.
- Following RNLI operating procedures with risk mitigations embedded in them.
- Not undertaking unauthorised or non-risk assessed activities.
- Informing the RNLI of any pre-existing conditions that may be affected.

Training criteria. State the possible effects of exposure and how to report it:

- Illness or injury.
- Report using Incident reporting system.



# Activity 1 - Safety and Induction

## Task 1.2.5

## Fire Safety

### a Fire safety

#### Training guidelines

##### • The fire triangle.

Training criteria. Identify the three elements that form the fire triangle:

- Fuel, oxygen, heat.

##### • Identifying types of fire.

Training criteria. Explain the type of fire identified by its class:

- A - Carboniferous: wood, paper, rubbish, material.
- B - Fuel (Liquid): petrol, diesel.
- C - Gases.
- D - Metals.
- ⚠ - Electrical: radar, SIMS, electronic navigation equipment.
- F - Cooking oils and fats: encountered on board casualty vessel.

##### • Raise or recognise fire alarms.

Recognising the fire alarm and the importance of communicating the existence of a fire to others.

Training criteria. Identify the sound of the fire alarm.

Training criteria. Explain how to raise the alarm if no fire alarm present.

Training criteria. State action by being alerted to the presence of a fire and the importance of this:

- Recognise the sound - test.
- Raise fire alarm in accordance with the current SOP.
- Actions as required in accordance with the current SOP.

##### • Portable firefighting appliances.

Locating a suitable extinguisher to fight each fire classification (if applicable).

Training criteria. State the types and location of the portable fire extinguishers available at the location. Training criteria. Describe the extinguisher best suited for each fire type:

- |                  |                       |
|------------------|-----------------------|
| • Water: type A. | • Foam: A, B.         |
| • CO2: B, ⚠.     | • Powder: A, B, C, ⚠. |

##### • Considerations for tackling a fire.

Training criteria. Identify considerations made before tackling a fire:

- Personal safety, safety of others, how effective the action is likely to be, safer to evacuate and call for assistance.

##### • Escape routes / fire exits.

Safe routes and exits. Training criteria. Identify safe routes and exits for your location and explain the importance of keeping these clear at all times:

- |                                            |                 |
|--------------------------------------------|-----------------|
| • Locate safe escape routes.               | • Signage used. |
| • Keep escape routes and fire exits clear. |                 |

##### • Muster points.

Safe muster point / fire assembly point. Training criteria. Identify the safe muster point or fire assembly point for your location:

- |                                                      |                                                 |
|------------------------------------------------------|-------------------------------------------------|
| • Identify muster point.                             | • Consider fire hazards on way to muster point. |
| • Furthest upwind point from fire where appropriate. |                                                 |

## Task 1.2.6

## Personal Safety and Wellbeing

### a Dealing with a safeguarding matter for all, including children and vulnerable adults

#### Training guidelines

##### Policies

- **Policies and process.**

Training criteria. Describe the safeguarding policy:

- RNLI safeguarding policy.
- Working with children and vulnerable adults guidelines.
- The safeguarding level associated to the role.

Training criteria. Identify the process of reporting on any safeguarding issue:

- As defined in the safeguarding policy.
- Those escalating to follow escalation reporting procedure.

##### Awareness

- **Safety.**

Training criteria. State why it is important to understand safeguarding:

- Best practice guidelines as described in the policy.
- Legal responsibility of the organisation and its people.

Training criteria. Describe why it is important to ensure the welfare of everyone including children and vulnerable adults:

- RNLI safeguarding policy.
- Working with children and vulnerable adults guidelines.

- **Identification.**

Training criteria. State who is classified as a child or vulnerable adult:

- As defined in the policy.

Training criteria. Explain what a potentially reportable safeguarding issue is:

- As defined in the policy.

- **Mandatory training.**

Training criteria. Demonstrate having conducted the online safeguarding training.

task continues

# Activity 1 - Safety and Induction

## Task 1.2.6

## Personal Safety and Wellbeing

### b Dealing with violence and conflict

#### Training guidelines

##### Policies

- **Policies and process.**

Training criteria. State the policies in place regarding violence and conflict:

- RNLI policy.
- In accordance with the current risk assessment for the area / location.
- In accordance with the current SOP and EOP.

Training criteria. Identify the process of reporting violent and conflict:

- As defined in the policy.
- Those escalating to follow escalation reporting procedure.

Training criteria. State the RNLI responsibilities regarding absconders:

- The RNLI is not a law enforcement organisation.
- Current guidelines on our parameters.
- When to request police presence.

##### Awareness

- **Safety.**

Training criteria. Explain how to protect yourself and others when encountering violence and conflict:

- RNLI policy.
- In accordance with the current risk assessment for the area / location.
- In accordance with the current SOP and EOP.
- Recognise the signs of violence and conflict.
- Awareness of how personal behaviours and responses can affect the situation.
- Remove yourself and avoid the situation and reduce the chance of escalation where possible.

task continues



## Task 1.2.6

## Personal Safety and Wellbeing

### c Mental health and wellbeing

#### Training guidelines

##### Awareness

###### • Causes.

Training criteria. State the common causes and triggers of mental ill health in relation to RNLI duties:

- As stated in the role specific handbook.
- In accordance with the current risk assessment for the area / location.
- Being sensitive to personal issues that may affect RNLI duties.
- In accordance with the current SOP.

###### • Identifying.

Training criteria. Explain the personal signs and symptoms of mental ill health:

- As stated in the role specific handbook.
- In accordance with the current SOP.

Training criteria. Explain the potential signs and symptoms of mental ill health in others:

- As stated in the role specific handbook.
- In accordance with the current SOP.

Training criteria. State the importance of engaging with mental ill health and ensuring support is made available:

- As stated in the role specific handbook.
- In accordance with the current SOP.

##### Support functions

###### • Supporting.

Training criteria. State the support functions available to support mental ill health:

- People services teams and Poole Support Centre.
- Trauma Risk Management (TRiM) practitioners – if present at station or place of work.
- Mental Health First Aiders – if present at station or place of work.
- Regional support staff.
- Support 24/7.
- Others - e.g. peers, friends, senior personnel and volunteers at your location.
- NHS and other mental health support providers.

task continues

# Activity 1 - Safety and Induction

## Task 1.2.6

## Personal Safety and Wellbeing

### d Personal protection

#### Training guidelines

#### Clinical

- **Infectious diseases control.**

Training criteria. Identify the current guidance of infectious diseases control.

Training criteria. State the PPE specific used to protect a carer and casualty.

Training criteria. Describe the steps to take to avoid contamination.

Training criteria. State the steps to take for suspected contamination cases.

Training criteria. State post exposure treatment and support:

- As described in the current clinical documentation.
- In accordance with the current reporting procedure and process.
- In accordance with the current SOP.

Training criteria. State any requirements for inoculations there may be for your role:

- As per current guidelines.
- Those that are advised and / or required by the role.

- **Decontamination.**

Training criteria. State the decontamination process for RNLI equipment:

- In accordance with the current SOP.
- When multi-use equipment is contaminated with biohazards and requires cleaning before re-use.

- **Clinical waste.**

Training criteria. State the Clinical waste procedure:

- In accordance with the current SOP and LOP.

task continues

## Task 1.2.6

## Personal Safety and Wellbeing

### e Body discovery and recovery

#### Training guidelines

##### Body discovery

- **Assessing.**

Training criteria. State the difference between casualty management and body recovery procedures:

- As per current guidelines.

- **Evidence protection.**

Training criteria. State the crime scene evidence considerations:

- In accordance with the current SOP.

##### Body recovery

- **Procedure.**

Training criteria. State the RNLI's role and responsibilities for body recoveries.

Training criteria. State the responsibilities of other emergency services.

Training criteria. Describe the equipment available to undertake body recovery.

Training criteria. State the process for body recovery:

- In accordance with the current SOP.

- **Dealing with the bereaved.**

Training criteria. Describe what to say / not to say to any bereaved on-site:

- As per dealing with vulnerable people task.
- Be kind and caring, just refrain from passing information and exposing them to the environment and remains.

- **Hazards.**

Training criteria. Describe the risk of infectious diseases during body recovery.

Training criteria. Describe the psychological effect that it can have on yourself and those around you:

- In accordance with the current SOP.
- See tasks on personal protection and wellbeing.

# Activity 1 - Safety and Induction

## Task 1.2.7

## Safe to the Shout

### a Safe driver awareness training

#### Training guidelines

#### Awareness training

Training criteria. Demonstrate successful completion of the online safe driver awareness training material "Safe to the Shout":

- Completed and recorded on the Learning Zone.



## Task 1.3.1

## SAR Unit Layout

### a Vessel layout and equipment

#### Training guidelines

#### Vessel terminology

- **Terms for parts of a vessel.**

Training criteria. Identify the parts of a vessel:

- |                 |              |
|-----------------|--------------|
| • Bow.          | • Stern.     |
| • Port.         | • Starboard. |
| • Beam.         | • Quarter.   |
| • Gunwale.      | • Transom.   |
| • Fore and Aft. |              |

#### Familiarity with stowages

- **Stowages and equipment for the SAR unit.**

Training criteria. Identify the main stowage areas and equipment for the SAR unit:

- |                                                                                       |                      |
|---------------------------------------------------------------------------------------|----------------------|
| • Walk through of SAR / shore unit identifying main stowages and equipment. Examples: |                      |
| - survivors lifejackets                                                               | - torch              |
| - pyrotechnics                                                                        | - handheld VHF radio |
| - deck tools                                                                          | - emergency aerial.  |

#### SAR unit compartments

- **Compartments of the SAR unit.**

Training criteria. Identify individual compartments of the SAR unit:

- |                    |                             |
|--------------------|-----------------------------|
| • Fore peak.       | • Fore cabin.               |
| • Survivors cabin. | • Tanks space.              |
| • Engine room.     | • Tiller flat.              |
| • Wheelhouse.      | • ILB POD and compartments. |

#### Deck fittings

- **Deck fittings and their uses. (appropriate to SAR unit).**

Training criteria. Identify the names and uses of main fittings found on deck of the SAR unit:

- |                                       |                                    |
|---------------------------------------|------------------------------------|
| • Cleat – making off lines.           | • Fairlead – guiding lines.        |
| • Bitts / bollard – making off lines. | • Capstan – winching lines.        |
| • D-rings – making off lines.         | • Catting davit - anchor recovery. |

#### Hazards

- **Hazards on the SAR unit.**

Training criteria. State the potential hazards posed by radar and aerals, working in a confined space, fuel, and slips, trips and falls:

- Radar: Radiation, spinning transceiver, wires and lines (esp. for helicopter operations).
- Aerals: Electrocution, eye injury, broken fibreglass, RF burns.
- Fuel: Fire, dermatitis, fume inhalation, ingestion.
- Confined / enclosed spaces: Entanglement, heat, entrapment, flooding, asphyxiation.
- Slips, trips and falls: Injury, damage to equipment.

task continues

# Activity 1 - Safety and Induction

## Task 1.3.1

## SAR Unit Layout

### a Vessel layout and equipment (continued)

#### Training guidelines

Training criteria. Explain how to minimise the risk of slips, trips and falls:

- Holding on .
- Door frames.
- Good housekeeping / husbandry.

#### Crew seating

##### • Safe crew positioning and seating.

Training criteria. Demonstrate the appropriate crew position / crew seating including seat adjustment if applicable:

- ILB - transit and dynamic crew positions.
- Safe body position with appropriate handholds.
- Correct feet and leg positioning.
- Resistance setting, harness and arm rests to suit individual and environment if applicable.

## Task 1.3.1

## SAR Unit Layout

### b Alarms

#### Training guidelines

#### Recognise alarm

- **Audible and visual alarms fitted to the SAR unit.**

Training criteria. Identify by type the audible and visual alarms for the SAR unit:

- Appropriate to SAR unit.
- MOB alarm.
- Fire alarm.
- Bilge alarm.
- Engine alarms - overheat, oil pressure etc.
- DSC alarm.
- Depth alarm.
- Navigation alarm.

# Activity 1 - Safety and Induction

## Task 1.3.2

## ALB Systems and Watertight Integrity

### a ALB systems

#### Training guidelines

#### Bilge system components

- **Bilge system layout.**

Training criteria. Identify the bilge system schematic diagram:

- As fitted to the SAR unit.

Training criteria. Identify the bilge system components external to the engine room:

- Bilge watch.
- Manual pump.
- Electrical pump switch.
- In compartments valves.
- Engaging the bilge system.

Training criteria. Identify the procedure for engaging the bilge system from outside the engine room:

- Tamar, Mersey, Shannon - follow Class SOP: CAN be done externally.
- Trent, Severn - follow Class SOP: CANNOT be done externally (mechanic operated).

#### Dangers / hazards whilst in the engine rooms

- **Dangers of working in the engine room and methods to minimise them.**

Training criteria. State the hazards found in the engine room and what can be done to minimise the risk:

- Hazards include:
  - noise
  - moving machinery
  - batteries
  - fluids: fuel, coolant, battery acid, hydraulic oil
  - flooding
  - contaminated bilge water
  - high pressure systems
  - fixed firefighting systems
  - electrical systems
  - heat
  - loose straps
  - slippery surfaces
  - confined spaces.
- Measures to minimise the risk include:
  - ear defenders
  - overalls
  - removal of lifejackets and jewellery
  - contain loose hair / clothing
  - good housekeeping / husbandry
  - if fitted, look through viewing panel in engine room hatch to ensure it is safe to open.

Training criteria. State who should be notified prior to entering the engine room:

- Notify person in command.

**task continues**



## Task 1.3.2

## ALB Systems and Watertight Integrity

### b Watertight integrity

#### Training guidelines

#### Watertight doors and hatches

- **Maintaining weather and watertight integrity.**

Training criteria. Describe what is meant by weather and watertight integrity and how it is maintained within the SAR unit:

- Keep sea and rain from entering and affecting internal compartments.
- Ensuring all hatches and doors are closed.
- Door / hatch indicator lights - position.
- Green - closed.
- Red - open.

- **Maintaining watertight integrity of the SAR unit compartments.**

Training criteria. Explain what precautions must be taken when opening and closing these compartments in order to maintain watertight integrity.

Training criteria. Describe the methods for ensuring weather and watertight integrity.

Training criteria. Describe how the status of watertight doors and hatches are indicated:

- Viewing panel, open with caution, close and secure or secure in open position.

#### Escape using escape hatches

- **Emergency Evacuation simulation test.**

Training criteria. Demonstrate the ability to access, release, open, secure and exit through the escape hatches:

- Locate the escape hatch(es).
- Unstow and secure ladder (where fitted).
- Open escape hatch(es).
- Secure escape hatch(es).
- Exit via escape hatch(es).

#### Self-righting

- **Watertight integrity and self-righting.**

Training criteria. Describe the effect on self-righting capability of leaving watertight doors and hatches open:

- Flooding:
  - instability may induce capsize (ALB - stability maintained with one compartment flooded)
  - may prevent self-righting once capsized.

# Activity 1 - Safety and Induction

## Task 1.4.1

## Roles and Responsibilities 1

### a Identify and communicate with personnel

#### Training guidelines

##### RNLI structure and key personnel

Training criteria. Identify key station personnel and their roles:

- LOM.
- Head launcher.
- Launch authority.
- LVDs.
- LPO.
- Coxswain / helm / commander.
- Water safety officer (WSO).
- Mechanics.
- Lifeboat trainer assessor.
- Lifeboat training coordinator.

##### RNLI communication structure

##### • Communications chain under normal and emergency situations.

Training criteria. State who to notify of normal exercise, service and maintenance issues:

- LOM / launch authority.
- Coxswain.
- Mechanic (including boat and vehicle).
- Coordinating authority.

Training criteria. State who to notify in emergency / incident situations:

- LOM / station manager.
- Central operations and information room (operations room).
- Coxswain / helm / commander.
- Head launcher.

##### • Operational Communication format.

Training criteria. Explain the briefing and debriefing format:

- Use recognised format for briefing / debriefing all operational duties including exercises and operational launches.
- Used with the exercise planning form and led by the command roles.
- Important to ensure all are briefed and aware of roles and responsibilities.

task continues

## Task 1.4.1

## Roles and Responsibilities 1

### b Roles, responsibilities and limitations

#### Training guidelines

##### Personal role relevant to development plan

###### • Role

Training criteria. Describe the role relevant to the position held in line with appropriate development plan:

- As identified in volunteer role profile.
- Command roles- lead and command in line with current guidance documents.
- All roles- assist and support operations and duties under the guidance of command roles.

###### • Attendance / Commitment Requirements.

Training criteria. State RNLI minimum attendance, station exercise routine:

- Reference volunteer commitment document.
- Reference attendance guidance.

###### • Competence Requirements.

Training criteria. Identify the tasks required by the training plan and explain how to achieve competence:

- Recognise framework.
- Training through station training and courses.
- Competence achieved through training and assessment.
- Tasks in training plan.

##### Limitations

###### • Personal limitations.

Training criteria. Describe the limitations of your role and possible consequences of exceeding them:

- Describe Limitations in line with competence status, training and role.
- Risks associated with exceeding them - confusion, conflict, unreal expectations.

##### Periods of non-availability

###### • Informing of non-availability.

Training criteria. Explain who, when and why to inform of periods of non-availability:

- Inform LOM / station manager / launch authority / SAR unit commander / head launcher as soon as possible.
- Use of RNLI callout and messaging system (RCAMS) if applicable.
- Important to ensure appropriate SAR unit-manning numbers.

##### Pager operations and response

###### • Operation, maintenance and response to pager.

Training criteria. Describe how the pager works, is maintained and the associated responsibilities whilst responding to a pager:

- Identify pager buttons.
- Obey road traffic act.
- Respond safely and appropriately.
- Care of pager.
- Ensuring battery is charged if applicable.

task continues

# Activity 1 - Safety and Induction

## Task 1.4.1

## Roles and Responsibilities 1

### c Operational safe working

#### Training guidelines

#### Policies

#### Safe working practices

- **Incident reporting.**

Training criteria. Identify who to report incidents to:

- LOM / launch authority or SAR unit commander.

- **Operating procedures.**

Training criteria. Explain the location and importance of standard, local and emergency procedures:

- Locate and explain the purpose of SOPs, LOPs and EOPs.
- The importance of adhering to and only using approved operating procedures.

- **Sea state and weather and their effect on personal safety.**

Training criteria. Explain the effects on personal safety of weather / sea state:

- Increased risk of injury.
- Importance of: securing equipment, securing PPE, appropriate positioning, appropriate hand holds.

- **Factors affecting safe SAR unit operations.**

Training criteria. Explain the factors that affect safe SAR unit operations:

- Afloat.
- Ashore.
- Weather.
- Members of the public.
- Tide.
- Traffic.
- Visibility.
- Ground conditions.
- Light level.
- Crew availability and experience.

- **Situational Awareness.**

Training criteria. State what information is important to pass to others:

- Other vessels.
- Dangerous occurrences.
- Hazards seen by you, where you are uncertain if the person in command has seen it.
- Anything that gives rise to concern / uncertainty.



## Task 1.4.2

## Roles and Responsibilities (Navigator)

### a Operational safe working

#### Training guidelines

##### Policy and guidance documents

- **Policy, procedures, and documents.**

Training criteria. State the relevant policies and procedures that apply to operational duties as a navigator:

- RNLI navigation safety policy, navigation (ALB and ILB) SOP, local knowledge LOP.
- Under direction from the person in command, ensure the craft is appropriately and safely navigated at all times.
- Maintain awareness of vessel location at all times with position fixing where and when appropriate.
- Safety of Life at Sea (SOLAS) guidelines.
- *International Aeronautical and Maritime Search and Rescue Manual (IAMSAR)* manual.
- Notice to mariners.
- MCA guidelines.

##### Up to date resources

Training criteria. Identify how to assess all resources are current and in date:

- Check with responsible person(s) for both paper and electronic charts.
- Chart corrections in bottom left hand corner on paper charts.
- Identified within system settings date of last update and chart license date.
- Understand that not all notice to mariners will be evident on charts as they may be temporary.

##### Position and intended movement (PIM)

Training criteria. Describe the purpose of a PIM and their importance. State the RNLI policy for your SAR unit:

- A position and intended movement (PIM) report is a transmission from a lifeboat to a coordinating authority, normally the coastguard, in which they report the current position and intended movement of a SAR unit.
- PIM reports are considered an essential part of the monitoring of safety for our lifeboat crew as well as allowing coordinating authorities to maintain visibility of assets.
- Frequency as per operational procedures.

# Activity 1 - Safety and Induction

## Task 1.4.3

## Roles and Responsibilities (Command)

### a Identify and communicate with personnel

#### Training guidelines

#### RNLI structure and key personnel

- **Key regional personnel.**

Training criteria. Identify key regional personnel:

- Know or know where to find the following (appropriate to own role).
- Head of region.
- Regional lifesaving lead.
- Area lifesaving manager.
- Regional technical staff.
- Regional support lead.
- Assessor Trainer.
- Regional lifesaving support coordinator.
- System support coordinator.

- **Function of the central operations and information room.**

Training criteria. Explain the duties carried out in the operations room and when you would contact the operations room:

- Central point of contact in emergency e.g. boat off-service, capsize, injury, equipment problems, passage reports, boat movements, charts.

task continues

## Task 1.4.3

## Roles and Responsibilities (Command)

### b Policies and Guidance Documentation

#### Training guidelines

#### Policy and guidance documents

- **Policy and procedures.**

Training criteria. State the relevant policies and procedures that apply to operational duties:

- RNLI policies and procedures.
- Administration procedures.
- Audit and Assurance processes.
- Safety, Health and Environment (SHE) policies and procedures.
- Maintenance processes and documentation.
- Competence framework documentation and database.

Training criteria. Demonstrate locating centralised communications regarding operations:

- Central updates and Horizon notices – discuss responsibility to ensure all have read and understood relevant information.

- **Operating Procedures.**

Training criteria. Explain the location, contents and importance of SOP:

- Follow to ensure the safety of people.
- Follow to ensure equipment is used as per design.
- The standard way of operating in normal circumstances.

Training criteria. Explain the location, contents and importance of LOP and how they are generated and approved:

- As per SOP but for a specific task and / or location.
- Developed locally in accordance with current guidance on Horizon.
- Not formally adopted until approved.

- **Risk assessments.**

Training criteria. Explain the location and importance of risk assessments:

- Follow to ensure the safety of people and equipment and part of process of producing an operating procedure.

task continues

# Activity 1 - Safety and Induction

## Task 1.4.3

## Roles and Responsibilities (Command)

### c Operational safe working

#### Training guidelines

#### Safe working practices

- **Situational awareness during operations.**

Training criteria. Explain the importance and level of situational awareness required:

- Perception - aware of what is happening around you.
- Comprehension - aware of the effect things happening around you will have.
- Projection - aware of what will happen in the future and / or the effect any actions taken will have on future events.

Training criteria. Explain how to carry out a dynamic risk assessment and the factors to consider:

- Dynamic risk assessment: a continuous process of identifying hazards, assessing risks and taking action in order to eliminate or reduce risk to personnel and vessel within operational circumstances.
- Continuous training, SOPS, knowledge and experience enable good decision making.

- **Safe operations.**

Training criteria. Describe the safety responsibilities of the role during SAR unit operations:

- Those in command have a responsibility for those they are overseeing.

- **Limitations of performance.**

Training criteria. Explain limitations on performance of both vessel and crew:

- Crew ability and experience, other SAR unit availability, SAR unit capability and limitations, proximity to hazards, duration of service, weather.
- Restricted service SAR units and ability to perform.

#### Communicate effectively

- **Briefing and debriefing.**

Training criteria. Demonstrate giving an exercise briefing to include appropriate use of the exercise planning form:

- Briefing using the SMEAC format.
- Appropriate use of the exercise planning document and recording.
- Debrief using current model.
- For exercise and services.



## Task 1.5.1

## SAR Unit Firefighting

### a Initial actions on discovering a fire

#### Training guidelines

#### Recognise fire types

- **Elements comprising the fire triangle.**

Training criteria. Identify the three elements that form the fire triangle:

- Fuel.
- Heat.
- Oxygen.

- **Identifying types of fire that may be encountered in a SAR unit:**

Training criteria. Explain the type of fire identified by its class:

- B - fuel (liquid): petrol, diesel.
- ⚡ - electrical: radar, SIMS, electronic navigation equipment.

#### Raise or recognise fire alarms

- **Recognising the fire alarm and the importance of communicating the existence of a fire to others.**

Training criteria. Identify the sound of the fire alarm.

Training criteria. Explain how to raise the alarm if no fire alarm present.

Training criteria. State the action to take on being alerted to the presence of a fire and the importance of this:

- Recognise the sound - test.
- Raise fire alarm in accordance with the current SOP.
- Actions as required in accordance with the current SOP.

#### Fire extinguishers

- **Locating a suitable extinguisher to fight each type of fire.**

Training criteria. State the types and location of the portable fire extinguishers available at the location. Training criteria. Describe the extinguisher best suited for each fire type:

- Foam: fuel / liquid fires.
- CO2: fuel / liquid fires, electrical fires.
- Powder: fuel / liquid fires, electrical fires.

#### Considerations when tackling a fire

Training criteria. Identify considerations made before tackling a fire:

- Personal safety, safety of others, how effective the action is likely to be, safer to evacuate and call for assistance.

Training criteria. Describe the hazards of enclosed spaces:

- Smoke inhalation.
- Fumes.
- Danger from explosions.

task continues

# Activity 1 - Safety and Induction

## Task 1.5.1

## SAR Unit Firefighting

### a Initial actions on discovering a fire (continued)

#### Training guidelines

##### Communication to emergency services

###### • Informing other emergency services.

Training criteria. State how to communicate the existence of a fire and to whom:

- Verbally.
- Radio to coastguard.
- Telephone.
- Alarms.
- Mayday and DSC.
- Fire and rescue service.

##### Isolation of fuel and air

###### • Locating and isolating fire flaps and fuel stops on the SAR unit (if fitted).

Training criteria. Demonstrate the correct procedure for isolating the fuel and air.

Training criteria. Explain when they should be isolated:

- Locate fuel stops / taps.
- Locate fire flaps / covers.

##### vClosing down engines

###### • Closing down engines.

Training criteria. State who should decide to close down the engines:

- Person in command.

task continues

## Task 1.5.1

## SAR Unit Firefighting

### b Evacuation procedures

#### Training guidelines

#### Escape routes / fire exits

- **Safe routes and exits.**

Training criteria. Identify safe exit routes:

- Locate safe escape routes.
- Keep escape routes and fire exits clear.

#### Muster points

- **Safe muster point / fire assembly point.**

Training criteria. Identify the safe muster point or fire assembly point for your location:

- Identify muster point.
- Consider fire hazards on way to muster point.
- Furthest upwind from fire.
- Assemble and await instructions.

- **Mustering equipment.**

Training criteria. State all equipment to be collected if afloat:

- Flares.
- Bucket.
- Radio.
- GPS.
- Water.
- Warm clothing.
- Casualty care kit.
- Consider ditching of O2, Entonox and unused extinguishers if safe to do so.

# Activity 1 - Safety and Induction

## Task 1.5.2

## Manage SAR Unit Firefighting

### a Immediate actions to be taken

#### Training guidelines

##### Closing down engines

Training criteria. Explain when closing down the engines would be appropriate:

- When safe to do so: shallows, lee shore, casualty transfer, alongside other vessels, SAR unit recovery, location of fire.

##### Dangers and precautions to be considered before accessing an enclosed space

##### • Accessing enclosed spaces, e.g. tank space, engine room.

Training criteria. Describe when it would be appropriate to access an enclosed space:

- In accordance with the current SOP.
- Very minor fire on board that can be tackled with extinguisher.
- Dangers of allowing more air into an enclosed space.

##### Portable and fixed firefighting systems

##### • Operating portable and fixed fire extinguishers (if applicable).

Training criteria. Identify and describe the firing mechanisms for the fixed firefighting system and state the actions to be taken prior to its use:

- In accordance with the current SOP.
- Consult person in command.

Training criteria. Describe when and how to operate portable fire extinguishers:

- When safe to do so.
- Likely to be effective.
- At discretion of user.
- In accordance with manufacturer's instructions.

Training criteria. Describe the dangers associated with the use and mis-use of all fire extinguishers whether fixed or portable:

- Frost burn.
- Mis-use (e.g. water on electrical).
- Inhalation of extinguishers output.
- Asphyxiation (e.g. CO2 in enclosed space).
- Limited capacity.

##### Next action

##### • Further actions.

Training criteria. Explain what further action may be required:

- Boundary cooling.
- Abandonment.
- Casualty care kit.
- Head count.

task continues



## Task 1.5.2

## Manage SAR Unit Firefighting

### b Further actions

#### Training guidelines

##### Personnel

- **Accounting for all personnel.**

Training criteria. Explain the purpose of carrying out a head count:

- Report total numbers to authorities and confirming present.

Training criteria. Describe actions to be taken if personnel are missing:

- Carry out a search.
- Report missing persons.
- Report last known location.

##### Boundary cooling

- **Uses of boundary cooling (if applicable).**

Training criteria. Describe how to carry out boundary cooling and its effect:

- Use of salvage pump, bucket etc.
- Take heat away from heat source.
- Keep adjacent / muster area cool.
- Dangers of flooding an enclosed space: stability - free surface effect.

##### Abandonment

- **Abandoning.**

Training criteria. Explain the considerations for abandonment:

- As a last resort at sea, first resort ashore.

Training criteria. Explain the abandonment procedure:

- Crew safety.
- Control of fire.
- Best option.

task continues

# Activity 1 - Safety and Induction

## Task 1.5.2

## Manage SAR Unit Firefighting

### c Fires on other vessels and vehicles

#### Training guidelines

#### Firefighting

- **RNLI firefighting policies.**

Training criteria. State the RNLI policy on fighting fires on other vessels:

- Firefighting policy on fighting fires on other vessels in accordance with the current policy:  
“It is the policy of the RNLI that the firefighting equipment carried on board the various SAR units is used solely for fighting fires aboard.”

Training criteria. State the RNLI policy on the carriage of firefighting equipment and fire fighters on board SAR units:

- Carriage of fire fighters and firefighting equipment on board RNLI SAR units in accordance with the current policy.

## Task 1.6.1

## Emergency and Survival Procedures

### a Capsize

#### Training guidelines

Completion of all parts of emergency and survival procedures is achieved by completion of the RNLI *Crew Emergency Procedures (CEP)* course, including all elements of the *STCW95 Personal Survival Techniques (PST)* certification.

#### Heavy weather considerations

- **Ensuring watertight integrity of the SAR unit.**

Training criteria. Explain the importance of maintaining watertight integrity:

- Keeping doors closed to ensure stability and self-righting capability are maintained (ALB).

#### Hazard awareness

- **Security of crew and equipment in order to minimise risk of injury during capsize.**

Training criteria. Explain why all crew should be correctly positioned / seated with correct seat adjustment (ALB):

- Assessment

Training criteria. Explain why all equipment should be correctly stowed:

- Risk of injury:
  - ALB: seat and harness correctly adjusted, use of hand holds, use of lifelines
  - ILB: correct position, posture and hand holds.
- Risk of injury from unsecured equipment.

#### Post capsize - immediate actions

- **Checks and actions required immediately post capsize.**

- **Training criteria. Demonstrate the immediate post capsize actions:**

- Checks:
  - head count
  - injury assessment
  - damage to SAR unit.
- Actions:
  - consider activating PLB
  - consider use of pyrotechnics
  - sea anchor deployed (if appropriate).

task continues

# Activity 1 - Safety and Induction

## Task 1.6.1

## Emergency and Survival Procedures

### b Righting the lifeboat

#### Training guidelines

ILBs:

#### Righting procedure

- **Lifeboat righting procedure and positioning of crew.**

Training criteria. Explain and, where appropriate, demonstrate the righting procedure of your SAR unit:

- Demonstrate ability to escape unaided from underneath capsized craft.
- Demonstrate righting procedure - in accordance with class specific SOP.
- All ILB crew to conduct practical capsize drills.

- **Reboarding procedure,**

Training criteria. Demonstrate the ability to reboard the ILB / IRH from the water:

- Reboard the ILB / IRH from the water unaided (in waves for ILB).

#### Post righting

- **Machinery restart procedure.**

Training criteria. Describe the engine re-start procedure for your SAR unit:

- In accordance with the current class specific SOP.

- **Emergency communications.**

Training criteria. Demonstrate operation of the emergency aerial:

- Locate and set up emergency aerial.

- **Considerations if unable to right the boat.**

- **Considerations before deciding whether to stand down or continue on service / exercise.**

- **Note that it is a Command responsibility to decide whether to continue or stand down.**

Training criteria. Explain what factors would be considered before continuing:

- |                                   |                                            |
|-----------------------------------|--------------------------------------------|
| • Crew welfare (delayed effects). | • Priority (self, crew, vessel, casualty). |
| • Condition of vessel.            | • Distance / location.                     |
| • Other asset availability.       |                                            |

task continues



## Task 1.6.1

## Emergency and Survival Procedures

### c Abandonment

#### Training guidelines

#### Decision process

- **Considerations for abandonment.**

Training criteria. Explain the factors which may contribute to a decision to abandon the SAR unit:

- Critical damage to vessel.
- Uncontainable fire.
- Risk to life if remain.

Training criteria. Describe abandonment options:

- "Abandonment is my last resort."
- To another vessel.
- Into the sea.
- To a liferaft.
- To a helicopter.
- Ashore.

#### Muster

- **Abandonment.**

Training criteria. Describe the equipment and actions necessary for abandonment:

- Muster in safe location, brief.
- Full PPE.
- Head count.
- Prepare liferaft (if applicable).
- Send distress.
- Equipment - answer must include the following and may include other equipment:
  - portable VHF
  - pyrotechnics and activate PLB
  - casualty care kit
  - torch.

Training criteria. Explain how to prepare the liferaft for deployment:

- Remove from stowage.
- Ensure painter secure.
- Do not deploy until instructed.

#### Evacuate

- **Deploying the liferaft.**

Training criteria. Explain when and demonstrate how to enter the liferaft:

- When instructed.
- Pull painter to activate.
- Enter feet first.
- Safe area.
- Right if inverted.
- Stay dry if possible.

Training criteria. Describe the hazards to be considered:

- Entanglement.
- Location for deployment (fire).
- Environment.
- Sharp objects.
- Consider other occupants.
- Crew separation.

task continues

# Activity 1 - Safety and Induction

## Task 1.6.1

## Emergency and Survival Procedures

### d Survival

#### Training guidelines

#### Liferaft management

##### • Managing the liferaft once on board.

Training criteria. Describe who will take command:

- Prompt discussion regarding the most able person to take command - most capable person.

Training criteria. Describe how to position yourself and others once on board.

Training criteria. Explain the factors to consider for a prolonged period prior to rescue:

- Equally spaced.
- Cut, stream, close, maintain.
- Casualty management.
- Crew welfare.
- Take sea sickness tablets.
- Set lookout.

##### • Importance of a lookout.

Training criteria. Explain the importance of maintaining a lookout at all times:

- Identify rescuer.
- Aware of hazards – big seas, wreckage, traffic, shoreline / rocks.

Training criteria. Explain methods to aid rescue:

- Torches.
- Visual fix from landmarks.
- Whistles.
- Use of portable GPS.
- Pyrotechnics.
- Position fix prior to abandonment.
- VHF.
- Electronic location devices.

Training criteria. Demonstrate Personal Locator Beacon (PLB) operation:

- Test.
- Limitations.
- Activation.
- When to use.

#### Survival techniques

##### • Surviving in cold water.

Training criteria. Demonstrate the correct body posture for cold-water survival:

- Heat Escape Lessening Position (HELP).
- Group together – use of harness.

Training criteria. Explain the methods used to increase the chances of survival:

- Correct use of PPE: visor down, cuffs closed, lifejacket inflated, dry suit vented, light positioned, whistle / flares to hand.
- Stay positive.
- Use of pyrotechnics, handheld or rocket from grab bag (when you can see your rescuer and they cannot see you, consider location).
- Group morale.
- Large target.
- Stay with flotsam.
- Consider all options.

task continues

## Task 1.6.1

## Emergency and Survival Procedures

### e Man overboard

#### Training guidelines

#### MOB

- **Initial actions to be taken for a MOB.**

Training criteria. State the actions in the event of a MOB:

- Raise alarm: shout, MOB alarm.
- Deploy Perry buoy (if applicable).
- Activate MOB function on electronic navigation equipment.
- Point if visual.
- Maintain communication to helm.

- **Further actions to be taken.**

Training criteria. State further actions that may be required, to include night and / or if MOB is lost from sight:

- Send DSC distress / mayday.
- Request other assets.

Training criteria. State additional equipment that may be used or prepared:

- Searchlights, night vision equipment, pyrotechnics.
- Rig recovery equipment.

- **MOB recovery simulation test.**

Training criteria. Demonstrate the recovery of a 35kg MOB dummy or test container from the water:

**ALB only:**

- An individual must demonstrate the ability to recover the MOB dummy from the water by using the rescue strop.

**ILB and Hovercraft only:**

- An individual must demonstrate the ability to recover the MOB dummy from the water.

task continues

# Activity 1 - Safety and Induction

## Task 1.6.1

## Emergency and Survival Procedures

### f Stabilise head to wind (ILB)

#### Training guidelines

ILB only

#### Sea anchor

- **Deploying the sea anchor.**

Training criteria. Explain a range of circumstances when the sea anchor may be deployed:

- Fouled propeller.
- Engine failure.
- Swamping.
- Propeller change.
- Helicopter transfer.

Training criteria. Demonstrate the deployment and recovery of the sea anchor:

- Inform helm.
- Awareness of the hazards of snagging and coiled line going underload.
- Deployed at full length.
- Does not independently secure to other point.
- Re-stowing - weight on top if fitted.
- Slack line.
- Communication.

## Task 1.6.2

## Manage Emergency Procedures

### a Emergency procedures

#### Training guidelines

**Note:** During a pass out trainees may or may not refer to check cards but must know that there are cards and where they are stowed without being prompted.

#### Stranding

- **Actions on stranding.**

- In accordance with the current EOP.
- Attempt to refloat from grounded / stranded position if possible.
- Actions to include:
  - sound alarm / alert all crew
  - close watertight doors (if fitted)
  - maintain VHF watch on 16 / 0 and confirm SAR unit position
  - exhibit lights / shapes and make appropriate sound signals
  - switch on deck lighting at night
  - check for hull damage
  - sound bilges and tanks
  - visually inspect compartments where possible
  - sound around SAR unit and determine which way deep water lies
  - determine the nature of seabed and local tidal information
  - reduce draught of SAR unit (if possible / needed)
  - contact coastguard using appropriate transmission (distress / urgency).

#### Collision

- **Actions on collision.**

Training criteria. Explain the immediate actions required by your role in the event of collision of the SAR unit:

- In accordance with the current EOP.
- Priority is to prevent loss of life or injury.
- Actions to include:
  - sound alarm / alert all crew
  - manoeuvre the ship so as to minimise effects of collision
  - close watertight doors (if fitted)
  - switch on deck lighting at night
  - maintain VHF watch on 16 / 0 and confirm SAR unit position
  - sound bilges and tanks for damage
  - check for fire / damage
  - offer assistance to other vessel - if object hit was another vessel
  - contact coastguard using appropriate transmission (distress / urgency).

**task continues**



# Activity 1 - Safety and Induction

## Task 1.6.2

## Manage Emergency Procedures

### a Emergency procedures (continued)

#### Training guidelines

#### Flooding

- **Actions on flooding.**

Training criteria. Explain the immediate actions required by your role in the event of flooding of the SAR unit:

- In accordance with the current EOP.
- Priority is to prevent further flooding and loss of buoyancy and stability.
- Actions to include:
  - sound alarm / alert all crew
  - close watertight doors (if fitted)
  - sound bilges and tanks
  - identify location of incoming water
  - cut of electrical supply running through area (if possible)
  - shore up area to stem flow - bungs / patches / improvised tools
  - check bilge pump for operation (if fitted)
  - check auxiliary pumps for operation (if fitted / used)
  - maintain VHF watch on 16 / 0 and confirm SAR unit position
  - contact coastguard using appropriate transmission (distress / urgency).

## Task 1.6.3

## Dynamic Water Survival

### a Hazards

#### Training guidelines

#### Physical hazards

Training criteria. Describe hazards and which may be present in fast flowing water:

- Debris.
- Eddies and whirlpools.
- Submerged obstacles.
- Entrapment.
- Collision with physical structures.

Training criteria. Identify the specific physical and structural hazards in your local area:

- May include:
  - bridges
  - jetties
  - weirs
  - strainers.
  - pontoons
  - moored craft
  - locks

#### Biological hazards

Training criteria. State the biological hazards which may be present in rivers and estuaries:

- Hazardous chemicals.
- Sewage.
- Effluent.
- Waterborne diseases.

Training criteria. State examples of diseases and illnesses you may be exposed to in the water:

- Weil's Disease (Leptospirosis).
- Hepatitis A.
- Hepatitis B.
- Tetanus.
- Typhoid.
- Poliomyelitis.
- Typhoid (Salmonella).

Training criteria. Describe possible symptoms:

- Vary greatly but include: flu-like symptoms, headaches, nausea, abdominal discomfort or diarrhoea, fevers, joint pains, muscle pains.

Training criteria. Describe precautions which must be taken:

- Hygiene precautions (washing hands and face), avoid eating, drinking and smoking, decontamination if immersed.

Training criteria. Describe actions if symptoms arise:

- Seek immediate medical help.
- Explain that it may be a result of contact with contaminated water. Inform station management.

task continues

# Activity 1 - Safety and Induction

## Task 1.6.3

## Dynamic Water Survival

### b Self-rescue techniques

#### Training guidelines

#### Self rescue and swimming in moving water

Training criteria. Defensive swimming. Describe the techniques and posture employed in defensive swimming:

- Feet up and pointed downstream.
- Keep hips up and use backward sweeps of arms for control.

Training criteria. Ferry glide. Describe the purpose of ferry gliding and the technique involved:

- Set body at an angle to the current to manoeuvre sideways across the flow.
- Up to 45 degrees.

Training criteria. Aggressive swimming. Describe the techniques of aggressive swimming and when aggressive swimming should be employed:

- Roll on to front and front-crawl with head upstream.
- Used to manoeuvre quickly to an area of safety or egress.

Training criteria. Breaking into an eddy. Describe when and how one would break into an eddy:

- To reach an area of still or calm water, possibly with exits from the water.
- Technique is to manoeuvre towards the eddy then aggressively swim and, at the eddy line, place an arm deep into the water.
- The current will turn the body into the eddy.

Training criteria. Avoiding Strainers. Describe actions to deal with strainers:

- Avoid altogether if possible.
- If it cannot be avoided, swim aggressively towards the strainer, gain momentum and use both arms to get over the strainer, keeping legs and feet high in the water.

Training criteria. Wave train. Explain what your breathing technique would be when negotiating a wave train:

- Swim defensively and take deep breaths in the troughs to avoid inhaling water.

task continues

## Task 1.6.3

## Dynamic Water Survival

### c Rescue techniques

#### Training guidelines

The following may be used along with, or instead of, normal person recovery to the SAR unit techniques.

#### Reach rescue

- **Uses of a reach rescue.**

Training criteria. Explain the benefits of a reach rescue:

- No need to enter the water.
- Minimal risk to rescuer.

Assessment criteria. Explain a potential disadvantage:

- Relies on the assistance of the casualty.
- Throw bag rescue

- **Deploying a throw bag.**

Training criteria. Demonstrate the effective use of a throw bag:

- Good line management; neat coils, no snag hazards.
- Good stance with good footing.
- Correct length of line for given rescue.
- Types of throw; over arm, under arm, sideways.
- Entrapment

Training criteria. Describe this risk associated with a body entrapment:

- Problems with maintaining airway.
- Other medical considerations, e.g. hypothermia etc.

Training criteria. Explain what actions can be taken to reduce the likelihood of entrapment:

- Don't put feet down if washed away.
- Take appropriate action when dealing with a strainer.

# Activity 1 - Safety and Induction

## Task 1.7.1

## Pyrotechnics

### a Safe use of pyrotechnics

#### Training guidelines

#### Characteristics

- **Range of pyrotechnics carried.**

Training criteria. Identify the range of pyrotechnics available to the SAR unit:

- Personal day / night distress signal.
- Red pinpoint hand flare.
- White parachute rocket (illuminating).
- Red parachute rocket.
- Line thrower.

- **Characteristics and applications (appropriate to SAR unit).**

Training criteria. Describe the features of a line thrower and when it would be used:

- Used as a messenger line, ship to ship or ship to shore.
- Fire in accordance with the current SOP.

Training criteria. Describe the characteristics of a red parachute rocket and when it would be used:

- Distress only for long range signalling.
- Fire in accordance with the current SOP.

Training criteria. Describe the characteristics of a white parachute rocket (illuminating) and when it would be used:

- Collision avoidance.
- Fire in accordance with the current SOP.
- Illumination for SAR operations.

Training criteria. Describe the characteristics of a red hand flare:

- Distress only.
- Fire in accordance with the current SOP.

Training criteria. Describe the characteristics of a handheld day / night distress signal:

- Day - orange smoke, 4 ribs, brown cap - helicopter and distress.
- Night - red pinpoint, 8 ribs, red cap - distress only.
- Fire in accordance with the current SOP.

#### Hazard awareness

- **Factors to consider before firing pyrotechnics.**

Training criteria. Explain who must be informed and the checks to be made prior to firing:

Training criteria. Identify the position to take prior to firing and describe appropriate PPE:

- Position in safe area - away from fuel sources, other persons.
- Inform coordinating authority.
- Helicopter and other overhead hazards.
- Full PPE, visor down.
- Fire in accordance with the current SOP.
- Back to wind.

#### Deployment

- **Firing methods.**

Training criteria. Describe how to fire all the pyrotechnic devices at the location relevant to the role:

- For all pyrotechnics at location appropriate to role.
- In accordance with the current SOP.

task continues



## Task 1.7.1

## Pyrotechnics

### b Misfire and out of date pyrotechnics

#### Training guidelines

#### Misfire procedure

- **Action to be taken in the event of a misfire and the information to be recorded.**

- **Training criteria. Explain the initial actions.**

Training criteria. State who must be informed.

Training criteria. State the information to be noted and who this should be passed to.

Training criteria. Describe the dangers that may be encountered should a pyrotechnic misfire:

- In accordance with the current SOP.
- Record: lot number, date of manufacture and date of expiry.
- Pass information to coordinating authority.
- Risk of injury.
- Fatality.

#### Out of date pyrotechnics

- **Disposing of out-of-date pyrotechnics.**

Training criteria. Describe the RNLI procedure for the disposal of out of date pyrotechnics:

- In accordance with the current SOP.

# Activity 1 - Safety and Induction

## Task 1.8.1

## Risk Assessment

### a Risk assessment awareness

#### Training guidelines

#### Reading and using

##### • Using risk assessments.

Training criteria. State the location and purpose of risk assessments relevant to role and location:

- Hard copies - beware that once printed there is the potential to go out of date as changes are implemented to the system.
- Centrally or location held and maintained risk assessments on RNLI systems.
- Used to measure the risks involved in RNLI duties to reduce and / or remove the risks that are reasonably foreseeable.

Training criteria. Explain the types and format of risk assessments and how to apply them to role and location:

- As per current format and guidelines.
- Generic risk assessments - held centrally for broad spectrum of tasks, activities and locations.
- Task / activity risk assessments - Held at location and / or centrally focused on specific tasks.
- Dynamic risk assessment - done by competent people faced with reasonable unforeseeable events or risks. An assessing process done and recorded per task, per event.

Training criteria. Describe the terminology used in risk assessments:

- In accordance with current format and terminology used.

Training criteria. Explain the importance of adhering to risk assessments:

- To reduce the risks to RNLI people and / or equipment and environment whilst undertaking RNLI duties.
- In the event there is no risk reduction measures in place, or the risk seems high, then stop and report.

## Task 1.9.1

## Restricted and Confined Spaces

### a Restricted spaces

#### Training guidelines

#### What is a restricted space?

Training Criteria. State what constitutes a restricted space:

- Restricted spaces are enclosed, or partially enclosed, spaces not intended for continuous human occupation. Restricted spaces may not be inherently dangerous but under certain circumstances can become dangerous.

#### Restricted or confined space?

Training Criteria. Give examples of restricted spaces and when they may become confined spaces:

- Restricted spaces may include engine rooms, machinery spaces or plant rooms. If conditions are such that any of the hazards associated with confined spaces are introduced, then a restricted space becomes a confined space.

#### Control measures

Training Criteria. Describe some control measures which may be in place:

- Control measures may include signs and locks, as well as procedural restrictions, for example seeking permission from the coxswain or mechanic before entering an engine room.

#### Casualty vessels

Training Criteria. State that the hazards and control measures described above apply equally to casualty vessels and that the same precautions are to be followed:

- Crew must seek permission from the SAR unit commander before going below decks on a casualty vessel and should only do so if absolutely necessary. They should also, if possible, ask the casualty vessel crew if there are any hazards or hazardous substances before doing so.

task continues

# Activity 1 - Safety and Induction

## Task 1.9.1

## Restricted and Confined Spaces

### b Confined spaces

#### Training guidelines

##### Confined spaces

Training criteria. Confirm having read and understood RNLI confined space policy:

- Confirm having completed *Confined Spaces* eLearning.

##### Hazards

Training criteria. Identify the hazards associated with confined spaces:

- Lack of oxygen.
- Gases, fumes and vapours.
- Flooding.
- Dust.
- Fire and explosion.
- Access restrictions.

##### Regulations

Training criteria. State where to find RNLI SHE confined space procedure:

- Horizon, SHE homepage, confined space policy.

##### Emergencies

Training criteria. Explain the emergency procedures for confined spaces:

- In line with current RNLI operating procedures.

## Task 1.10.1

## Working at Height

### a Working at Height

#### Training guidelines

#### Working at height

Training criteria. Confirm having read and understood RNLI working at height policy (Horizon):

- Confirm having completed *RNLI Working at Height* eLearning.



## Activity 3 - Launch and Recovery

Photo: RNLI / Nigel Millard



### Tasks

|            |                                          |
|------------|------------------------------------------|
| <b>3.1</b> | <b>Launch and Recovery (Afloat Crew)</b> |
| 3.1.1      | Launch Procedures (Afloat Crew)          |
| 3.1.2      | Recovery Procedures (Afloat Crew)        |
| 3.1.3      | Manage Launch and Recovery (Afloat Crew) |
| <b>3.2</b> | <b>Launch and Recovery (Shore Crew)</b>  |
| 3.2.1      | Launch Procedures (Shore Crew)           |
| 3.2.2      | Recovery Procedures (Shore Crew)         |
| 3.2.3      | Manage Launch and Recovery (Shore Crew)  |
| <b>3.3</b> | <b>Plant Operator</b>                    |
| 3.3.1      | Plant Operator                           |
| <b>3.4</b> | <b>Authorising Launches</b>              |
| 3.4.1      | Authorising Launches                     |

## Activity 3 - Launch and Recovery

### Task 3.1.1

### Launch Procedures (Afloat Crew)

#### a Embarkation and launch

##### Training guidelines

##### Prepare for launch

- **Carrying out pre-launch checks.**

Training criteria. Demonstrate appropriate checks to be carried out prior to launching:

- In accordance with the current class specific SOP and LOP including:
  - location of launch
  - equipment prepared
  - briefing received on role and hazards.

##### Embarkation

- **Embarking the SAR unit.**

Training criteria. Identify the embarkation points and explain the reasons for their use:

- In accordance with the current class specific SOP and LOP.

Training criteria. Demonstrate embarkation of the boat:

- Board the boat safely and unaided.

Training criteria. Describe the hazards associated with embarkation and the precautions to be taken:

- Falls, slips.
- Height.
- Entanglement.
- Footing and securing of the ladder.

Training criteria. Demonstrate the correct signals for launching procedures:

- In accordance with the current SOP and LOP.
- As stated in the role handbook and equipment specific manual.
- To include:
  - hand signal
  - VHF
  - whistle
  - horn.

##### Use of launch equipment

- **Using authorised station equipment.**

Training criteria. Demonstrate the correct procedure and follow instruction for launching the SAR unit:

- In accordance with the class specific SOP and LOP.
- Demonstrate correct launch procedures under the direction of the coxswain / helm / commander.

Training criteria. State the risks associated with the launching procedure:

- Moving equipment.
- Overhead hazards.
- Lines under load.

Training criteria. Explain the safe location for transiting pre-launch on the SAR unit:

- Away from hazards and visible to those overseeing operations.
- Approved position.
- Ideally seated.
- Always with good hand holds.
- In accordance with the class specific SOP and LOP.

task continues

### Task 3.1.1

### Launch Procedures (Afloat Crew)

#### a Embarkation and launch (continued)

##### Training guidelines

- **Launch equipment failure.**

Training criteria. Describe actions to be taken for a range of equipment failure:

- Notify head launcher / coxswain / helm / commander.
- Carry out any other tasks as directed by the coxswain / helm / commander.

##### Secure equipment following launch of the SAR unit

- **Securing launch equipment.**

Training criteria. Demonstrate the correct procedures for securing launch equipment:

- Secure lines / tools.
- In accordance with the class specific SOP and LOP.

task continues

## Activity 3 - Launch and Recovery

### Task 3.1.1

### Launch Procedures (Afloat Crew)

#### **b** Shore crew competences for afloat crew

##### Training guidelines

Afloat crew who do not hold a shore crew plan should be competent in the following:

##### **Prepare for launch**

- **Carrying out pre-launch checks (shore crew).**

Training criteria. Demonstrate appropriate checks to be carried out prior to launching:

- In accordance with the current class specific SOP and LOP including:
  - location of launch
  - equipment prepared
  - briefing received on role and hazards.

##### **Use of launch equipment**

- **Using authorised station equipment.**

Training criteria. Demonstrate the correct procedure and follow instruction for launching the SAR unit:

- In accordance with the current class specific SOP and LOP.
- Shore crew - demonstrate correct procedures and following instructions from head launcher.

Training criteria. State the hazards associated with the launching:

- Moving equipment.
- Lines under load.
- Overhead hazards.

- **Launch equipment failure.**

Training criteria. Describe actions to be taken for a range of equipment failure:

- Notify head launcher / SAR unit commander.
- Carry out any tasks as directed by the person in charge.
- In accordance with the class specific SOP and LOP.

##### **Secure equipment following launch of the SAR unit**

- **Securing launch equipment.**

Training criteria. Demonstrate the correct procedures for securing launch equipment:

- Secure lines / tools.
- Re-couple trailer / carriage to launch plant.
- In accordance with the class specific SOP and LOP.



## Task 3.1.2

## Recovery Procedures (Afloat Crew)

### a Recovery and disembarkation

#### Training guidelines

#### Recovery preparation

- **Carrying out pre-recovery checks.**

Training criteria. Demonstrate the appropriate checks to be carried out prior to recovery:

- In accordance with the current class specific SOP and LOP including:
  - location of recovery
  - equipment prepared
  - briefing received on role and hazards.

- **Preparing the equipment for recovery.**

Training criteria. Identify the recovery equipment.

Training criteria. Demonstrate rigging the recovery equipment:

- In accordance with the current class specific SOP and LOP including:
  - positioning of equipment
  - preparing skids
  - lines and wires
  - haul up spans
  - quarter stoppers
  - davit
  - arrester springs
  - positioning of crew.

#### Recovery

- **Recovering SAR unit.**

Training criteria. Demonstrate your role in recovering the SAR unit:

- In accordance with class specific SOP and LOP.

Training criteria. State the hazards associated with the recovery procedure:

- Moving equipment.
- Lines under load.
- Overhead hazards.

- **Recovery equipment failure.**

Training criteria. Describe actions to be taken for a range of equipment failure:

- Notify head launcher / coxswain / helm / commander.
- Carry out other tasks as directed by the person in charge.

#### Disembark

Training criteria. Identify the disembarkation points and explain the reasons for their use:

- In accordance with the current class specific SOP and LOP.
- Safe and controlled manner.

Training criteria. Describe the hazards associated with disembarkation and the precautions to be taken:

- Falls.
- Slips.
- Entanglement.
- Height.
- Foot and secure the ladder.
- Ensure equipment is not moved until crew safely disembarked.

task continues

## Activity 3 - Launch and Recovery

### Task 3.1.2

### Recovery Procedures (Afloat Crew)

#### a Recovery and disembarkation (continued)

##### Training guidelines

Training criteria. Demonstrate the correct signals for recovery procedures:

- In accordance with the current SOP and LOP.
- As stated in the role handbook and equipment specific manual.
- To include:
  - hand signal
  - VHF
  - whistle
  - horn.

Training criteria. Demonstrate disembarkation of the boat:

- Disembark the boat safely and unaided.

##### Secure recovery equipment

##### • Securing recovery equipment.

Training criteria. Demonstrate the correct procedures for securing recovery equipment:

- As directed by the person in command:
  - secure lines
  - re-couple carriages
  - in accordance with the current class specific SOP and LOP.

##### Post recovery and operational readiness

##### • Post recovery checks.

Training criteria. Demonstrate the correct procedures for post recovery:

- Follow post recovery checks for SAR unit.
- Report defects.
- Clean and re-stow equipment used.
- Replace equipment used.
- Report refuelled and ready for service to coordinating authority.

task continues

### Task 3.1.2

### Recovery Procedures (Afloat Crew)

#### b Operational readiness

##### Training guidelines

##### Fuelling

###### • Procedure.

Training criteria. Explain the correct fuelling procedure for the SAR unit:

- In accordance with the current SOP and LOP.
- Identify required fuelling point on the SAR unit.
- Have all necessary equipment associated with the task to hand.

Training criteria. State the PPE required for fuelling:

- Eye protection.
- Coveralls.
- Disposable gloves.

###### • Fire precautions.

Training criteria. Explain the precautions to be taken when refuelling the SAR unit:

- Fire.
- Naked lights.
- Spillage.
- Restricted access.
- Vapour.
- Have suitable fire extinguishers to hand.

###### • Limiting fuel spillage.

Training criteria. Explain the procedure for dealing with fuel spillages:

- Isolate source of spillage.
- Locate spill kit.
- Follow spill kit manufacturer's user instructions.
- Dispose of contaminated items in accordance with RNLI waste management procedure.

##### Post recovery

###### • Post recovery checks.

Training criteria. Demonstrate the correct procedures for post recovery:

- Follow post recovery checks for SAR unit.
- Report defects.
- Clean and re-stow equipment used.
- Replace equipment used.
- Report refuelled and ready for service to coordinating authority.

task continues

# Activity 3 - Launch and Recovery

## Task 3.1.2

## Recovery Procedures (Afloat Crew)

### c Shore crew competences for afloat crew

#### Training guidelines

Afloat crew who do not hold a shore crew plan should be competent in the following:

#### Recovery preparation

- **Carrying out pre-recovery checks.**

Training criteria. Demonstrate the appropriate checks to be carried out prior to recovery:

- In accordance with the current class specific SOP and LOP including:
  - location of recovery
  - equipment prepared
  - briefing received on role and hazards.

- **Preparing the equipment for recovery.**

Training criteria. Identify the recovery equipment.

Training criteria. Demonstrate rigging the recovery equipment:

- In accordance with the current class specific SOP and LOP including:
  - positioning of equipment
  - lines and wires
  - quarter stoppers
  - arrester springs
  - preparing skids
  - haul up spans
  - davit
  - positioning of crew.

#### Recovery

- **Recovering SAR unit.**

Training criteria. Demonstrate your role in recovering the SAR unit:

- In accordance with class specific SOP and LOP.

Training criteria. State the hazards associated with the recovery procedure:

- Moving equipment.
- Overhead hazards.
- Lines under load.
- Risk of injury.

- **Recovery equipment failure.**

Training criteria. Describe actions to be taken for a range of equipment failure:

- Notify head launcher / coxswain / helm / commander.
- Shut down and secure all equipment.
- In accordance with the class specific SOP and LOP.
- Aware of safety, health and environment responsibilities.

#### Secure recovery equipment

- **Securing recovery equipment.**

Training criteria. Demonstrate the correct procedures for securing recovery equipment:

- As directed by the person in command.
- Secure lines.
- Re-couple carriages.
- In accordance with the current class specific SOP and LOP.

### Task 3.1.3

### Manage Launch and Recovery (Afloat Crew)

#### a Initiate launch

##### Training guidelines

##### Launch procedures

- **Authority to launch.**

Training criteria. State who has the authority to initiate the launch of the SAR unit:

- Qualified launch authority.
- Coxswain / helm / commander when self-launching.

- **Authorising launch.**

Training criteria. Describe when a launch can be authorised:

- Training exercise.
- Passage.
- Machinery trials.
- Service.
- PR launch.

- **Self-launching.**

Training criteria. Explain the considerations before self-launching:

- Delay would result in a loss of life.
- SAR unit within operational limits.
- Crew experience / ability.
- Launch authority informed - even if afloat on exercise through coastguard.

- **Informing relevant personnel prior to launch.**

Training criteria. State who should be informed prior to launching under normal circumstances:

- Lifeboat operations manager.
- Launch authority.
- Coordinating authority.

##### Choice of SAR unit

- **Limitations of the SAR unit.**

Training criteria. State the considerations to be made when deciding if a launch is appropriate:

- Is a more suitable asset available?
- Weather / sea state.
- Day / night.
- Crew ability.
- Casualty type.
- Casualty numbers.
- *RNLI Policies and Guidelines*
- Tidal conditions.
- Ability to put casualties ashore.

Training criteria. State when a launch authority is required to be at station for launches:

- Night time launches for D class lifeboats.

task continues



## Activity 3 - Launch and Recovery

### Task 3.1.3

### Manage Launch and Recovery (Afloat Crew)

#### a Initiate launch (continued)

##### Training guidelines

##### Selection of crew

- **Limitations of the crew.**

Training criteria. State the considerations to be made when selecting the appropriate crew:

- All required roles filled.
- Physical ability.
- Experience.
- Crew competence.
- Task specific competence.
- Sea worthiness (sea sickness).

Training criteria. State the minimum and maximum crewing levels required for the relevant SAR unit:

- Knowledge of minimum and maximum crewing levels.

##### Choice of launch and recovery site

- **Methods.**

Training criteria. State all the launch and recovery methods available:

- In accordance with the current SOP and LOP.

- **Locations.**

Training criteria. State all of the locations approved for launch and recovery including alternative sites:

- In accordance with the current LOP.
- Normal launch / recovery site or alternative launch / recovery site / method, e.g. net recovery.

Training criteria. Explain local limitations and restrictions affecting safe launch and recovery at approved sites:

- In accordance with the current LOP.
- Weather.
- Tide / sea state.
- Beach conditions.
- Required crew numbers.
- Public access and control.
- Location of alternative launch sites.

task continues

### Task 3.1.3

### Manage Launch and Recovery (Afloat Crew)

#### b Safe working practices

##### Training guidelines

##### Safe working practices

##### • Responsibilities

Training criteria. Explain roles and responsibilities during launch and recovery operations:

- As stated in the role handbook.

##### Safety measures

##### • Embarking / disembarking the SAR unit

Training criteria. Explain the embarkation / disembarkation points and the reason for their use:

- In accordance with the current class specific SOP and LOP.
- with due regard to safety and hazards at all times.
- Hazards include:
  - falls
  - slips
  - entanglement
  - height.

##### • Safety of crew.

Training criteria. Explain the importance of ensuring crew don appropriate PPE:

- Protection.
- Visibility and identification during operations.

Training criteria. Demonstrate briefing and ensuring crew stay clear of hazards:

- As per current briefing model.
- Raise awareness of hazards.
- Raise awareness of hazardous and no-go areas.

task continues

## Activity 3 - Launch and Recovery

### Task 3.1.3

### Manage Launch and Recovery (Afloat Crew)

#### c Manage launch and recovery

##### Training guidelines

##### Pre and post launch

- **Checks.**

Training criteria. Demonstrate managing pre and post launch checks:

- In accordance with the current class specific SOP and LOP.

##### Launch and recovery

- **Managing.**

Training criteria. Demonstrate managing launch and recovery operations:

- In accordance with the current class specific SOP and LOP.
- Procedure followed in a safe manner.
- Safely and appropriately managing appropriate roles involved.

Training criteria. Explain the actions to be taken for a range of emergencies, including use of available resource:

- In accordance with the current class specific SOP, LOP and relevant check cards.
- Use only authorised launch and recovery equipment, e.g. in bog down.

##### Post recovery

- **Post recovery checks.**

Training criteria. Explain the post recovery procedures:

- Post recovery check list.
- Wash down SAR unit and all used equipment.
- Correctly re stow any used equipment.
- Refuel.
- Report any defective equipment.
- Report ready for service to launch authority..

Training criteria. Explain the importance of post recovery procedures:

- Preparedness for future services.
- Defect identification.
- Equipment maintenance.

Training criteria. Demonstrate managing post recovery procedures:

- In accordance with the SOP and LOP.
- In accordance with the current post recovery checklist.
- To include:
  - debrief
  - TRiM and welfare
  - incidents
  - reports.

task continues

### Task 3.1.3

### Manage Launch and Recovery (Afloat Crew)

#### d Alternative launch and recovery sites

##### Training guidelines

##### Alternative launch and recovery sites

- **Locating alternative launch sites.**

Training criteria. State the location and distance from station to the alternative launch and recovery sites:

- State the approved alternative launch site(s) (if applicable).

Training criteria. State the restrictions on using them:

- People - RNLI personnel required and trained, members of the public.
- Traffic - on route and at the launch site.
- Plant - the use of specific plant at that site.
- Environment - weather / sea state, access, ground.

- **Using alternative launch sites.**

Training criteria. State the considerations for operating from alternative launch and recovery sites:

- In accordance with the current SOP and LOP.
- Safety of transit: control vehicles, communications, access routes, permission.
- Due regard for Road Traffic Act.
- Crew briefing.
- Inform coordinating authority.

##### Alternative launch procedure

- **Alternative procedures for launching.**

Training criteria. State the procedures for alternative launch and recovery of SAR units:

- In accordance with the current LOP.
- Launch from within boathouse.
- Alternative vehicles.
- Alternative launch in the event of mechanical / electrical failure.

## Activity 3 - Launch and Recovery

### Task 3.2.1

### Launch Procedures (Shore Crew)

#### a Launch procedure

##### Training guidelines

##### Prepare for launch

- **Carrying out pre-launch checks.**

Training criteria. Demonstrate appropriate checks to be carried out prior to launching:

- In accordance with the current class specific SOP and LOP including:
  - location of launch
  - equipment prepared
  - briefing received on role and hazards.

##### Use of launch equipment

- **Using authorised station equipment.**

Training criteria. Demonstrate the correct procedure and follow instruction for launching the SAR unit:

- In accordance with the current class specific SOP and LOP.
- Shore crew - demonstrate correct procedures and following instructions from head launcher.

Training criteria. State the hazards associated with the launching:

- Moving equipment.
- Lines under load.
- Overhead hazards.
- Risk of injury.

- **Launch equipment failure.**

Training criteria. Describe actions to be taken for a range of equipment failure:

- Notify head launcher / coxswain / helm / commander.
- Carry out any tasks as directed by the person in charge.
- In accordance with the class specific SOP and LOP.

##### Secure equipment following launch of the SAR unit

- **Securing launch equipment.**

Training criteria. Demonstrate the correct procedures for securing launch equipment:

- Secure lines / tools.
- Re-couple trailer / carriage to launch plant.
- In accordance with the class specific SOP and LOP.



## Task 3.2.2

## Recovery Procedures (Shore Crew)

### a Recovery procedure

#### Training guidelines

#### Recovery preparation

- **Carrying out pre-recovery checks.**

Training criteria. Demonstrate the appropriate checks to be carried out prior to recovery:

- In accordance with the current class specific SOP and LOP including:
  - location of recovery
  - equipment prepared
  - briefing received on role and hazards.

- **Preparing the equipment for recovery.**

Training criteria. Identify the recovery equipment.

Training criteria. Demonstrate rigging the recovery equipment:

- In accordance with the current class specific SOP and LOP including:
  - positioning of equipment
  - preparing skids
  - lines and wires
  - haul up spans
  - quarter stoppers
  - davit
  - arrester springs
  - positioning of crew.

#### Recovery

- **Recovering SAR unit.**

Training criteria. Demonstrate your role in recovering the SAR unit:

- In accordance with class specific SOP and LOP.

Training criteria. State the hazards associated with the recovery procedure:

- Moving equipment.
- Lines under load.
- Overhead hazards.
- Risk of injury.

- **Recovery equipment failure.**

Training criteria. Describe actions to be taken for a range of equipment failure:

- Notify head launcher / coxswain / helm / commander.
- Shut down and secure all equipment.
- In accordance with the class specific SOP and LOP.
- Aware of safety, health and environment responsibilities.

#### Secure recovery equipment

- **Securing recovery equipment.**

Training criteria. Demonstrate the correct procedures for securing recovery equipment:

- As directed by the person in command.
- Secure lines.
- Re-couple carriages.
- In accordance with the current class specific SOP and LOP.

task continues

## Activity 3 - Launch and Recovery

### Task 3.2.2

### Recovery Procedures (Shore Crew)

#### a Recovery procedure (continued)

##### Training guidelines

##### Disembark

- **Disembarking the SAR unit.**

Training criteria. Identify the disembarkation points and explain the reasons for their use:

- In accordance with the current class specific SOP and LOP.
- Safe and controlled manner.

Training criteria. Describe the hazards associated with disembarkation and the precautions to be taken:

- Falls, slips.
- Entanglement.
- Height.
- Foot and secure the ladder at all times.
- Ensure equipment is not moved until crew safely disembarked.

Training criteria. Demonstrate the correct signals for recovery procedures:

- In accordance with the current SOP and LOP.
- As stated in the role handbook and equipment specific manual.
- To include:
  - hand signal
  - VHF
  - whistle
  - horn.

task continues

### Task 3.2.2

### Recovery Procedures (Shore Crew)

#### b Operational readiness

##### Training guidelines

##### Fuelling

- **Managing fuelling.**

Training criteria. Explain the correct fuelling procedure for the SAR unit:

- In accordance with the current SOP and LOP.
- Identify required fuelling point on the SAR unit.
- Have all necessary equipment associated with the task to hand.

Training criteria. State the PPE required for fuelling:

- Eye protection.
- Coveralls.
- Disposable gloves.

- **Fire precautions.**

Training criteria. Explain the precautions to be taken when refuelling the SAR unit:

- Fire.
- Naked lights.
- Spillage.
- Restricted access.
- Vapour.
- Have suitable fire extinguishers to hand.

- **Limiting fuel spillage.**

Training criteria. Explain the procedure for dealing with fuel spillages:

- Isolate source of spillage.
- Locate spill kit.
- Follow spill kit manufacturer's user instructions.
- Dispose of contaminated items in accordance with RNLI waste management procedure.

##### Post recovery

- **Post recovery checks.**

Training criteria. Demonstrate the correct procedures for post recovery:

- Follow post recovery checks for SAR unit.
- Report defects.
- Clean and re-stow equipment used.
- Replace equipment used.
- Report refuelled and ready for service to coordinating authority.

## Activity 3 - Launch and Recovery

### Task 3.2.3

### Manage Launch and Recovery (Shore Crew)

#### a Initiate launch

##### Training guidelines

##### Launch procedures

- **Authority to launch.**

Training criteria. State who has the authority to initiate the launch of the SAR unit:

- Lifeboat operations manager (if a qualified launch authority).
- Any other qualified launch authority.
- Coxswain / helm / commander when self-launching.

- **Authorising launch.**

Training criteria. Describe when a launch can be authorised:

- Training exercise.
- Service.
- Passage.
- PR launch.
- Machinery trials.

- **Self-launching.**

Training criteria. Explain the considerations before self-launching:

- Delay would result in a loss of life.
- SAR unit within operational limits.
- Crew experience / ability.
- Launch authority informed - even if afloat on exercise through coastguard.

- **Informing relevant personnel prior to launch.**

Training criteria. State who should be informed prior to launching under normal circumstances:

- Lifeboat operations manager.
- Launch authority.
- Coordinating authority.

Training criteria. State when a launch authority is required to be at station for launches:

- Night time launches for ILBs.

##### Selection of shore crew

- **Limitations of the crew.**

Training criteria. State the considerations to be made when selecting the appropriate crew:

- All required roles filled.
- Physical ability.
- Experience.
- Crew competence.
- Task specific competence.

task continues

### Task 3.2.3

### Manage Launch and Recovery (Shore Crew)

#### a Initiate launch (continued)

##### Training guidelines

##### Choice of launch and recovery site

- **Methods.**

Training criteria. State all the launch and recovery methods available:

- In accordance with the current SOP and LOP.

- **Locations.**

Training criteria. State all of the locations approved for launch and recovery including alternative sites:

- In accordance with the current LOP.
- Normal launch / recovery site or alternative launch / recovery site / method, e.g. net recovery.

Training criteria. Explain local limitations and restrictions affecting safe launch and recovery at approved sites:

- In accordance with the current LOP.
- Weather.
- Tide / sea state.
- Beach conditions.
- Required crew numbers.
- Public access and control.
- Location of alternative launch sites.

task continues

## Activity 3 - Launch and Recovery

### Task 3.2.3

### Manage Launch and Recovery (Shore Crew)

#### b Safe working practices

##### Training guidelines

##### Safe working practices

- **Responsibilities.**

Training criteria. Explain roles and responsibilities during launch and recovery operations:

- As stated in the role handbook.

##### Safety measures

- **Embarking / disembarking the SAR unit.**

Training criteria. Explain the embarkation / disembarkation points and the reason for their use:

- In accordance with the current class specific SOP and LOP.
- with due regard to safety and hazards at all times.
- Hazards include:
  - falls, slips
  - height
  - secure ladder at all times.
  - entanglement
  - secure footing

- **Safety of crew.**

Training criteria. Explain the importance of ensuring crew don appropriate PPE:

- Protection.
- Visibility and identification during operations.

Training criteria. Demonstrate briefing and ensuring crew stay clear of hazards:

- As per current briefing model.
- Raise awareness of hazards.
- Raise awareness of hazardous and no-go areas.

task continues



### Task 3.2.3

### Manage Launch and Recovery (Shore Crew)

#### c Manage launch and recovery

##### Training guidelines

##### Pre and post launch

- **Checks.**

Training criteria. Demonstrate managing pre and post launch checks:

- In accordance with the current class specific SOP and LOP.

##### Launch and recovery

- **Managing.**

Training criteria. Demonstrate managing launch and recovery operations:

- In accordance with the current class specific SOP and LOP.
- Procedure followed in a safe manner.
- Safely and appropriately managing appropriate roles involved.

Training criteria. Explain the actions to be taken for a range of emergencies, including use of available resource:

- In accordance with the current class specific SOP and LOP.
- Use only authorised launch and recovery equipment, e.g. in bog down.

##### Post recovery

- **Post recovery checks.**

Training criteria. Explain the post recovery procedures:

- Post recovery check list.
- Wash down SAR unit and all used equipment.
- Correctly restow any used equipment.
- Refuel.
- Report any defective equipment.
- Report ready for service to launch authority and coordinating authority.

Training criteria. Explain the importance of post recovery procedures:

- Preparedness for future services.
- Defect identification.
- Equipment maintenance.

Training criteria. Demonstrate managing post recovery procedures:

- In accordance with the SOP and LOP.
- In accordance with the current post recovery checklist.

task continues

## Activity 3 - Launch and Recovery

### Task 3.2.3

### Manage Launch and Recovery (Shore Crew)

#### d Alternative launch sites and procedures

##### Training guidelines

##### Alternative launch and recovery sites

- **Locating alternative launch sites.**

Training criteria. State the location and distance from station to the alternative launch and recovery sites:

- State the approved alternative launch site(s) (if applicable).

Training criteria. State the restrictions on using them:

- People - RNLI personnel required and trained, members of the public.
- Traffic - on route and at the launch site.
- Plant - the use of specific plant at that site.
- Environment - weather / sea state, access, ground.

- **Using alternative launch sites.**

Training criteria. State the considerations for operating from alternative launch and recovery sites:

- In accordance with the current SOP and LOP.
- Safety of transit: control vehicles, communications, access routes, permission.
- Due regard for Road Traffic Act.
- Crew briefing.
- Inform coordinating authority.

##### Alternative launch procedure

- **Alternative procedures for launching.**

Training criteria. State the procedures for alternative launch and recovery of SAR units:

- In accordance with the current LOP.
- Launch from within boathouse.
- Alternative vehicles.
- Alternative launch in the event of mechanical / electrical failure.

### Task 3.3.1

### Plant Operator

#### a Safe working practices and communication

##### Training guidelines

##### Safe working practices

- **Various types of machinery and systems fitted.**

Training criteria. Explain the safe working practices associated with the various types of machinery:

- As stated in the role handbook.
- Engine:
  - noise - use correct PPE
  - heat - be aware of hot surfaces, e.g. radiator, exhaust
  - fumes - dangers from starting a vehicle in the boathouse.
- Winches:
  - rope handling - use gloves, keep hands away from the winch
  - risk of lines under load - lines parting and whipping hazards
  - risk of crush injury - enforce the 'Restricted Zone'.
- Electrical:
  - shock hazard of mains and low voltage electrical equipment
  - ensure all equipment switches are turned OFF before removing plugs
  - always refit cable / socket covers.

Training criteria. Explain roles and responsibilities during launch and recovery operations:

- As stated in the role handbook.

##### Communication

- **Communication used during operations.**

Training criteria. Explain the appropriate approved methods of communication used during operations:

- As stated in the equipment specific manual.
- Radio:
  - operation
  - working channels
  - emergency use on channel 16
  - headsets.
- Hand signals:
  - recognised signals
  - approved local variations.
- Audible signals:
  - use of horn
  - whistle.

task continues

## Activity 3 - Launch and Recovery

### Task 3.3.1

### Plant Operator

#### a Safe working practices and communication (continued)

##### Training guidelines

##### Changes in attitude

- **Effects of attitude or emotional state on operator ability.**

Training criteria. Describe the effects changes in attitude or emotional state may have on operating ability:

- Driving to launch and recovery site:
  - loss of concentration
  - clouded judgement
  - higher temptation to speed.
- Tiredness:
  - inability to spot hazards
  - slower reaction times.
- Anger:
  - lack of courteous driving
  - dangerous driving.

##### Driving factors

- **Factors to be considered when operating.**

Training criteria. Explain what factors must be considered whilst operating:

- Vehicle safety:
  - positioning of vehicle
  - appropriate speed
  - traffic density.
- Vehicle occupants:
  - positioning in vehicle
  - use of seat belts.
- Other road users:
  - proximity to vehicle
  - pedestrians
  - awareness of the vehicle and its purpose.
- RNLI image:
  - operate in a calm and considerate manner
  - the driver represents the face of the RNLI.

task continues

## Task 3.3.1

## Plant Operator

### b Operating techniques

#### Training guidelines

#### Operating position

##### • Adjusting.

Training criteria. Explain what adjustments can be made to the operating position:

- Longitudinal, height and lumbar adjustment of seats - to enable safe and correct operating position.
- Steering column adjustment - to enable safe and correct driving position.
- General operating position.

#### Safe operating skills

##### • Observations made prior to moving.

Training criteria. State the observations to be made before the plant is manoeuvred:

- Check blind spots.
- Check security of trailer and safety attachments.
- Ensure adequate clearance.
- Establish communications with a banksperson.
- Only move off when it is safe to do so.

##### • Reversing, close manoeuvring and parking.

Training criteria. State the considerations when reversing, close manoeuvring and parking:

- Obstructions.
- Use of chocks.
- Use a banksperson.
- Walk around plant.

##### • Operating plant in launch and recovery environments.

Training criteria. Explain the considerations to take when operating in different environments:

- The law.
- Own vehicle size and type.
- Tidal state and sea conditions.
- Ground conditions.
- Awareness of other vehicles and pedestrians.
- Weather conditions.
- Limitations of the plant.

#### Appropriate speed

##### • Appropriate speed.

Training criteria. State the factors that determine an appropriate speed:

- Drivers of RNLI vehicles must drive in accordance with road traffic regulations and comply with all traffic signs and speed limits.
- Operators must know the speed limits for the vehicle with / without a trailer.
- Crossing rough terrain.
- Proximity of pedestrians.
- Traffic density.
- Safety of crew and equipment.

Training criteria. Describe factors that may make a change of speed necessary:

- Crew comfort and safety.
- Rough terrain.
- Traffic density.
- Public safety.
- Unknown ground conditions.
- Limitations of equipment.

task continues

## Activity 3 - Launch and Recovery

### Task 3.3.1

### Plant Operator

#### b Operating techniques (continued)

##### Training guidelines

##### Safety

###### • Controlling risks.

Training criteria. Explain the measures in place to control risks:

- Be able to identify appropriate risk assessments.
- In accordance with the current SOPs and LOPs for the launch and recovery equipment.

###### • Injury sustained on the plant.

Training criteria. Explain the procedures for dealing with injury sustained on the plant:

- Make the plant safe if possible.
- Alert the boathouse.
- Seek medical help if required.

##### Public awareness

###### • Using lights and audible signals.

Training criteria. State the appropriate time to use lights and audible signals:

- Always switch on headlights when the plant is moving.
- Always sound the horn before moving the plant / operating a winch.
- Blue lights, amber lights and sirens, where their use is permitted, must be used strictly in accordance with current policy.
- Route planning

###### • Planning a route to and from a launch site. (if applicable)

Training criteria. State the factors to consider when planning a route to and from a launch site:

- Ground conditions.
- Tidal constraints.
- Public access.
- Traffic density.

##### De-bog

###### • De-bogging. (if applicable)

Training criteria. Explain techniques and resources available for vehicle recovery:

- Inform station, ask for assistance.
- Be able to identify the appropriate bogging down procedure.
- Use the towing points on the vehicle as described in the equipment specific manual.
- In accordance with the current SOP and LOP.

##### Enter the water

###### • Entering the water. (if applicable)

Training criteria. State the factors to consider prior to and after a vehicle enters the water:

- Only if the vehicle is marinised.
- Correct operator PPE.
- Ground conditions.
- Tidal conditions.
- Sea state.
- Vehicle protection hatches.
- Depth of water
- Maximum wading depth of vehicle.

task continues



### Task 3.3.1

### Plant Operator

#### b Operating techniques (continued)

##### Training guidelines

##### Abandonment

- **Abandoning the vehicle. (if applicable)**

Training criteria. Explain the factors that determine when a vehicle is to be abandoned:

- After an accident.
- Vehicle bog down.
- Drive train failure.
- Power unit failure.

- **Actions to take prior to abandonment (if applicable).**

Training criteria. State the actions to be taken prior to abandonment:

- Inform station / coordinating authority.
- Remove or secure loose or vulnerable equipment.
- Secure vehicle doors or hatches.
- Note the position of abandonment.
- Prepare vehicle for abandonment and attach position marker if necessary.

##### Post operational checks

- **Carrying out post operational checks.**

Training criteria. Describe the post operational checks for the plant:

- As required by the station post operational check list and the plant planned maintenance document.

## Activity 3 - Launch and Recovery

### Task 3.4.1

### Authorising Launches

#### a Operational readiness

##### Training guidelines

##### Operational readiness

###### • Readiness Procedures.

Training criteria. Explain the states of readiness:

- Off service.
- Restricted service (personnel or material defects).
- Ready for service.

Training criteria. State who and when to advise in a change of operational readiness:

- Central Operations Information Room (COIR)
- Station management.
- Coastguard.
- Flank stations.

Training criteria. Explain the additional considerations for authorising launches of SAR units on restricted service:

- The effects of the personnel or material defect(s) on the operation of launch and recovery equipment and lifeboats.

###### • Ensuring Readiness.

Training criteria. Explain the importance of having all facilities and plant operationally ready for service:

- To ensure operational readiness.
- To ensure the strategic performance standards are met where possible.

Training criteria. Explain the importance of defect reporting in a timely fashion:

- To ensure that the station mechanic as well as coastal and regional support staff are fully informed.
- To ensure that defects are rectified as soon as possible.

Training criteria. State how to follow up and ensure a speedy repair of defects to regain state of readiness:

- Collaboration with station mechanics and coastal and regional support staff.
- Updates to defect reporting system.

task continues

## Task 3.4.1

## Authorising Launches

### b Launch procedures

#### Training guidelines

#### Coastguard

##### • Procedures.

Training criteria. Explain the coastguard requests used:

- Launch request - request for launch for service.
- Immediate readiness - at immediate readiness to launch / proceed, with crew on board or in the immediate vicinity of the boat.
- Stand-by - crew warned of probable SAR mission and brought to an agreed notice for launch / proceeding.

Training criteria. Describe the coastguard priority system:

- Distress phase: a situation where there is reasonable certainty that a vessel or other craft, including an aircraft or a person, is threatened by grave and imminent danger and requires immediate assistance.
- Alert phase: a situation where apprehension exists as to the safety of an aircraft or marine vessel, and of the person on board.
- Uncertainty phase: a situation where doubt exists as to the safety of an aircraft or a marine vessel, and of the person on board.

Training criteria. Demonstrate receiving, recording and undertaking a launch request:

- Using the SMEAC model.
- Based on a coastguard launch format.
- Passed to station command roles.

#### RNLI assets

##### • Limitations.

Training criteria. State the limitations of the SAR units:

- For launch and recovery plant and vessels at location as per current guidelines.

Training criteria. Explain the limitations of the crew:

- Physical - exposure to conditions and long duration services.
- Mental - exposure to traumatic events.
- Importance of crew rotations and debriefs.
- Competence of trainees.

Training criteria. Explain how meteorological and tidal factors can limit operations:

- Meteorological factors that can limit launching and recovery at location e.g. high winds, lightning, ice / snow, wave damage.
- Tidal factors that can limit launching and recovery at location e.g. spring low tide, tidal surge.

task continues

## Activity 3 - Launch and Recovery

### Task 3.4.1

### Authorising Launches

#### b Launch procedures (continued)

##### Training guidelines

##### • Launches.

Training criteria. Explain when it is appropriate to authorise a launch:

- Within the capabilities of the SAR unit as defined in limitations.
- When requested to by the SAR coordinating authority and likely within the capabilities of the SAR unit and personnel.
- When there are enough competent personnel available to undertake the service.

Training criteria. Explain when it is not appropriate to authorise a launch:

- Outside the capabilities of the SAR unit as defined in limitations.
- A lack of competent personnel to undertake a service.
- When equipment is off service / restricted service and not appropriate or safe to use.
- When there is a more appropriate asset or organisation to undertake service.
- When the tasking is not appropriate for the RNLI or outside our scope.

Training criteria. Explain when a launch authority must attend for launch and recovery:

- ILB, specifically D class, launches during hours of darkness.

Training criteria. State the Coastguard procedure for bypassing the launch authority:

- Extreme cases only.
- As per the current procedure.

Training criteria. Explain the self-launching procedure:

- In all cases, the launch authority and SAR coordinating authority are to be informed and to approve at the earliest possible time and the self-launch guidelines are to be followed.
- Coastguard to crew - a direct launch request when delay would result in death.
- Crew to coastguard - when already launched or incident is observed directly or reported through a third party.
- Launch authority to crew - incident is observed directly or reported through a third party.

Training criteria. Explain when a launch can be authorised for non-operational launches:

- Passage.
- Trials.
- Maintenance.
- Training and assessments.
- Fundraising events.
- Public relations events.

task continues

### Task 3.4.1

### Authorising Launches

#### c Alerting system

##### Training guidelines

##### Alerting systems

- **System.**

Training criteria. Explain the alerting system:

- As per current call out and communications system.
- Call out and communications systems codes used and their purpose.
- The scope and purpose of the alerting system.
- Allocation and maintenance of associated pagers to crew.

Training criteria. Demonstrate testing the alerting system:

- As per current call out and communications system.

Training criteria. Explain alternative alerting systems in the event of system failure:

- As per current local systems and RNLI guidelines.

task continues

## Activity 3 - Launch and Recovery

### Task 3.4.1

### Authorising Launches

#### d Reporting

##### Training guidelines

##### Operational administration

##### • Exercises.

Training criteria. Demonstrate entering an exercise return:

- As per current RNLI database system.

Training criteria. Explain the importance of the exercise planning form.

Training criteria. Explain the importance and procedure of signing off exercise planning documentation:

- Exercise planning form.
- Ensure safe, appropriate exercises are conducted.
- Ensure the right crew attend at the right time and regularly.
- To authorise and document exercise launches.

##### • Services.

Training criteria. Demonstrate entering a Return of Service:

- As per current RNLI database system.

Training criteria. Demonstrate entering a crew attendance report:

- As per current RNLI database system.





## Activity 4 - Seamanship

Photo: RNL / Erik Woolcott



## Tasks

|             |                                             |
|-------------|---------------------------------------------|
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## Activity 4 - Seamanship

### Task 4.1.1

### Lookout

a

#### Lookout

##### Training guidelines

##### Maintain lookout

- **Effective lookout.**

Training criteria. Describe how to achieve an effective lookout:

- Position as directed and report any sightings to SAR unit commander.
- Maintain night vision.
- Reduce level of background lighting.

- **Equipment.**

Training criteria. State the equipment available to assist with lookout:

- Eyes and ears.
- Binoculars.
- Searchlights.
- Night scope.

- **Use of binoculars and search lights.**

Training criteria. Demonstrate the set-up, adjustment and use of binoculars and search lights:

- Adjust binoculars for personal use.
- Set up searchlights and focus.

- **Distress signals.**

Training criteria. State what visual distress signals may be detected:

- Red parachute or handheld flares.
- Orange smoke.
- Raising and lowering of arms.

## Task 4.1.2

## Watchkeeping

a

### International Regulations for Preventing Collisions at Sea (IRPCS)

#### Training guidelines

#### IRPCS

- **Basic knowledge of IRPCS.**

Training criteria. Describe how to assess if a risk of collision exists:

- Constant bearing, decreasing range.

- **Working knowledge of the steering and sailing rules, lights, shapes and sound signals appropriate to the SAR unit.**

Training criteria. Explain the rule and its application for the following:

- Rule 2: Responsibility.
- Rule 5: Lookout.
- Rule 6: Safe speed.
- Rule 7: Risk of collision.
- Rule 10: Traffic separation schemes.
- Rule 19: Conduct of vessels in restricted visibility.

- **Navigation lights**

Training criteria. Identify navigation lights for a power-driven vessel (PDV):

- Port, starboard, masthead and stern to include:
  - <7m <7knots
  - PDV <50m
  - PDV >50m.

Training criteria. State the aspect of the vessel from lights:

- Correctly assess the aspect of the vessel from its navigation lights.

task continues

## Activity 4 - Seamanship

### Task 4.1.2

### Watchkeeping

#### b Buoyage

##### Training guidelines

##### Buoyage

- **Demonstrate a good working knowledge of International Association of Marine Aids to Navigation and Lighthouse Authorities (IALA) (Region A) buoyage.**

Training criteria. Explain all types of buoyage, their purpose and how to identify them by day and night:

- To include the shape, colour, top mark, flashing sequence and light colour of the following:
  - cardinal
  - lateral
  - preferred channel
  - special
  - isolated danger
  - safe water / fairway buoy
  - temporary wreck.
- Local buoyage variants where appropriate.

##### Inland waters (Republic of Ireland only)

- **Demonstrate a working knowledge of CEVNI code for inland waterways, including buoyage and signage (appropriate to location).**

Training criteria. Explain the signs, markings and buoyage used in your location:

- In accordance with the European Code for Inland Waterways (CEVNI).

task continues



## Task 4.1.2

## Watchkeeping

c

### Radar lookout (where fitted)

#### Training guidelines

#### Purpose of a radar lookout

Training criteria. Explain the purpose of radar lookout:

- To ensure a lookout is maintained...by all available means ...appropriate to the conditions... (i.e. Rule 5).

#### Understanding the radar display

Training criteria. Explain what is meant by range and bearing:

- To include relative and true bearing and appropriate units (degrees and NM).
- Explain the difference between head up or north up display.
- Demonstrate use of range rings and VRM to determine the range of a target.
- Demonstrate the use of EBL to determine the bearing of a target.

Note: Turning on an initial set up of the radar is to be conducted by a competent crew member who has completed the radar tasks 7.3.1.

#### Reporting targets

Training criteria. Demonstrate giving a report of a target for a target which is:

- Decreasing in range.
- On or close to the SAR unit's heading.

Note: A crew member acting as radar lookout does not need knowledge of radar theory or the use of radar as a navigation instrument.

# Activity 4 - Seamanship

## Task 4.1.3

## IRPCS (Command)

a

### International Regulations for Preventing Collisions at Sea (IRPCS) and buoyage

#### Training guidelines

#### IRPCS

- **Working knowledge of the steering and sailing rules, lights, shapes and sound signals appropriate to the SAR unit.**

Training criteria. Explain the rule and its application for the following:

- Rule 8: Actions to avoid collision.
- Rule 13: Overtaking.
- Rule 14: Head on.
- Rule 15: Crossing.
- Rule 16: Action by give-way vessel.
- Rule 17: Action by stand-on vessel.
- Rule 18: Responsibilities between vessels.
- Rule 21: Definitions.
- Rule 23: Power-driven vessel.
- Rule 25: Sailing.
- Rule 26: Fishing.
- Rule 27: Vessels not under command or restricted in their ability to manoeuvre.
- Rule 28: Vessels constrained by their draught.
- Rule 30: Anchored vessels and vessels aground.
- Rule 34: Manoeuvring and warning signals.
- Rule 35: Sound signals in restricted visibility.
- Additionally Rules 9, 10, 24, 29, 31 if appropriate to location.

Note, these rules are in addition to those required to be learned for earlier stages of development.

#### Buoyage

- **Demonstrate a comprehensive knowledge of IALA (Region A) buoyage.**

Training criteria. Explain all types of buoyage, their purpose and how to identify them by day and night.

## Task 4.2.1

## Rope Handling

a

### Cordage and usage

#### Training guidelines

#### Cordage

##### • Properties and applications of each type.

Training criteria. Explain the properties and applications for each type of rope carried on your SAR unit:

- Polyamide (Nylon): smooth exterior, strong, stretches (approx a quarter), sinks and loses some strength when wet:
  - tow rope (ILB)
  - casualty drogue
  - heaving line.
  - anchor rope
  - seaward drogue
- Polyester: stretches (approx a third), sinks, maintains properties when wet:
  - a frame topping lift
  - berthing / mooring line.
  - lines on breeches buoy block
- Polypropylene: hairy exterior, stretches (by almost half), floats:
  - catting rope
  - veering lines
  - fender rope
  - rescue strop
  - rope stopper
  - scramble net
  - grapnel line
  - breasting lines.
- Polyethylene: stretches (approx a third), floats:
  - life ring.
- High Modulus Polyethylene (HMPE): ropes are commonly referred to in the RNLI by the trade names Dyneema® or Dynex®:
  - tow rope (ALB) and bridle
  - haul up spans
  - stoppers.
  - winch ropes (as applicable)
  - quarter stoppers (slip way)
- Steel wire rope (where applicable):
  - boathouse / vehicle
  - davits.

Training criteria. Explain why they are appropriate for that application:

- Nylon - stretch capabilities minimises snatching.
- Polyester - strength with minimum stretch even when wet.
- Polypropylene - floating line aids pickup and visibility.
- Polyethylene - floating line aids pickup and visibility.
- HMPE - cleanliness, high strength, light weight.

task continues

# Activity 4 - Seamanship

## Task 4.2.1

## Rope Handling

### b Knots, bends and hitches

#### Training guidelines

#### Knot terminology

##### • Parts of a rope.

Training criteria. Explain the terms used when describing parts of a rope:

- Working end: the active end of the rope.
- Standing end: the inactive end of the rope.
- Bight: any slack section of a line forming a partial loop.
- Loop: a bight where the lines cross.
- Standing part: the part of the rope between the ends.

##### • Using bends and hitches.

Training criteria. Explain when to use bends and hitches:

- A bend is a method of temporarily joining two ropes.
- A hitch is a method of joining a rope to a structure or ring.

#### Knot tying

##### • Tying a range of knots.

Training criteria. Demonstrate the ability to tie the range of knots listed below.

Training criteria. Describe the appropriate applications for each:

- Bowline: a temporary eye in a rope. Can be used as a lifeline or to join two ropes together.
- Round turn and two half hitches: securing a rope to a ring shackle or spar.
- Reef Knot: joining two ropes of equal thickness.
- Sheet Bend: joining two ropes of unequal thickness.
- Double sheet bend: a more secure version of a sheet bend.
- Clove hitch: securing a rope to a rail, post or ring.
- Rolling hitch: to take temporary weight off a line. Holds fast in one direction, slides in the other.
- Figure of eight: stopper knot to prevent a rope unreeving through a block. Used to mark the end of a rope.

#### Making fast

##### • Securing to a range of fittings.

Training criteria. Demonstrate securing to a range of fittings, including cleats, bollards, bitts, rings and posts:

- What knot, bend or hitch and why.
- Alongside (scope) – rising / falling tide.
- Best lead for line.
- Chafe.
- Ease of slipping.

task continues

## Task 4.2.1

## Rope Handling

### c Working ropes

#### Training guidelines

#### Hazards

- **Dangers of working with ropes under load.**

Training criteria. Explain the danger of working with ropes under load:

- Fingers, hands and legs may be trapped or injured.
- Recoil of rope parting under load.
- Precautions.

Training criteria. Explain the precautions to be taken when working with ropes under load:

- Stand clear of ropes under load.
- Stand out of direct line of ropes under load.
- Awareness of Snatch loading.
- Never step over or cross any rope under load.
- Keep working turns on bitts, bollards.
- PPE in place for working with lines under load.
- PPE requirements for handling steel wire ropes.

- **Slipping a rope under load.**

Training criteria. Demonstrate slipping a rope under load:

- Never take all turns off bitts, bollards.
- Ease rope out under control until weight is off, then cast off.

#### Use of heaving line / throw bag

- **Throwing a heaving line / throw bag.**

Training criteria. Demonstrate how to throw a heaving line / throw bag:

- Prior to throwing observe weather, surrounding hazards, persons.
- Secure, coil and throw.
- Demonstrate re-packing and stowing.

#### Coil loose ends

- **Coiling loose ends.**

Training criteria. Demonstrate how to tidy up loose ends:

- Coiling - from the appropriate end.

- **Safety.**

Training criteria. Explain the routine for securing the deck:

- Trip hazards.
- Stowing deck tools.
- Stowages / lockers secure.

task continues

# Activity 4 - Seamanship

## Task 4.2.1

## Rope Handling

### d Rope stowage and maintenance

#### Training guidelines

##### Stowage

- **Checks prior to stowing.**

Training criteria. Describe the checks to be carried out prior to stowing:

- Check the rope for any damage or defects.
- Wash in fresh water.
- Dry prior to stowing.

- **Stowing a rope.**

Training criteria. Demonstrate stowing of rope:

- Coil needs to fit into stowage / locker.
- Coil / fake directly in stowage / locker.
- Coil should stream out of stowage / locker freely.
- Coil to hand.

##### Maintenance

- **Defects and causes.**

Training criteria. Describe the defects and common causes of wear in rope:

- Defects:
  - stretching
  - wear
  - flattening
  - crows footing
  - fraying
  - UV damage
  - corrosion and kinking in steel wire rope.
- Causes:
  - general use
  - rust
  - ultraviolet
  - crushing
  - chafing.

- **Actions on finding a defect.**

Training criteria. Explain the actions to be taken when a defect is found:

- Report defect to person responsible immediately.
- Do not use ropes with defects.
- Ropes with defects must be replaced as soon as possible.



### Task 4.2.2

### Capstans

#### a Using a capstan

##### Training guidelines

##### Capstan (if fitted)

##### • Operating.

Training criteria. Demonstrate how to operate capstan for:

- Anchor recovery.
- Breasting ropes – for berthing and recovery.
- Taking weight on a line.

Training criteria. Describe factors to consider when using a capstan:

- Aware of loose items, clothing, lifejacket straps, hair.
- Keep hands clear.
- Appropriate number of clockwise turns.
- Aware of riding turns.
- Stand clear.
- Two-way communication and warnings to other crew.

##### • Hazards.

Training criteria. Explain the hazards to consider when operating, including riding turns, loaded lines and standing in bights:

- Loaded line.
- Riding turns.
- Entrapment of hands.
- Standing in bights.

# Activity 4 - Seamanship

## Task 4.3.1

## Anchoring

### a Anchor deployment

#### Training guidelines

##### Precautions

###### • Precautions.

Training criteria. Explain the safety precautions to be considered:

- Crew must be aware of standing / placing hands in bights and coils.
- Warps and lines under load.
- Use of correct PPE and safety lines.
- Manual Handling - heavy anchors.

##### Anchor deployment

###### • Rigging the anchor warp and chain.

Training criteria. Demonstrate the method of rigging the anchor on the SAR unit:

- In accordance with the current class specific SOP.

###### • Communication.

Training criteria. Describe communication required whilst anchoring:

- Two-way communication.
- Clear, concise and timely communication, e.g. when ready, when to let go, problems arising.

###### • Removing and deploying the anchor.

Training criteria. Demonstrate the removal and deployment of the anchor from its stowage:

- Removal In accordance with the current class specific SOP.
- Deployment:
  - safety - e.g. crew must be aware of standing / placing hands in bights and coils
  - chain and warp clear of fouling
  - warp under control - (e.g. round post).

###### • Paying out and securing the anchor warp.

Training criteria. Demonstrate paying out the anchor warp and secure as instructed.

Training criteria. Explain how the anchor warp is marked if appropriate:

- Safety – e.g. fingers clear of cleats / bollards / bitts, secure position.
- Warp paid out under control when instructed.

##### Anchor holding

###### • Recognising when an anchor is holding.

Training criteria. Describe the methods used to tell if the anchor is holding:

- Transits.
- Depth.
- Tension on warp.
- GPS.
- Radar.

###### • Actions to take if anchor fails to hold.

Training criteria. Describe what actions can be taken if an anchor fails to hold or drags:

- Pay out more warp.
- Redeploy anchor.

task continues

## Task 4.3.1

## Anchoring

### b Anchor recovery

#### Training guidelines

#### Anchor recovery

- **Communication required whilst recovering the anchor.**

Training criteria. Describe communication required whilst recovering the anchor:

- Initial brief.
- Two-way communication.
- Clear, concise and timely communication, e.g. when ready, when to haul, problems arising.

- **Using the capstan (if fitted).**

Training criteria. Demonstrate the use of a capstan if fitted during anchor recovery:

- Keep hands at safe distance.
- Secure stance.
- Apply turn in correct direction.
- Adequate number of turns.
- Be aware of riding turns.

- **Recovering and stowing the anchor warp and chain.**

Training criteria. Demonstrate how the anchor and associated equipment is recovered and placed in its stowage:

- In accordance with the current class specific SOP.

- **ILB Anchor recovery.**

Training criteria. Demonstrate the ability to recover 50m of anchor warp as quickly as possible:

- Warp should be recovered to the boat within approximately ninety seconds.

- **Precautions.**

Training criteria. Demonstrate the additional safety checks to be carried out once the anchor and associated equipment is on board the SAR unit:

- Anchor, anchor warp and deck tools secured
- Hatches closed.
- Foredeck clear.
- Manual handling.

## Activity 4 - Seamanship

### Task 4.3.2

### Manage Anchoring

#### a Identify and plan safe anchorage

##### Training guidelines

##### Safe anchorage

- **Identifying a safe anchorage.**

Training criteria. Describe how to identify a safe anchorage:

- Weather now and forecast.
- Shelter.
- Sea bed.
- Foul ground.
- Tidal information - charted depth.

- **Information from the chart.**

Training criteria. Identify, using a local chart, both safe and unsafe / prohibited anchorages and / or a symbol marking a safe anchorage:

- Identify safe anchorage on a chart to include symbols:
  - safe and unsafe
  - foul ground
  - prohibited areas
  - submarine cables.

- **Allowances to be made in poor weather.**

Training criteria. State additional considerations that must be made in choosing an anchorage for poor weather:

- Sea room and scope.
- Shelter.
- Other vessels.

##### Scope

- **Calculating scope.**

Training criteria. Explain the ratio between depth of water and the anchor warp required:

- Minimum 6x depth of water (warp and chain) or 4x depth of water (chain).

- **Factors to be considered for an extended stay.**

Training criteria. Explain the factors for an extended stay:

- Increase the scope.
- Weather forecast.
- Rise / fall of tide.
- Swinging circle.
- Day shape / light (if appropriate).

task continues

## Task 4.3.2

## Manage Anchoring

### b Deploy anchor

#### Training guidelines

#### Anchor deployment

- **Safety checks to be carried out.**

Training criteria. State the safety checks required:

- Initial brief.
- Deck tools available.
- Task allocation.
- Scope expected.
- Importance: injury prevention and clear expectation of role.
- Anchor rigged correctly.
- Crew in safe positions.
- Anchor type.

- **Communication.**

Training criteria. Describe the importance, and content, of a briefing to the anchor party:

- Two-way communication - crew and person responsible.
- When ready.
- When to let go.
- Warp given.
- Direction of lead.
- Pay out / hold.
- Make fast.

- **Managing deployment.**

Training criteria. State what you would expect whilst deploying the anchor.

Training criteria. Demonstrate how to manage the deployment of the anchor:

- In accordance with the current class specific SOP.

#### Anchor holding

- **Recognising when an anchor is holding.**

Training criteria. Describe the immediate indicators that an anchor is holding:

- Transits.
- Tension on warp.

- **Monitor holding.**

Training criteria. State what can be used to monitor holding:

- Transits.
- Radar ranges.
- GPS.
- Bearings.
- Depth alarms.

- **Actions to take if anchor fails to hold.**

Training criteria. State what actions can be taken if an anchor fails to hold or drags:

- Pay out more warp.
- Redeploy anchor.

task continues

## Activity 4 - Seamanship

### Task 4.3.2

### Manage Anchoring

c

#### Recover anchor

##### Training guidelines

##### Anchor recovery

- **Managing anchor recovery.**

Training criteria. Demonstrate the management of anchor recovery including communication and safety brief:

- In accordance with the class specific SOP, to include:
  - Correct terminology.
  - Clear initial brief - to include stowage and securing of equipment.
  - Communication with crew.
  - Safe use of capstan.
  - Lead of warp.
  - Warp remaining.

- **Post recovery checks.**

Training criteria. State the post recovery checks that must be carried out:

- Anchor stowed.
- Warps and lines stowed.
- Tools secured.
- Foredeck clear.
- Information regarding equipment defects.

task continues



## Task 4.3.2

## Manage Anchoring

### d Fouled anchor

#### Training guidelines

#### Fouled anchor

- **Fouled anchor and its possible causes.**

Training criteria. State possible causes of an anchor becoming fouled:

- Foul ground.
- Fishing gear.
- Submarine cable.
- Other vessel anchor / mooring.

Training criteria. Describe how you may identify a fouled anchor:

- Unable to retrieve it.
- Additional load.
- Visual indications.

- **Safety precautions and hazards.**

Training criteria. Describe the safety precautions and hazards that must be considered:

- Brief given following dynamic risk assessment.
- Crew / vessel safety.
- Load on warp.
- Use of capstan and fairleads.
- Power cables.
- Parting of the warp.

- **Recovering a fouled anchor.**

Training criteria. Describe methods for recovering an anchor that has become fouled:

- Reverse or change angle of pull.
- Shorten / lengthen the scope as appropriate.

- **Actions to be taken if an anchor cannot be recovered.**

Training criteria. State what actions may be taken if the anchor cannot be recovered:

- Buoy the anchor for later recovery.
- Record position.
- Inform appropriate personnel - coordinating authority, LOM / launch authority.

## Activity 4 - Seamanship

### Task 4.4.1

### Veering Down

#### a Veering down

##### Training guidelines

##### Veering down

- **Appropriate use of veering down.**

Training criteria. Describe veering down:

- Is a process of manoeuvring the SAR unit astern under anchor in order to maintain the vessel's head to sea.

Training criteria. State when it is appropriate to veer down:

- When judged to be the safest and most suitable method to reach a casualty by the helm.

Training criteria. Explain the risks associated with veering down:

- The SAR unit may be operating in a dangerous area with confused seas.
- Crew operating warps and lines under tension.
- Proximity to additional hazards, potential for error, limited manoeuvrability.

- **Veering down.**

Training criteria. Demonstrate crew roles during veering down operations:

- In accordance with the current class specific SOP.
- Operating the veering line as directed.
- Operating dip stick as directed.
- Maintaining a lookout for hazards.
- Line / casualty management on deck.

- **Recovering the anchor warp.**

Training criteria. Demonstrate recovery of the anchor warp in adverse conditions following veering down:

- Clear communications.
- Coordinate hauling and manoeuvring.
- Stow warp (temporarily if needed).
- Cut and run procedure if directed by the helm.

## Task 4.4.2

## Manage Veering Down

### a Veering down

#### Training guidelines

#### Veering down

- **Appropriate use of veering down.**

Training criteria. Describe the factors when considering whether to veer down:

- When it is the only safe method to reach a casualty.
- When it is the most appropriate or effective way to reach a casualty.
- Consider experience of crew.
- Consider weather and other environmental conditions.

Training criteria. Explain the risks associated with veering down:

- The SAR unit may be operating in a dangerous area with confused seas and surf.
- Crew operating warps and lines under tension.
- Proximity to additional hazards, potential for error, limited manoeuvrability.

Training criteria. Explain the necessity of having a clear plan:

- All crew will have a common understanding of roles and procedures.
- Reduces scope for misunderstanding.

- **Veering down.**

Training criteria. Demonstrate managing veering down operations:

- Brief - crew and coordinating authority.
- Two-way communication.
- Maintain vessel aspect.
- In accordance with the current class specific SOP.

- **Recovering the anchor warp.**

Training criteria. Describe how to recover the anchor warp in adverse conditions following veering down:

- Crew brief.
- Crew safety.
- Clear communications.
- Coordinate hauling and manoeuvring.
- Stow warp (if possible).
- Consider cut and run.

# Activity 4 - Seamanship

## Task 4.5.1

## Towing

### a Rig and pass tow

#### Training guidelines

#### Prepare and rig a tow

- **Communication prior to rigging.**

Training criteria. Describe the communications needed from the coxswain / helm prior to rigging for a tow:

- Crew brief: nature of problem, hazards, use of the towing assessment checklist.
- Brief casualty vessel on conduct of tow.

- **Safety precautions during rigging.**

Training criteria. State the safety precautions that should be followed:

- Crew given specific tasks.
- Considerations for rough weather - harnesses, hand holds.
- Keep clear of bights and coils on deck.

- **Rigging the towing equipment.**

Training criteria. Demonstrate how to rig the tow rope prior to it being passed to the casualty:

- Sufficient tow rope to be faked out on deck.
- Passed through fairlead.
- Turned on bitts / descender.
- Attach heaving line.
- Allocate one crew per task.

- **Rigging a bridle on the casualty.**

Training criteria. Explain why and how you would rig a bridle for the casualty vessel:

- May be best means of attaching tow rope due to lack of suitable attachments on bow of casualty vessel.
- Identify suitable strong points and distribute load evenly.

#### Pass tow

- **Passing the tow.**

Training criteria. Describe the different methods of passing a tow available to the SAR unit:

- Alongside if weather / sea state permits.
- Pass tow rope directly.
- Heaving line / throw bag.
- Line thrower.
- Use daughter craft or other vessel.

- **Equipment to be taken when boarding a casualty vessel.**

Training criteria. State the equipment to take on board the casualty vessel:

- Portable VHF.
- Casualty care kit.
- Casualty drogue (If carried).
- Knife.
- Torch.

task continues

## Task 4.5.1

## Towing

### b Secure and monitor tow

#### Training guidelines

#### Secure tow

##### • Securing the tow.

Training criteria. Demonstrate how to make fast on both the casualty vessel and the SAR unit:

- Only pass and make fast when order given by the SAR unit commander.
- In accordance with SOP.
- Make fast to casualty vessels strongest point (bitts, anchor winch, cleats, mast) - discuss problems with deck stepped / keel stepped.
- Tow made fast to bitts, bollards on SAR unit, keeping working turns.

##### • Deploying the casualty drogue.

Training criteria. Explain the use of the casualty drogue and how it is deployed:

- Used to slow the forward movement of a vessel being towed.
- Assists in preventing broaching.
- Assists a vessel without steerage.
- Deploy from stern of casualty in accordance with SOP.

##### • Towing dangers.

Training criteria. Describe the potential dangers during a tow:

- Lines under load.
- Tow parts.
- Stepping across the tow rope.
- Casualty fenders.
- Standing in bights or coils.

Training criteria. State the precautions that can be taken to minimise the dangers:

- Keep a safe distance.
- Those not involved must keep clear (No Go area).

##### • Monitor tow.

Training criteria. Explain what factors you should take into account when towing:

- Communicate to SAR unit commander.
- Weather considerations.
- Chafe: freshening the nip.
- Monitoring the tow.
- Communication with casualty.
- Casualty comfort.
- Good line management.

##### • Chafe and damage.

Training criteria. Describe how to reduce chafe on the casualty vessel and the SAR unit:

- Monitoring of both vessels and tow rope.
- Using chafing covers.
- Protect line on casualty.
- Freshening the nip.
- Lead towline through fairleads.
- Lead through fairlead / bow roller.

task continues

# Activity 4 - Seamanship

## Task 4.5.1

## Towing

### c Transfer tow

#### Training guidelines

#### Transfer tow alongside SAR unit

- **Transferring the tow alongside the SAR unit.**

Training criteria. Describe the communication given prior to transfer:

- Brief crew and casualty prior to transfer.
- Preparation of deck, e.g. fenders, springs.

- **Safety factors.**

Training criteria. Explain the safety factors that should be considered before transferring the tow:

- Awareness of those on deck.
- Weather conditions.
- Size of vessel to be brought alongside.
- Lines under load.
- Tripping over lines.
- Snatch.
- Loose lines.
- Crew / casualty safety.

- **Positioning of casualty vessel.**

Training criteria. Describe how the casualty vessel should be positioned alongside the SAR unit:

- Under coxswain / helm instruction.
- When safe to do so.
- Stern of SAR unit aft of casualty.

- **Fendering arrangements.**

Training criteria. State the requirement for fenders:

- Preventing damage to vessels.
- Free board.
- Vessel stanchions.
- Compatibility of vessels.
- Hull shape and projections.

- **Securing the tow alongside.**

Training criteria. Describe the method of securing the casualty alongside:

- Bow and stern lines.
- Springs: importance of loading on springs.

#### Transfer tow to another vessel / mooring

- **Transferring the tow to another vessel / mooring.**

Training criteria. Describe the sequence of transferring the tow to another vessel or mooring:

- Brief given prior to transfer.
- Fendering and lines on casualty.
- Wait until given the order for appropriate sequence of release.
- Moor casualty.

task continues

## Task 4.5.1

## Towing

### d Release tow

#### Training guidelines

#### Release tow

- **Releasing the tow.**

Training criteria. Describe how and when to release the tow:

- When instructed to do so.
- With care when vessels alongside.
- Sequence of release: weather / sea state, location.
- Good line management.

Training criteria. State the safety considerations:

- Crush injuries.
- Lines under load.
- Keep lines clear of propellers / thrusters.

#### Emergency release

- **Releasing the tow in an emergency.**

Training criteria. State reasons for emergency release:

- Fire.
- MOB.
- Sinking.
- Grounding.

Training criteria. Explain methods of release:

- Cut tow rope closest to securing point (ALB only).
- Slip and let all line pay out from the drum.

Training criteria. Describe safety precautions:

- Keep clear.
- Be aware of recoil.



# Activity 4 - Seamanship

## Task 4.5.2

## Manage Towing

### a Evaluation

#### Training guidelines

#### Decision to tow

- **Evaluating the factors affecting the decision to tow.**

Training criteria. Explain the factors affecting the decision to tow:

- Use of the towing assessment checklist.
- Is life at risk and is towing the best or only method of saving life?
- Advice of coastguard.
- Other vessels available?
- Vessel in danger.
- Crew (both) competence to establish and manage tow.
- SAR unit limitations.
- Weather / sea state: now and later.
- Vessel size / displacement.
- Condition of casualty vessel.
- Windage.
- Has the casualty vessel master requested a tow.
- Suitable safe haven identified.
- Distance to safe haven.

- **RNLI policy on towing.**

Training criteria. State the RNLI policy on towing:

- See policy ***PO1026 Policy on Salvage and Towing***. See also ***GU1028 Lines Under Load***.
- A tow must only be undertaken as part of an attempt to save life.
- Exceptionally, a tow may be undertaken if an abandoned vessel would pose a significant risk to navigation or could be mistaken in the future as a casualty vessel.
- A casualty vessel must be taken to the nearest safe haven – agreed prior to towing
- Towing by lifeboat must not be conducted if a tug or other suitable vessel can be summoned without undue delay.

#### Communications

- **Establishing communications.**

Training criteria. Describe the communications needed between the casualty vessel and the SAR unit:

- Discussion with master / skipper of casualty vessel.
- Nature of problem.
- Injury to persons / POB.
- Destination - nearest position of safety and security as determined by SAR unit.
- Ability of casualty to establish tow.
- Requirement to put crew on board casualty.
- Brief casualty and crew on methods for establishing tow.

task continues

## Task 4.5.2

## Manage Towing

### a Evaluation (continued)

#### Training guidelines

#### Casualty vessel stability

- **Assessing other vessels.**

Training criteria. State the common signs of casualty vessels having stability or buoyancy related issues:

- Pitching and rolling effects.
- Heel and angle.
- History of or currently grounded.
- Sitting lower than standard waterline.

Training criteria. Explain the importance of assessing the hazards of boarding a casualty vessel showing signs of stability or buoyancy issues:

- The safety aspects of boarding a casualty where stability is considered an issue.
- If in doubt don't do it, don't go below decks.
- Risk assessment, risk vs benefit.

Training criteria. Explain the hazards of towing a vessel regarding the SAR unit and casualty vessels stability:

- A grounded vessel and the possible effects of towing it clear – is it holed or damaged.
- Towing effects – shifting gear / cargo and flooding on casualty vessel.
- Towing effects – effects on handling and stability of SAR unit of a wallowing vessel and danger of vessel sinking under tow.

#### Casualty destination

- **Consideration of where to tow casualty vessel**

Training criteria. State the factors to be considered when determining where to tow a casualty vessel to:

- Nearest place of safety.
- Permission of harbour master may be required.
- Coastguard may direct a harbour master to take a casualty vessel.

task continues

## Activity 4 - Seamanship

### Task 4.5.2

### Manage Towing

#### b Pass tow

##### Training guidelines

##### Pass the tow

- **Transferring crew to the casualty vessel.**

Training criteria. Explain when a member of the SAR unit crew may be placed on board the casualty:

- Casualty unable to rig tow.
- Casualty support - casualty care, communications, risk vs benefit.
- State of casualty vessel - flooding, fire.

Training criteria. Demonstrate transferring a crew member to casualty vessel:

- Appropriate to class, conditions and casualty vessel.

- **Positioning the SAR unit to pass the tow.**

Training criteria. Demonstrate where the SAR unit should be positioned to pass the tow.

Training criteria. Demonstrate how to pass the tow:

- Where - safe location.
- How - within range of heaving line / rocket line / tow rope.
- Alongside:
  - casualty lying head to sea
  - at anchor or sea anchor.
- Crossing the T - casualty lying beam on to sea.
- Downwind / sea - casualty running with wind / sea.

- **Using the casualty drogue.**

Training criteria. Explain when and why the casualty drogue would be used:

- Vessels with steering problems:
  - loss of rudder
  - jammed rudder
  - jet drive.
- Vessels affected by sea / weather:
  - prevent surfing
  - broaching.
- Vessels with shallow draft.

task continues

## Task 4.5.2

## Manage Towing

c

### Conduct of tow

#### Training guidelines

#### Secure the tow

- **Determining the length of tow.**

Training criteria. Explain the factors that determine the length of tow:

- Vessel size and displacement.
- Sea conditions.
- Sea room.
- Both vessels should be in the same position relative to the wave pattern.

- **Communications.**

Training criteria. Describe the communications required whilst securing the tow:

- Communications between both vessels.
- Communications between SAR unit commander and crew - make fast, payout etc.
- Update coordinating authority.
- Keep casualty briefed and updated.
- Use crew onboard casualty for communications.

- **Safety precautions.**

Training criteria. Describe hazards and safety precautions:

- |                                  |                                        |
|----------------------------------|----------------------------------------|
| • Tension on tow rope.           | • Recoil.                              |
| • Standing in bights.            | • Safe use of towing points .          |
| • Correct use of descender.      | • Use of towing bridles.               |
| • Making fast.                   | • Correct PPE.                         |
| • Line management (monitor tow). | • Keep towing area clear (No Go area). |

#### Towing

- **Considerations.**

Training criteria. Explain the considerations during towing, in relation to the vessels and crew members involved:

- Monitoring the crew:
 

|                       |                        |
|-----------------------|------------------------|
| - limitations of crew | - exposure to elements |
| - fatigue.            |                        |
- Monitoring the tow:
 

|                                   |                                   |
|-----------------------------------|-----------------------------------|
| - safe towing speed for casualty  | - monitor casualty condition      |
| - freshen the nip / prevent chafe | - changes in weather / sea state. |
- Communications with casualty.
- Communication with coordinating authority.
- Communication with harbour authority (pollution risk, fire, taking in water, permission to enter harbour).
- Actions to take if declined entry to harbour.

task continues

# Activity 4 - Seamanship

## Task 4.5.2

## Manage Towing

### d Transfer tow

#### Training guidelines

#### Transfer tow alongside SAR unit

Training criteria. Explain the considerations, including safety factors, for alongside towing.

Training criteria. Demonstrate the procedures used to bring a vessel alongside the SAR unit:

- Distance to tow.
- Other vessels.
- Confined space manoeuvres.
- Condition of casualty vessel.
- Line management.
- Sea state / weather conditions.
- Crew brief / safety.
- Fendering.

#### • Positioning the casualty vessel.

Training criteria. Demonstrate how the casualty vessel should be positioned alongside the SAR unit:

- Alongside: choice of side dependent on casualty damage, destination / berth, weather conditions.
- Stern of SAR unit aft of casualty vessel, better manoeuvrability.

#### • Fendering requirements.

Training criteria. Explain the requirement for fendering and where they would be deployed:

- To prevent damage to both vessels.
- To keep vessels clear of one another.
- Positioning:
  - freeboard
  - size / displacement
  - flair
  - stanchions.

#### • Securing the tow.

Training criteria. Demonstrate how the tow is secured:

- Bow and stern lines.
- Springs: loading of springs to prevent movement.

#### Transfer tow to another vessel / mooring

#### • Considerations prior to transferring the tow.

Training criteria. Describe the considerations prior to the transfer of the tow:

- Crew briefing.
- Casualty briefing (owner responsible for mooring their own vessel).
- Preparation of lines, tow ropes, fenders.
- Location of tow transfer or mooring.

Training criteria. Describe what additional fenders and lines may be required:

- Casualty's lines and fenders rigged as required, heaving line on messenger.

#### • Transfer tow.

Training criteria. Demonstrate the ability to manoeuvre the SAR unit into position on a mooring / alongside with a casualty vessel in alongside tow:

- To a predetermined position alongside.
- To a predetermined mooring.

task continues

### Task 4.5.2

### Manage Towing

e

#### Release tow

##### Training guidelines

##### Release tow

- **Considerations for releasing the tow.**

Training criteria. Explain the considerations when releasing the tow under normal conditions:

- Safety factors:
  - tow rope to be kept clear of propellers
  - keep clear of bights and coils
  - release when safe to do so
  - crew brief
  - casualty brief
  - release tow - SAR unit or casualty end.

##### Emergency release

- **Releasing the tow in an emergency.**

Training criteria. Explain how to release the tow in an emergency situation:

- Crew briefed.
- Cut with knife - close to bitts / bollards to minimise on board recoil.
- Keep out of line of tow rope.
- Keep as far away as possible.

# Activity 4 - Seamanship

## Task 4.6.1

## Stability Introduction

a

### Stability and buoyancy

#### Training guidelines

#### Vessel stability

**Note:** this section can be completed by conducting the eLearning on Learning Zone.

- **Buoyancy and stability.**

Training criteria. Describe the effect and generation of buoyancy in vessels:

- How buoyancy is generated and the effect.

Training criteria. Describe the effect and importance of stability in vessels:

- The principles of angle of vanishing stability.
- The principles of how to make a vessel stable.

- **Hazards.**

Training criteria. Explain the free surface effect:

- The hazards of having water / liquids onboard a vessel - inc. sea water and fuel.
- The hazards associated with water / liquid moving freely around vessel compartments affecting vessel stability.

Training criteria. Describe how uneven load or shifting load can affect stability:

- Even load to reduce heel and listing effects on stability.
- Moving load amplifying normal vessel movement creating instability.
- Important to evenly load, and secure loads to aid vessel stability.

Training criteria. Explain the causes and actions in the event of flooding:

- Watertight integrity.
- Sea state and direction of travel.
- In accordance with the current EOP.

Training criteria. Explain the causes and actions in the event of being holed:

- Collision.
- Grounding.
- In accordance with the current EOP.



## Task 4.6.2

## Hydrology and Hull Types

### a Hydrology

#### Training guidelines

#### Hydrology

##### • Wave formation.

Training criteria. Explain the factors affecting the size and formation of waves:

- Wind strength, direction, duration.

Training criteria. Explain the hazards associated with breaking waves and the changes that take place:

- Moving water, increased buoyancy and reduction of displacement, loss of control, weight and impact of water, potential for capsize, potential for becoming airborne.

Training criteria. Explain the effect of headland, beaches and shoals upon wave shape.

Training criteria. Explain the effect of current and wind upon wave shape:

- Headland:
  - compression of flow
  - increase of current
  - steep breaking waves
  - shoaling
  - overfalls,
  - wind over tide.
- Shoals:
  - overfalls
  - increase in current
  - breaking waves
  - wind over tide.
- Beaches:
  - interaction with seabed
  - wave size increases
  - wave breaking at depth equal to 1.3 x height of wave
  - shallow beach gradual process
  - steep beach sudden change
  - lateral currents and rip currents exist.

task continues

## Task 4.6.2

## Hydrology and Hull Types

### b Hull types and propulsion

#### Training guidelines

#### Hull types

##### • Types of hull.

Training criteria. State the benefits and limitations of various hull types:

- Planing hull:
  - higher speed in good conditions
  - shallow V-hull may slam in heavy waves
  - twin engines on a planing hull give improved turning.
- Displacement hull:
  - reduced top speed
  - good sea-keeping
  - inclined to roll.
- Semi-displacement hull:
  - good sea-keeping with moderate speed
  - needs a lot of power to go fast
  - long keel / small rudders give reduced turning.

#### Propulsion systems

##### • Types of drive system.

Training criteria. Describe the benefits and limitations of various drive systems:

- Twin and single screw:
  - simple reliable system
  - loss of drive in aerated water.
  - can be fouled
- Jets:
  - good manoeuvrability and shallow water operation.
- Outboard motors:
  - steerable power
  - easily changed
  - self-contained
  - exposed.

##### • Dangers of drive systems.

Training criteria. Describe the dangers associated with the different types of drive system:

- Exposed propellers:
  - easily damaged
  - fouling
  - moving parts hazardous for persons in the water.
- Jets:
  - moving control parts
  - high pressure flow.
- Outboard motors:
  - petrol fire risk
  - moving parts hazardous for persons in the water.

## Task 4.7.1

## Helming 1

### a Controls and alarms

#### Training guidelines

#### Helm position layout

- **Controls and instruments.**

Training criteria. Identify the controls and instruments:

- Wheel / tiller.
- Throttles.
- Gear control.
- Compass.
- Depth sounder.
- MOB alarm.
- Thruster.
- Trim tabs.
- Rev counters.
- Helm indicator.
- Water ballast.
- Pressure and temperature gauges.
- Intercom.
- Control stick.
- Elevator pedals.
- Lights.
- Horn.
- Auto pilot.
- Weather instruments.

Training criteria. Demonstrate the operation of the controls and instruments:

- Helm.
- Gears / throttles / jets.
- Instruments to monitor status.

#### Actions on alarms

- **Alarms relevant to crew helming.**

Training criteria. State the actions to be taken when an alarm is activated:

- Inform SAR unit commander / mechanic.
- Be ready to manoeuvre / slow down / stop.

task continues

# Activity 4 - Seamanship

## Task 4.7.1

## Helming 1

### b Steer a course

#### Training guidelines

##### Maintain a course

- **Maintaining a course.**

Training criteria. Explain how to maintain a course:

- As fitted:
  - compass
  - visual reference
  - transit
  - auto pilot.

##### Manoeuvring

- **Altering speed / course.**

Training criteria. Demonstrate methods available for altering speed / course:

- As fitted:
  - as instructed
  - tiller
  - trim tabs
  - control stick
  - wheel
  - engines
  - auto pilot
  - elevators.

Training criteria. State the possible dangers associated with using the auto pilot (if fitted):

- Be aware of rapid course changes using auto pilot.

- **Prior to altering speed / course.**

Training criteria. Explain the actions to be taken prior to a speed / course alteration:

- Look around especially astern.
- Check along new heading.
- Consider the effects of actions on others.
- Warn the crew.
- Only alter course if safe to do so.

- **Communications.**

Training criteria. State the required communications and their importance:

- Repeat helm orders.
- Confirm when complete.
- Awareness of lookouts.
- Awareness of hazards and communicating any identified to person in command.

task continues

## Task 4.7.1

## Helming 1

c

### Transfer control

#### Training guidelines

#### Transfer control

- **Transferring control.**

Training criteria. State when and how controls are transferred:

- When:
  - to ALB USP for overview when manoeuvring, berthing or in search area
  - to ALB wheelhouse for rough weather, extended passage
  - transfer only when safe to do so e.g. clear astern to stop in channels
  - transfer only when ILB stopped and in neutral.
- How:
  - ensure helm in new position
  - communications established
  - course, hazard, position information passed in hand over
  - throttle setting or neutral.

- **Communications.**

Training criteria. State the required communications during the transfer of control:

- Two-way communication.
- Via Intercom to confirm engine / helm / steering order.
- Clear concise commands repeated and confirmed.
- Set throttles and take station (if applicable).
- Confirm control.

- **Precautions.**

Training criteria. Explain the precautions to be taken during the transfer of control:

- Sufficient sea room.
- Clear communications.
- Maintain control until successfully transferred.

task continues

## Task 4.7.1

## Helming 1

### d Man overboard procedure

#### Training guidelines

##### Initial actions

- **Immediate actions in the event of MOB.**

Training criteria. Demonstrate the immediate actions to be taken in the event of a MOB:

- Reduce speed.
- Prepare to manoeuvre.

Training criteria. State what equipment must be activated and the actions to be expected of the crew:

- Shout and point.
- Maintain visual contact.
- Press MOB on navigation equipment.
- In accordance with the current SOP.

##### Recovery

- **Manoeuvring the SAR unit for recovery.**

Training criteria. Demonstrate recovery of the MOB:

- Communication.
- Consider weather / sea state.
- Direction of approach.
- Speed of approach.

- **Positioning the SAR unit for recovery of the casualty.**

Training criteria. Demonstrate how to manoeuvre the SAR unit to best aid recovery of a casualty from the water:

- In accordance with the current SOP.
- Consider all safe options allowing for wind and sea direction minimising risk to casualty and crew.
- Proceed dead slow, ideally upwind, to approach MOB.
- Reduced speed.
- Stop prop if safe to do so.

### Task 4.7.2

### Helming 2

#### a Manoeuvring in confined areas

##### Training guidelines

##### Confined areas

- **Controlling the SAR unit in confined areas.**

Training criteria. Demonstrate how to manoeuvre the SAR unit in confined areas using the helm, both engines and forward and astern propulsion. Must take proper account of wind, tide and any other external forces.

##### Manoeuvring alongside

Where location allows, demonstrate manoeuvring alongside and departing safely in a variety of conditions.

- Use of both engines and forward and astern power.
- Manoeuvre the SAR unit in confined areas allowing for wind and tide.
- Crew brief / fenders / escape route as applicable.
- Using available locations and associated methods possible at location.

task continues



# Activity 4 - Seamanship

## Task 4.7.2

## Helming 2

### b Boat handling in heavy weather

#### Training guidelines

#### Heavy weather

- **Restrictions and limitations in heavy weather.**

Training criteria. State the weather / sea limitations for your SAR unit:

- See **GU 1041 – Powered Rescue Craft Operational Limitations:**
- In accordance with the current guidance, normally:
  - ALB:
    - no weather restriction - unless local conditions are unmanageable.
  - B-Class:
    - daylight – up to rough sea state (F 6/7)
    - darkness – up to moderate / rough sea state (F 5/6).
  - D-class:
    - daylight – up to moderate sea state (F 5)
    - darkness - Slight to moderate sea state (F 4/5). In darkness to remain in sight of shore or supporting ALB or B Class.
  - IRB / RWC:
    - no night capability; Within 400 metres of the shore or within visual range of supporting lifeboat / shore unit.
  - IRH:
    - force 5; Significant wave height not in excess of 600 mm; Wind speed 25 knots.

- **Manoeuvring the SAR unit.**

Training criteria. Explain the actions to take to limit damage and to prevent injury to the personnel onboard:

- Appropriate speed.
- Change heading.
- Avoid overfalls.
- Adjust trim.
- Crew positioning.
- Ballast (if applicable).

- **Manoeuvring the SAR unit at appropriate speed.**

Training criteria. Explain how to maintain the appropriate speed:

- Appropriate use of helm and throttle.
- Consider alternative course. Carry out any other tasks as directed by the coxswain / helm / commander.

task continues

## Task 4.7.2

## Helming 2

### c Going alongside another vessel

#### Training guidelines

#### Manoeuvre alongside

##### • Manoeuvring alongside another vessel.

Training criteria. Demonstrate how to manoeuvre the SAR unit alongside another vessel, using engine and helm movements to achieve this:

- Complete the manoeuvre using engine and helm movements to achieve this.
- Minimum steerage speed of target vessel.

##### • Risks to the SAR unit, crew and casualty.

Training criteria. Describe the risks associated with coming alongside another vessel for your crew, the SAR unit and the other vessel:

- Minimise risk by communication with other vessel, crew brief, safe speed and fendering.
- Injury / entrapment / MOB.
- Equipment overhanging the side of the other vessel.
- Sea state.
  - Risk of collision.
- Propeller fouling.
  - Damage to vessel.
- Size relationship between vessels.

##### • Effects of interaction.

Training criteria. Explain what interaction is and what effect it will have on the SAR unit and the other vessel:

- Interaction is the force between moving vessels.
- Bow and stern areas will push apart, in-between will draw together.

Training criteria. Describe other forms of interaction and how these can be identified and their effects reduced:

- Other forms include squat where the vessel is drawn down in shallow water, reducing speed reduces squat.

#### Manoeuvre close to another vessel

##### • Manoeuvring close to another vessel.

Training criteria. Demonstrate how to hold station with another vessel without making contact with it:

- Complete the manoeuvre using engine and helm movements to achieve this.

#### Manoeuvre away from another vessel

##### • Manoeuvring away from another vessel.

Training criteria. Demonstrate how to manoeuvre away from another vessel to avoid damage to both vessels and risk of injury to crew and casualty:

- Control of speed.
- Correct use of helm.
- Consider crew brief.
- Consider weather / sea state / tide.

task continues

# Activity 4 - Seamanship

## Task 4.7.2

## Helming 2

### d Emergency steering

#### Training guidelines

#### Rig emergency steering

- **Preparing the emergency steering.**

Training criteria. Explain the preparations prior to rigging the equipment:

- Reduce speed.

Training criteria. Locate the bypass valve and all associated equipment:

- Locate bypass valve.
- Identify and gather equipment from stowage.

- **Precautions and limitations.**

Training criteria. Describe the precautions and limitations of rigging and using emergency steering:

- Precautions:
  - no persons should enter the steering gear compartment without the coxswain's approval
  - the helm should not be used when any person is in the steering gear compartment SAR unit commander
  - the hydraulic steering must not be bypassed without the approval of the coxswain / helm
  - the hydraulic system bypass valve must be opened before fitting the tiller arm in position on rudder stock
  - when de-rigging the emergency steering, the tiller arm must be removed before bypass valves are closed
  - keep clear of moving tiller and gear.
- Limitations:
  - may result in limited manoeuvrability or reduced speed capability
  - poor communications.

- **Rigging the emergency steering.**

Training criteria. Demonstrate rigging the emergency steering equipment:

- In accordance with the current class specific SOP.
- Report when ready.

#### Use of emergency steering

- **Emergency steering.**

Training criteria. Demonstrate the use of emergency steering:

- In accordance with the current class specific SOP.
- Helm communicates required steering action using voice / intercom or agreed hand signals.

#### De-rig emergency steering

- **De-rigging the emergency steering.**

Training criteria. Demonstrate de-rigging the emergency steering equipment:

- In accordance with the current class specific SOP.
- Equipment to be stowed securely.

task continues

### Task 4.7.2

### Helming 2

#### e Propeller change (ILB)

Training guidelines

ILB Only

#### Change a propeller due to damage

Training criteria. Change a propeller whilst afloat:

- Carry out propeller change correctly in accordance with current emergency operating procedures.

# Activity 4 - Seamanship

## Task 4.7.3

## Mooring and Berthing

### a Berthing alongside

#### Training guidelines

##### Prepare

###### • Preparing to berth.

Training criteria. Describe or demonstrate preparation for berthing:

- Crew briefed.
- Fenders rigged at the correct height.
- All lines on deck, faked out ready for use as directed.

###### • Communications prior to berthing.

Training criteria. Describe the communication to expect from the coxswain / helm / commander:

- Order of lines.
- Crew position on deck.
- Position of fenders.
- Position of bollards / bitts.
- Effects of tide / weather.
- Position of embarkation / disembarkation points.

###### • Berthing for a prolonged period.

Training criteria. Explain the factors to consider if berthing for a prolonged period:

- Awareness of tidal rise and fall.
- Prevention of chafe.
- Security.
- Access.

##### Secure

###### • Factors to be considered.

Training criteria. State factors to be considered when securing the vessel:

- Trips and slips.
- Entrapment of hands and fingers.
- Getting ashore safely.
- Correct use of lines and knots.
- Keep lines clear of bights, props etc.
- Lines under tension.
- Throwing heaving lines.
- Securing the SAR unit.

Training criteria. Describe how to secure the SAR unit:

- Bow and stern line.
- Springs.
- Breasting ropes.

##### Release

###### • Slipping and releasing lines.

Training criteria. Explain rigging lines for slipping and the order of release:

- Run bight ashore.
- Lines to be released as instructed.
- Lines to be recovered on board.
- Be aware of lines under load.
- All lines and fenders to be stowed.
- Decks cleared ready for sea.
- Keep lines clear of bights, propellers etc.

###### • Communications whilst casting off.

Training criteria. Describe the instructions to expect from the coxswain / helm / commander during the release procedure:

- Coxswain / helm / commander to pass instructions on when to let go.
- Inform coxswain / helm / commander if there is a problem.

task continues

## Task 4.7.3

## Mooring and Berthing

### b Mooring to a buoy

#### Training guidelines

#### Prepare

##### • Preparing to moor.

Training criteria. Describe or demonstrate preparation for mooring to a buoy:

- Crew briefed.
- Boat hook.
- Line ready.
- Illumination.

##### • Communications prior to mooring.

Training criteria. Describe the communication to expect from the coxswain / helm / commander:

- Two-way communication.
- Specific tasks on deck.
- Terminology.
- Crew brief with method.
- Effect of tide and weather.
- Hand signals may be used (e.g. distance to buoy).

##### • Mooring for a prolonged period.

Training criteria. Explain the factors to consider if mooring for a prolonged period:

- Access.
- Coxswain / helm / commander is responsible for safe mooring.
- Additional line for security.
- May show an anchor light.
- Prevention of chafe.
- Security.

#### Secure

##### • Factors to be considered.

Training criteria. State factors to be considered when securing the vessel:

- Safety lines.
- Getting ashore safely.
- Lines under tension.
- Trips and slips, hands and fingers.
- Correct use of lines and knots.
- Keep lines clear (e.g. of bights, props).

##### • Securing the SAR unit.

Training criteria. Describe how to secure the SAR unit:

- Mooring band with chain.
- Line attached.
- Use additional lines for a prolonged period.
- Use of SAR unit lines.

#### Release

##### • Slipping and releasing lines.

Training criteria. Explain rigging lines for slipping and the order of release:

- Brief to single up / rig as slip back to vessel.
- Slip the end without an eye or knot.
- Report when clear.
- Keep lines clear (e.g. of bights, props).
- Release as instructed.
- Engines may be used to ease tension.
- Decks cleared ready for sea.

##### • Communications whilst releasing.

Training criteria. Describe the instructions to expect from the coxswain / helm / commander during the release procedure:

- Coxswain / helm / commander to pass instructions on when to let go.
- Inform coxswain / helm / commander if there is a problem.

## Activity 4 - Seamanship

### Task 4.7.4

### Hovercraft Handling

#### a Manoeuvring the IRH in confined areas

##### Training guidelines

##### Confined areas

- **Controlling the Hovercraft on land.**

Training criteria. Demonstrate the ability to manoeuvre the Hovercraft through a land slalom:

- To the requirements of the "IRH - Land Test Circuit".
- For two complete circuits of the course.
- On a course set in the Up / Down wind direction.
- The SAR unit must not travel a distance more than twice its own length away from the slalom cones.
- At a speed appropriate to the prevailing conditions.

Training criteria. Demonstrate the ability to maintain a number of specified speeds in a variety of wind conditions:

- To the requirements of the "IRH - Land Test Circuit".
- 4 knots: - Up-wind, Down-wind, Across-wind.
- 12 knots: - Up-wind, Down-wind, Across-wind.

**Note: weather-cocking must not occur during any of these manoeuvres.**

Training criteria. Demonstrate the ability to maintain a number of specified speeds through 180° turns to port and starboard:

- To the requirements of the "IRH - Land Test Circuit".

Training criteria. Demonstrate how to carry out emergency stop in a straight line and through 180°:

- To the requirements of the "IRH - Land Test Circuit".

Training criteria. Demonstrate the ability to park the SAR unit in the suitable position for carrying out a land rescue:

- To the requirements of the "IRH - Land Test Circuit"

task continues



## Task 4.7.4

## Hovercraft Handling

### b Hovercraft handling at sea

#### Training guidelines

##### Heavy weather

- **Restrictions and limitations in heavy weather.**

Training criteria. State the weather / sea limitations for your SAR unit:

- In accordance with the current guidance, normally:
  - IRH: Force 5; Significant wave height not in excess of 600 millimetres; Wind speed 25 knots.

- **Manoeuvring the SAR unit.**

Training criteria. Explain the actions to take to limit damage and to prevent injury to the personnel onboard:

- Appropriate speed.
- Change heading.
- Adjust trim.
- Crew positioning.

##### Above hump speed

- **Manoeuvring the Hovercraft at appropriate speed.**

Training criteria. Demonstrate the ability to maintain a number of specified speeds in a variety of wind conditions:

- To the requirements of the "IRH (Afloat) Test Circuit":
  - 12 knots: - Up-wind, Down-wind, Across-wind
  - 20 knots: - Up-wind, Down-wind, Across-wind.

**Note: Weather cocking must not occur during any of these manoeuvres.**

Training criteria. Demonstrate the ability to maintain a number of specified speeds through turns to port and starboard:

- To the requirements of the "IRH (Afloat) Test Circuit"
- Keep the angles within specifications of the Yaw Graph:
  - 12 knots: - to PORT
  - 12 knots: - to STARBOARD
  - 20 knots: - to PORT
  - 20 knots: - to STARBOARD.

Training criteria. Demonstrate how to carry out emergency stop in a straight line:

- To the requirements of the "IRH (Afloat) Test Circuit".

Training criteria. Demonstrate how to carry out a displacement turn:

- To the requirements of the "IRH (Afloat) Test Circuit".
- Describe the effects of the wind during the turn.

task continues

### Task 4.7.4

### Hovercraft Handling

#### c Recovery of person from water

##### Training guidelines

##### Man overboard

- **Manoeuvring the SAR unit to the MOB position.**

Training criteria. Demonstrate the manoeuvres that are available to return to the position where the man overboard occurred:

- To the requirements of the "IRH (Afloat) Test Circuit".

##### Recovery of casualty

- **Positioning the SAR unit for recovery of the casualty.**

Training criteria. Demonstrate how to manoeuvre the SAR unit to best aid recovery of a casualty from the water:

- In accordance with the current SOP.

## Task 4.8.1

## Breeches Buoy

### a Prepare and deploy

#### Training guidelines

#### Preparation

- **Reasons for deployment.**

Training criteria. Explain what the breeches buoy can be used for:

- To recover casualties from otherwise inaccessible locations.
- To transfer equipment / personnel.

- **Factors to be taken into consideration.**

Training criteria. Describe the factors that should be considered before deploying the breeches buoy:

- How to best position the SAR unit.
- Placing the SAR unit within range.
- Passing the messenger line.
- Sea state / wind / tide.
- Other assets, e.g. cliff team, helicopter.

#### Deployment

- **Precautions to be taken during set-up.**

Training criteria. Explain what precautions need to be taken when setting up the breeches buoy:

- Ensure all lines are free to run out and cannot foul during deployment.
- Crew must be aware of trip hazards on deck.
- Hand hazards.
- Full PPE.
- Personal injury.
- Line management.

- **Set-up and deploy.**

Training criteria. Demonstrate set-up and deployment of breeches buoy:

- In accordance with the current SOP.
- Deploy as directed.

- **Alternative uses of breeches buoy equipment.**

Training criteria. Describe alternative uses for breeches buoy equipment:

- Use to send salvage pump.
- Send daughter boat / liferaft under control of veering lines.

task continues

# Activity 4 - Seamanship

## Task 4.8.1

## Breeches Buoy

### b Operate and recover

#### Training guidelines

#### Effective operation

- **Communications required.**

Training criteria. Explain the communication required:

- Two-way communication between the lifeboat and casualty.

Training criteria. Describe additional communication that may be required and how this is achieved:

- When to get into breeches buoy.
- When crew to haul.
- May use loud hailer / radio / etc.

- **Hazards associated with deployment and operation.**

Training criteria. State the problems / hazards that might be experienced during deployment and operation of the breeches buoy:

- Heavy weather hazards, to crew and casualty.
- Entanglement.
- Rope burn.
- Confusion.
- Anchor dragging.
- Tide / wind changing.
- Lines fouling / twisting.

#### Recovery

- **Recovering the breeches buoy.**

Training criteria. Demonstrate the recovery procedure:

- In accordance with the current SOP.
- Considerations for recovery of tail block.
- Line management.
- Consider "cut and run".

- **Stowing the breeches buoy.**

Training criteria. Describe the procedures for cleaning and stowing equipment after use:

- Check for damage.
- Rinse and dry.
- Coil into correct stowage.
- Secure as appropriate.

## Task 4.9.1

## Daughter Craft

### a Prepare and deploy

#### Training guidelines

#### Initial preparation

##### • Intended uses.

Training criteria. Describe the intended uses of the daughter craft:

- Shoreline search.
- Crew and casualty transfer.
- Rescue.

Training criteria. Describe safety considerations and limitations of the daughter craft:

- In accordance with the current SOP.
- Towing capabilities.
- Night operations.
- Endurance.
- Swell / surf size.
- Load capacity (people).
- Speed / power / control.
- Maximum operating sea state.

Training criteria. Explain the concept of risk versus benefit relating to the use of daughter craft:

- Primary safety concerns:
  - crew
  - casualty's equipment.
  - casualty and other water users
- Hazards affecting risk to include:
  - environmental hazards
  - physical hazards
  - examples provided relating to the use of the daughter craft
  - procedures to follow if the risk outweighs the benefit
  - dynamic risk assessment: Stop, Assess, Plan, Execute, Review
  - applies principles of risk versus benefit in any given situation and throughout assessment.
  - human hazards
  - equipment hazards

##### • Preparing for launch.

Training criteria. Demonstrate preparation for launch:

- In accordance with the current SOP.
- Muster and check associated equipment.
- Check pressures (if appropriate).
- Check fuel (if appropriate).

##### • Additional equipment.

Training criteria. State the additional equipment to be carried:

- Radio.
- Flares.
- Survivor lifejackets.
- As relevant to the task.

task continues

## Activity 4 - Seamanship

### Task 4.9.1

### Daughter Craft

#### a Prepare and deploy (continued)

##### Training guidelines

##### Launch

- **Hazards with the launching.**

Training criteria. Describe potential hazards when launching:

- Injury / manual handling.
- Man overboard.
- Loss / capsize.
- Engine swamping.

- **Prior to launching.**

Training criteria. Demonstrate the actions to be taken immediately prior to launch:

- Crew ready in full PPE - SAR unit / daughter craft / boarding boat.
- Launch on lee side of SAR unit.
- Ensure launch area clear.
- Check drains are closed.
- Secure painter.
- Tilt engine fully up.
- Check with coxswain before launching.

- **Launching.**

Training criteria. Demonstrate the actions to be taken once the daughter craft / boarding boat is in the water:

- Move alongside.
- Secure fore and aft.
- Remove launching strops (if appropriate).

task continues

## Task 4.9.1

## Daughter Craft

### b Manoeuvre

#### Training guidelines

#### Embark crew to daughter craft / boarding boat

Training criteria. Demonstrate how to embark:

- From mother vessel:
    - daughter craft properly secured
    - embark one at a time
    - stow foot pump and safety equipment.
  - Embarking (from deep water):
    - embark unaided
    - embark on windward side.
- embark on lee side
  - ship and secure paddles / oars
  - embark one at a time

#### Engine start (if fitted)

Training criteria. Demonstrate how to start the engine:

- In accordance with current SOP.
- Prime fuel.
- Consider need for choke.
- Check air vent open.
- Check in neutral.

#### • Carrying out checks immediately after starting.

Training criteria. Explain the checks to be carried out immediately after starting the engine:

- Check cooling water.
- Gear movement.
- Free movement of tiller.

#### Handle the daughter craft / boarding boat

#### • Row / Motor Manoeuvring.

Training criteria. Demonstrate the ability to manoeuvre:

- Holding off.
- Proceeding to shore.
- Row or motor the boat up, down and across sea.
- Come alongside.
- Turning in a confined space.
- Returning to mother vessel.

#### • Limitations.

Training criteria. Explain the limitations:

- To be in communication at all times.
- Weather and sea limitations in accordance with SOP.
- Consider veering down from SAR unit.

#### • Emergency procedures.

Training criteria. Explain actions to be taken in the event of a sponson / keel failure:

- Crew / casualty position / ballast.
- Attempt recovery using minimum speed required.
- Communicate with SAR unit / coord authority.

Training criteria. Explain actions to be taken in the event of engine failure (if fitted):

- In open sea.
- On the shoreline.
- In surf zone.

task continues



## Activity 4 - Seamanship

### Task 4.9.1

### Daughter Craft

#### **b** Manoeuvre (continued)

##### Training guidelines

Training criteria. Explain actions to be taken in the event of a capsize of the daughter craft / boarding boat:

- Stay with craft if safe to do so.
- Ensure persons accounted for.
- Attempt recovery.
- Communicate with SAR unit / coordinating authority.

task continues

## Task 4.9.1

## Daughter Craft

c

### Recover and secure

#### Training guidelines

#### Disembark

##### • Disembarking.

Training criteria. Demonstrate disembarkation:

- Daughter craft / boarding boat properly secured.
- Disembark the daughter craft appropriately.
- Disembark one at a time.
- Recover paddles / oars, and equipment.
- Request assistance from SAR unit crew.
- Ensure deck clear prior to recovery.

#### Recovery

##### • Prior to recovery.

Training criteria. Demonstrate actions to be taken prior to recovery of the daughter craft / boarding boat:

- Remove and secure loose equipment.
- Open drains or bail excess water before lifting.
- Isolate fuel (if applicable).
- Tilt up engine (if applicable).
- Ensure SAR unit deck clear for daughter craft / boarding boat.
- Prepare recovery equipment.
- Appropriate PPE - SAR unit crew.

##### • Recovering.

Training criteria. Demonstrate the recovery:

- In accordance with the current SOP.

#### Securing of craft

##### • Stowing / post recovery.

Training criteria. Demonstrate the post recovery routine:

- Wash down craft and PPE.
- Refuel.
- Deflate and stow in accordance with the current SOP.
- Lash securely (if applicable) or restow.
- Notify of defects.
- Replenishment of kit.

## Activity 4 - Seamanship

### Task 4.10.1

### Helicopter Operations

#### a Prepare for helicopter operations

##### Training guidelines

##### Preparation

- **Preparing for the arrival of the helicopter.**

Training criteria. State the PPE that must be worn by the SAR unit crew:

- Lifejackets and protective headgear.

- **Additional equipment.**

Training criteria. Explain how a hi-line is stowed on the SAR unit and what additional PPE is required:

- Hi-line gloves.
- Place into a bucket or similar container clear of obstructions.

##### Night time operations

- **Actions that will NEVER be performed at night.**

Training criteria. State what must be avoided when working with helicopters at night:

- Flares are not to be fired or searchlights to be shone directly at the helicopter.
- No flash photography.

##### Landings and take-off

- **Preparing for the arrival and departure of the helicopter.**

Training criteria. Describe the hazards associated during aircraft landing and take-off:

- In accordance with the current SOP.
- Pilot in command of aircraft has overall responsibility and command.
- To include – hot landing, shut down landing techniques.
- To include the hazards of – rotor wash, blade sail, rotor angle on uneven terrain, flying debris.

Training criteria. Describe the procedure of preparing for and your role during landing and take-off:

- In accordance with the current SOP.
- Preparation to include removing loose objects, members of the public, marking of landing area (by day and night if required), briefing from air crew.
- RNLI responsibilities during aircraft manoeuvres.
- When it is safe to approach the aircraft.

task continues

## Task 4.10.1

## Helicopter Operations

### b Carry out helicopter transfer

#### Training guidelines

#### SAR unit position

- **Transferring between helicopter and SAR unit.**

Training criteria. Explain the different methods for transfer between a helicopter and SAR unit:

- Hi-line.
- Double lift.
- Single lift.
- Stretcher.

#### Safety factors

- **Dangers.**

Training criteria. Explain why the helicopter crewman, strop or helicopter winch wire must not be touched until it has made contact with either the SAR unit or water:

- A static discharge may occur if not earthed.

Training criteria. Describe the areas the SAR unit must avoid in relation to helicopter position, and explain why:

- Forward - acceleration area.
- Directly underneath - not visible.
- To Port - pilot unable to see lifeboat.
- Avoid down wash - capsize risk, unsafe working area.

- **Actions taken by a person being winched into a helicopter.**

Training criteria. State the actions that must be taken by a person being winched into a helicopter:

- ALB crewmember - MUST remove the auto inflation head from the lifejacket and to be replaced IMMEDIATELY when back on lifeboat.
- ILB crewmember - MUST remove ILB jacket and don an inflatable survivors lifejacket or helicopter's own.
- Follow instructions from helicopter crew, e.g. to keep arms down.

Training criteria. Describe the procedure for wet and dry winching:

- PPE requirements for RNLI personnel and casualties.
- In accordance with the current SOP.
- Briefing from aircrew.

#### Emergency procedures

- **Risk during helicopter operations.**

Training criteria. Describe the actions to take in an emergency:

- If a helicopter suffers a major mechanical failure that forces the pilot to 'ditch' the aircraft then the pilot will always attempt to move the aircraft away to port prior to ditching in the sea.

Training criteria. Describe the actions to take in the event of a fouled winch line / hi-line:

- If hi-line is knotted, leave alone.
- Helicopter crew will cut the wire if fouled on superstructure - crew to haul in.

Training criteria. Describe the actions to take if there is a problem with the SAR unit:

- Abort to starboard.
- Contact helicopter on VHF with the words "break off, break off, break off".

# Activity 4 - Seamanship

## Task 4.10.2

## Manage Helicopter Operations

### a Prepare for helicopter operations

#### Training guidelines

#### Liaising

- **Planning helicopter operations.**

Training criteria. Describe the requirements of planning and liaising with SAR Helicopter organisations in advance of any operations:

- In accordance with the current procedures.
- Liaison with organisations for knowledge sharing / ways of working.
- Planning for future exercises and role demonstrations.
- Passing of current RNLI personnel competence records.

#### Preparation

- **Preparing for the arrival of the helicopter.**

Training criteria. Describe how a plan is agreed between helicopter and SAR unit:

- If exercise via telephone or visit prior to event.
- Via VHF between SAR unit coxswain / helm and helicopter.

Training criteria. Describe how the SAR unit is prepared for the arrival of the helicopter:

- Plan must be agreed prior to starting, to include:
  - agreed course and speed
  - lookout
  - crew briefing
  - deck cleared
  - hi-line handling equipment ready
  - prepare sea anchor (if required).

Training criteria. Describe the required actions for both rotating and fixed aerals:

- MF switched off (ALB).
- Radar switched to stand by (scanner rotation OFF).

- **Maintaining communication with the helicopter.**

Training criteria. Describe how communications will be maintained:

- Agree working channel.
- Limited communications due to noise.

- **Factors affecting the positioning of the SAR unit.**

Training criteria. Explain the conditions affecting positioning of the SAR unit:

- Prevailing wind direction / speed.
- Direction / height of the sea.
- Tidal stream effect.
- Sea room available.

task continues

### Task 4.10.2

### Manage Helicopter Operations

#### b Carry out helicopter transfers

##### Training guidelines

##### SAR unit position

- **Positioning the SAR unit.**

Training criteria. Explain the options for positioning the SAR unit:

- Up wind – clear view of the lifeboat, helicopter flying preference.
- Cross-wind – sea room, hazard.
- Static – Sea state, casualty condition, location , mechanical failure.

- **Winching Procedures.**

Training criteria. Explain the procedures and options for winching operations with SAR helicopters:

- In accordance with the current SOP.
- Including winching whilst underway.
- Including winching whilst static.

- **Hi-line Procedures.**

Training criteria. Explain the procedures and options for Hi-line operations with SAR helicopters:

- In accordance with the current SOP.

# Activity 4 - Seamanship

## Task 4.11.1

## SAR Unit Drogue

### a Prepare

#### Training guidelines

#### Principles

- **Effects of a drogue.**

Training criteria. Explain how a drogue works and the effect it has on a SAR unit:

- By causing a drag on the stern.
- Slowing boat.
- Reducing yaw.
- Running.

Training criteria. Explain what tripping means:

- Reversing the drogue in the water prior to recovery.

- **When it is used.**

Training criteria. Describe in what circumstances a drogue should be used:

- Single engine failure.
- Prevent broaching.
- Following seas.
- Heavy weather.
- Crossing bars.
- Entering harbour, beaching.

#### Preparation

- **Rigging a drogue.**

Training criteria. Demonstrate how to rig the drogue:

- In accordance with the current SOP.

- **Precautions.**

Training criteria. Identify the risks involved in rigging the drogue:

- Lines clear to pay out.
- Deck clear.
- Good communication.
- Lines under heavy load.
- Lines near propellers.
- Bights.

task continues



## Task 4.11.1

## SAR Unit Drogue

### b Deploy and recover

#### Training guidelines

#### Deployment

##### • Deploying a drogue.

Training criteria. Demonstrate the correct sequence for deploying the drogue:

- In accordance with the current SOP.
- Deploy capsized / tripped.
- Drogue line made off at first mark.

Training criteria. Explain the actions required following deployment:

- Drogue must be controlled and monitored at all times.

##### • Adjusting once deployed.

Training criteria. Demonstrate how to use a tripping line (if fitted) and how to adjust the length of the drogue and tripping lines:

- In accordance with the current SOP.
- Drogue line followed by trip line.

Training criteria. Explain under what circumstances the drogue would need to be adjusted:

- Changing sea conditions.
- Breaking the surface.
- Chafe on warp.
- Fouled drogue due to spinning.

#### Recovery

##### • Recovering a drogue.

Training criteria. Explain the sequence for recovering the drogue:

- In accordance with the current SOP.

Training criteria. Describe how the coxswain may manoeuvre the SAR unit to aid recovery:

- SAR unit may be manoeuvred to the drogue.
- To take length off line.

##### • Communications.

Training criteria. State the communication to expect from the coxswain:

- Trip and recover when directed by the coxswain.

Training criteria. State the information the coxswain would require from the crew:

- When tripped (if appropriate).
- Position of drogue.
- Recovered.
- Stowed.

## Activity 5 - Search and Rescue

Photo: RNLI David Edwards



### Tasks

#### 5.1 Locate and Assist Casualty

5.1.1 Locate and Assist 1

5.1.2 Locate and Assist 2

5.1.3 Mud Rescue

#### 5.2 Plan Search and Rescue (SAR)

5.2.1 Plan SAR

#### 5.3 Manage Search and Rescue (SAR)

5.3.1 Manage SAR

5.3.2 Manage Mud Rescue

#### 5.4 Manage Operational Services

5.4.1 Manage Operational Services

# Activity 5 - Search and Rescue

## Task 5.1.1

## Locate and Assist 1

### a Searching

#### Training guidelines

#### Search methods

##### • Types of search equipment.

Training criteria. Identify the visual search equipment carried on the SAR unit:

- If applicable:
  - white parachute flares
  - search lights
  - binoculars
  - image Intensifier.

##### • Operating visual search equipment.

Training criteria. State how the equipment is used to assist with locating a casualty.

Training criteria. Demonstrate how to operate binoculars, image intensifier and search lights:

- White parachute flares: see pyrotechnics task.
- Binoculars:
  - further inspection of identified target
- Image Intensifier:
  - turn on
  - re-intensify.
- Search lights:
  - set up and adjust for use

- adjusted to suit operator.

- adjust to suit operator

- panning of light.

##### • Limitations of search equipment.

Training criteria. Identify the limitations of the equipment available:

- White parachute flares:
  - see pyrotechnics task
  - limited numbers
- Binoculars:
  - narrow field of vision
- Image intensifier:
  - battery life
  - danger to operator's eyes.
- Search lights:
  - power supply
  - disrupts night vision

- disrupts night vision

- short burn time.

- maintaining stability.

- limited field of vision

- glare in poor weather

- back scatter of shore lighting.

##### • Visual scanning techniques.

Training criteria. Explain the techniques and limitations when conducting a visual search:

- 10 - 15° sectors.
- Fatigue / boredom.
- No binoculars.
- SCAN - FOCUS - SCAN.
- Relative to speed of SAR unit.
- Rotation of lookout posts.

Training criteria. Explain why it is important and how to maintain night vision for effective night searching:

- Recovery time.
- How eyes work at night e.g. colour identification and peripheral vision.
- Current night scanning guidelines.

task continues

## Task 5.1.1

## Locate and Assist 1

### b Recover casualty

#### Training guidelines

#### Recovery techniques and equipment

- **Types of recovery equipment.**

Training criteria. Identify and explain the use of the recovery equipment available on the SAR unit:

- Jason's cradle.
- Scramble net.
- Lifting strop.
- Throw bag.
- All in accordance with equipment manuals and current SOP.
- A frame.
- Ambulance pouch.
- Mud lance.
- Hypohoist.

- **Precautions when using the recovery equipment.**

Training criteria. Explain the precautions to be taken when using the recovery equipment:

- State of casualty.
- Casualty brief if appropriate.
- Equipment to be used correctly and in accordance with SOP.
- Crew briefing.
- Hydrostatic squeeze.

- **Survivor life jacket.**

Training criteria. State the location of survivor life jackets.

Training criteria. Demonstrate the correct fitting of a survivor life jacket.

#### Crew welfare

- **Hazards.**

Training criteria. State the physical and environmental hazards you may be exposed to:

- Manual handling.
- Bights in lines.
- Fatigue.
- Hazardous materials, substances and environments.
- Lines under load.
- Entrapment.
- Anti-social behaviour.

task continues

# Activity 5 - Search and Rescue

## Task 5.1.1

## Locate and Assist 1

### c Casualty management

#### Training guidelines

#### Evaluation

##### • Evaluating the casualty - person.

Training criteria. Describe the considerations to be made during the evaluation of the casualty - person:

- Is further assistance required?
- Is evacuation required?
- Risk to crew (e.g. aggression, size, incapacity of individual).
- Communications.
- Medical care advice / support.

##### • Evaluating the casualty - vessel.

Training criteria. Describe the considerations to be made during the evaluation of the casualty vessel and assessing the need for recovery:

- Is further assistance required?
- Water ingress.
- Whether to assist casualty e.g. danger to shipping / others.
- Condition of vessel.
- Position of vessel.

##### • Evaluating the casualty - equipment.

Training criteria. Describe the considerations, hazards and techniques for the recovery of casualty equipment:

- Further assistance required.
- Awareness of recovering - scuba divers, windsurfers, kite surfers, kayaks and canoes, personal watercraft, sailing dinghies.
- Recover equipment or leave and record position?
- Techniques for making safe.
- Prioritise casualties over equipment recovery.
- Room onboard SAR unit or ability to tow.
- Hazards associated with casualty vessel: unsecured loads, dangerous cargoes.
- Hazards associated with recovering the equipment: wind, stability, lines, capsized equipment.
- Evacuation

##### • Evacuating the casualty.

Training criteria. Explain how the casualty may be prepared and evacuated:

- Person:
  - helicopter transfer
  - transfer ashore
  - transfer to own SAR unit
  - no live stretcher transfers for training.
- Vessel:
  - transfer casualties to SAR unit
  - request equipment transferred to vessel.
  - take vessel under tow
- Preparation for transfer:
  - wear lifejacket, adjusted and inflated
  - casualty brief.
  - secure in stretcher

task continues

## Task 5.1.1

## Locate and Assist 1

### d Operate the salvage pump

#### Training guidelines

#### Salvage pump

##### • Pre-start precautions.

Training criteria. Explain the safety precautions you would take prior to starting the salvage pump:

- Check operating environment.
- Minimise lift height.
- Keep suction clear.
- Manual handling.
- Run in well-ventilated area.
- Location relative to SAR unit (if applicable).

##### • Operating the salvage pump.

Training criteria. Demonstrate the starting procedure for the salvage pump:

- In accordance with the current SOP.

Training criteria. State the hazards whilst using the pump:

- Heat.
- Fumes – Do not use in a confined space.
- Noise.

Training criteria. Demonstrate how to adjust the controls of the salvage pump to give best performance and smooth running:

- Adjust revs to pumping requirement.
- Duration of pump use.

##### • Stopping the salvage pump.

Training criteria. Demonstrate the stopping routine for the salvage pump:

- In accordance with the current SOP.

##### • Post recovery.

Training criteria. Demonstrate the post recovery routine for the salvage pump:

- Remove hoses and drain.
- Flush through with fresh water.
- Re-stow.
- Allow pump to cool.
- Check fuel / oil levels.

##### • Refuelling.

Training criteria. Identify where the spare fuel for the pump is located on the SAR unit (if carried):

- As appropriate to SAR unit.

task continues



# Activity 5 - Search and Rescue

## Task 5.1.1

## Locate and Assist 1

### e Animal rescue

#### Training guidelines

#### Rescuing animals

- **Hazards associated with the rescue of animals.**

Training criteria. Explain the dangers associated with and precautions to be taken when rescuing animals both large and small:

- Fight or flight reaction.
- Kicks.
- Biohazards / infection.
- Full PPE helmet with visor down.
- Bites.
- Crush.

Training criteria. Describe considerations to be made during the rescue of a large animal:

- Whether to rescue or to stabilise and leave at scene.
- Consider requesting specialist animal rescue assets.
- Permission may be needed to land the animal.
- Recognise that arrival on scene will not be calming to the animal.
- If possible, contact a professional and / or ISPCA / RSPCA / SSPCA / USPCA.

Training criteria. State the action to be taken if a crew member becomes ill in the days immediately following an animal rescue:

- Seek medical advice from either a GP or hospital.

## Task 5.1.2

## Locate and Assist 2

### a Electronic search aids

#### Training guidelines

#### Search methods

- **Types of search equipment.**

Training criteria. Identify the electronic search equipment carried on the SAR unit:

- If applicable:
  - Radar
  - VHF
  - Automatic identification system (AIS)
  - DF.

- **Operating the search equipment.**

Training criteria. Describe how electronic search equipment is used to assist with locating a casualty.

Training criteria. Demonstrate how to operate the VHF / DF / AIS:

- Radar:
  - identifying targets (inc SARTs)
  - assessment of visibility.
  - multi vessel management
- VHF:
  - speak with casualty to gather further information about location.
- DF:
  - casualty to transmit a long count on VHF
  - vessel direction indicated on equipment by light or relative bearing from SAR unit
  - correctly select the frequency for picking up a crew member PLB or EPIRB homing signal.
- AIS:
  - identify casualty vessels by MMSI.

- **Limitations of search equipment.**

Training criteria. Identify the limitations of electronic search equipment available:

- Radar:
  - small targets in poor weather
  - snow / heavy rain
  - target material / construction
  - height of antenna.
- VHF / DF:
  - quality of transmission
  - available channels.
  - line of sight
- AIS:
  - casualty needs to have AIS function.

# Activity 5 - Search and Rescue

## Task 5.1.3

## Mud Rescue

### a Mud rescue

#### Training guidelines

##### Equipment

Training criteria. Identify and explain the mud rescue equipment carried on the SAR unit:

- Mud lance, extinguisher, spare Co2 cylinders, water pump.
- A frame, block and tackle, strop.
- Casualty refuge, air pump.
- Loud hailer.
- Stretcher.
- Plastic shovels.
- Mud mats and boards.
- Survivor lifejackets.

##### Techniques

Training criteria. State the five principles of mud rescue:

- Locate.
- Stabilise.
- Transport.
- Access.
- Extricate.

Training criteria. Explain methods to access the casualty:

- SAR unit positioning alongside casualty.
- Pepper potting, with mud boards / matts.
- Use of casualty refuge.

Training criteria. Demonstrate stabilising a casualty using mud mats and boards:

- As per current SOP.

Training criteria. State the different methods of extraction:

- Digging.
- Mud lance.

Training criteria. Demonstrate preparing / using / refilling and disconnecting the mud lance:

- As per current use of mud lance SOP.

Training criteria. Demonstrate self-rescue techniques:

- When one or boot feet are trapped (do not exceed knee depth).

Training criteria. Demonstrate decontamination routine carried out after mud rescue:

- As per current SOP.

##### Hazards and precautions

Training criteria. State the PPE which must be worn during mud rescues (crew and casualty):

- Crew:
  - dry suit
  - eye protection (visor or safety glasses)
  - gloves.
- Casualty:
  - eye / face protection
  - ear defenders (if appropriate)
  - lifejacket (if appropriate).

task continues

### Task 5.1.3

### Mud Rescue

a

#### Mud rescue (continued)

##### Training guidelines

Training criteria. State hazards associated with mud rescue:

- Manual handling.
- Exhaustion / fatigue.
- Tide.
- Entrapment.
- Exposure.
- Hidden objects.
- Contamination.

Training criteria. State the hazards when approaching a stuck casualty in the hovercraft:

- Running over the casualty.
- Gradient.
- Noise.
- Spray / mud.
- Training criteria. Explain additional casualty care considerations during mud rescues:
- Decontamination for casualty (if appropriate).

# Activity 5 - Search and Rescue

## Task 5.2.1

## Plan SAR

### a Plan SAR

#### Training guidelines

#### Operational information

- **Obtaining all relevant incident information.**

Training criteria. State all required incident and environmental information and identify the sources of this information:

- |                                   |                           |
|-----------------------------------|---------------------------|
| • Description of incident.        | • Search target.          |
| • Last known position.            | • Drift start position.   |
| • Commence search position.       | • Time.                   |
| • Casualty type.                  | • Tide.                   |
| • Weather.                        | • POB.                    |
| • SARTs and EPIRBs.               | • Other available assets. |
| • Sources of information include: |                           |
| - coordinating authority          | - first informant         |
| - casualty themselves             | - internet                |
| - weather on scene.               |                           |

#### SAR planning skills

- **Plotting the position of a casualty.**

Training criteria. Demonstrate the ability to plot on charts and electronic navigation system:

- Drift start position.
- Commence search position.
- Plot the position on paper chart and electronic navigation aids as appropriate.

- **Plotting the datum position.**

Training criteria. Demonstrate the plotting of a datum position on paper / electronic charts:

- Calculate the correct tide and wind information to create the correct vector and apply to drift start position to give datum position (ALB).
- Ensure time noted as delay will alter the datum position.
- Transfer this to a paper or electronic chart.

- **Developing search plans.**

Training criteria. Describe when you may need to develop a search plan:

- Is the position of the casualty known?
- Tasking information incomplete.
- As requested by the coordinating authority.

- **Choosing a search pattern.**

Training criteria. Describe the variety of search patterns that may be employed in various circumstances:

- |                                                                     |                    |
|---------------------------------------------------------------------|--------------------|
| • Parallel track.                                                   | • Creeping line.   |
| • Expanding square.                                                 | • Sector search.   |
| • Multi-vessel.                                                     | • Ship / aircraft. |
| • River / estuary – keyhole.                                        |                    |
| • Note that coastguard may recommend or stipulate a search pattern. |                    |

task continues

## Task 5.2.1

## Plan SAR

### a Plan SAR (continued)

#### Training guidelines

Training criteria. State the factors that determine the choice of search pattern:

- Age of datum.
- Other assets available.
- Sea state / tides.
- Nature of search targets.
- Casualty survival times.
- Sun / moon position.
- Accuracy of initial information.
- Weather conditions.
- Geography.
- Equipment available.
- Available daylight.

Training criteria. State further factors that might affect a successful search:

- Crew positioning and scanning techniques; crew fatigue; equipment available.

#### • Plotting / setting-up a search pattern.

Training criteria. Demonstrate how to plot / set-up a search pattern on paper / electronic charts using information from the SAR check cards:

- Plot the correct pattern utilising the information available in conjunction with the SAR cards.
- Limitations of carrying out datum search on the electronic chart / navigation system.

#### • Planning a search pattern.

Training criteria. Explain the checks that should be made prior to starting a search:

- Depth throughout search area and identify minimum expected.
- Hazards including wrecks, over falls, traffic separation schemes (TSS), and vessel traffic services (VTS).
- Report to person in command.
- Time to complete.

Training criteria. Explain how weather may affect your search planning:

- Sweep width calculations.
- Search orientation for area searches.
- Search speeds.
- Type of search pattern.
- Additional crew.

#### • Planning a search pattern.

Training criteria. Explain what factors will affect the success of the search plan:

- Appropriateness of search plan and SAR unit.
- Management by on scene coordinator (OSC) and SMC.
- Local environmental conditions.
- Request and use of other SAR assets.
- Equipment used.
- Effect of crew fatigue.

# Activity 5 - Search and Rescue

## Task 5.3.1

## Manage SAR

### a Manage search plan

#### Training guidelines

#### Search plan

##### • Adjusting a search plan.

Training criteria. State factors that determine the need to adjust to the search plan:

- Changes in weather / sea conditions.
- Other assets available.
- Tidal conditions.
- New casualties.
- Day / night.
- Updated information.
- Changed circumstances.

Training criteria. Describe different ways of adjusting the search plan:

- Search speed.
- Search pattern type.
- Track spacing.
- Orientation.
- Leg length.
- Area covered.
- Sweep width.
- Addition or removal of assets.

#### Communications

##### • Communications required.

Training criteria. State what communications are made and with whom:

- Comprehensive crew briefing at outset.
- Periodic crew sitreps.
- Communication with other search assets, casualty and SMC.

#### Managing SAR unit crew

##### • Assigning crew to search.

Training criteria. Describe where and how crew should be positioned:

- In a safe position, using best height of eye suited to the conditions to cover allocated sector as briefed.
- Effects of fatigue and complacency.

Training criteria. Explain the effects of fatigue and long periods of tasking:

- Poor, ineffective searching. Lack of focus and demotivation.

Training criteria. Describe different methods for combatting these effects:

- Rotation.
- Appropriate PPE.
- Information updates.
- Crew changes.
- Refreshments.

Training Criteria. Explain how to report search effectiveness and the use of the terminology:

- Ideal.
- Normal.
- Reduced.

task continues



## Task 5.3.1

## Manage SAR

### b SAR coordination

#### Training guidelines

#### Role of on-scene coordinator

##### • Responding to a request for an On-scene Coordinator (OSC).

Training criteria. Explain when and by whom the OSC could be designated:

- When multiple search assets are deployed.
- Mutual agreement between SAR units.
- Designated by SMC.
- Ability to decline request to be OSC.

##### • Duties of the OSC.

Training criteria. State the duties of the OSC:

- Coordinate search as directed by SMC.
- Communication between SAR units.
- Liaise with SMC.
- Capability of other units.
- Speed and manoeuvrability of other vessels.
- Manpower.
- Modify search plan.
- Sitreps.
- Keeps records.
- Report casualties / survivors.
- Speed of search.

##### • Selecting the OSC.

Training criteria. Describe the factors that determine the selection of the OSC:

- Communication limitations.
- Navigational equipment.
- Crew available.
- Size of vessel.
- Stability of platform.
- Local knowledge.
- Weather conditions.
- Position of vessel.

##### • Using the available assets.

Training criteria. State the factors that determine the use of the available assets:

- Size of search area.
- Draught.
- Height of eye
- Capability of asset.
- Search speed available.

##### • Implications of the OSC role.

Training criteria. Explain the implications of the OSC role:

- Crew fatigue.
- Monitoring of all units.
- Search capability of other SAR units.
- Monitoring own vessel safety.
- Ability to efficiently manage assets on scene.
- Own search capability may be degraded.

task continues

# Activity 5 - Search and Rescue

## Task 5.3.1

## Manage SAR

### c

### Casualty management

#### Training guidelines

#### SAR unit stability

- **SAR unit maximum casualty carrying limit.**

Training criteria. State the limitations in accordance with type of SAR unit:

- As stated in the class specific manual.

#### Casualty transfer

- **Transferring / recovering a person / vessel to the SAR unit.**

Training criteria. Explain the factors affecting the decision to transfer / recover a person / vessel to the SAR unit:

- Damage to the SAR unit.
- Risk versus benefit in transferring casualties.
- Exacerbating injuries.
- Risk to traffic if vessel abandoned.
- Consideration when attending vessels on fire.
- Stability of vessel.

#### Plan evacuation of casualty from SAR unit

- **Evaluating the need for casualty evacuation.**

Training criteria. State the considerations to be made when planning to evacuate casualty(s):

- |                       |                                   |
|-----------------------|-----------------------------------|
| • Crew safety.        | • Number of casualties.           |
| • Nature of injuries. | • Triage.                         |
| • Time to evacuate.   | • Distance to medical assistance. |
| • Weather conditions. | • Other assets.                   |

#### Evacuation

- **Evacuating the casualty.**

Training criteria. State the factors to be considered during the evacuation of the casualty(s):

- |                                 |                     |
|---------------------------------|---------------------|
| • Prevention of further injury. | • Promote recovery. |
| • Crew safety.                  |                     |

Training criteria. Describe actions that may be taken during an evacuation:

- Evacuate as appropriate to situation, e.g. daughter craft, stretcher.
- Request attendance of more suitable asset to evacuate the casualty.

## Task 5.3.2

## Manage Mud Rescue

### a Manage mud rescue

#### Training guidelines

#### Planning

Training criteria. State the tasking information required:

- Description of incident.
- Time and location of incident.

Training criteria. State what additional information may be required:

- Tide.
- Weather.
- Number of casualties.
- Other assets.

Training criteria. Explain where additional information may be obtained:

- Coordinating authority.
- First informant.
- Casualty themselves.
- Other assets on scene.

Training criteria. Explain how this information will affect rescue planning:

- Number and type of assets required.
- Number and type of assets available.
- Time limitations due to weather.
- Time limitations due to tide.
- Which service(s) is best placed to conduct the rescue.

Training criteria. Explain the five stages of mud rescue and the commanders involvement in each:

- Locate.
- Access.
- Stabilise.
- Extricate.
- Transport.

#### Execution

Training criteria. Demonstrate briefing the crew. Brief to include:

- Casualty description.
- Weather.
- Hazards.
- Equipment required.
- Location.
- Crew roles.
- Scanning / action on sighting casualty.
- Additional PPE requirements.
- Interaction with other agencies.

Training criteria. Describe access and egress from rescue site:

- Stop in a safe location to assess situation.
- Communicate with other agencies.
- Must be able to depart without flying over casualty.
- Consider use of mud boards / casualty refuge.
- Plan safest egress and casualty handover site.

Training criteria. Demonstrate (as commander) approaching a casualty who is stuck and successful use of escape route:

- Discussion with pilot.
- Decide approach and escape routes.
- Brief crew.

Training criteria. Explain when you may work with other agencies:

- Coastguard and / or fire and rescue if first on scene.
- If required as safety cover for other agencies.
- If require additional resources from other agencies.

task continues

## Activity 5 - Search and Rescue

### Task 5.3.2

### Manage Mud Rescue

#### a Manage mud rescue (continued)

##### Training guidelines

Training criteria. State who to liaise with and how they may be identified (locally and remotely):

- Coastguard officer in charge – tabard / helmet / callsign.
- Coastguard team leader – callsign.
- Fire and rescue officer in charge – tabard / helmet.
- Coordinating authority – callsign.

Training criteria. Explain why the hovercraft should be shut down alongside the casualty and identify limitations:

- To prevent casualty sinking any further.
- So as not to create a bigger danger area.
- Battery life of hovercraft.

Training criteria. Explain what hazards may affect your crew and other agencies:

- |                    |                         |
|--------------------|-------------------------|
| • Manual handling. | • Exhaustion / fatigue. |
| • Tide.            | • Entrapment.           |
| • Exposure.        | • Hidden objects.       |
| • Contamination.   |                         |

Training criteria. Demonstrate a full mud rescue, including use of the mud lance. To include:

- |                           |                          |
|---------------------------|--------------------------|
| • Brief to crew.          | • Safety considerations. |
| • Correct PPE.            | • LASET 5 principles.    |
| • In accordance with SOP. | • Debrief on completion. |

Training criteria. State the considerations for casualty transport:

- |                       |                                   |
|-----------------------|-----------------------------------|
| • Crew safety.        | • Number of casualties.           |
| • Nature of injuries. | • Triage.                         |
| • Time to evacuate.   | • Weather.                        |
| • Other assets.       | • Distance to medical assistance. |

Training criteria. Describe known casualty landing points in your area:

- Refer to chart or OS map as well as local knowledge LOP.

Training criteria. Demonstrate correct decontamination for hovercraft, crew and equipment:

- As per SOP.

## Task 5.4.1

## Manage Operational Services

### a Operational procedures

#### Training guidelines

#### Operational procedures

##### • SAR Procedures.

Training criteria. Explain the full scope of operational services and the management input and actions required:

- Preparing for recovery.
- Ensuring resilience.
- Debriefing and ready for service.
- Recording / monitoring activities.
- Ensuring personal roles are fulfilled.
- Coordination with other agencies.
- Reporting and crew welfare.

Training criteria. Explain the required communications and procedures between the SAR unit, boathouse, and the coordinating authority:

- Not through boathouse. Boathouse to contact SAR unit through coordinating authority.
- Operations normal transmissions to be made and maintained as per current procedures.

##### • Station Management.

Training criteria. Explain the management of the remaining crew at station during services:

- Rotation - crews afloat / cover ashore.
- Welfare - food, drink, sleep, rest.
- Roles - radio log, shore crew roles.

Training criteria. Describe the importance of controlling access to station during services:

- General safety and security.
- Media presence and issues.
- Appropriateness of individuals' presence (casualty family and friends, onlookers).

##### • Other Agencies and Procedures.

Training criteria. Explain the other SAR agencies that may be involved in RNLI operations:

- SAR helicopter assets.
- Independent lifeboats.
- Emergency services.
- Others through the SAR coordinating authority.
- Coastguard teams.
- Local maritime SAR organisations.
- General public.

Training criteria. Describe any local emergency plans in place:

- Mass casualty.
- Terror.
- RNLI personnel injury / incidents.
- Pollution.
- Hazard to wildlife.
- Other.

task continues

# Activity 5 - Search and Rescue

## Task 5.4.1

## Manage Operational Services

### b Operations afloat

#### Training guidelines

#### SAR operations afloat

##### • Procedures.

Training criteria. Describe the process of a maritime SAR mission:

- Full process end to end of a typical service.
- To include: request, briefing, launch, search, rescue, recovery, debrief and recording phases.

Training criteria. State the terminology used during SAR tasking:

- As per current IAMSAR and SAR agreement documents.
- To include all terminology used during SAR services.

Training criteria. State the various methods of operational communications used:

- Radio (afloat) with the coastguard.
- Land-line direct to coastguard.
- Operations room.

Training criteria. Explain the Launching Authorities' role during services:

- Attendance as required.
- Overview of the service.
- Providing support / welfare.
- Providing direction, focus, motivation.
- Consider prompts for crew change.
- Consider request for additional assets.
- Plan for return of vessel and crew.

##### • Resilience.

Training criteria. Describe the importance of crew rotation during lengthy service:

- Coordinate with coxswain / helm / commander, preparing and rotating crews.

Training criteria. State the on-station fuel reserves and what affect, if any, this has on endurance:

- Station fuel reserves.



## Activity 6 - Operational Communications

Photo: RNLI / Laura Lewis





### Tasks

#### 6.1 Short Range Communications

6.1.1 Operate VHF

6.1.2 SAR Communications

#### 6.2 Long Range Communications

6.2.1 Operate MF Radio

6.2.2 LRC Qualification

6.2.3 LRC Satellite Module

#### 6.3 Tetra Radio Communications

6.3.1 Tetra Communications

# Activity 6 - Operational Communications

## Task 6.1.1

## Operate VHF

### a Operate VHF radio and handheld radio

#### Training guidelines

All crew are to complete the Introduction to Lifeboat Radio and SAR Communications eLearning on Learning Zone followed by practical exercises afloat to demonstrate the skills on the SAR Unit.

#### Operate VHF radio

- **Switch on the VHF radio and identify the main controls.**

Training criteria. Demonstrate turning on and operating the main functions of the VHF radio:

- Switch on / off and adjust volume.
- Adjust squelch.
- Change channel to select channel 16, 0 and 31.
- Dual watch / Tri watch.
- Scan.
- Adjust power output.
- Adjust illumination settings.

#### Operate handheld VHF radio

- **Switch on the handheld VHF radio and identify the main controls.**

Training criteria. Demonstrate turning on and operating the main functions of the VHF radio:

- Switch on / off volume.
- Adjust squelch.
- Change channel to select channel 16, 0 and 31.
- Dual watch / Tri watch.
- Adjust power output.
- Adjust illumination settings.

#### Post-operational checks for handheld VHF radio

- **Describe post-operational checks for handheld radios.**

- Turn off.
- Check overall condition.
- Clean in fresh water (if required).
- Return to charging cradle (dry).

task continues

## Task 6.1.1

## Operate VHF

### b Emergency procedures

#### Training guidelines

#### Emergency aerial

- **Locating the emergency aerial.**

Training criteria. Identify the location of the emergency aerial on the SAR unit:

- As appropriate to SAR unit.

- **Rigging the emergency aerial (ILB only).**

Training criteria. Demonstrate the procedure for rigging the emergency aerial:

- Power off.
- As appropriate to SAR unit.

- **Limitations.**

Training criteria. State the limitations of using an emergency aerial:

- ILB only:
  - low gain antenna (poor range)
  - loose wires.
- ALB only:
  - fixed low down (poor range)
  - only transmits effectively port and starboard.

- **Transmitting on the emergency aerial.**

Training criteria. Demonstrate the procedure for connecting the emergency radio to the SAR unit radio:

- Establish connection between radio and emergency aerial (via connectors or SIMS).

Training criteria. Demonstrate making a test transmission:

- A test transmission to the coastguard.

#### Undesignated distress alert

- **Locating the undesignated distress alert button(s).**

Training criteria. Identify the distress alert button(s) on the VHF DSC radio fitted to the SAR unit:

- As appropriate to SAR unit.

- **Operating the undesignated distress alert button(s).**

Training criteria. Explain the operation of the distress alert button(s):

- Dependent on the DSC type; lift, press and hold or press both buttons and hold.

#### Mayday format

- **Formatting a Mayday call and message.**

Assessment criteria. Describe how to send a mayday in the event of distress:

- MIRPDANIO

task continues

# Activity 6 - Operational Communications

## Task 6.1.1

## Operate VHF

### c Communications and direction finding

#### Training guidelines

#### Communications

- **Procedures.**

Training criteria. State the organisations and callsigns used for search and rescue communications.

Training criteria. State search and rescue communication responsibilities and capabilities.

Training criteria. Describe the channels and messages used for search and rescue communications.

Training criteria. Explain the communications procedures for operations.

- **Operational communications.**

Training criteria. Demonstrate how to send launch on service messages.

Training criteria. Demonstrate communications when on service, exercise, trials or passage.

- **Lost communications.**

Training criteria. Explain the lost / failed communications procedure.

- **Testing.**

Training criteria. Explain routine testing of communications equipment.

#### Direction finding

- **DF equipment.**

Training criteria. Explain the purpose and use of direction finding equipment:

- In accordance with the current SOP.

### Task 6.1.2

### SAR Communications

#### a SRC qualification

##### Training guidelines

##### VHF radio operations

- **SRC syllabus appropriate for the national administration.**

Training criteria. Demonstrate possession of an original Short Range Certificate (SRC), Restricted Operator Certificate (ROC), Long Range Certificate (LRC) or General Operator's Certificate (GOC):

- The identified qualifications (originals) must be sighted and confirmed in date.

# Activity 6 - Operational Communications

## Task 6.2.1

## Operate MF Radio

### a Operate MF radio

#### Training guidelines

#### Switch on and adjust MF radio

- **Switch on the MF radio and set up for optimum performance.**  
Training criteria. Demonstrate how to switch on the MF radio equipment and adjust controls for optimum performance.
- **Switches and controls on a MF radio.**  
Training criteria. Identify the controls and describe the effect of the switches on a MF radio.  
Training criteria. Explain how to tune the MF radio and to ensure that the radio is in the correct mode.

#### Switch on the MF DSC

- **Switch on the MF DSC (if separate from MF radio).**  
Training criteria. Demonstrate how to switch on the MF DSC unit.
- **MF DSC scanning.**  
Training criteria. Explain the importance of scanning the correct MF / HF distress frequencies.
- **Manual and automatic position input of SAR unit position into the MF DSC.**  
Training criteria. Demonstrate how to find the automatic input of data from the SAR unit GPS into the MF DSC and how to manually enter this information.

task continues

## Task 6.2.1

## Operate MF Radio

### b SAR communication

#### Training guidelines

#### Frequencies for radio communication

- **MF frequencies for communication.**

Training criteria. Explain the frequencies used for communications, to include distress, urgency, safety and calling, ship to ship, DSC and search and rescue.

#### Routine messages

- **Sending and receiving routine messages.**

Training criteria. Demonstrate how to make a routine call on MF radio and how to receive a call on MF radio.

Training criteria. State the MF frequencies to use.

#### Communication

- **Establishing communications.**

Training criteria. Demonstrate how to establish communications with the coastguard.

- **Maintain communication.**

Training criteria. Demonstrate communications when on service, exercise, trials or passage.

task continues

# Activity 6 - Operational Communications

## Task 6.2.1

## Operate MF Radio

### c Emergency procedures

#### Training guidelines

#### Mayday, Pan Pan and Sécurité

- **Sending Mayday, Pan Pan and Sécurité.**

Training criteria. Demonstrate how to send a Mayday, Pan Pan and Sécurité and explain when each would be used. Training criteria. State the MF frequencies to use.

- **Receiving and responding to Mayday, Pan Pan and Sécurité.**

Training criteria. Explain the required actions on receipt for these messages.

#### Distress, urgency, safety and routine messages by MF DSC

- **Distress. Initiate Distress communications by DSC and RT.**

Assessment criteria. Describe how to send an undesignated / designated DSC distress alert.

Assessment criteria. Describe how to format a Mayday call and message.

Assessment criteria. Describe how to cancel a false distress alert.

- **Urgency. Initiate Urgency communications by DSC and RT.**

Assessment criteria. Describe how to send a DSC urgency announcement.

Assessment criteria. Describe how to format a Pan Pan call and message.

- **Safety. Initiate Safety communications by DSC and RT.**

Assessment criteria. Describe how to send a DSC safety announcement.

Assessment criteria. Describe how to format a Sécurité call and message.

Assessment criteria. Explain the frequencies that should be used.

- **Test Transmissions.**

Assessment criteria. Demonstrate sending an external MF DSC Test to the coastguard

#### Relevance of HF DSC alerts

- **Responding to HF DSC distress alerts.**

Training criteria. Describe the procedure for responding to HF DSC distress alerts.



### Task 6.2.2

### LRC Qualification

#### a LRC qualification

##### Training guidelines

##### MF radio operations

- **Long range certificate or higher.**

Training criteria. Demonstrate possession of an original Long Range Certificate (LRC) or General Operator's Certificate GOC:

- The identified qualifications (originals) must be sighted and confirmed in date.

## Activity 6 - Operational Communications

### Task 6.2.3

### LRC Satellite Module

#### a LRC qualification

##### Training guidelines

##### Satellite communications

- **Long range certificate satellite module or higher.**

Training criteria. Demonstrate possession of an original Long Range Certificate (LRC) with satellite endorsement or General Operator's Certificate (GOC):

- The identified qualification (originals) must be sighted and confirmed in date.

## Task 6.3.1

## Tetra Communications

### a Tetra

#### Training guidelines

#### Operate Tetra radio system

- **Turning on and main controls.**

Training criteria. Demonstrate turning on and operating main controls of the radio:

- Turn unit on / off procedure, volume control set, adjust backlight.
- Input PIN and / or PUC code.
- Check service availability:
  - RED = no service
  - GREEN = service available
  - AMBER = TXI activated.

Training criteria. Explain radio features:

- Battery care, charging, and life.
- Handhelds not waterproof.
- Range of buttons, functions and display.

- **User modes.**

Training criteria. Explain Trunk Mode Operation (TMO) and Direct Mode Operation (DMO):

- TMO - handset works over third party network like a mobile phone system.
- DMO - direct connection to another unit. Use's VHF mode with 'line of sight' limitation.

Training criteria. Demonstrate changing from TMO to DMO mode:

- Use appropriate 'hot' key, mode successfully changes.

Training criteria. Explain 'TXI activated':

- Prevents the radio TX in environments where to transmit is a risk (e.g. hospitals and fuel).

Training criteria. Demonstrate turning TXI mode on and off:

- Use appropriate 'hot' key, mode successfully changes.

task continues

# Activity 6 - Operational Communications

## Task 6.3.1

## Tetra Communications

### a Tetra (continued)

#### Training guidelines

#### Transmit and receive voice messages

- **Voice calling.**

Training criteria. Demonstrate how to transmit and receive a voice message:

- Pause for beep when using push to talk (PTT), correct voice procedure, clarity, brevity, terminology.

- **Point to point calling.**

Training criteria. Demonstrate making a point to point call:

- Ensure set is in correct mode, enter handset Individual Short Subscriber Identity (ISSI) number.

- **Mobile telephone calling.**

Training criteria. Demonstrate how to make a mobile telephone call:

- Input number, select C type option, call.

- **Fleet map channels.**

Training criteria. Explain the primary talk groups available to RNLI / MCA:

- As per current structure available to the set.

Training criteria. Demonstrate the selection of talk groups:

- As per current structure available to the set.

#### Security of Tetra radio equipment

- **Radio unit security.**

Training criteria. Explain the security measures in place for controlling use of Tetra equipment:

- In accordance with the current SOP.

- **PIN codes.**

Training criteria. Explain the result of incorrect PIN entries:

- Three unsuccessful attempts locks the unit.

- **Loss, damage or locked equipment.**

Training criteria. Explain the procedure to be followed when equipment is damaged, lost or stolen:

- In accordance with the current SOP.



## Activity 7 - Navigation

Photo: Gavin Jones



## Tasks

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# Activity 7 - Navigation

## Task 7.1.1

## Navigation Introduction

### a Introduction to tides and weather

#### Training guidelines

#### Tidal information

- **Obtaining tidal information.**

Training criteria. Identify sources of tidal information:

- *Admiralty Almanac.*
- Local tide tables.
- Tidal stream atlas.
- Internet.
- Local sources.

- **Awareness.**

Training criteria. Describe spring / neap tides and the effect on local area:

- Spring = larger range, faster stream.
- Neap = smaller range, slower stream.
- Effects may include:
  - access
  - launch / recovery window
  - launch / recovery sites.

Training criteria. Describe safety implications of tides and tidal streams on local sites:

- Hazard of immersion to people and equipment.
- Hazard of being cut off.
- Hazard of bogging launch vehicle.

#### Meteorology

- **Forecast.**

Training criteria. Describe the sources and terminology used in shipping / inshore forecast:

- Online or other local means such as on a notifications board.
- Understand basic terminology used for:
  - wind speed and direction
  - precipitation
  - visibility
  - sea state and wave height.

task continues



## Task 7.1.1

## Navigation Introduction

### b Introduction to charts

#### Training guidelines

#### Charts

Training criteria. Identify the common features of charts and chart symbols:

- For charts used at location, to include:
  - title
  - scale
  - colours
  - depth soundings / contours
  - latitude / longitude
  - compass rose
  - notices
  - common symbols
  - numbers
  - buoyage
  - hazards
  - local information.

Training criteria. Demonstrate the ability to plot a position on a chart given:

- Latitude and longitude.
- Range and bearing from a fixed point.
- Bearings from 3 fixed points.

Training criteria. Given two positions on a chart, demonstrate the ability to determine:

- The distance between the two positions.
- The bearing between the two positions (both directions).
- Identify any hazards between the two positions.

## Task 7.1.2

## Navigation

### a Use of charts

#### Training guidelines

#### Interpret chart information

- **Selecting the appropriate chart.**

Training criteria. Identify the most appropriate chart for your area of operation.

Training criteria. Explain the method(s) of identifying further charts:

- Identify the chart for your location.
- Magenta outline with number.
- Adjoining chart margin number.
- Chart catalogue.
- *Admiralty Almanac*.
- Use of notes on chart.

- **Chart updates.**

Training criteria. Identify where chart updates are recorded:

- Bottom left hand corner.

Training criteria. State who is responsible for the updating of charts:

- COIR - Central Operations and Information Room.
- LOM.
- Coxswain / helm / commander.

Training criteria. Explain the dangers of using an uncorrected chart:

- Inaccuracy of information.
- E.g. buoy positions, light present / not present, sequence changes, new hazards, changes to depth.

- **Variation and deviation.**

Training criteria. Explain variation and deviation:

- Difference between true and magnetic north.
- Effect of own SAR unit and other objects on compass.

Training criteria. Identify where variation can be found on the chart and where information on deviation can be found aboard the SAR unit:

- Centre of the compass rose.

Training criteria. State current variation in local area:

- In relation to variation reference and current year.

task continues

## Task 7.1.2

## Navigation

### b Position fixing

#### Training guidelines

#### Plotting

##### • Position fixing

Training criteria. Demonstrate fixing the position of the SAR unit using visual and electronic means. State the inaccuracies that could arise:

- Plotting latitude / longitude.
- Radar fix.
- Distance and bearing: accuracy of charted position compared with visual sighting.
- Apply variation / deviation.
- Sounding.

##### • Frequency of fixing.

Training criteria. State required frequency of position fixing and when you would increase the frequency of fixing:

- In accordance with SOP.
- Reduced visibility.
- Close proximity to coast or dangers and in pilotage waters.
- On service.
- Failure of electronic equipment.

#### Situational awareness

##### • Identifying.

Training criteria. Explain dangers of not understanding navigational position and actions to be taken position is not known:

- Grounding or collision.
- Inform the person in command.
- Inability to conduct SAR activities.

#### Routes

##### • Planning.

Training criteria. Explain how to plan a simple route:

- Plan a simple route between two locations, to include:
  - hazards
  - IRPCS considerations
  - STD calculations and monitoring of progress.

##### • Undertaking.

Training criteria. Demonstrate how to follow a planned route:

- Using a pre-planned route to follow between two locations; done under the guidance of a person in command.

#### High speed navigation

Training criteria. Explain the considerations for navigating at higher speeds:

- Need increased situational awareness (less time to notice or react to hazards).
- Planning for next actions further ahead of time due to reduced thinking time.
- Clear communications and procedures between person in command and navigating / helming crew.
- Increased monitoring with reduced fixing times.
- Electronic navigational equipment set up to best aid high speed navigation.

task continues

# Activity 7 - Navigation

## Task 7.1.2

## Navigation

### c Tides and tidal streams

#### Training guidelines

##### Tides

Training criteria. Identify from the chart the standard port for tidal information:

- Located above the tidal diamond information box.

Training criteria. Demonstrate the use of tide tables, tidal curves or other suitable reference tools to determine the height of tide:

- In accordance with standard navigational practice.

##### Tidal streams

Training criteria. State where to find tidal stream information:

- Tidal diamond tables.
- Tidal stream atlas.
- Electronic sources as appropriate.

Training criteria. Demonstrate the use of tidal diamonds to determine tidal stream speed and direction:

- In accordance with standard navigational practice.

Training criteria. Demonstrate the use of a tidal stream atlas to determine tidal stream speed and direction:

- In accordance with standard navigational practice.

## Task 7.1.3

## Navigation Advanced

### a Passage planning and execution

#### Training guidelines

#### Planning a route

##### • Planning.

Training criteria. State the appropriate tidal information.

Training criteria. Explain the chart features that may influence your planned route.

Training criteria. Explain any other factors that may be considered.

Training criteria. Explain the considerations when planning waypoints:

- To include:
  - low Water (standard port)
  - high Water (standard port)
  - range (spring / neap)
  - direction
  - over falls
  - wrecks
  - buoyed channel
  - hazards to shipping
  - weather
  - safe havens
  - in date almanac
  - waypoint convergence
  - clearance heights
  - notices to mariners
  - rate
  - limiting danger lines
  - shoal ground
  - depths
  - traffic separation scheme (TSS)
  - no go areas
  - sea state
  - contingency plans
  - proximity to hazards

#### Plotting a route

##### • Plotting a route with significant information.

Training criteria. Demonstrate how to construct a route consisting a minimum of three waypoints:

- Creating a route to include 3 waypoints in accordance with the current SOP.

##### • Applying variation and deviation (ALB)

Training criteria. Demonstrate the ability to convert a true course to a magnetic course and a magnetic course to a true course:

- Convert true to magnetic and magnetic to compass (both directions).

#### Following a route

##### • Determining progress on route.

Training criteria. Explain the methods used to determine position:

- Dead reckoning.
- Visual 3-point fix.
- Other fixing methods.
- Estimated position.

##### • Determining future position.

Training criteria. Explain how to plot a projected position on the route (appropriate to SAR unit):

- Dead reckoning.
- Course to steer.
- Estimated position.
- Course made good.

task continues

# Activity 7 - Navigation

## Task 7.1.3

## Navigation Advanced

### a Passage planning and execution (continued)

#### Training guidelines

- **Amending a route.**

Training criteria. Demonstrate how to amend the plan to reflect a route change:

- Re-plot on chart: course, distance, dead reckoning (DR).
- Re-check sounding and navigational hazards.

#### Pilotage

- **Piloting.**

Training criteria. Demonstrate safe pilotage of a planned route using a variety of techniques by day and night:

- Demonstrated using the full range of SAR unit operating speeds where possible.
- In accordance with the current SOP.

#### Passage

- **Passaging.**

Training criteria. Demonstrate safe passage of a planned route using a variety of techniques by day and night:

- Demonstrated using the full range of SAR unit operating speeds where possible.
- In accordance with the current SOP.

#### Situation Awareness

Training criteria. Explain the importance of understanding the environment:

- Comprehension – fully appreciating the environment.
- Recognition of when a plan is not on track.
- Recognition of when a plan is not appropriate.
- Recognition of when circumstances have changed.
- Recognition of the need to stop and re-evaluate.

task continues

## Task 7.1.3

## Navigation Advanced

### b Navigational duties

#### Training guidelines

#### Command navigation

##### • Managing.

Training criteria. Demonstrate effective and safe management of navigating the SAR unit in a variety of conditions to include at least two of the following:

- Daylight.
- Darkness.
- Restricted visibility.
- To include:
  - Confirming personal navigational skills.
  - Timely and accurate application of all navigation aids.
  - Speed, distance and time awareness.
  - Adapt management style in relation to the prevailing conditions.
  - Dynamic pre-planning and risk assessment.

Training criteria. Demonstrate effective and safe management of crew during navigational duties to include:

- Allocation of crew roles against competence.
- Effective briefing and communications.
- Appropriate positioning onboard to manage.
- Confirmation and correction of information and data.
- In accordance with the current navigation policy, SOP, IRPCS and local by-laws and regulations.

#### Crew input to safe passage and navigation

Training criteria. Explain the importance of utilising and validating crew input:

- Ensure optimal use of crew to ensure a full understanding of circumstances.
- Ensure all crew understand they have a role in ensuring safe passage.
- Acknowledging hazards identified by other crew members.

## Task 7.2.1

## E Navigation

### a Operate electronic navigation equipment

#### Training guidelines

#### Equipment

- **Electronic navigation equipment.**

Training criteria. Describe the function and use of the electronic navigation equipment available on the SAR unit:

- GPS: bearing, distance, accurate position, time, estimated time of arrival (ETA), course over ground (COG), speed over ground (SOG), can feed other equipment with information.
- Chart plotter: indicates SAR unit position, plotting and following routes, search pattern plotting (if appropriate).
- Depth sounder: provides indication of depth of water below the keel of the SAR unit.
- Radar: collision avoidance, vessel tracking, casualty location, general navigation and pilotage
- Ships log: Indicates speed and distance travelled.

#### Start up

- **Switching on and adjusting equipment.**

Training criteria. State the order in which the electronic navigation equipment must be powered up:

- GPS (as it feeds other equipment) followed by chart plotter and radar as appropriate.

Training criteria. Describe how to recognise correctly functioning equipment:

- All navigational aids must be checked to ascertain their accuracy. Reference can be made to known.
- Bearings, depths and position fixes. Respond to error messages and alarms.

Training criteria. Demonstrate how to start, set up and adjust the equipment for optimum performance:

- Use appropriate range scales.
- Adjust brightness.
- Operate the day / night function.

#### Alarms

- **Recognising equipment specific alarms.**

Training criteria. Explain the alarms present on the equipment:

- GPS / chart plotter: XTE, waypoint proximity, anchor watch, lost signal, MOB.
- Echo sounder: shallow water alarm.

Training criteria. Describe how to acknowledge and cancel the alarms:

- Check which alarm has sounded then cancel using the appropriate button / function.

Training criteria. Demonstrate how to adjust the alarm settings:

- Adjust settings for XTE, waypoint proximity, anchor watch, shallow water.

task continues



## Task 7.2.1

## E Navigation

### a Operate electronic navigation equipment (continued)

#### Training guidelines

##### Limitations

##### • Limitations of equipment.

Training criteria. Explain the limitations of the electronic equipment:

- GPS / chart plotter: operator error, power failure, poor sited or shielded aerial, poor satellite reception.
- Echo sounder: disturbed water, false echoes, sediment.

##### Action in the event of failure

##### • Actions to take on failure of navigational equipment.

Training criteria. Describe the actions to take when the navigation equipment fails:

- Acknowledge / cancel / record alarms.
- Fix position by other available means on paper chart.
- Reduce speed and adapt plan and resources to account for electronic aid failure.
- Inform the person in command of the SAR unit.

Training criteria. Demonstrate use of secondary navigation system (if fitted):

- Stand-alone GPS / chart plotter / handheld GPS as appropriate to the SAR unit.

##### Shutdown

##### • Pre-shutdown procedure.

Training criteria. Explain the actions to be carried out prior to shut down:

- Cancel route.
- Clear sail plan.
- Delete unwanted waypoints and tracks.

##### • Shutdown procedure.

Training criteria. Demonstrate the shutdown procedure in accordance with specific equipment instructions:

- Shutdown as stated in the equipment specific manual.

task continues

## Task 7.2.1

## E Navigation

### b Routes and waypoints

#### Training guidelines

##### Create waypoint

- **Entering, editing and deleting waypoints.**

Training criteria. Demonstrate entering, editing, and deleting a waypoint / casualty position and mark.  
Training criteria. Demonstrate using the boundary function (if fitted):

- Use the boundary function to highlight hazard areas and no go zones.
- Using latitude / longitude, range and bearing to enter, edit and delete waypoints or casualty position and marks.

##### Create route

- **Entering and editing routes.**

Training criteria. Demonstrate constructing and editing a route:

- Construct and edit a route including moving waypoints.

- **Pre-programmed routes.**

Training criteria. Explain the important of checking pre-made routes:

- Confirming they are current and safe.

##### Navigating using waypoint or route

- **Following routes and waypoints.**

Training criteria. Demonstrate how to follow a route or go to waypoint:

- Follow a route or waypoint as appropriate to equipment fitted to SAR unit.
- Accessing different displays and navigational information.

Training criteria. Demonstrate the various displays available, and describe the navigational information displayed:

- Present latitude / longitude, course over ground (COG), speed over ground (SOG), bearing, range and time to waypoint, cross track error (XTE), velocity made good (VMG).

- **Altering the route direction.**

Training criteria. Demonstrate the options available for altering route direction:

- Forward, reverse and advance.

- **Monitoring track.**

Training criteria. Explain what visual checks can be made to monitor the position relative to the route:

- XTE data and distance to waypoint, indicated position on electronic chart display.

Training criteria. Explain how to confirm the position:

- Compare with radar, echo sounder, compass, visual.

- **Deleting routes.**

Training criteria. Demonstrate how to delete a route:

- Delete a route as appropriate to equipment fitted to the SAR unit.

task continues

## Task 7.2.1

## E Navigation

### c MOB function

#### Training guidelines

#### Man overboard function

- **Operating MOB function.**

Training criteria. Demonstrate the MOB function:

- Activate MOB function appropriate to equipment fitted to the SAR unit.

Training criteria. Explain the navigational information provided:

- Range and bearing to MOB.

Training criteria. Demonstrate how to cancel the MOB:

- Cancel MOB function appropriate to equipment fitted to the SAR unit.
- Reactivate MOB function as required.

## Task 7.2.2

## E Navigation Advanced

a

### Search patterns

#### Training guidelines

#### Create a search pattern

- **Entering, editing and deleting a search pattern.**

Training criteria. State the search patterns that may be used on electronic navigation equipment:

- Only area searches should be plotted on electronic navigation equipment.

Training criteria. Demonstrate how to create and edit a search pattern:

- Create, edit and delete a viable and appropriate area search pattern.

Training criteria. Demonstrate how to delete a search pattern:

- Create, edit and delete a viable and appropriate area search pattern.

#### Follow a search pattern

- **Following a search pattern.**

Training criteria. Demonstrate use of electronic navigation equipment to aid completion of a planned search pattern:

- Follow a planned route to undertake an effective search.

task continues

## Task 7.2.2

## E Navigation Advanced

**b Automatic Identification System (AIS)**

## Training guidelines

**AIS**

- **System use and limitations.**

Training criteria. Explain the purpose of AIS.

Training criteria. State which system type is fitted:

- ALB - type A.
- ILB - type B.
- Used to aid ship identification and pass basic details such as course and speed on other vessels.
- Type A is prioritised on receiving ships' displays. Type B information may not be displayed if there is too little bandwidth or may be de-selected by receiving vessels, for example to reduce cluttered displays.

Training criteria. Explain the limitations and hazards of AIS:

- Not all vessels are fitted with AIS.
- AIS positions may not correspond with radar targets.
- It is important that information is cross-checked with other sources.
- Collision avoidance must be carried out in accordance with IRPCS.
- Decisions under IRPCS must be based on visual and / or radar information.

Training criteria. Demonstrate use of AIS to identify other vessels and their information:

- Use in accordance with the equipment specific manual.
- To include Identifying other vessels' position, movement and identification .

task continues

## Task 7.2.2

## E Navigation Advanced

### c Electronic navigation and situational awareness

#### Training guidelines

#### Display

- **Chart type.**

Training criteria. Identify the chart type used by the system.

Training criteria. Explain the effect this can have on the image:

- Raster – exact copies of paper charts, hence all chart information is retained. Scaling and resolution when zooming in can be a problem.
- Vector – may show less information depending on version. Constant scale for symbols, figures and words when zooming.

#### Situation awareness

- **Awareness.**

Training criteria. Explain situational awareness whilst using electronic navigation equipment:

- Importance of looking up and not fixating on the screen.
- Comparing the display screen image with real world surroundings.

Training criteria. Explain the benefits and hazards of the zoom function:

- Zoomed-in to include:
  - benefit:
    - can see more detail
    - can steer a more precise course.
  - disadvantage:
    - cannot see bigger picture
    - cannot see what is coming up (especially at high speed).
- Zoomed-out to include:
  - benefit:
    - can see bigger picture
    - can plan for longer distances.
  - disadvantage
    - minor detail may be lost
    - not accurate for precise navigation in channels.

## Task 7.3.1

## Radar

### a Operate radar

#### Training guidelines

#### Hazards

##### • Hazards posed by radar.

Training criteria. State the hazards posed by radar:

- Radiation.
- High voltage.
- Rotating antenna.

#### Start up

##### • Switching on and setting up the radar.

Training criteria. State the checks to be made prior to switch on.

Training criteria. Demonstrate how the radar is switched on and placed into transmit mode.

Training criteria. Demonstrate how to enter navigational information on set up of the radar.

Training criteria. Demonstrate the sequence for setting up for optimum performance:

- Breaker / capsized reset made (if applicable).
- Brightness.
- Tuning (not all radars).
- Ships heading.
- Turn on and transmit as stated in equipment manual.
- Antenna clear of obstructions.
- Gain.
- GPS latitude / longitude.
- Checking accuracy of VRM and EBL.

Training criteria. Demonstrate the procedure for checking the VRM and the EBL:

- VRM checked against the fixed range rings before use.
- EBL checked against a compass bearing taken.

#### Adjustment

##### • Adjusting for weather and sea conditions.

Training criteria. Demonstrate how to adjust the radar to allow for the prevailing weather:

- Anti-clutter sea – reduces the reception of wave echoes at short range.
- Anti-clutter rain – reduces the size of large block echoes (rain / snow) across the whole display.
- Use of clutter controls could cause the loss of targets.

##### • Orientation.

Training criteria. Explain the different orientations of the radar display:

- Head up – heading marker always at top of display.
- North up – heading marker aligned with vessel's compass.
- Course up – stabilizes picture on vessels current heading.

task continues

# Activity 7 - Navigation

## Task 7.3.1

## Radar

### a Operate radar (continued)

#### Training guidelines

##### Additional controls

###### • Availability of additional controls.

Training criteria. Demonstrate how to adjust the radar for day / night operation.

Training criteria. Demonstrate how and explain why the pulse length can be altered.

Training criteria. Demonstrate the use of other functions:

- Pulse length: long pulse for better target detection / short pulse for better target discrimination.
- Trails.
- Parallel index lines (if applicable).
- Range rings.
- Casualty waypoint.
- Route.
- Off centering.
- Guard zones.
- Overlays with chart plot / AIS.
- DF.
- Day / night control.

##### Limitations

###### • Limitations of radar.

Training criteria. Explain the limitations of radar for target identification.

Training criteria. Explain the effect of incorrect radar set up:

- Size.
- Aspect.
- Radar horizon.
- Shadow areas.
- Misinterpretation.
- Shape.
- Material (metal, wood, GRP).
- Weather.
- Poor detection.
- Unreliable information.

##### Alarms

###### • Radar alarms.

Training criteria. Identify the alarms available on the radar (if fitted):

- CPA.
- Guard zone.
- System errors.
- Lost target.
- Lost GPS data.

##### Standby mode

###### • Stopping radar transmission.

Training criteria. Demonstrate the radar standby function:

- Stop radar transmissions using TX / standby function as per equipment in use.

Training criteria. Explain the situations where the radar would be placed on standby:

- During helicopter winching.
- Alongside (quay or another vessel).
- In a lock.
- When mast is lowered on SAR unit.

##### Close down

###### • Closing down the radar.

Training criteria. Demonstrate the close down procedure for the radar:

- As stated in the equipment specific manual and in accordance with SOP and LOP.

task continues



## Task 7.3.1

## Radar

## b Radar lookout

## Training guidelines

## Radar echoes

- **The primary use of radar.**

Training criteria. State the three main purposes of radar on the SAR unit:

- Collision avoidance.
- Safe navigation.
- Casualty location.

- **Operating the radar.**

Training criteria. Identify the radar functions that could be used to optimise target detection and location.

Training criteria. Explain why the range scale should be periodically adjusted.

Training criteria. Explain when you would adjust the picture / controls.

Training criteria. Explain the use of VRM and EBL.

Training criteria. Explain the use of automatic radar plotting aids:

- Range.
- Tuning (as appropriate).
- VRM - Variable Range Marker.
- EBL - Electronic Bearing Line.
- Gain control.
- Clutter controls.
- Pulse length.
- True motion.
- Relative motion.
- To aid and identify targets at various distances.
- Frequently and following any change of range.
- VRM – measures the range to a target by placing the VRM on the inside edge of the target.
- EBL – a line that can be rotated around the screen to determine the bearing of the target.
- Acquire a target, wait for a period then analyse data, to assist in collision avoidance.
- Parallel indexing.
- Advantages / disadvantages of plotting.

- **Determining if a radar echo is a target.**

Training criteria. Explain how cross-referencing can be used to assess and confirm a target:

- Prioritise targets.
- Cross check targets on chart plotter.
- Compare target position to chart.
- Confirm targets using visual lookouts.

task continues

# Activity 7 - Navigation

## Task 7.3.1

## Radar

### c Plot and monitor targets

#### Training guidelines

##### Assess target

- **Obtaining the basic target information.**

Training criteria. Describe the target information that is required.

Training criteria. Explain where this information can be obtained:

- Range and bearing.
- Relative motion from MARPA.
- Using VRM and EBL displays.

- **Benefits and limitations of mini automatic radar plotting aid (MARPA).**

Training criteria. Demonstrate how to acquire a target.

Training criteria. Explain the information that is displayed.

Training criteria. Explain who should be informed.

Training criteria. Explain the dangers posed if target information is mis-interpreted.

Training criteria. Demonstrate how to cancel a target:

- Acquire target manually.
- Person in command of the SAR unit.
- Select target and cancel.
- Target range, heading, speed, CPA, TCPA.
- Risk of collision or grounding.

##### Target information

Training criteria. List the various types of target:

- Vessels.
- Land.
- SARTs.
- Man made structure.
- Navigation marks.
- Racons.
- Aircraft.

Training criteria. What information will aid target identification:

- Position.
- Course / speed.
- Closest Point of Approach (CPA) / Time to Closest Point of Approach (TCPA).
- Range / bearing.
- Size.

##### Monitor target

- **Determining the risk of collision.**

Training criteria. State the two factors that identify a risk of collision exists.

Training criteria. Demonstrate how the radar can be used to determine this risk:

- Constant bearing.
- Decreasing range.
- Use of VRM / EBL and / or MARPA.

##### Collision avoidance

- **Actions to avoid a collision.**

Training criteria. State who must be informed that a risk of collision has been identified:

- Inform the person in command.

task continues

### Task 7.3.1

### Radar

#### d Command responsibilities

##### Training guidelines

##### Responsibility

- **Actions.**

Training criteria. State the action to be taken in both clear weather and in restricted visibility.

Training criteria. Explain when Rule 19 applies and the obligations it places on vessels:

- Must have a thorough knowledge of the conduct of vessels in clear weather and in restricted visibility (rule 19).

- **Situational awareness.**

Training criteria. Demonstrate the correct interpretation of radar information.

Training criteria. Demonstrate responding correctly to radar information:

- Comprehension – correct appreciation of the situation.
- Action – correct actions taken for situations.

task continues

# Activity 7 - Navigation

## Task 7.3.1

## Radar

### e Radar as a navigation aid

#### Training guidelines

##### Position fixing

- **Fixing the SAR unit position using radar.**

Training criteria. Describe the radar functions that are available to fix a position.

Training criteria. Describe what precautions should be considered.

Training criteria. Demonstrate a position fix using the radar:

- Fix using VRM.
- Clear identification of targets used to fix position.
- Demonstrate position fix.

##### Pilotage

- **Pilotage using radar.**

Training criteria. Explain the choice of orientation for pilotage including the ranges that would be used.

Training criteria. Demonstrate the use of the VRM, EBL and parallel index lines during pilotage (if fitted):

- Normally head up (but personal preference) and lower ranges.
- Demonstrate use of VRM, EBL and parallel index lines during pilotage.

task continues

## Task 7.3.1

## Radar

## f

## Radar as an aid to SAR

## Training guidelines

## Search and rescue transponder (SART)

- **Recognising a SART.**

Training criteria. State when a SART would be used and how it would show on the radar display:

- A distress signal and primarily an aid to the location of survivors.
- A line of 12 blips along the bearing of the SART with the first echo as the target.

Training criteria. Describe how the signal will alter as the range to the SART decreases:

- As range decreases the blips will become arcs at about 1 mile the arcs become concentric circles.

## Small craft

- **Improving small target detection.**

Training criteria. Demonstrate the function on the radar that can be altered to enhance the return echo from small craft:

- Use long pulse function, increase gain.

## Chart plotter integration

- **Integrating the radar and the chart plotter.**

Training criteria. Describe the information that can be displayed on the radar from the plotter (if appropriate).

Training criteria. Describe the radar information which can be shown on the chart plotter (if appropriate):

- Waypoint and route (if appropriate).
- Latitude / longitude.
- Automatically displayed on radar from electronic chart system (if appropriate).
- MARPA targets.

## Task 7.4.1

## Local Knowledge (Ashore)

### a Ashore

#### Training guidelines

#### Local knowledge

- **Local knowledge ashore.**

Training criteria. Identify relevant navigation aids and local geography relevant to shore based activities:

- In accordance with the current LOP - to a level where local knowledge can be used to see briefed hazards or aids / references during operations.
- To include:
  - navigational aids
  - geographical markers
  - points of interest / reference
  - hazards
  - no go areas
  - names of relevant locations
  - any other information required to operate safely / effectively in the area.

Training criteria. Identify emergency extraction points:

- In accordance with the current LOP.

- **Local regulations.**

Training criteria. Demonstrate a working knowledge of relevant local regulations:

- In accordance with the current LOP.

- **Updates.**

Training criteria. State the location of updated local information and regulations:

- Locations communications board / folder or similar.
- In accordance with the current LOP.

## Task 7.4.2

## Local Knowledge (Afloat)

### a Afloat

#### Training guidelines

#### Local knowledge

##### • Local knowledge afloat.

Training criteria. Identify all relevant navigation aids and local geography relevant to afloat based activities:

- In accordance with the current LOP - to a level where local knowledge can be used to see briefed hazards or aids / references during operations.
- To include:
  - navigational aids
  - geographical marker
  - points of interest / reference
  - hazards
  - no go areas
  - names of relevant locations
  - any other information required to operate safely / effectively in the area.

Training criteria. Identify the emergency extraction points:

- In accordance with the current LOP.

##### • Local regulations.

Training criteria. State the location of the local regulations:

- In accordance with the current LOP.

Training criteria. Demonstrate a working knowledge of the appropriate regulations:

- In accordance with the current LOP.

##### • Updates.

Training criteria. State the location of updated local information and regulations:

- Locations communications board / folder or similar.
- In accordance with the current LOP.

### Task 7.4.3

### Local Knowledge (Command)

#### a Command local knowledge

##### Training guidelines

- **Updates.**

Training criteria. Describe the process to update local information and regulations:

- Feed from Poole, national and local sources.
- In accordance with the current Local Operating Procedures (LOPs).

- **LOPs.**

Training criteria. Describe the process to produce and maintain LOPs:

- In accordance with current LOP template and guidelines.
- LOPs to be approved centrally.

- **Keeping station personnel current.**

Training criteria. Explain the importance of communicating updated information to all personnel:

- Ensuring all crew are aware of new hazards, features, regulations.
- Enables them to maintain a proper lookout and have effective situational awareness.



### Task 7.4.4

### Local Knowledge (Endorsement)

#### a Local endorsement

##### Training guidelines

##### Local endorsements

- **Regulation requirements.**

Training criteria. Demonstrate possession of an original local knowledge endorsement:

- In accordance with local regulations.
- Originals must be sighted.
- In accordance with the current LOP.

Activity 8 - Clinical

Photo: RNLI / Nigel Millard



## Tasks

|       |                                        |
|-------|----------------------------------------|
| 8.1   | First Aid 1                            |
| 8.1.1 | First Aid 1                            |
| 8.2   | First Aid 2                            |
| 8.2.1 | First Aid 2                            |
| 8.3   | Casualty Care                          |
| 8.3.1 | Casualty Care                          |
| 8.4   | Automated External Defibrillator (AED) |
| 8.4.1 | Automated External Defibrillator (AED) |
| 8.9   | Registered Medical Professional        |
| 8.9.1 | Registered Medical Professional        |

### Task 8.1.1

### First Aid 1

#### a First Aid 1

##### Training guidelines

Successful completion of the approved First Aid 1 eLearning course.

First Aid 1 eLearning course is valid for 3 years from the date of completion.

## Task 8.2.1

## First Aid 2

a

## First Aid 2

## Training guidelines

Successful completion of the approved First Aid 2 Course or demonstrate possession of a valid standards of training, certification, and watchkeeping for seafarers (STCW) compliant elementary First Aid or higher qualification.

First Aid 2 qualification (or STCW compliant equivalent) is valid for 5 years from the date of completion.

An in-date First Aid 2 qualification supersedes the requirement for an in-date First Aid 1 qualification.

### Task 8.3.1

### Casualty Care

#### a Casualty Care

##### Training guidelines

Successful completion of the approved **Casualty Care Course**. Casualty Care Course is valid for 3 years from the date of completion.

An in-date Casualty Care qualification supersedes the requirement for First Aid 1 and First Aid 2.

##### Course syllabus to include:

- Assessment and triage.
- Airway management and choking.
- Cardiopulmonary resuscitation (CPR) and oxygen therapy.
- Injury and pain relief.
- Illness.
- Immersion.

## Task 8.4.1

## Automated External Defibrillator (AED)

## a AED

## Training guidelines

Designated casualty carers at locations with AED equipment to have passed the approved AED course. AED qualification remains valid for 12 months from successful completion of AED training.

## Syllabus to include:

- Correct use of AED.
- Use of AED combined with CPR.

### Task 8.9.1

### Registered Medical Professional

#### a Registered Medical Professional

##### Training guidelines

Demonstrate possession of an appropriate current medical profession registration.

Possession of appropriate registration supersedes the requirement for First Aid 1 and First Aid 2.

##### Appropriate Registrations include (but are not limited to):

- General Medical Council (GMC).
- Immediate Medical Care (IMC).
- Health and Care Professions Council (HCPC).





## Activity 9 - Propulsion and Auxiliary Systems

Photo: Stephen Duncombe



## Activity 9 - Propulsion and Auxiliary Systems

### Tasks

|            |                                   |
|------------|-----------------------------------|
| <b>9.1</b> | Operate Main Machinery            |
| 9.1.1      | Operate ALB Main Machinery        |
| 9.1.2      | Operate ILB Main Machinery        |
| 9.1.3      | Operate IRH Main Machinery        |
| <b>9.2</b> | Operate Auxiliary Systems         |
| 9.2.1      | Operate Auxiliary Systems         |
| <b>9.3</b> | Mechanical Faults and Emergencies |
| 9.3.1      | Diagnose Faults                   |
| 9.3.2      | Rectify Faults                    |
| 9.3.3      | Mechanical Emergencies            |

# Activity 9 - Propulsion and Auxiliary Systems

## Task 9.1.1

## Operate ALB Main Machinery

### a Alarms

#### Training guidelines

#### Engine and machinery alarms

- **Responding to alarms.**

Training criteria. Describe the different visual and audible alarms fitted to the SAR unit:

- Oil pressures.
- Temperature alarms.
- Fuel alarms.
- Alternator charging.
- Software alarms (sensors).
- Level alarms.

- **Actions on detecting an alarm.**

Training criteria. Explain the actions to be taken in the event of activation of an alarm:

- Carry out procedure in accordance with current documentation.
- Inform person in command of the SAR unit.
- Visual check of gauges (where fitted) to confirm alarm.
- Visual check of engine tell-tales.
- Stop the engine if required and carry out single engine running.
- Monitor the engines - if not operationally safe to stop.

Training criteria. Describe the implications these alarms may have on engine operation:

- Power restriction.
- Running time restriction.
- Additional monitoring.
- Engine inoperative and loss of auxiliaries.
- Boat manoeuvrability affected.

task continues

## Activity 9 - Propulsion and Auxiliary Systems

### Task 9.1.1

### Operate ALB Main Machinery

#### b Fuelling

##### Training guidelines

##### Fuelling

- **Managing fuelling.**

Training criteria. Explain the correct fuelling procedure for the SAR unit:

- In accordance with SOP and LOP.
- Identify required fuelling point on the SAR unit.
- Have all necessary equipment associated with the task to hand.

Training criteria. State the PPE required for fuelling:

- State PPE listed in SOP and LOP.

- **Fire precautions.**

Training criteria. Explain the precautions to be taken when refuelling the SAR unit:

- Fire.
- Spillage.
- Vapour.
- Naked lights.
- Restricted access.
- Have suitable fire extinguishers to hand.

- **Limiting fuel spillage.**

Training criteria. Explain the procedure for dealing with fuel spillages:

- Isolate source of spillage.
- Locate spill kit and follow spill kit manufacturer's user instructions.
- Dispose of contaminated items in accordance with RNLI waste management procedure.

task continues

# Activity 9 - Propulsion and Auxiliary Systems

## Task 9.1.1

## Operate ALB Main Machinery

### c Start engines and conduct running checks

#### Training guidelines

#### Pre-start checks

- **Pre-start routine.**

Training criteria. Explain the sequence in which the checks must be carried out and why certain checks must be carried out prior to starting:

- State the sequence specified in the pre-start and start checks SOP.
- Purpose is to ensure engine is fit for service and to prevent damage to machinery systems.

Training criteria. Demonstrate the pre-start checks appropriate to the SAR unit:

- Main battery isolators.
- Engine alarm cancel switch.
- Battery box fan running.
- Fluid levels and pressures.
- Fluid types available to top up as necessary.
- Fuel supply and return valve configuration.
- Sea water inlet cooling valves .
- Other sea water system cooling / lubrication valves.
- Stop button / methods.
- Engine start isolation switch.
- Checks to be carried out around the engine to be started.
- Checks / operations to ancillary systems or machinery attached to the engine.
- Engine control systems as fitted.
- Throttles set to neutral position.
- Shore power disconnected.

#### Start engines

- **Starting the engines.**

Training criteria. Demonstrate the procedure for starting the engine appropriate to the SAR unit:

- Local start and remote start buttons.

#### Running checks

- **Carrying out running checks.**

Training criteria. State the running checks appropriate to the SAR unit:

- Correct oil pressure range.
- Engine idle speed.
- Abnormal mechanical vibration.
- Correct engine operating temperature.
- Operation of ancillary systems or machinery attached to the engine.
- Describe the running checks dictated by operational or weather conditions.
- Raw water coolant flow.
- Abnormal mechanical noises and operation.
- Presence of fluid leaks.
- Correct charge rate parameter.

**task continues**

# Activity 9 - Propulsion and Auxiliary Systems

## Task 9.1.1

## Operate ALB Main Machinery

### d Monitor engines

#### Training guidelines

#### Monitor engines

- **Operating parameters of the engines.**

Training criteria. Identify the normal operating parameters in accordance with RNLI operator manual:

- Engine:
  - oil pressure
  - coolant temperature
  - idle speed
  - maximum revolutions per minute (RPM).
- Ancillary equipment:
  - gearbox oil pressure
  - alternator charge / discharge rate
  - hydraulic system pressures.

- **Monitoring engine parameters.**

Training criteria. Explain the importance of monitoring engine parameters:

- For early indication of a potential problem.
- To avoid a breakdown.
- For signs of normal operating parameters being exceeded.
- For trend analysis.

#### Record engine readings

Monitor and record the appropriate information.

Training criteria. Demonstrate the correct method of recording the readings taken whilst engines are running:

- Machinery logbook:
  - pressures
  - temperatures
  - running time / engine hours
  - charge rates
  - defects arising during operation.

task continues

# Activity 9 - Propulsion and Auxiliary Systems

## Task 9.1.1

## Operate ALB Main Machinery

### e Stop engines

#### Training guidelines

#### Stop engines

- **Stopping the engines.**

Training criteria. Demonstrate the correct procedure for stopping the engines:

- In accordance with the current SOP.
- Obtain permission from person in command.
- Engine sufficiently cooled down.
- Turbo charger sufficiently run down to prevent premature bearing wear.

#### Emergency shut down

- **Shutting down the engines in an emergency.**

Training criteria. Describe when the engines may need to be shut down in an emergency:

- |                              |                                  |
|------------------------------|----------------------------------|
| • Abnormal mechanical noise. | • Abnormal mechanical vibration. |
| • Mechanical failure.        | • System failure.                |
| • Ancillary failure.         | • Engine fire.                   |
| • Compartment fire.          | • Compartment flood.             |

Training criteria. Explain the procedure and methods for shutting down engines in an emergency as appropriate to SAR unit:

- In accordance with the current SOP.
- Obtain permission from person in command.
- Normal stop buttons at all locations.
- Remote mechanical stop locations.
- Cover air intake / discharge CO2 fire extinguisher into air intake.



## Activity 9 - Propulsion and Auxiliary Systems

### Task 9.1.2

### Operate ILB Main Machinery

#### a Alarms

##### Training guidelines

##### Engine alarms

- **Responding to engine alarms.**

Training criteria. Describe the different visual and audible alarms fitted to the SAR unit:

- Oil pressures.
- Coolant temperature.
- Fuel alarms.
- Alternator charging.

- **Actions on detecting an alarm.**

Training criteria. Explain the actions to be taken in the event of activation of an engine alarm:

- In accordance with the current SOP.
- Inform person in command of the SAR unit.
- Visual check of gauges (if fitted) to confirm alarm.
- Visual check of tell-tales.
- Stop the engine if required and carry out single engine running.
- Monitor the engines - if not operationally safe to stop.

Training criteria. Describe the implications these alarms may have on engine operation:

- Power restriction - including inbuilt protection.
- Running time restriction.
- Additional monitoring.
- Engine inoperative and loss of auxiliaries.
- Manoeuvrability affected.

task continues

## Activity 9 - Propulsion and Auxiliary Systems

### Task 9.1.2

### Operate ILB Main Machinery

#### b Fuelling

##### Training guidelines

##### Fuelling

- **Managing fuelling.**

Training criteria. Explain the correct fuelling procedure for the SAR unit:

- In accordance with SOP and LOP.
- Identify required fuelling point on the SAR unit.
- Have all necessary equipment associated with the task to hand.

Training criteria. State the PPE required for fuelling:

- State PPE listed in SOP and LOP

- **Fire precautions.**

Training criteria. Explain the precautions to be taken when refuelling the SAR unit:

- Fire.
- Spillage.
- Vapour.
- Naked lights.
- Restricted access.
- Have suitable fire extinguishers to hand.

- **Limiting fuel spillage.**

Training criteria. Explain the procedure for dealing with fuel spillages:

- Isolate source of spillage.
- Locate spill kit.
- Follow spill kit manufacturer's user instructions.
- Dispose of contaminated items in accordance with RNLI waste management procedure.

task continues

## Activity 9 - Propulsion and Auxiliary Systems

### Task 9.1.2

### Operate ILB Main Machinery

#### c Start engines and conduct running checks

##### Training guidelines

##### Pre-start checks

- **Pre-start routine.**

Training criteria. Explain and demonstrate the correct sequence of pre-start checks:

- Explain the purpose of pre-start checks and how they ensure an engine is fit for service and is not damaged.
- Demonstrate the correct sequence of pre-start and start checks in accordance with SOP.

##### Start engines

- **Starting the engines.**

Training criteria. Demonstrate the procedure for starting the engine appropriate to the SAR unit:

- Local start button.
- Recoil hand start (if fitted).

##### Running checks

- **Carrying out running checks.**

Training criteria. State the running checks appropriate to the SAR unit:

- Oil low pressure indication.
- Raw water coolant flow.
- Engine idle speed.
- Abnormal mechanical noise.
- Abnormal mechanical vibration.

Training criteria. Explain the importance of recognising changes in engine performance:

- Including loss of power and loss of an engine.

task continues

# Activity 9 - Propulsion and Auxiliary Systems

## Task 9.1.2

## Operate ILB Main Machinery

### d Monitor engines

#### Training guidelines

#### Monitor engines

- **Operating parameters of the engines.**

Training criteria. Identify the normal operating parameters in accordance with the manufacturer's operator manual:

- Engine:
  - idle speed
  - maximum RPM
  - coolant temperature.

- **Monitoring engine parameters.**

Training criteria. Explain the importance of monitoring engine parameters:

- For indication of potential problem.
- To avoid a breakdown.
- Signs of operating parameters being exceeded.
- For trend analysis.

#### Record engine readings

- **Monitor and record the appropriate information.**

Training criteria. Demonstrate the correct method of recording the readings taken whilst engines are running:

- Defect log - defects arising during operation.
- Engine hour logbook - running hours.

task continues

## Activity 9 - Propulsion and Auxiliary Systems

### Task 9.1.2

### Operate ILB Main Machinery

#### e Stop engines

##### Training guidelines

##### Stop engines

- **Stopping the engines.**

Training criteria. Demonstrate the correct procedure for stopping the engines:

- In accordance with SOP.
- Obtain permission from person in command.
- Engine sufficiently cooled down.
- Turbo charger sufficiently run down to prevent premature bearing wear.

##### Emergency shut down

- **Shutting down the engines in an emergency.**

Training criteria. Describe when the engines may need to be shut down in an emergency:

- Abnormal mechanical noise.
- Abnormal mechanical vibration.
- Fuel leak.
- Mechanical failure.
- Engine fire.
- Fouled propeller / impeller.
- Overheat (no tell-tale).

Training criteria. Explain the procedure and methods for shutting down engines in an emergency as appropriate to SAR unit:

- In accordance with SOP.
- Obtain permission from person in command.
- Cover air intake / discharge CO2 fire extinguisher into air intake.

# Activity 9 - Propulsion and Auxiliary Systems

## Task 9.1.3

## Operate IRH Main Machinery

### a Alarms

#### Training guidelines

#### Engine alarms

- **Responding to engine alarms.**

Training criteria. Describe the different visual and audible alarms fitted to the SAR unit:

- Oil pressures.
- Coolant temperature.
- Fuel alarms.
- Alternator charging.

- **Actions on detecting an alarm.**

Training criteria. Explain the actions to be taken in the event of activation of an engine alarm:

- In accordance with the current SOP.
- Inform person in command of the SAR unit.
- Visual check of gauges (if fitted) to confirm alarm.
- Visual check of tell-tales.
- Stop the engine if required and carry out single engine running.
- Monitor the engines - if not operationally safe to stop.

Training criteria. Describe the implications these alarms may have on engine operation:

- Power restriction - including inbuilt protection.
- Running time restriction.
- Additional monitoring.
- Engine inoperative and loss of auxiliaries.
- Manoeuvrability affected.

task continues

## Activity 9 - Propulsion and Auxiliary Systems

### Task 9.1.3

### Operate IRH Main Machinery

#### b Fuelling

##### Training guidelines

##### Fuelling

- **Managing fuelling.**

Training criteria. Explain the correct fuelling procedure for the SAR unit:

- In accordance with SOP and LOP.
- Identify required fuelling point on the SAR unit.
- Have all necessary equipment associated with the task to hand.

Training criteria. State the PPE required for fuelling:

- State PPE listed in SOP and LOP.

- **Fire precautions.**

Training criteria. Explain the precautions to be taken when refuelling the SAR unit:

- Fire.
- Spillage.
- Vapour.
- Naked lights.
- Restricted access.
- Have suitable fire extinguishers to hand.

- **Limiting fuel spillage.**

Training criteria. Explain the procedure for dealing with fuel spillages:

- Isolate source of spillage.
- Locate spill kit.
- Follow spill kit manufacturer's user instructions.
- Dispose of contaminated items in accordance with RNLI waste management procedure.

task continues

## Activity 9 - Propulsion and Auxiliary Systems

### Task 9.1.3

### Operate IRH Main Machinery

#### c Start engines and conduct running checks

##### Training guidelines

##### Pre-start checks

- **Pre-start routine.**

Training criteria. Explain and demonstrate the correct sequence of pre-start checks:

- Explain the purpose of pre-start checks and how they ensure an engine is fit for service and is not damaged.
- Demonstrate the correct sequence of pre-start and start checks in accordance with SOP.

##### Start engines

- **Starting the engines.**

Training criteria. Demonstrate the procedure for starting the engine appropriate to the SAR unit:

- Local start button.

##### Running checks

- **Carrying out running checks.**

Training criteria. State the running checks appropriate to the SAR unit:

- Oil low pressure indication.
- Engine idle speed.
- Abnormal mechanical noise.
- Abnormal mechanical vibration.

Training criteria. Explain the importance of recognising changes in engine performance:

- Including loss of power and loss of an engine.

task continues



## Activity 9 - Propulsion and Auxiliary Systems

### Task 9.1.3

### Operate IRH Main Machinery

#### d Monitor engines

##### Training guidelines

##### Monitor engines

- **Operating parameters of the engines.**

Training criteria. Identify the normal operating parameters in accordance with the manufacturer's operator manual:

- Engine:
  - idle speed
  - maximum RPM
  - coolant temperature.

- **Monitoring engine parameters.**

Training criteria. Explain the importance of monitoring engine parameters:

- For indication of potential problem.
- To avoid a breakdown.
- Signs of operating parameters being exceeded.
- For trend analysis.

##### Record engine readings

- **Monitor and record the appropriate information.**

Training criteria. Demonstrate the correct method of recording the readings taken whilst engines are running:

- Defect log - defects arising during operation.
- Engine hour logbook - running hours.

task continues

# Activity 9 - Propulsion and Auxiliary Systems

## Task 9.1.3

## Operate IRH Main Machinery

### e Stop engines

#### Training guidelines

#### Stop engines

- **Stopping the engines.**

Training criteria. Demonstrate the correct procedure for stopping the engines:

- In accordance with SOP.
- Obtain permission from person in command.
- Engine sufficiently cooled down.
- Turbo charger sufficiently run down to prevent premature bearing wear.

#### Emergency shut down

- **Shutting down the engines in an emergency.**

Training criteria. Describe when the engines may need to be shut down in an emergency:

- Abnormal mechanical noise.
- Abnormal mechanical vibration.
- Fuel leak.
- Mechanical failure.
- Engine, machinery or electrical fire.
- Fouled propeller.
- Overheat (no tell-tale).

Training criteria. Explain the procedure and methods for shutting down engines in an emergency as appropriate to SAR unit:

- In accordance with SOP.
- Obtain permission from person in command.
- Cover air intake / discharge CO2 fire extinguisher into air intake.

# Activity 9 - Propulsion and Auxiliary Systems

## Task 9.2.1

## Operate Auxiliary Systems

### a Run and monitor auxiliary systems

#### Training guidelines

#### Pre-start checks

Training criteria. Demonstrate the pre-start checks associated with the systems fitted to the SAR unit:

- Explain and demonstrate the correct pre-start and start checks associated with each of the SAR unit systems, correctly in accordance with SOP.

#### Operate systems

Training criteria. Describe and explain the layout, component and function of SAR unit systems:

- In accordance with SOP and EOP:
  - demonstrate the correct procedure for operating the systems fitted to the SAR unit.
  - explain the running checks associated with the systems fitted to the SAR unit.
  - explain the frequency in which the systems should be checked.
- System: fuel. To include, but not limited to:
  - transfer / stripping pump operation; fuel filter change over
  - isolation, supply / return, cross connection and stripping valves
  - filter bowl / tank contamination.
- System: cooling (fresh and sea water). To include, but not limited to:
  - system layout, shaft / strainer, cross-connection valve
  - strainers, intake and flush valves, sea water pumps.
- System: hydraulics. To include, but not limited to:
  - isolation valves, bypass valves, fluid levels, header tank air pressure
  - pump drive belts, control systems, physical obstructions
  - pump engagement, control system operation, manual overrides / equipment
  - system function, pressures, temperatures, leakages.
- System: electrical system. To include, but not limited to:
  - isolators, circuit breakers, power supply, earth leakage check
  - system function, charge rates, discharge rates, earth leakages
  - emergency power supply.
- System: bilge and general service pump. To include, but not limited to:
  - system layout, configuration and operation
  - location of electric and general service pumps
  - location of hydrants and discharges.
- System: HVAC. To include, but not limited to:
  - isolation valves, pumps, control systems, fluid levels and pressures, reset buttons
  - fan operation and emergency stop; system function; temperature control; leakage of fluids and gasses.
- System: gearbox and shafting. To include, but not limited to:
  - gearbox operation, controls and back-up operation. Gearbox lubrication system.
- System: air system. To include, but not limited to:
  - aftercoolers, turbo, turbo bypasses, run away flaps.
- System: firefighting. To include, but not limited to:
  - layout of heat and flame detection and fixed firefighting system
  - system operation and checks.

task continues

# Activity 9 - Propulsion and Auxiliary Systems

## Task 9.2.1

## Operate Auxiliary Systems

### b Generator and shore charge system

#### Training guidelines

#### Generator

- **Operating the generator.**

Training criteria. Demonstrate the operation of the onboard generator:

- Pre-start checks:
  - in accordance with SOP: Fluid levels, cooling / exhaust valves, electrical configuration, obstruction free
  - applying load.
- Starting:
  - in accordance with SOP.
- Running checks:
  - raw water coolant discharge, running speed, abnormal noise, leaks, charge rate, panel faults.
- Stopping the generator:
  - in accordance with SOP, close down associated systems electrical configuration.

- **Operating the switching to charge the various battery banks.**

Training criteria. Demonstrate the operation of the isolators and circuit breakers fitted to the system:

- Locate main and navigation / communication battery isolators and cross connectors, generator control breaker, generator charge breaker.
- Configure system to charge stated battery banks.

- **Refuelling the generator.**

Training criteria. Explain the procedure for refuelling the generator:

- Fuel type - diesel.
- Spare fuel location - containers on board, SAR unit's main tank, transfer system.
- Hazards - fire, spillage, contamination.
- Spillage control - spill kit location, instructions, contaminated material disposal.

#### Operate shore charge system

- **Charging the SAR unit battery banks from external charging source.**

Training criteria. Demonstrate the procedure for charging the batteries using shore mounted or onboard battery chargers:

- In accordance with SOP.
- Shore charger power - locate mains electrical distribution panel and supply isolation.
- SAR unit - charging socket, battery bank selector switch / circuit breakers, charge fuses.
- Operation - correct sequence of configuring and switching on the battery charger and use of battery box fans.

task continues

# Activity 9 - Propulsion and Auxiliary Systems

## Task 9.2.1

## Operate Auxiliary Systems

### c Bilge and general service pumps

#### Training guidelines

#### Pump system

- **Layout of the pump system.**

Training criteria. Describe the layout of the pump system including the system components:

- Compartments - which have suction points, where the discharge overboard locators are, where the pumps are housed, wandering suctions, remote pump controls, auto pumps.
- Components - pumps, control switches / panels, suction valves, discharge valves, cross connection valves, non-return valves, flow indicators, strum boxes, wandering suction hoses, auto pumps, bilge filters.

- **Preparing and operating the bilge pumps.**

Training criteria. Identify the valves to correctly open and close to enable individual compartments to be pumped out:

- Suction.
- Discharge.
- Hydrant.

Training criteria. Demonstrate the method of engaging the pumps:

- Manual operation - local handle, remote handle.
- Electrical operation - switch isolation, local operation arrangement.
- Auto pump - control panel, local operation arrangement.

#### General service pump

- **Preparing and operating the general service pump.**

Training criteria. Identify the valves to correctly open and close to enable water to be discharged from the deck hydrants:

- Suction - through hull and bilge.
- Discharge.
- Hydrant.

Training criteria. Explain the use of the general service pump with a wandering hose:

- System limitations.
- Maximum allowed engine revolutions whilst pump is engaged.
- Pumping rates.

Training criteria. Demonstrate the correct method of engaging the pumps:

- Manual clutch - local handle, remote handle.
- Electrical clutch - switch isolation, local operation arrangement.

## Activity 9 - Propulsion and Auxiliary Systems

### Task 9.3.1

### Diagnose Faults

#### a Fault finding techniques

##### Training guidelines

##### Fault diagnosis

- **Carrying out fault diagnosis.**

Training criteria. Explain various methods of diagnosing faults or abnormalities in equipment / system operation / performance:

- Mechanical - performance, vibration, abnormal noise, abnormal exhaust smoke, fluid leaks, broken components, smell, comparison, gauges, touch.
- Electrical - performance. operation, earth leakage, measuring instrumentation, tripped breakers or blown fuses.

Training criteria. Demonstrate the use of diagnostic tools available:

- Multimeter: voltage, resistance, current.
- Thermometer: temperature.
- Tachometer: engine revolutions.
- Engine gauges: reference between engines.
- Built in diagnostics: warning lights / alarms, diagnostic codes (if fitted).

##### Role limitations

- **Understanding what faults can be diagnosed and rectified afloat and which cannot.**

- **Training criteria. State, for own SAR unit, the range of faults which can be addressed afloat and which cannot:**

- In accordance with EOPs and mechanic's handbook.
- Your core competency level.

- **When and who to call for additional assistance.**

Training criteria. Explain the system in place to call for advice or assistance:

- During working day: station technical management - regional base.
- Out of hours: duty technical manager - available through COIR.
- When afloat: launch authority and COIR.

task continues

# Activity 9 - Propulsion and Auxiliary Systems

## Task 9.3.1

## Diagnose Faults

### b Identify causes of faults and actions required

#### Training guidelines

#### Equipment failure

##### • Types of equipment failure.

Training criteria. Describe types of equipment failures that may occur on the SAR unit:

- Engine:
  - low oil pressure
  - component failures
  - fluid leaks
  - unexpected shutdown
  - catastrophic engine failure.
  - high coolant temperature
  - fuel restriction
  - mechanical fault
  - air starvation
- Electrical:
  - power failure
  - switching / distribution
  - starter motor
  - charging fault.

Training criteria. Explain the reasons these types of failures may occur:

- Symptom : low oil pressure. Possible cause:
  - low level
  - oil temperature
  - oil dilution
  - sensor failure.
- Symptom : high coolant temperature. Possible cause:
  - low level
  - blocked sea strainers
  - worn water pump impeller
  - insufficient direct / indirect cooling flow
  - slipping or snapped water pump drive belts
  - sensor failure.
- Symptom : component failures. Possible cause:
  - vibration
  - lack of lubrication
  - loose fittings.
  - stress
  - friction
- Symptom : fuel restriction. Possible cause:
  - fuel contamination
  - pipe work blockage
  - filter blockage.
  - valve configuration
  - suction strainer blockage
- Symptom : fluid leaks. Possible cause:
  - loose fittings / flanges, cracks
  - worn / damaged seals.
  - damaged pipe work
- Symptom : electrical power failure. Possible cause:
  - short circuits
  - earth leakages
  - component failures
  - broken / loose terminals or wires
  - no charge
  - incorrect charging / distribution.

task continues

## Activity 9 - Propulsion and Auxiliary Systems

### Task 9.3.1

### Diagnose Faults

#### b Identify causes of faults and actions required (continued)

##### Training guidelines

DAMAGE LIMITATION ACTIONS AFLOAT MAY ONLY BE CONDUCTED WITH THE PERMISSION OF THE SAR UNIT COMMANDER.

##### Damage limitation

- **Actions to limit damage on mechanical and electrical equipment and systems.**

Training criteria. Describe the various actions to be taken to limit the amount of damage caused to equipment in the event of a failure:

- Mechanical failures:
  - stop the equipment
  - limit engine revolutions
  - opening cooling cross connections
  - isolate systems
  - cover leaks
  - change over fuel supplies
  - monitor and top up fluids
  - carry out temporary repairs if safe and practical.
- Electrical failures:
  - stop the equipment
  - isolate circuits
  - isolate shorted cables
  - make battery breaker cross connection
  - carry out temporary repairs if safe and practical.



## Activity 9 - Propulsion and Auxiliary Systems

### Task 9.3.2

### Rectify Faults

#### a Rectify faults

##### Training guidelines

**FAULT RECTIFICATION AFLOAT MAY ONLY BE CONDUCTED WITH THE PERMISSION OF THE SAR UNIT COMMANDER.**

##### Temporary repairs

##### • Carrying out temporary repairs.

Training criteria. Describe the techniques that can be used to carry out repairs to the systems to enable the SAR unit to remain operational:

- Mechanical failures:
  - replenish fluids
  - isolate systems
  - tighten fittings / flanges
  - replace filters
  - apply lubrication
  - clear blockages
  - opening cooling cross connections
  - disconnect pump drives / belts
  - replace propeller
  - stop / reduce leaks.
- Electrical failures:
  - stop the equipment
  - isolate shorted cables
  - by-pass faulty switches
  - carry out temporary repairs to cables / joints / tighten loose connections.
  - isolate circuits
  - replace blown fuses / reset breakers
  - replace spark plugs

##### • Materials available to effect temporary repairs.

Training criteria. Identify materials available and explain how they can be used to carry out repairs:

- Fuel filters - fuel restriction, loss of power.
- Spark plugs - misfire, loss of power.
- Fluids / lubricants - topping up.
- Outboard engine prop - damage, drive bush slipping.
- Cable ties - various uses.
- Air filters - air restriction , loss of power.
- Rubber sheeting - covering leaking pipe work.
- Drive belts - replacing damaged items.
- Fuses - replacing blown items.
- Bulbs - replacing blown items.
- Jubilee clips - replacing broken clips, holding rubber sheeting on to leaking pipe work.
- Insulation tape - insulating exposed electrical cables / connections.

**task continues**

## Activity 9 - Propulsion and Auxiliary Systems

### Task 9.3.2

### Rectify Faults

#### a Rectify faults (continued)

##### Training guidelines

##### Role limitations

- **Understanding what faults can be diagnosed and rectified afloat and which cannot.**

Training criteria. State, for own SAR unit, the range of faults which can be addressed afloat and which cannot:

- In accordance with EOPs and mechanic's handbook.
- Your core competency level.

- **When and who to call for additional assistance.**

Training criteria. Explain the system in place to call for advice or assistance:

- During working day: station technical management - regional base.
- Out of hours: duty technical manager - available through COIR.
- When afloat: launch authority and COIR.

task continues

# Activity 9 - Propulsion and Auxiliary Systems

## Task 9.3.2

## Rectify Faults

### b Monitor temporary repairs

#### Training guidelines

#### Limit of temporary repairs

- **Limitations of temporary repairs.**

Training criteria. Explain the limitations that may be present in the machinery or systems following a temporary repair:

- Engine power restrictions:
  - machinery maximum revs may need to be reduced
  - some auxiliary equipment may be unavailable.
- Engine use:
  - engine may only be available to be used for short periods of time.
  - System use restriction:
    - fluid consumption or loss, power consumption, system may only be used for short periods of time
    - some auxiliary equipment may be unavailable.
- Leaks:
  - may need regular topping up.

- **Monitoring temporary repairs.**

Training criteria. Explain the importance of monitoring temporary repairs:

- Temporary repairs to machinery or systems may be subject to conditions beyond the capability of the temporary repair.
- The repair may mask an underlying problem that may cause more damage.
- Fluid loss may still be present, creating hazards (e.g. fire, slip, contamination).
- Machinery or system failure may occur despite repairs.
- Repairs may fail without warning.

- **Frequency of monitoring.**

Training criteria. State the intervals in which a repair must be monitored:

- Checks should be made as frequently as possible, taking into account:
  - operational commitments
  - weather conditions
  - potential to cause further damage or loss
  - potential to cause fire, contamination or other hazard.

#### Permanent repairs

- **Reporting of temporary repairs**

Training criteria. Explain how faults and temporary repairs are recorded and reported:

- Faults, defects and temporary repairs must be entered into the appropriate log at the station and the appropriate reporting mechanism with engineering and supply.

# Activity 9 - Propulsion and Auxiliary Systems

## Task 9.3.3

## Mechanical Emergencies

### a Carry out emergency procedures

#### Training guidelines

RESPONSES TO MECHANICAL EMERGENCIES MAY ONLY BE CONDUCTED WITH THE PERMISSION OF THE SAR UNIT COMMANDER.

#### Emergency steering system

- **Locating the emergency steering system valves and components.**

Training criteria. Identify the steering emergency bypass valves and emergency operation equipment:

- Steering system emergency bypass valves.
- Steering system emergency operation equipment.
- Emergency tiller and ropes.
- Emergency steering wheel.
- Emergency steering and jet hydraulic systems.

- **Putting the steering system into emergency operation mode.**

Training criteria. Demonstrate the procedure for putting the steering system into emergency mode:

- In accordance with EOP:
  - operation - obtain permission from SAR unit commander, awareness of person in steering compartment
  - valves - identify, operate in correct sequence as fitted
  - rig emergency tiller and ropes
  - fit emergency steering wheel
  - emergency steering and jet hydraulic systems
  - transfer control / use.

- **Resuming normal operation.**

Training criteria. Demonstrate returning the system to normal operation:

- Return system to normal operation in accordance with EOP.
- Carry out function test.

#### Trim tab manual operation

- **Locating the trim tab manual operating override buttons and hand pump unit.**

Training criteria. Identify the manual override buttons and hand pump:

- Raise button.
- Hand pump.
- Lower button.
- Hand pump handle.

- **Operating the trim tabs manually.**

Training criteria. Demonstrate the procedure for manually raising and lowering the trim tabs:

- Raise and lower the trim tabs.
- Pump the trim tabs through a full up and down cycle.

task continues

## Activity 9 - Propulsion and Auxiliary Systems

### Task 9.3.3

### Mechanical Emergencies

#### a Carry out emergency procedures (continued)

##### Training guidelines

##### Hydraulic systems

- **Actions to be taken in the event of loss of pressure in a hydraulic system.**

Training criteria. Describe the actions to be taken in the event of loss of hydraulic pressure:

- Inform the person in command of the SAR unit.
- Investigate cause of pressure loss – fluid loss, pump drive, pump failure, valve failure, valve incorrectly configured, electrical failure.
- Instigate emergency action – rig alternative system of operation if available.
- Remedial action – cut belts, disconnect drive, check valve configuration.

- **Actions to be taken in the event of a loss of fluid in a hydraulic system.**

Training criteria. Describe the actions to be taken in the event of loss of hydraulic fluid:

- Inform the person in command of the SAR unit.
- Identify problem – cause of leak.
- Remedial action – tighten / replace fitting / hose, cover leak to minimise fluid loss, isolate leaking pipe / component.
- Monitor – top up if loss is slight.
- Clean – ensure leaked fluid is not a hazard.
- Instigate emergency action if required – rig alternative system of operation if available.

- **Actions to be taken in the event of a loss of electrical control of a hydraulic system.**

Training criteria. Describe the actions to be taken in the event of loss of electrical control to the hydraulic system:

- Inform the person in command of the SAR unit.
- Check fuses, breakers and connections.
- Operate systems manually if required:
  - mast
  - transom door
  - pumps
  - crane
  - trim tabs
  - capstans
  - jet hydraulic solenoids.

task continues

# Activity 9 - Propulsion and Auxiliary Systems

## Task 9.3.3

## Mechanical Emergencies

### a Carry out emergency procedures (continued)

#### Training guidelines

#### Main engine failure

- **Actions to be taken in the event of the loss of a main engine.**

Training criteria. Describe the actions to be taken in the event of main engine failure:

- In accordance with SOP / EOP.
- Inform the person in command of the SAR unit.
- Change battery bank coupler to maintain electrical supplies.
- Open shaft lubrication cross connection (if fitted).
- Monitor gearbox temperatures.
- Identify problem:
  - mechanical failure
  - loss of cooling
  - fuel supply / quality
  - control failure.
- Remedial action:
  - clear strainers
  - change fuel supply
  - change filters
  - top up fluids.
- Monitoring:
  - electrical consumption
  - fuel consumption (even out fuel tank usage if not contaminated)
  - power restriction (if engine is still able to run without causing additional damage).

#### Main engine / gearbox controls

- **Actions to be taken in the event of the loss of main engine / gearbox control systems.**

Training criteria. Describe the actions to be taken and demonstrate the rigging and operation of emergency controls:

- Inform the person in command of the SAR unit.
- In accordance with EOP:
  - rig emergency throttles (if available)
  - carry out emergency gear box operation procedure.
- Identify the problem:
  - power supply
  - plugs disconnected
  - mechanical cables sheared / disconnected.



## Activity 10 - Planned Maintenance

Photo: RNLI / Tom Kerley





### Tasks

|      |                                |
|------|--------------------------------|
| 10.1 | Maintain and Service Equipment |
|------|--------------------------------|

|        |                                |
|--------|--------------------------------|
| 10.1.1 | Maintain and Service Equipment |
|--------|--------------------------------|

# Activity 10 - Planned Maintenance

## Task 10.1.1

## Maintain and Service Equipment

### a Identify the planned maintenance schedule

#### Training guidelines

#### Maintenance

##### • The planned maintenance system.

Training criteria. Explain the current planned maintenance system in place:

- Crew - post recovery checks.
- Station mechanics - two-weekly, monthly, three monthly, six monthly, annual checks.
- Station technical support - six monthly, annual checks.
- Contractor.

Training criteria. Explain why the maintenance system is in place:

- Operational readiness.
- Equipment / system test.
- Trend analysis.
- Equipment serviceability.
- Service history.

Training criteria. Explain why and how it is updated:

- Annual review.
- PM amendment form - station, technical input.
- Still appropriate to the equipment / systems.
- Address shortfalls / over maintenance.

#### Documentation

##### • Planned maintenance documentation.

Training criteria. Identify the planned maintenance documentation at station:

- Planned maintenance hard copy and on MySAR.

##### • Specific documentation.

Training criteria. Identify the tasks and maintenance schedules that are appropriate to the role:

- Electrical, hull, machinery and electronic tasks.
- Post recovery, two weekly, monthly, three monthly, six monthly, annually.
- SAR unit, launch and recovery equipment.

#### Record planned maintenance

##### • Recording the maintenance carried out.

Training criteria. Explain the system in place for recording the planned maintenance appropriate to the role:

- Hard copy of planned maintenance report retained as per current system.
- Update the RNLI database system.

task continues

## Task 10.1.1

## Maintain and Service Equipment

### b Carry out planned maintenance

#### Training guidelines

#### Planned maintenance

- **Carrying out planned maintenance.**

Training criteria. Describe the planned maintenance tasks appropriate to the equipment:

- Electrical - as per the tasks in the appropriate maintenance document.
- Hull - as per the tasks in the appropriate maintenance document.
- Machinery - as per the tasks in the appropriate maintenance document.
- Electronic - as per the tasks in the appropriate maintenance document.

- **Identifying and sourcing spares and equipment required for tasks.**

Training criteria. Identify the parts list for the equipment and locate the station held spares and tools:

- Planned maintenance document.
- Equipment specific parts list - hard copy.
- Equipment specific parts list - soft copy.
- RNLI database system.
- Tools - hand tools, special tools, test tools.
- Lubrication - oils, greases, applicators.
- Spares - anodes, filters, gaskets, heat detectors, spark plugs, impellers, consumables.

Training criteria. Demonstrate the correct use of hand tools and power tools as they would be employed for common planned maintenance tasks appropriate to the SAR unit.

Training criteria. Demonstrate how to select the correct job cards for common maintenance tasks and how to correctly complete tasks in accordance with the job card.

#### E Nav system

- **Type.**

Training criteria. Explain how and when to change the system Idents (if fitted):

- Relief fleet.
- Updates.

- **Updating.**

Training criteria. Explain how and when to update the navigational software and chart updates:

- When informed of required navigational changes centrally or locally and when software updates required.

- **Saving.**

Training criteria. Demonstrate downloading data between an external computer and the SAR unit (if fitted):

- Saving workspace to station computer.
- Saving workspace to SAR unit system.

task continues

# Activity 10 - Planned Maintenance

## Task 10.1.1

## Maintain and Service Equipment

### c Identify and record defects and equipment shortfalls

#### Training guidelines

##### Defects

- **Identifying defective equipment.**

Training criteria. Explain the ways of identifying if a piece of equipment is defective:

- Visual - damage, leaks, discoloration.
- Function - vibration, noise, movement, exhaust discharge.
- Performance - idle speed, maximum engine speed, power, gauge readings.

##### Record defects and deficiencies

- **Recording defects and deficiencies.**

Training criteria. Explain the system in place for the reporting of defects and any deficiencies:

- Machinery log.
- Station defect log.
- Defect notification on database system.
- Report to regional base - station technical management.
- Report to COIR.

- **Ordering stores against a defect.**

Training criteria. Explain the system in place to order stores against a defect:

- Request via regional base - station technical management.
- Request via duty technical manager (available via operations room).
- Direct stores order through database system.

- **Recording completed defects.**

Training criteria. Explain the system for recording a defect on equipment that has been cleared:

- Station defect log.
- Defect notification on database system.
- Report to regional base.



## Activity 15 - The Media

Photo: Nicholas Leach



## Tasks

|              |                                    |
|--------------|------------------------------------|
| <b>15.1</b>  | <b>Media Awareness</b>             |
| 15.1.1       | Media Awareness                    |
| <b>15.2</b>  | <b>Camera Operations</b>           |
| 15.2.1       | Camera Operations                  |
| <b>15.3</b>  | <b>Understanding the Media</b>     |
| 15.3.1       | Understanding the Media            |
| <b>15.4</b>  | <b>Key Messages</b>                |
| 15.4.1       | Key Messages                       |
| <b>15.5</b>  | <b>Press Releases</b>              |
| 15.5.1       | Press Releases                     |
| <b>15.6</b>  | <b>Incident Handling</b>           |
| 15.6.1       | Incident Handling                  |
| <b>15.7</b>  | <b>Media Interview Logistics</b>   |
| 15.7.1       | Media Interview Logistics          |
| <b>15.8</b>  | <b>Basic Photography and Video</b> |
| 15.8.1       | Basic Photography and Video        |
| <b>15.9</b>  | <b>Interview Skills</b>            |
| 15.9.1       | Interview Skills                   |
| <b>15.10</b> | <b>Video Production</b>            |
| 15.10.1      | Video Production                   |
| <b>15.11</b> | <b>Social Media</b>                |
| 15.11.1      | Social Media                       |

# Activity 15 - The Media

## Task 15.1.1

## Media Awareness

### a Base knowledge

#### Assessment guidelines

#### Media awareness

- **News stories.**

Assessment criteria. Identify potential positive and negative stories:

- Good news stories including interesting rescues, human interest stories, landmarks or records, celebrity involvement and major fundraising achievements.
- Bad news stories including rescues gone wrong, health and safety incidents, public criticism of the RNLI.

- **Interacting.**

Assessment criteria. Describe the procedure of interacting with the media:

- Only those members of staff or volunteers authorised to speak to the media as part of their role should do so.
- All other staff or volunteers should contact their line manager; regional media manager (RMM) / regional media officer (RMO) / lifeboat press officer (LPO) or duty press officer before interacting with the media.
- Ensure you have support and guidance from a member of the communications team or LPO before media interaction / interviews.

- **Reputation.**

Assessment criteria. Describe RNLI reputation:

- Strong.
- Lifesaving organisation.
- Volunteer based.
- Well-respected charity.
- Historical.

Assessment criteria. Describe risks to reputation:

- Unprofessional behaviour.
- High risk.
- Judgemental / non caring.
- Mis-use of funds.
- Follow procedure to avoid these risks.

task continues



## Task 15.1.1

## Media Awareness

### a Base knowledge (continued)

#### Assessment guidelines

#### Media tools

##### • Recording.

Assessment criteria. Describe the importance of recording footage:

- Important tool for showing what we do.
- Effective fundraising tool.
- Internal training and reviewing tool.
- Helps achieve media coverage and so raises awareness of the RNLI.

Assessment criteria. Describe the sensitivity required when recording footage:

- Granting of permission where appropriate.
- Personal property.
- Do not pass direct to the media.
- Individuals in distress.
- Must be uploaded to the Source, (the RNLI's video library) and moderated by member of the media team.
- Must meet GDPR requirements, including download footage from the camera's memory card onto the RNLI station PC / laptop / RNLI encrypted external hard drive.
- Understand the informed consent process.

##### • Social media.

Assessment criteria. Explain the use of social media in the RNLI:

- Advantages:
  - raises profile
  - free
  - direct channel of interactive communication to supporters and public.
- Disadvantages:
  - open, seen by anybody
  - demanding on time to manage
  - direct channel of interactive communication to supporters and public
  - risk to reputation applies.

#### Key messages

##### • RNLI key messages.

Assessment criteria. Identify RNLI key messages:

- Volunteers; charity; RNLI; saves lives at sea (VCRS).
- Supporting messages as specified in Loud and Clear.

## Task 15.2.1

## Camera Operations

### a Camera operations

#### Assessment guidelines

#### Camera operation

- **Operating the camera.**

Assessment criteria. Demonstrate how to operate the relevant camera:

- Power to camera:
  - ILB camera and ALB camera
  - helmet camera
  - record unit must be put on charge after use and left on trickle charge.
- Inserting memory card:
  - remove and install the memory card into protective case for all cameras except cameras using micro SD cards.
- Making a recording:
  - location of record button
  - how much footage does an 8GB card hold
  - each time record button is pressed a file is generated on the card.

#### Camera maintenance

- **Maintaining the camera.**

Assessment criteria. Explain the post-recovery maintenance procedure:

- Refer to RNLI *Camera Planned Maintenance Document*.

## Task 15.3.1

## Understanding the Media

### a LPO roles and responsibilities

#### Assessment guidelines

#### LPO role

##### • Roles and responsibilities.

Assessment criteria. Describe the LPO role:

- To raise awareness of the RNLI by publicising rescues and other operational activity.
- Main contact for local media enquiring about the RNLI.
- Advise and inform station management and wider crew about media issues.
- Guardian of the RNLI's reputation.

Assessment criteria. Explain the LPO's position at the lifeboat station and within the RNLI media team:

- Member of the lifeboat management group (LMG).
- Works closely with the LOM and other members of the LMG.
- Key part of RNLI media team – staff and volunteers.
- Liaison between the station and RMM / RMO / Poole media team.

Assessment criteria. Describe the LPO's main responsibilities:

- As stated in the current LPO volunteer role profile.

task continues

## Task 15.3.1

## Understanding the Media

### b Understanding the media

#### Assessment guidelines

#### Understanding the media

##### • Different types of media.

Assessment criteria. Describe different types of media:

- AREA COVERAGE: local / regional / national.
- FOR PRINT: newspapers / magazines / journals.
- FOR BROADCAST: radio / television news / documentaries / film company requests.
- INTERNET: websites / blogs / social media.

Assessment criteria. Explain the relevance, benefits and pitfalls of social media:

- Examples of social media to include: Facebook, Twitter (or 'X'), other relevant social networks.
- Benefits:
  - cost-effective (usually free) way to raise awareness
  - requires little resource
  - effective substitute for a website
  - journalists use it as a news gathering source
  - supporters can view then pass on RNLI news and content to their own social networks, potentially reaching a very large audience
  - receiving news or a personal recommendation about the RNLI from a friend or family member can be extremely compelling.
- Pitfalls:
  - risk to reputation if volunteers / staff share inappropriate, sensitive or offensive content (photographs, video and text)
  - public and very immediate forum for grievances to be aired, either by the public or by volunteers
  - complaints and other comments need to be dealt with promptly and professionally
  - not always possible to control or delete content.

Assessment criteria. Identify which type of media to target for major / minor incidents:

- Examples to include:
  - television or radio for major incidents
  - local press for minor rescues
  - when a story may have national potential / specialist media interest.

##### • Requirements of the media.

Assessment criteria. Explain what the media requires from a Lifeboat Press Officer (LPO):

- As a first point of contact for local RNLI news.
- For a timely, accurate news release with pictures / video when possible.
- As an interviewee.
- For dramatic, quirky news stories, human interest stories, opportunity to go afloat, features.
- Availability of news release author to respond to requests.

Assessment criteria. State the most appropriate way to contact the media:

- Email as soon as realistically possible.
- With a phone call follow up to the email if applicable.
- The importance of building personal contact with journalists.

Assessment criteria. Explain relevant media deadlines:

- Summary of deadlines for local media.
- Rolling nature of broadcast media.
- Long lead times for feature publications.

## Task 15.4.1

## Key Messages

### a Understanding and using RNLI key messages

#### Assessment guidelines

#### Effective key messaging

##### • RNLI key messages.

Assessment criteria. Explain the importance of key messages:

- As important pieces of information we want everyone to know and understand about the RNLI.
- To help convey the RNLI's charity and operational messages clearly and succinctly.
- To give consistency to RNLI communications.
- To be integral to all RNLI communications.

Assessment criteria. State where key messages are found:

- Loud and Clear.
- RNLI website.
- *Media Skills Handbook*.
- Communications from RMO / Poole media team.

Assessment criteria. Identify RNLI key messages:

- Volunteers; charity; RNLI; saves lives at sea (VCRS).
- Supporting messages as specified in Loud and Clear.

##### • Communicating RNLI key messages.

Assessment criteria. Explain how to use RNLI key messages:

- By including in all press releases, interviews, conversations, communications.
- By inserting naturally into flowing sentences so that key messaging does not seem forced.
- By encouraging everyone giving media interviews to use key messages supported by suitable examples.

Assessment criteria. Demonstrate the use of RNLI key messages in verbal and written communications:

- In a written press release and in social media posts.

## Task 15.5.1

## Press Releases

### a Writing a news release

#### Assessment guidelines

#### Gathering information

- **Information gathering for a news release.**

Assessment criteria. Identify circumstances when a news release should be issued:

- As soon as possible after a relevant rescue.
- Distinguish which shouts should not lead to news release, for example:
  - suicides
  - sensitive subject
  - crime
  - false alarm.
- Other subjects to write a news release about include:
  - training
  - new crew
  - awards
  - fund raising event.

Assessment criteria. State how details of an incident are sourced:

- Following pager, call to LOM / station.
- Reference to coxswain / helm's report prior to service return.
- Reference to service return as a basis for information following its submission.
- Post incident interview with coxswain / helm / crew.

Assessment criteria. State essential information required to write a rescue news release:

- Basic info - 5Ws+H.
- Other information such as weather / sea conditions.
- Quote from relevant person, including the key message.

Assessment criteria. State information which would not be included in a rescue news release:

- Name of rescuee / casualty – unless permission given.
- Criticism of rescuee or third party.
- Acronyms, technical language, RNLI jargon.

#### Writing

- **Writing a news release.**

Assessment criteria. Explain key points of news release writing:

- Headline, intro, main text, quote, conclusion, contact details, photo caption, notes to editors.
- Keep it short and simple; stick to facts.
- House style / key messages.
- Check spelling and grammar - 4 eyes rule.

Assessment criteria. Demonstrate ability to write a news release:

- By showing examples of their recently written press releases.
- By writing an appropriate press release from details of fictional shout.

task continues

## Task 15.5.1

## Press Releases

### b Issuing a news release

#### Assessment guidelines

##### Distribution

- **Distributing a news release.**

Assessment criteria. State the benefits of using the RNLI news centre:

- Great media resource.
- Immediacy.
- Standardised output.
- Widespread circulation.
- Targeting supporters.

Assessment criteria. Demonstrate uploading news to the RNLI news centre:

- In accordance with the current SOP.

Assessment criteria. Explain how a news release should be issued:

- Emailed to relevant contacts, with a link to the relevant news releases from the RNLI news centre.
- Group email addresses.

##### Audience

- **Audience for a news release.**

Assessment criteria. Explain where to send a news release:

- Relevant local and regional media.
- Internal RNLI contacts, to include:
  - RMO.

## Task 15.6.1

## Incident Handling

### a Dealing with major incidents

#### Assessment guidelines

#### Lines of communication

##### • Establishing communication with the media.

Assessment criteria. Identify the type of incidents the duty press officer / RMO / RMM should be alerted to:

- Examples include:
  - large passenger vessel in trouble
  - multi-casualty lifeboat rescue
  - rescue in extreme weather
  - animal rescue
  - fatality (not suicide).
  - multi-lifeboat rescue / search
  - aircraft crash
  - celebrity rescuee
  - unusual rescue

Assessment criteria. Explain how communications would be co-ordinated for an incident of wide media-interest:

- RMO / RMM / duty press officer to co-ordinate communications.
- They will agree plan of action with all involved, including LPO and LOM (or delegated manager) where required.
- All parties should stay in regular contact.
- Agree who will write news release; source photographs / video; contact media; respond to media requests; carry out interviews; brief interviewees.

Assessment criteria. Describe how PR for multi-lifeboat rescues and joint lifeguard / lifeboat rescues should be handled:

- For multi-lifeboat rescues:
  - RMO / RMM / duty press officer should be informed and relevant LPOs should stay in touch
  - one release to be issued
  - course of action to be agreed as above.
- For joint lifeguard / lifeboat rescues:
  - PRM / PO / duty press officer should be informed, and agreement reached about who writes and issues release and deals with media enquiries.

#### Media handling

##### • Dealing with the media.

Assessment criteria. Explain how to satisfy the demands of the media during a major incident:

- Issue a news release and where possible photographs and video as soon as possible, with follow up releases as the incident progresses if appropriate.
- Respond to enquiries and requests for interviews as soon as possible, calling on support from RMO / RMM / duty press officer when appropriate and by following course of action previously described.
- If more than one person is dealing with media enquiries, they must stay in regular contact.
- If multiple requests for interviews are received, consider briefing more than one person to carry out interviews – in consultation with RMO / RMM / duty press officer.

Assessment criteria. Describe how to deal with media at a lifeboat station during a sensitive incident, such as a fatality:

- Keep media outside the lifeboat station.
- Ensure LPO discusses what they want and agree on a course of action.
- Offer a spokesperson for interview, which may satisfy them and encourage them to leave.



## Task 15.6.1

## Incident Handling

### b Reputation management

#### Assessment guidelines

#### RNLI reputation

- **Protecting the RNLI's reputation.**

Assessment criteria. Identify incidents / situations when the reputation of the RNLI could be put at risk:

- Operational situations, for example:
  - injuries / death of crew
  - crew missing during rescue / exercise
  - lifeboat breakdown / damaged / capsize
  - rescue that goes wrong
  - injury to member of public at station or during RNLI event.
- Non-operational situations, for example:
  - volunteer accused of criminal activity
  - disputes within a crew / branch
  - financial impropriety
  - inappropriate discussions / photographs / video posted on social media.

Assessment criteria. State the course of action to take to protect the reputation of the RNLI:

- Alert RMO / RMM / duty press officer, who would coordinate appropriate action and response.
- Keep in regular contact with LOM / RMO.
- Monitor traditional and social media.
- Stress importance of confidentiality to crew and other volunteers who are aware.

# Activity 15 - The Media

## Task 15.71

## Media Interview Logistics

### a Managing Interview Requests

#### Assessment guidelines

#### Interview formats

- **Different types of media interview.**

Assessment criteria. Describe the different types of media interview and the correct response to them:

- Press.
- Radio.
- Television.
- Internet.
- Over the phone.
- Face to face.
- Live.
- Pre-recorded.
- Response:
  - show understanding of deadlines
  - appreciate sometimes better to provide a written quote / unnecessary to provide answer instantly.

Assessment criteria. State which interview requests should be referred to the RMO / RMM / duty press officer:

- Any national request.
- TV request; controversial issues, such as MCA; political; RNLI policy; sensitive.
- Anything LPO has concerns about.

- **Negotiating an interview request.**

Assessment criteria. Explain how to negotiate with a journalist to agree beneficial interview arrangements:

- Discuss timings, location to ensure mutually beneficial. Don't be pushed into following their deadlines if it doesn't suit.
- Explain crew are volunteers so they are not always available 9-5.
- Negotiate news angle to the best advantage of the RNLI, for example: ask to include charity messaging in rescue / operations coverage.
- Do not be afraid to explain importance of this to journalists.

task continues

## Task 15.71

## Media Interview Logistics

### b Media Interview Logistics

#### Assessment guidelines

##### Interviewee selection

- **Selecting an interviewee.**

Assessment Criteria: State how an interviewee should be chosen:

- To be a suitable spokesperson because they are:
  - a subject expert
  - an eyewitness
  - a station representative.
- Consideration as to the interviewee's media experience; willingness; confidence; availability.
- Endeavour to consult LOM.

- **Briefing an interviewee.**

Assessment Criteria: Explain how to brief an interviewee:

- Share arrangements of interview, discuss what key points and messages should be included.
- Suggest the interviewee rehearses key points and help them with this if appropriate.
- Discuss possible stumbling blocks / difficult questions / what not to say.
- Ensure the interviewee is dressed appropriately in operational kit or branded clothing.

## Task 15.8.1

## Basic Photography and Video

### a Photography

#### Assessment guidelines

#### Basic photography

- **Different types of media.**

Assessment criteria. Explain why images are so important to secure quality coverage:

- Meeting media demands - colour, drama, action, human interest.
- Helps tell the story and conveys role played by RNLI.
- Gives the RNLI an edge in terms of media coverage.
- Maximises branding / portrays the RNLI in a positive light / draws attention to our work, increasing propensity to give.

Assessment criteria. Describe what makes a good RNLI photograph:

- Drama / colour / scale / summing up a story / people.
- Branding.
- Sharp, well-focussed.
- Filling the frame.

- **Producing a good photograph.**

Assessment criteria. Identify points to consider when taking a photograph:

- Keep camera steady.
- Fill the frame.
- Background - avoid busy, distracting backdrop.
- Take it outside for best light.
- Keep horizon level.
- Place the subject slightly off centre.
- Avoid large groups of people.

Assessment criteria. Identify how photos can be improved by photo editing software:

- Depends on software available but should be familiar with auto-correct / enhance functions - cropping, resizing, red-eye removal.
- Additionally straightening horizon; removing blemishes – (do not over-fix and alter the image to distort the truth).

task continues

## Task 15.8.1

## Basic Photography and Video

### b Video

#### Assessment guidelines

#### Video

##### • Recognising the impact of film.

Assessment criteria. Explain how video can have an impact on media coverage:

- More likely to get TV coverage with footage of a rescue - TV demands moving pictures. Also vital for social media and used on growing number of websites.
- Also use in documentaries, factual series etc. All crucial to raising awareness and propensity to give.

Assessment criteria. Describe what should be considered during the editing process:

- Aim for 1-2 minute clip which encapsulates the rescue. A few long shots rather than several short ones.
- Include shots which show the lifeboat or / and the crew, do not only concentrate on shots of the casualty.
- Try to include branding.
- Check everything operationally correct. Get edit checked by coxswain, mechanic or helm before uploading to video library.
- Check audio for bad language etc.
- Check all rescuees clearly identified have given permission to use the video.

##### • Uploading procedure and distribution.

Assessment criteria. Explain the process involved in ensuring the video appears on the RNLI press centre:

- Upload to RNLI video library via station log on. Logo and title screen added during this process.
- Check spelling on title template as spell check is not available.
- Communications team alerted via email that video has been uploaded.
- Communications team watch and moderate video to make it visible on the RNLI news centre.
- Links are attached to our social media pages: Facebook and Youtube.
- If footage is required quickly or out-of-hours (OOH) by the media, call the RMO or duty PO to ensure it is uploaded to the news centre asap.

Assessment criteria. Describe how video is distributed and made available to the media:

- Media can download from the website.
- A video can be embedded in the press release on the news centre, and a link to the press release sent to journalists.

task continues

## Task 15.8.1

## Basic Photography and Video

c

### Ethics and consent

#### Assessment guidelines

#### Ethics and consent

- **Consenting to image use.**

Assessment criteria. State when consent would be required before issuing a photograph or video:

- If a rescuee is clearly identifiable.
- If children are pictured - parent's permission required.
- Written consent is required.
- If it is not possible to obtain consent, close-up images that identify the casualty can be edited out of the photograph / video.
- Copyright - who took the photo? If not an RNLI photographer, do we know who took it and can we use it with a credit to the photographer?
- Is there a restriction on using the photos more widely in the RNLI (for example, advertorial, marketing)?

Assessment criteria. Describe when a photograph / video should not be used:

- If permission has not been granted by rescuee who is clearly identifiable or by parents of a child photographed / videoed.
- If the photograph / video illustrates grief; death; pain, is indecent in any way or potentially brings the RNLI into disrepute.
- Anything that shows incorrect operational practices.
- Breaches with the GDPR.

- **Using images on social media:**

Assessment criteria. Describe the positive and negative impact of sharing photographs and video on social media sites:

- YouTube, Facebook, Twitter (or 'X') et al can be great ways to share photographs / videos, engage with supporters and raise awareness.
- All videos shared must have been uploaded to The Source and moderated.
- Guidelines described above must be followed. Always assume ANYONE can see what is posted - not just 'friends'.
- Risk to reputation posed by posting photographs / videos that are indecent; show incorrect operational procedures; are in poor taste etc.
- Anything on crew's / station's own personal social media pages can potentially be used by media.

## Task 15.9.1

## Interview Skills

### a Preparing and conducting an interview

#### Assessment guidelines

#### Interview preparation

##### • Preparing for an interview.

Assessment criteria. Describe how to prepare for a media interview:

- Ensure the brief / subject / angle the journalist is taking is fully understood.
- Think about what needs to be said in the interview and how to incorporate key messages supported with relevant examples.
- Research information where appropriate.
- Rehearse key points which need to be communicated.
- Consider possible difficult questions and how to work around them.
- Ensure appropriate clothing is worn during the interview.

Assessment criteria. Explain what helps make a successful interview:

- Delivery of key messages and key facts are eloquently conveyed.
- Sticking to the facts - do not express personal opinion.
- Avoiding jargon or technical language.
- Talking slowly and clearly.
- Smiling if appropriate - even on radio!
- Well branded images / good visual broadcast piece.
- Being seen as a positive ambassador for the RNLI.

#### Interview techniques

##### • Interview techniques.

Assessment criteria. Demonstrate ability to carry out a proficient broadcast interview:

- With evidence of previous broadcast interview, if available.
- In the form of a mock interview.

## Task 15.10.1

## Video Production

### a Video editing

#### Assessment guidelines

##### Footage transfer

- **Transferring the footage.**

Assessment criteria. Demonstrate how to transfer footage from the camera to the PC:

- Insert memory card into card reader / PC.
- Locate recordings from card in 'MY COMPUTER > correct drive'
- Copy and paste the files into media folder on the PC.
- Delete files from card, format card and return to camera.

##### Video creation

- **Editing a video.**

Assessment criteria. State key points to consider when editing RNLI video footage:

- Creation of a 1-2 minute edit that illustrates the point of rescue.
- Pictures that best represent the RNLI - branding, shots of the RNLI in action.
- Removal of any inappropriate audio or swearing.
- Only the inclusion of shots that are operationally correct.

Assessment criteria. Demonstrate use of editing software:

- 'Capture', 'Edit' and 'Make Movie' tabs.
- Location of recently saved recordings via 'Show Videos'. Select 'Files' at top of page.
- The timeline.
- Edit footage using the 'Razor Blade' and 'Dustbin' icons.
- Removal of inappropriate audio or swearing.
- Watch edit with volume increased to check for mistakes.
- 'Make Movie' and create edit.
- Save edit to correct folder on PC.
- Watch completed edit to recheck for glitches / mistakes before uploading to the Source (RNLI's video library) for moderation.
- Check operational content with coxswain, mechanic, helm, LOM, lifeguard manager etc.

Assessment criteria. Demonstrate understanding of filing system on PC to store edited and unedited videos:

- Video files kept in one folder system called 'MEDIA' on the 'C:Drive'.
- Each of the folders contains the following videos:
  - Edited videos - a library of all the short edits that have been created.
  - Services - raw unedited rescue footage.
  - Exercises - raw unedited exercise footage.
  - Frames - screengrabs.

task continues



## Task 15.10.1

## Video Production

### b Video distribution

#### Assessment guidelines

#### Video usage

- **Video upload procedure and distribution.**

Assessment criteria. Demonstrate how to upload video to the Source:

- Use Internet browser to access the Source via RNLI2.
- Station Login: station email address (e.g. poole@rnli.org.uk).
- Password: 'lifeboat£' (unless changed by the station).
- For upload user guide refer to *The Source How To Guide*.

Assessment criteria. Explain the process involved in ensuring the video appears on the RNLI press centre:

- Upload to the Source.
- Alert the communications team via email or phone that video has been uploaded.
- Communications team watch and moderate video to make it visible on the RNLI press centre.
- Links are attached to our social media pages: Facebook and Youtube.
- If footage is required quickly or outside office hours by the media, call the newsdesk team, the RMO or RMM to ensure it is uploaded to the press centre asap.

Assessment criteria. Describe how video is distributed and made available to the media:

- Media can download from the website.
- A link can be emailed as part of a press release (preferable) by the LPO or separately if the release has already been issued.

task continues

## Task 15.10.1

## Video Production

### c

### Ethics and consent

#### Assessment guidelines

#### Ethics and consent

- **Consenting to video use.**

Assessment criteria. State when consent would be required before issuing a video:

- GDPR restrictions if a rescuee is clearly identifiable.
- If children are pictured - parent's written permission required.
- Written consent is required for people who might be recognised in our videos and / or photos.
- If it is not possible to obtain consent, close-up images that identify the casualty can be edited out of the video.
- Copyright - who took the video? If not an RNLI filmmaker, do we know who took it and can we use it with a credit to the filmmaker?
- Is there a restriction on using the video more widely in the RNLI (for example, advertorial, marketing)?

Assessment criteria. Describe when a video should not be used:

- If permission has not been granted by rescuee who is clearly identifiable or by parents of a child videoed.
- If the video illustrates grief / death / pain, is indecent in any way or potentially brings the RNLI into disrepute.
- Anything that shows incorrect operational practices.

- **Using video on Social Media:**

Assessment criteria. Describe the positive and negative impact of sharing videos on social media sites:

- YouTube, Facebook, Twitter (or 'X') et al can be great ways to share videos, engage with supporters and raise awareness.
- It is better for the RNLI if videos used on these sites comes from the central video library link, so that the total number of hits can be monitored.
- Guidelines described above must be followed. Always assume ANYONE can see what is posted - not just 'friends'.
- Risk to reputation posed by posting videos that are indecent / show incorrect operational procedures / are in poor taste etc.
- Anything on crew's / station's own personal social media pages can potentially be used by media.

## Task 15.11.1

## Social Media

### a Social media

#### Assessment guidelines

#### Social media

##### • Understanding social media.

Assessment criteria. Explain the relevance of social media:

- Show a basic understanding of social media, Facebook, Twitter (or 'X') and other relevant social media networks - eg YouTube, Instagram, Snapchat.
- Facebook - community / sharing information pictures and video.
- Twitter (or 'X') - breaking news / brief / building relationships with local media.
- Identify the person responsible for station social media activity.

Assessment criteria. Explain the benefits / pitfalls:

- Benefits - free / relatively simple / widespread / raises profile / direct channel of interactive communication to supporters and public.
- Pitfalls - open / seen by anybody / demanding on time to manage / direct channel of interactive communication to supporters and public / risk to reputation applies.

##### • Managing reputation.

Assessment criteria. Describe how a negative post could pose a reputational risk for the RNLI:

- Show an understanding that social media is not private and can be used by the media.
- Repercussions of inappropriate images on social media - offensive / operationally incorrect / inappropriate.
- Recognise if and when a post should be ignored or deleted and recognise possible repercussions.
- Reply and correct incorrect posts.

Assessment criteria. Explain possible responses to a negative post:

- Deal with complaints promptly and professionally. Consult RMO / duty PO if unsure. Confirm facts, acknowledge feelings, offer a solution / take action.
- Avoid defensive language.
- Third party posts should only be deleted as last resort. Can appear to be censoring. Attempt to respond and turn it to our advantage by explaining facts.
- Delete if offensive / abusive or inappropriate / insensitive nature – particularly if potentially libellous or posted by RNLI personnel.

Training Criteria Describe the differences between personal profiles and RNLI profile:

- Personal profile - has friends, even with privacy settings in place, never completely private, shouldn't share confidential / sensitive RNLI info. Refer to social media guidelines.
- Station page - has 'likes'. Represents the station and wider RNLI content - link to news releases, moderated video. Safety messages, station news etc. need to be managed by agreed station reps.

task continues

# Activity 15 - The Media

## Task 15.11.1

## Social Media

### a Social media (continued)

#### Assessment guidelines

#### Using social media

- **Ability to using social media.**

Assessment criteria. Demonstrate how to compose a social media post:

- Clear, concise, avoid jargon. Make it compelling. Include link to website.
- Show an awareness of hash tags and using @ to directly target someone.

Assessment criteria. State possible activity on station which would make suitable social material:

- Discuss some of the local RNLI stories which would be of interest socially.

Assessment criteria. Demonstrate how to link to a News Centre release on social media:

- Link to an LPO news release / video on social media.

Assessment criteria. Demonstrate how to physically remove a social media post:

- Edit option on Facebook - Hide / delete.
- Delete posts - but reposts and replies still visible. Undo repost / favourite.

#### Ethical obligations

- **Understanding when social media is not advisable.**

Assessment criteria. Demonstrate an understanding of the RNLI *Social Media Operations Circular*:

- Show knowledge of the operations circular which requires crew not to use social media during an incident.

Assessment criteria. Explain why it is inappropriate to 'live post' during incidents:

- Don't know the outcome of the rescue until it's over - may not want to publicise.
- Something could go wrong. Friends / family of rescuee may not be aware.

Assessment criteria. Describe other potential incidents which should not be highlighted on social media:

- Inappropriate to post about body recovery; suicide.
- Don't criticise or ridicule rescuees, other agencies.
- Take care not to inadvertently name or identify casualties.
- Human suffering and tragedy; we don't want to use someone's grief, pain or embarrassment for PR.
- Don't enter political debate.



Activity 19 - Qualifications for Audit

Photo: RNLI\David Edwards



### Tasks

|             |                               |
|-------------|-------------------------------|
| <b>19.3</b> | External Medical Certificates |
|-------------|-------------------------------|

|               |             |
|---------------|-------------|
| <b>19.3.1</b> | ML5 Medical |
|---------------|-------------|

|               |              |
|---------------|--------------|
| <b>19.3.2</b> | ENG1 Medical |
|---------------|--------------|

|             |             |
|-------------|-------------|
| Task 19.3.1 | ML5 Medical |
|-------------|-------------|

|   |                |
|---|----------------|
| a | Qualifications |
|---|----------------|

Training guidelines

ML5 medical

- **Possess an ML5 medical certificate.**  
Training criteria. Demonstrate possession of an original ML5 medical certificate:
  - Original certificate to be sighted.
  - Record on database.
  - 5 year validity.



### Task 19.3.2

### ENG1 Medical

a

#### Qualifications

##### Training guidelines

#### ENG1 medical

- **Possess an ENG1 medical certificate.**

Training criteria. Demonstrate possession of an original ENG1 medical certificate:

- Original certificate to be sighted, confirmed in date and a copy taken.
- Record on database.
- 2 year validity.

## Activity 30 - Training, Assessment and Verification

Photo: Tracie Tabor



### Tasks

|             |                         |
|-------------|-------------------------|
| <b>30.1</b> | <b>RNLI Trainer</b>     |
| 30.1.1      | RNLI Trainer Theory     |
| 30.1.2      | RNLI Trainer Practical  |
| <b>30.2</b> | <b>RNLI Assessor</b>    |
| 30.2.1      | RNLI Assessor Theory    |
| 30.2.2      | RNLI Assessor Practical |

# Activity 30 - Training, Assessment and Verification

## Task 30.1.1

## RNLI Trainer Theory

### a Learning Theory

#### Training guidelines

##### RNLI trainer

##### • Roles and responsibilities

Training criteria. Identify the attributes of the competent trainer:

- Personable.
- Good rapport.
- Knowledgeable.
- Encourages appeal.
- Skilled.
- Organised.
- Vigilant.
- Neutral (not biased).
- Constructive.
- Good administrative skills.
- Engaging.
- Empathetic.
- Safe.
- Give feedback.
- Professional.
- Good listener.
- Objective.
- Open.
- Pro-active.

Training criteria. State the role of the assessor:

- Key in the role of quality assurance in the competence framework system.
- Confirm competence and standardisation.

Training criteria. State the role of the verifier:

- Key in the role of quality assurance in the competence framework system.
- Confirm the training, assessment process and policy requirement is met.

Training criteria. Describe the purpose of training:

- Enable learning through direction, structure and support.
- Create a safe environment.
- Ensure the candidate and organisational needs and demands are met.

##### Defining learning

##### • What is learning?

Training criteria. Identify the components that formulate learning:

- Knowledge.
- Skills.
- Attitudes.
- Formal and informal learning.

task continues

## Task 30.1.1

## RNLI Trainer Theory

### a Learning Theory (continued)

#### Training guidelines

##### Learning styles

- **Different learning styles.**

Training criteria. Describe the importance of understanding peoples different learning styles:

- Helps tailor the delivery of information to varying candidates.
- Allows an adaptive and flexible approach.

Training criteria. Explain the different learning styles:

- Activist.
- Theorist.
- Pragmatist.
- Reflector.

Training criteria. Describe your own learning style and that of your candidates:

- In accordance with results from L&OD approved learning style assessment.

##### Identifying training needs

- **Identification and analysis of skills and knowledge gaps.**

Training criteria. Identify reasons why training needs may be identified:

- Learning new skills.
- Succession planning.
- Ascertain return on investment.
- Poor work performance.
- Expanding or diversifying into new areas.
- Introduction of new equipment or procedures.
- Legislative.

Training criteria. Describe conducting a training needs analysis:

- In accordance with current L&OD 'Training Needs Analysis' model.

##### Training outcomes

- **Desired outcomes of training.**

Training criteria. Describe how to write training outcomes:

- Training outcomes / schemes of work are clearly specified in terms of:
  - PERFORMANCE: what the learner will be able to do as a result of what has been learned
  - STANDARDS: minimum acceptable performance level student must demonstrate to be considered competent
  - ENVIRONMENT: the condition under which the learning will take place.

task continues

# Activity 30 - Training, Assessment and Verification

## Task 30.1.1

## RNLI Trainer Theory

### b Structure

#### Training guidelines

#### Structure

##### • Session Planning.

Training criteria. State the typical structure of a training session:

- Introduction for example: briefing, scene setting, scope, roles.
- Main event for example: session, activity, task, lecture.
- Conclusion for example: debrief, recap, review, test.
- General guide or 10 - 80 - 10 balance.

Training criteria. Explain the concept and benefit of blended learning:

- As per the current blended learning approach.

Training criteria. Describe how to structure a training session using a lesson plan:

- In accordance with current approved lesson plan template.

#### Delivery

##### • Methods.

Training criteria. Identify a range of delivery methods.

Training criteria. Describe appropriate times to use each method:

- To include:
  - Lecture - teaching 'en masse', one way.
  - Demonstration - practical tasks or process.
  - Discussion - no set answer, structure, open.
  - Role-play / simulation - safe simulation of real life tasks / scenarios.
  - Question and answer sessions - interactive.
  - Scenarios - real life application.
  - Group tasks - team work based process or problem solving.
  - Webinar - distance en masse learning.
  - Brainstorming - problem solving.

Training criteria. Explain effective communication in training:

- Clear message.
- Listening skills.
- Appropriate language.
- Clear language.
- Jargon free.

Training criteria. Describe the importance of being inclusive and diverse:

- Appropriate behaviour, presence for audience.
- Training sessions are as inclusive as possible.
- In accordance with the current Inclusion and diversity policy / statement.

task continues

## Task 30.1.1

## RNLI Trainer Theory

### b Structure (continued)

#### Training guidelines

#### Risk

- **Managing.**

Training criteria. Describe the risk reduction strategies used in training:

- To include the current range of signage and equipment.
- Training session risk assessment process.
- Linked to risk assessments.

Training criteria. Explain how to complete a training session risk assessment:

- Training risk assessment sheet.
- Dynamic risk assessment.

Training criteria. Explain how to action identified mitigating actions:

- Move, change, remove, cancel.
- Re assessment to confirm mitigation and record.

Training criteria. Describe the accident and incident reporting system:

- In accordance with the current RNLI database system.

task continues

# Activity 30 - Training, Assessment and Verification

## Task 30.1.1

## RNLI Trainer Theory

### c Training Standards

#### Training guidelines

#### Competence Framework

- **The structure and contents.**

Training criteria. Identify the key areas of the competence standard:

- Explains activities, tasks, skill base, knowledge base, training criteria and training guidelines.

Training criteria. Identify training criteria:

- To include current training criteria and training guidelines appropriate to role.

Training criteria. Identify key training and assessment activities:

- To include pass out and revalidation handbook.

#### Compliance

- **Policies.**

Training criteria. State training policies and procedures including health safety and welfare.

Training criteria. State requirements for equality and diversity:

- Training and assessment policy.
- L&OD governance.
- Equality policies.
- Inclusion and diversity.
- Safeguarding.
- Appeals procedure.
- Complaints procedure.

#### Documentation

- **Recording development.**

Training criteria. Explain the purpose of recording learning:

- Individual - a record of their efforts, outcomes, compliance, and competence.
- Organisation - ensures it is evidenced, recorded, auditable, verified, defensible, development.

Training criteria. State the current RNLI data systems:

- Microsoft Dynamics AX.
- SharePoint.
- Course booking.
- Learning Zone.

task continues



## Task 30.1.1

## RNLI Trainer Theory

### d Learning Environment

#### Training guidelines

##### Resources

###### • Learning resources.

Training criteria. Identify the different types of resource / teaching aids used for training:

- Standard Operating Procedures (SOPs).
- Emergency Operating Procedures (EOPs).
- Local Operating Procedures (LOPs).
- Learning Zone eLearning.
- Scenario planning sheets.
- Training risk assessments.
- Administration procedures.
- Handouts.
- Manuals and handbooks.
- Pre prepared training products eg. lesson plans.
- Webinars.
- Videos.

Training criteria. Explain how these can assist in the training event:

- Used as reference material for candidates.
- Used to reinforce and structure content of training.

##### Environment

###### • Consulting.

Training criteria. Describe the need to consult with the department / location prior to training:

- To ensure sufficient cover has been arranged for the candidate.
- To ensure all resources and locations are available.
- For travel and accommodation requirements.
- To ensure appropriate permissions have been granted for resources use if required.

###### • Learning environments.

Training criteria. Describe the variables that create an effective learning environment:

- Preparation.
- Presentation.
- Physical space.
- Timing and pacing of group activities.
- Training room / venue standards.
- Recording information.
- Venue.
- Resources / equipment / teaching aids.
- Evaluation methods.

task continues

# Activity 30 - Training, Assessment and Verification

## Task 30.1.1

## RNLI Trainer Theory

### e Evaluation

#### Training guidelines

##### Feedback

###### • Evaluating.

Training criteria. Explain the methods and structure of feedback:

- Timely.
- Face to face.
- Formal.
- Holistic.
- End of session / section / skill.

Training criteria. Explain the importance of giving feedback:

- Aids candidate development.
- Accelerates learning curve.
- Testing method.
- Confirming outcome.

##### Evaluation

###### • Evaluating.

Training criteria. State the Kirkpatrick's evaluation model:

- Level 1 - 4.

##### Reviewing

###### • Reviewing.

Training criteria. Describe how to review and modify training sessions:

- Modifying in relation to feedback received.
- Modifying in relation to reviews of content and standard changes.

Training criteria. Describe how to review and develop trainer skills:

- Collate, assess and act on feedback from candidates.
- Collate, assess and act on feedback from peers.
- Through verification.
- Standardisation meetings.
- Maintaining continuing professional development (CPD) portfolio.

## Task 30.1.2

## RNLI Trainer Practical

### a Practical Training

#### Training guidelines

##### Delivery

##### • Preparing.

Training criteria. Demonstrate the checks to be made prior to training regarding your equipment, resources and environment:

- All equipment is checked, tested and prepared where appropriate.

##### • Delivery.

Training criteria. Demonstrate delivery of a session in line with the competence framework:

- In accordance with session delivered.

Training criteria. Demonstrate delivery of a session using a delivery method suitable to the subject and candidates:

- Presents information in small chunks.
- Makes session interesting.
- Accounts for differing learning styles.
- Checks for understanding.
- Allows for breaks.

Training criteria. Demonstrate the use of a range of different teaching aids / resources:

- They should be:
  - simple and easy to understand
  - brief and concise
  - able to emphasis essential points
  - appropriate in size
  - clearly visible
  - interesting.

Training criteria. Demonstrate use the of practical experiences to reinforce learning points:

- Uses practical experiences relevant to subject.

Training criteria. Demonstrate good time management throughout the session:

- Manages time in relation to the prepared training session plan.
- Adaptable to changing conditions and parameters.

##### • Managing Candidates.

Training criteria. Demonstrate managing group dynamics:

- Manages group dynamics to ensure involvement from the whole group.

Training criteria. Demonstrate dealing with questions effectively:

- Responds to questions, in a professional manner.
- Answer questions in a direct and thorough manner.

Training criteria. Demonstrate effective delivery of feedback:

- In accordance with the current RNLI feedback model.

# Activity 30 - Training, Assessment and Verification

## Task 30.2.1

## RNLI Assessor Theory

### a Principles

#### Training guidelines

##### RNLI Assessor

###### • Roles and responsibilities.

Training criteria. Identify the attributes of the competent assessor:

- Personable.
- Good rapport.
- Good admin skills.
- Give feedback.
- Safe.
- Organised.
- Vigilant.
- Neutral (not biased).
- Constructive.
- Knowledgeable.
- Engaging.
- Empathetic.
- Encourages appeal.
- Skilled.
- Professional.
- Good listener.
- Objective.
- Open.
- Proactive.

Training criteria. Explain the role of the assessor:

- Key in the role of quality assurance in the competence framework system.
- Confirm competence and standardisation.

Training criteria. Describe the role of the verifier:

- Key in the role of quality assurance in the competence framework system.
- Confirm the training, assessment process and policy requirement is met.

##### Defining Assessment

###### • What is assessment.

Training criteria. Describe the purpose of assessment:

- To check required level of competency has been achieved against a set of pre-determined criteria.

##### Methodology

###### • Assessment types.

Training criteria. Identify different types of assessment:

- Formative assessment.
- Diagnostic assessment.
- Summative assessment.
- Accreditation of prior learning (APL).

Training criteria. Describe the principles of assessment:

- Validity.
- Practical.
- Fairness.
- Reliability.
- Flexibility.
- Equitable.

Training criteria. Describe the advantages and disadvantages of holistic assessment:

- ADVANTAGES: easy to create a real life situation through a scenario, looks at the bigger picture.
- DISADVANTAGES: possible to miss parts if too much is assessed at the same time, large topic for candidate to cover.

task continues

## Task 30.2.1

## RNLI Assessor Theory

a

### Principles (continued)

Training guidelines

#### Competence

- **Defining.**

Training criteria. State the definition of a competent person:

- Someone who can apply the relevant knowledge and / or skills of an occupation or job role in the environment.

task continues

# Activity 30 - Training, Assessment and Verification

## Task 30.2.1

## RNLI Assessor Theory

### b Planning

#### Training guidelines

#### Planning

##### • Assessment plan.

Training criteria. Describe how to plan an assessment:

- Define the tasks to be assessed.
- Work out the steps.
- Develop procedures.
- Consult with others.
- Reflect on your plan.
- Evaluate the process.

##### • Variety of assessment methods.

Training criteria. Describe different ways of collecting evidence:

- Direct evidence.
- Supplementary evidence.
- Indirect evidence.

Training criteria. Describe the different assessment methods used:

- Observation.
- Oral tests / questioning.
- Simulation / role play / case study / scenario.
- Written / online tests.
- Portfolio.

##### • Policies.

Training criteria. Describe legal issues, policies and procedures including health safety and welfare in assessment.

Training criteria. State requirements for equality and diversity:

- Training and assessment policy.
- Equality policies.
- Safeguarding.
- Complaints procedure.
- L&OD governance.
- Inclusion and diversity.
- Appeals procedure.

##### • Consulting.

Training criteria. Describe the need to consult with the department / location prior to assessment:

- To ensure sufficient cover has been arranged for the candidate.
- To ensure all resources and locations are available.
- For travel and accommodation requirements.
- To ensure appropriate permissions have been granted for resources use if required.

#### Documentation

##### • Recording.

Training criteria. Describe the purpose for recording assessments:

- To protect the candidate, trainer, assessor, managers, organisation from litigation.
- To be used for appeal and auditing purposes.
- Evidence of competency.

Training criteria. Describe the current RNLI Data systems:

- Operations forms.
- SharePoint.
- Learning Zone.
- AX.
- Course booking.

task continues

## Task 30.2.1

## RNLI Assessor Theory

### c Structure

#### Training guidelines

##### Briefing

- **Pre assessment briefing.**

The purpose of pre-assessment briefing.

Training criteria. State the reasons why candidates are briefed prior to their assessment:

- So the candidate understands exactly what is expected of them before, during and after the assessment.
- To identify any specific needs of the candidate.
- To put the candidate at ease.
- To explain the purpose, context and process of the assessment.
- To explain legal and ethical considerations where appropriate.
- To explain the potential outcomes of eg. competent, action plan, re-training.
- To ensure the process is conducted safely, appropriately, and in a timely manner.

##### Assessment

- **Structure.**

Training criteria. Explain the structure of an assessment:

- Briefing.
- The assessment of skill / knowledge / behaviour.
- Debriefing including feedback and outcome.
- Further action / next step.
- Recording.

##### De briefing

- **Feedback.**

Training criteria. State the importance of feedback:

- Feedback forms an essential part of effective learning.
- It is vital for candidates to know what they did well and where they need to improve in order to learn from their experiences.

Training criteria. Describe key considerations for giving feedback as assessors:

- Be positive, without raising any false expectations.
- Be precise about gaps in the student's performance.
- Identify whether only certain parts of the assessment need to be repeated and, if so, which parts.
- Suggest strategies to help the student overcome gaps.
- Arrange another opportunity for the student to be assessed (if they are not competent).

task continues

## Task 30.2.1

## RNLI Assessor Theory

### d Evaluation and Documentation

#### Training guidelines

#### Decision making skills

- **Final decision.**

Training criteria. Explain how to reach a final decision on a candidates assessment:

- Weighing the evidence collected at the time of assessment against the training criteria.

- **Analysing reasons for non-competence.**

Training criteria. Describe how to analyse reasons for non-competence:

- Evaluates the evidence against a number of factors:
  - the competence framework
  - the training criteria
  - the skills, knowledge, and behavioural requirements
  - the technical principles of assessment
  - not meeting prerequisite / compliance requirements.

- **Appeal and review processes.**

Training criteria. Explain the appeal and review process:

- In accordance with the current review and appeal process.

#### Evaluation

- **Reviewing assessment.**

Training criteria. Describe how to review and modify assessments:

- Modify in relation to feedback received.
- Modify in relation to reviews of content and standard changes.
- Modify in relation to verification outcomes.

Training criteria. Describe how to review and develop assessor skills:

- Collate, assess and act on feedback from candidates.
- Collate, assess and act on feedback from peers.
- Through verification.
- Standardisation meetings.
- Maintaining CPD portfolio.



## Task 30.2.2

## RNLI Assessor Practical

### a Conducting an Assessment

#### Training guidelines

#### Assessment tools

- **Range of assessment tools available.**

Training criteria. Describe a range of assessment tools available:

- Log books.
- Performance check lists.
- Online tests.
- Written question papers.

- **Collecting evidence.**

Training criteria. Demonstrate observation skills:

- Scanning.
- Focusing.
- Tracking.
- On task / off task.
- Note taking.
- Alternatives to workplace observations.

Training criteria. Describe the range of questioning skills used when assessing:

- Open.
- Closed.
- Specific.
- Leading.
- Probing.
- Management.
- Hypothetical.
- Reflective / clarification.

Training criteria. Describe when the assessment method may need adjusting:

- Language, literacy and numeracy.
- Practical and theoretical sections.
- A change in environment that it is being conducted in.

#### Brief

- **Pre-assessment briefing.**

Training criteria. Demonstrate a pre-assessment briefing:

- Explains assessment clearly.
- Describes what is involved in the assessment.
- Puts candidates at ease.
- Addresses any specific needs of candidate where appropriate.
- Describes what will happen if the candidate fails.
- Gives candidate opportunity to ask questions about the assessment.

task continues

# Activity 30 - Training, Assessment and Verification

## Task 30.2.2

## RNLI Assessor Practical

### a Conducting an Assessment (continued)

#### Training guidelines

##### Conduct assessment

- **Conduct assessment in accordance with RNLI standards and the scope of respective assessments.**

Training criteria. Demonstrate assessment of a candidate in accordance with competence standards:

- Plans evidence gathering activities.
- Organises the assessment (pre-assessment brief to identify and explain the context).
- Gathers the evidence.
- Provides feedback to the student during the assessment.
- Makes the assessment decision.
- Records assessment results.
- Provides feedback on performance to the student.
- Reports on the conduct of the assessment.

##### Debriefing

- **Feedback.**

Training criteria. Demonstrate giving appropriate and timely feedback:

- In accordance with the current RNLI feedback model.



## Activity 39 - Lifeboat Trainer Assessor Qualifications for Audit

Photo: RNLI / Nigel Millard



## Tasks

### 39.1 Powerboat Qualifications

39.1.1 RYA National Powerboat Instructor

39.1.2 RYA Advanced Powerboat Instructor

39.1.3 RYA Powerboat Trainer

39.1.4 RYA Powerboat Examiner

### 39.2 PWC Qualifications

39.2.1 RYA Personal Water Craft (PWC) Instructor

39.2.2 RYA Personal Water Craft (PWC) Trainer

### 39.3 Shore Based Qualifications

39.3.1 RYA Shore Based Instructor

39.3.2 RYA Radar Instructor

39.3.3 RYA Short Range Certificate (SRC) Assessor

### 39.4 Cruising Qualifications

39.4.1 RYA Cruising Instructor - Power

### 39.5 Yachtmaster Qualifications

39.5.1 RYA Yachtmaster Instructor - Power

39.5.3 RYA Yachtmaster Instructor - Ocean

39.5.4 RYA Yachtmaster Examiner - Power

39.5.6 RYA Yachtmaster Examiner - Ocean

## Task 39.1.1

## RYA National Powerboat Instructor

### a Qualifications

#### Training guidelines

#### Powerboat Instructor

- **Possess the RYA National Powerboat Instructor qualification.**

Training criteria. Demonstrate current possession of the RYA national powerboat instructor qualification:

- Equivalent qualifications by similar bodies can be used that are comparable.
- The identified qualification (original) must be sighted by the assessor and confirmed in date.

## Task 39.1.2

## RYA Advanced Powerboat Instructor

### a Qualifications

#### Training guidelines

#### Advanced Powerboat Instructor

- **Possess the RYA advanced powerboat instructor qualification.**

Training criteria. Demonstrate current possession of the RYA advanced powerboat instructor qualification:

- Equivalent qualifications by similar bodies can be used that are comparable.
- The identified qualification (original) must be sighted by the assessor and confirmed in date.

## Activity 39 - Lifeboat Trainer Assessor Qualifications for Audit

### Task 39.1.3

### RYA Powerboat Trainer

#### a Qualifications

##### Training guidelines

##### Powerboat Trainer

- **Possess the RYA powerboat trainer qualification.**

Training criteria. Demonstrate current possession of the RYA powerboat trainer qualification:

- Equivalent qualifications by similar bodies can be used that are comparable.
- The identified qualification (original) must be sighted by the assessor and confirmed in date.



## Task 39.1.4

## RYA Powerboat Examiner

### a Qualifications

#### Training guidelines

#### Powerboat Examiner

- **Possess the RYA powerboat examiner qualification.**

Training criteria. Demonstrate current possession of the RYA powerboat examiner qualification:

- Equivalent qualifications by similar bodies can be used that are comparable.
- The identified qualification (original) must be sighted by the assessor and confirmed in date.

## Task 39.2.1

## RYA Personal Water Craft (PWC) Instructor

a

### Qualifications

#### Training guidelines

#### Personal Water Craft (PWC) Instructor

- **Possess the RYA PWC instructor qualification.**

Training criteria. Demonstrate current possession of the RYA PWC instructor qualification:

- Equivalent qualifications by similar bodies can be used that are comparable.
- The identified qualification (original) must be sighted by the assessor and confirmed in date.

## Task 39.2.2

## RYA Personal Water Craft (PWC) Trainer

### a Qualifications

#### Training guidelines

#### Personal Water Craft (PWC) Trainer

- **Possess the RYA PWC trainer qualification.**

Training criteria. Demonstrate current possession of the RYA PWC trainer qualification:

- Equivalent qualifications by similar bodies can be used that are comparable.
- The identified qualification (original) must be sighted by the assessor and confirmed in date.

## Task 39.3.1

## RYA Shore Based Instructor

### a Qualifications

#### Training guidelines

#### Shore based Instructor

- **Possess the RYA Shore based Instructor qualification.**

Training criteria. Demonstrate current possession of the RYA shore based instructor qualification:

- Equivalent qualifications by similar bodies can be used that are comparable.
- The identified qualification (original) must be sighted by the assessor and confirmed in date.

## Task 39.3.2

## RYA Radar Instructor

a

### Qualifications

#### Training guidelines

#### Radar Instructor

- **Possess the RYA radar instructor qualification.**

Training criteria. Demonstrate current possession of the RYA radar instructor qualification:

- Equivalent qualifications by similar bodies can be used that are comparable.
- The identified qualification (original) must be sighted by the assessor and confirmed in date.

## Task 39.3.3

## RYA Short Range Certificate (SRC) Assessor

### a Qualifications

#### Training guidelines

#### SRC Assessor

- **Possess the RYA SRC Assessor qualification.**

Training criteria. Demonstrate current possession of the RYA SRC assessor qualification:

- Equivalent qualifications by similar bodies can be used that are comparable.
- The identified qualification (original) must be sighted by the assessor and confirmed in date.

## Task 39.4.1

## RYA Cruising Instructor - Power

### a Qualifications

#### Training guidelines

#### Cruising Instructor - Power

- **Possess the RYA Cruising Instructor - Power qualification.**

Training criteria. Demonstrate current possession of the RYA cruising instructor - power qualification:

- Equivalent qualifications by similar bodies can be used that are comparable.
- The identified qualification (original) must be sighted by the assessor and confirmed in date.

## Task 39.5.1

## RYA Yachtmaster Instructor - Power

a

### Qualifications

#### Training guidelines

#### Yachtmaster Instructor - Power

- **Possess the RYA yachtmaster instructor - power qualification.**

Training criteria. Demonstrate current possession of the RYA yachtmaster instructor - power qualification:

- Equivalent qualifications by similar bodies can be used that are comparable.
- The identified qualification (original) must be sighted by the assessor and confirmed in date.



## Task 39.5.3

## RYA Yachtmaster Instructor - Ocean

a

### Qualifications

#### Training guidelines

#### Yachtmaster Ocean Instructor

- **Possess the RYA yachtmaster ocean instructor qualification.**

Training criteria. Demonstrate current possession of the RYA yachtmaster ocean instructor qualification:

- Equivalent qualifications by similar bodies can be used that are comparable.
- The identified qualification (original) must be sighted by the assessor and confirmed in date.

## Task 39.5.4

## RYA Yachtmaster Examiner - Power

### a Qualifications

#### Training guidelines

#### Yachtmaster Examiner - Power

- **Possess the RYA yachtmaster examiner - power qualification.**

Training criteria. Demonstrate current possession of the RYA yachtmaster examiner - power qualification:

- Equivalent qualifications by similar bodies can be used that are comparable.
- The identified qualification (original) must be sighted by the assessor and confirmed in date.

## Task 39.5.6

## RYA Yachtmaster Examiner - Ocean

a

### Qualifications

#### Training guidelines

#### Yachtmaster Ocean Examiner

- **Possess the RYA yachtmaster ocean examiner qualification.**

Training criteria. Demonstrate current possession of the RYA yachtmaster ocean examiner qualification:

- Equivalent qualifications by similar bodies can be used that are comparable.
- The identified qualification (original) must be sighted by the assessor and confirmed in date.

# Further Reading

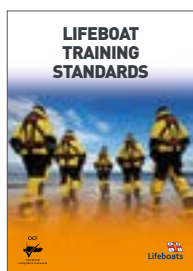
There are four OCF resources available for operational lifeboat crew, including this handbook. These will be provided as printed resources to your station, but can also be accessed via your mobile device.

You are encouraged to become familiar with these standards and other supporting documentation.

Use the camera on your mobile device with the QR codes to access the following resources listed on this page.



**TP-OCF-01**  
**An Introduction to the Operational Competence Framework**



**TP-OCF-02**  
**Lifeboat Training Standards**



**TP-OCF-03**  
**Lifeboat Currency Activity and Training Handbook**



**TP-OCF-04**  
**Lifeboat Pass Out and Revalidation Handbook**

|                 |                                                                                   |
|-----------------|-----------------------------------------------------------------------------------|
| <b>AED:</b>     | Automated external defibrillator                                                  |
| <b>AIS:</b>     | Automatic identification system                                                   |
| <b>ALB:</b>     | All-weather lifeboat                                                              |
| <b>ALM:</b>     | Area lifesaving manager                                                           |
| <b>APL:</b>     | Accreditation of prior learning                                                   |
| <b>ARPA:</b>    | Automatic radar plotting aid                                                      |
| <b>AT:</b>      | Assessor trainer                                                                  |
| <b>ATV:</b>     | All-terrain vehicle                                                               |
| <b>CEP:</b>     | Crew emergency procedures                                                         |
| <b>CEVNI:</b>   | European code for inland waterways                                                |
| <b>COG:</b>     | Course over ground                                                                |
| <b>COIR:</b>    | Central operations and information room                                           |
| <b>COSHH:</b>   | Control of substances hazardous to health                                         |
| <b>CPA:</b>     | Closest point of approach                                                         |
| <b>CPD:</b>     | Continuing professional development                                               |
| <b>CPR:</b>     | Cardiopulmonary resuscitation                                                     |
| <b>DF:</b>      | Direction finder                                                                  |
| <b>DMO:</b>     | Direct mode operation                                                             |
| <b>DR:</b>      | Dead reckoning                                                                    |
| <b>DSC:</b>     | Digital selective calling                                                         |
| <b>E&amp;S:</b> | Engineering and supply                                                            |
| <b>EBL:</b>     | Electronic bearing line                                                           |
| <b>EOP:</b>     | Emergency operating procedure                                                     |
| <b>EPIRB:</b>   | Emergency position indicating radio beacon                                        |
| <b>ETA:</b>     | Estimated time of arrival                                                         |
| <b>GDPR:</b>    | General data protection regulation                                                |
| <b>GOC:</b>     | General operator's certificate                                                    |
| <b>GMC:</b>     | General medical council                                                           |
| <b>GPS:</b>     | Global positioning system                                                         |
| <b>IALA:</b>    | International Association of marine aids to navigation and Lighthouse Authorities |
| <b>HCPC:</b>    | Health and care professions council                                               |
| <b>HMPE:</b>    | High modulus polyethylene                                                         |
| <b>ILB:</b>     | Inshore lifeboat                                                                  |
| <b>IAMSAR:</b>  | International Aeronautical and Maritime Search and Rescue                         |
| <b>IMC:</b>     | Immediate medical care                                                            |
| <b>IRH:</b>     | Inshore rescue hovercraft                                                         |

# Abbreviations

|                  |                                                            |
|------------------|------------------------------------------------------------|
| <b>IRPCS:</b>    | International regulations for preventing collisions at sea |
| <b>ISSI:</b>     | Individual short subscriber identity                       |
| <b>kg(s):</b>    | Kilogramme(s)                                              |
| <b>L&amp;OD:</b> | Learning and organisational development                    |
| <b>LMG:</b>      | Lifeboat management group                                  |
| <b>LOM:</b>      | Lifeboat operations manager                                |
| <b>LOP:</b>      | Local operating procedure                                  |
| <b>LPO:</b>      | Lifeboat press officer                                     |
| <b>LRC:</b>      | Long range certificate                                     |
| <b>LSAR:</b>     | Lifesaving activity reporting                              |
| <b>LSO:</b>      | Lifesaving operations                                      |
| <b>LTA:</b>      | Lifeboat trainer assessor                                  |
| <b>LTC:</b>      | Lifeboat training coordinator                              |
| <b>LVD:</b>      | Launch vehicle driver                                      |
| <b>m:</b>        | Metre                                                      |
| <b>mm:</b>       | Millimetre                                                 |
| <b>MARPA:</b>    | Mini automatic radar plotting aid                          |
| <b>MBD:</b>      | Machinery breakdown drill                                  |
| <b>MCA:</b>      | Maritime and Coastguard Agency                             |
| <b>MOB:</b>      | Man overboard                                              |
| <b>NHS:</b>      | National Health Service                                    |
| <b>OCF:</b>      | Operational competence framework                           |
| <b>OOH:</b>      | Out-of-hours                                               |
| <b>OS:</b>       | Ordnance survey                                            |
| <b>OSC:</b>      | On scene coordinator                                       |
| <b>PDV:</b>      | Power-driven vessel                                        |
| <b>PLB:</b>      | Personal locator beacon                                    |
| <b>PIM:</b>      | Position and intended movement                             |
| <b>PPE:</b>      | Personal protective equipment                              |
| <b>PTT:</b>      | Push to talk                                               |
| <b>PWL:</b>      | Personal water craft                                       |
| <b>RCAMS:</b>    | RNLI callout and messaging system                          |
| <b>RMM:</b>      | Regional media manager                                     |
| <b>RMO:</b>      | Regional media officer                                     |
| <b>RNLI:</b>     | Royal National Lifeboat Institution                        |
| <b>RYA:</b>      | Royal Yachting Association                                 |

|               |                                                                         |
|---------------|-------------------------------------------------------------------------|
| <b>ROC:</b>   | Restricted operator certificate                                         |
| <b>RPM:</b>   | Revolutions per minute                                                  |
| <b>RWC:</b>   | Rescue watercraft                                                       |
| <b>SAR:</b>   | Search and rescue                                                       |
| <b>SART:</b>  | Search and rescue transponder                                           |
| <b>SHE:</b>   | Safety, health and environment                                          |
| <b>SIMS:</b>  | Systems and information management system                               |
| <b>SMEAC:</b> | Situation, mission, execution, administration, command / communications |
| <b>SOG:</b>   | Speed over ground                                                       |
| <b>SOLAS:</b> | Safety of life at sea                                                   |
| <b>SOP:</b>   | Standard operating procedure                                            |
| <b>SRC:</b>   | Short range certificate                                                 |
| <b>SST:</b>   | Safe system of training                                                 |
| <b>STCW:</b>  | Standards of training, certification and watchkeeping                   |
| <b>STD:</b>   | Speed, time, distance                                                   |
| <b>TCPA:</b>  | Time to closest point of approach                                       |
| <b>TMO:</b>   | Trunk mode operation                                                    |
| <b>TRiM:</b>  | Trauma risk management                                                  |
| <b>TSS:</b>   | Traffic separation scheme                                               |
| <b>VHF:</b>   | Very high frequency                                                     |
| <b>VRM:</b>   | Variable range marker                                                   |
| <b>WSO:</b>   | Water safety officer                                                    |
| <b>XTE:</b>   | Cross track error                                                       |

## The RNLI is the charity that saves lives at sea

The Royal National Lifeboat Institution, a charity registered in England and Wales (209603), Scotland (SC037736), the Republic of Ireland (CHY 2678 and 20003326), the Bailiwick of Jersey (14), the Isle of Man (1308 and 006329F), the Bailiwick of Guernsey and Alderney.

## TP-OCF-02

Version: 1.0

Date produced: November 2023

RNLI Classification: Protected

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Department: Learning and Organisational Development

Produced by: Learning Experience